

PEARSON

DEMYSTIFYING NUMBER SYSTEM

(Practical Concepts and Their Applications)

For the CAT and Other MBA Exams

Nishit K. Sinha

Demystifying

NUMBER SYSTEM

(Practical Concepts and their Applications)

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PEARSON

Chandigarh • Delhi • Chennai

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Dedicated to
the Little Angels in my Life
Aadya and Aarna and counting....

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Preface

How to Prepare for Common Admission Test (CAT): Story of Moves

Let me start with the famous story "Garry Kasparov Vs IBM Deep Blue computer":

During 1996–1997, computer Deep Blue developed by IBM played two series of matches with the then world chess champion Garry Kasparov. While the 1st series of 6 matches played in 1996 were won by Garry Kasparov (Wins–3, Draw–2, Loss–1), disapproving the IBM's supremacy claim of machine over man, the second series played in May 1997 was won by IBM's machines. After this loss, Kasparov said that he sometimes saw deep intelligence and creativity in the machine's moves, suggesting that during the second game, human chess players, in violation of the rules, intervened. As expected IBM vehemently denied that it cheated, saying that the only human intervention occurred between games and not during the games. The rules provided for the developers to modify the program between games, an opportunity they said they used to shore up weaknesses in the computer's playing prowess that were revealed during the course of the match.

Cut to the CAT Preparation Now

What are the typical responses/reactions we have when we see a question? Some of those most frequently occurring are listed below:

	Responses/Reactions
1.	Solution is clear end to end
2.	Solution is somewhat clear
3.	Not at all having any idea of the solution

Response/Reaction 1 I know where to start and how to reach to the final solution. So, it is just a matter of few seconds/minutes that I will have the solution.

Response/Reaction 2 Solution is somewhat clear—knowing how to start, but not absolutely sure if it will lead to the end solution.

Response/Reaction 3 Not at all having any idea of the solution—How to approach or proceed is not clear?

You may wonder why I started with Kasparov Vs IBM story and how is it linked to the CAT preparation and the possible reactions. To answer this, I raise another question—Human beings have the thinking capabilities, and so s/he can decide the moves on the basis of the changing situations and so s/he plays chess. How does a computer actually play chess?

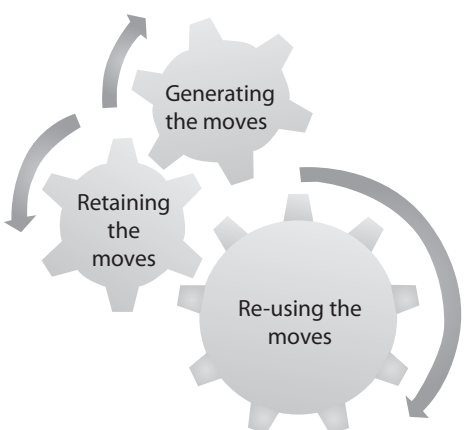
If we go through the Kasparov Vs Deep Blue story in flashback, we get to know that there were some chess grandmasters who were the part of IBM Deep Blue manufacturing team and supplying the most crucial information—the database of moves played across thousands of games. So while these moves created a repository of information inside the 'brain' of computer, a complex modelling was done to decide which counter move is best for what move—something on the lines of decision tree analysis through probabilistic calculations. And here was the need for faster calculation. So, for 'X' move placed by Kasparov, computer used to analyze how many times this move has been put and respective countermoves are having what probability of leading to winning position.

Cut Back to CAT Preparation

This might have answered the most universal query of the students—why are we able to solve a particular question and not been able to solve another question? So its just about the moves – solution to every question involves some moves (loosely speaking, steps are having the same connotation as moves). If we have the moves ready with us from the starting to the end of the solution, we have 1st response/reaction. And so on 2nd or 3rd response that how many moves we have.

Summarizing, we can solve a question if either we have the moves already installed inside our brain, or else we would be required to generate it on our own at that very moment inside the examination hall. Probably this is the single most important reason that why a question from "Function Chapter" seems comparatively easier to a student who has gone through maths in 10+2 with respect to a student who has not gone through maths in his/her 10+2.

So, how can a student who has not gone through maths can come at par with a student who is a maths graduate?



And how moves will be generated (the base of this triad)—By going through the concepts, and solving not just quantity of questions, but quality of questions too.

Looking upon the pattern of CAT QA for the past years, one can be reasonably sure that fetching 40% marks in QA is sufficient to get approximate 97–98 percentile. This translates into 10 questions (as per CAT 2008). The idea is—if we can have the moves ready for 6 questions, for an example, then we would be required to generate moves only for 4 questions inside the examination hall, and this saves your considerable time that can be invested in other sections, including QA too.

For example, there was a question in CAT 2008—What is the number of terms in the expansion of $(a + b + c)^{20}$?

I have put almost the same question in my QA book, which was published in 2006, in Miscellaneous chapter–What is the number of terms in the expansion of $(a + b + c)^{10}$?

I am not saying that our objective should be to match the question—that as a student I should have done exactly the same question appearing in CAT of that year before sitting in CAT. Rather, its about thought process—retaining the moves used earlier while solving a question—and the ability to use the same in even slightly different situation. And what is true for QA is mostly true for other sections too.

About this Book

Number system being one of the most important areas under Quantitative Aptitude rightly deserves a separate book. Following table emphasises upon the same:

CAT Year	CAT 08	CAT 07	CAT 06	CAT 05	CAT 04	CAT 03	CAT 02	CAT 01	CAT 00	CAT 99
No. of Qs	4	2	5	6	2	11	9	12	9	8

Roughly this translates to approximate 20% of QA paper every year, on an average.

A mention of online CAT 2009 CBT (Computer Based Test) has not been done in the above table because there were 20 papers (distributed across 10 days, 2 slots per day) as per the actual planning of IIMs and weightage of the areas under QA section was not entirely same across all the slots/papers. Still, there were 3-5 questions from number system in QA (out of 20 questions) in almost every CAT 2009 paper.

This book is designed at the idea stage, to structure stage to content stage—keeping in mind the requirement of students who might not have been in touch with their studies or the fundamentals required to solve the questions in Number

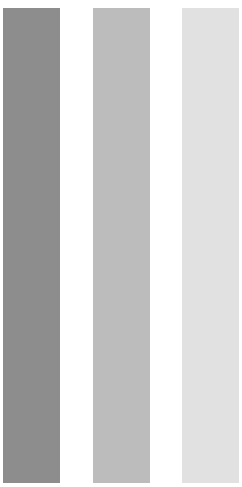
System. In addition to these, while writing this book, I have kept in mind the following points:

1. Focus is upon how to build an approach to solve a question—so that even if one has never seen a question like the asked before one in his/her life, one should be able to solve that under testing conditions.
2. Concepts have been dealt in an easy way so that you can understand those, and then application of those concepts under different contexts—to stimulate your thoughts that "Why I never thought like this?"
3. The questions that you are going to have in this book will challenge you enough to keep attempting it unless you have solved it.
4. There are six section tests to help you better understand your conceptual clarity and application of those.

Lastly, CAT questions from last 10 years (CAT 1999 – CAT 2008) have been appended so as to give you a simulated feeling of CAT.

Best of Luck to all the future CEOs, for life.

I can be reached at–nishitkumarsinha@rediffmail.com



Chapter 1

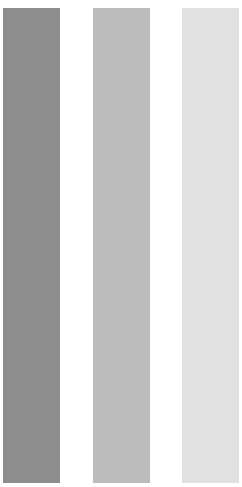
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DEMYSTIFYING NUMBER SYSTEM

ALL ABOUT NUMBERS

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- Types of numbers and their properties
- Methods of counting—digits and numbers
- VBODMAS rule—Precedence of the operations
- Types of questions asked from this chapter in the CAT and other exams

Number is "a numeral or group of numerals". In other words, it is a word or symbol, or a combination of words or symbols, used in counting several things.

TYPES OF NUMBER

Broadly, numbers can be defined to be of two types: Real Number and Imaginary Number.

Real Number Real number is a number that can be expressed on the number line. In other words, all the real numbers can be felt or experienced in the real world.

Imaginary Number In contrast to Real numbers, Imaginary numbers cannot be plotted on the number line, or, they cannot be felt or experienced in the real world.

Types of Real Number

Rational Number

A rational number is a number that can be expressed as a fraction (p/q) where p and q are integers and $q \neq 0$. The term 'rational' comes from other the word ratio, because the rational numbers are the ones that can be written in the format of a ratio.

For example, 3/5, 4/8, 22/7 etc., are rational numbers. Rational numbers can be positive as well as negative.

In a rational number p/q, p is known as numerator and q is known as denominator.

We also understand that,

If numerator is more than denominator, then ratio is known as improper ratio.

Example: 5/3, 7/4.

If the numerator is less than the denominator, then the ratio is known as proper ratio.

Example: 3/7, 7/18.

So what kind of numbers are Rational Numbers?

(i) All the integers—positive or negative or zero—are rational numbers.
For example: (–2), (–100), 10 etc., are rational numbers.

(ii) Is 0.5555555…(5 repeating till infinity) a rational number?
Yes, it is. Because 0.5555…(5 repeating till infinity) $= \frac{5}{9}$.
All the repeating decimal numbers with a fixed period are rational.
For example, 0.543543543543.....(543 repeating till infinity) $= \frac{543}{999}$.

(iii) All finite decimals like 0.48 are rational. In other words, all the terminating decimals are rational numbers.

For example, 0.5678345678 is a rational number because it can be written in the form of p/q.

Methods to convert decimals into fractions:

1. If a number is finite decimal like 0.89.

$$0.89 = \frac{89}{100}$$

$$1.27 = \frac{127}{100}$$

$$0.7 = \frac{7}{10} = \frac{70}{100}$$

2. If the number is repeating decimal number like 0.454545.....

$$0.454545..... = \frac{45}{99}$$

$$0.444444..... = \frac{4}{9}$$

$$0.678678678678..... = \frac{678}{999}$$

$$1.23232323..... = 1\frac{23}{99} = \frac{122}{99}$$

So dividing by 9 repeats the digits once, dividing by 99 repeats the digits twice, dividing by 999 repeats the digits thrice and so on. In other words, number of 9's in the denominator will be equal to the number of repeating digits in the number.

1/9	2/9	3/9	4/9	5/9	8/9	25/99	214/99
0.1111....	0.2222.....	0.3333....	0.444....	0.5555...	0.888....	0.252525....	2.161616...

Worked-out Example 1 Convert 3.1232323.....into rational form.

☝ ***Solution*** Digits are not repeating after the decimal, rather digits are repeating after one digit from decimal.

So we can write 3.1232323.....= 3.1 + 0.0232323....

We just have to convert 0.0232323....into rational form. Remaining part = 3.1 can be converted to rational part very easily.

$$0.02323232..... = 0.1 \times 0.232323..... = \frac{1}{10} \times \frac{23}{99} = \frac{23}{990}$$

$$\text{So, } 3.1232323..... = \frac{31}{10} + \frac{23}{990} = \frac{3092}{990}$$

Alternatively, we can do this question in the following way too:

Let P = 3.1232323..... $\Rightarrow$ 10 P = 31.232323..... $= 31\frac{23}{99} = \frac{3092}{99}$

Hence P $= \frac{3092}{990}$

Worked-out Example 2 Convert 3.15474747.....into rational form.

☝ ***Solution*** Assume P = 3.15474747.....

We can see that digits repeat two digits after decimal.

100 P = 315.474747..... $= 315\frac{47}{99} = \frac{31232}{99}$

Hence, P $= \frac{31232}{9900}$

Irrational Numbers

Any real number that is not rational is irrational. An irrational number is a number that cannot be expressed as a fraction p/q for any integer p and q. Irrational numbers have decimal expansions that neither terminate nor become periodic.

The most famous irrational number is $\sqrt{2}$, also known as Pythagoras's constant.

Real numbers are also defined as the set of Rational and Irrational numbers.

Worked-out Example 3 Identify which of the following are rational numbers?

(a) $\frac{5}{9}$

(b) 0.33333 (3 repeating till infinity)

(c) $\sqrt{2}$

(d) $\frac{22}{7}$

(e) π

(f) 0.573573573 (573 repeating till infinity)

(g) 5.637962460677264967783465 21 (till infinity)

(h) $\sqrt{5}^{100}$

(i) $\sqrt{5}^{100} + \frac{1}{\sqrt{5}^{100}}$

☝ ***Solution***

(a) Yes

(b) Yes. 0.3333333333333(3 repeating till infinity) $= \frac{3}{9} =$ Rational number

(c) No. $\sqrt{2}$ cannot be written in the form of $\frac{p}{q}$ where p and q are integers ($q \neq 0$). Hence irrational number.

(d) Yes. $\frac{22}{7}$ is in the form of $\frac{p}{q}$. Hence rational number.

(e) Irrational number $\pi(22/7)$ is only an approximate value, and not the exact value. Hence, No.

(f) 0.573573753 (573 repeating till infinity) $= \frac{573}{999} =$ Rational number

(g) Irrational number. Despite 5.63796246067726496778346521.....goes till infinity, but it does not show any pattern (unlike previous question), hence cannot be presented as a ratio = $\frac{p}{q}$.

(h) $\sqrt{5}^{100} = 5^{50}$ = rational number

(i) $\sqrt{5}^{100} \frac{1}{+\sqrt{5}^{100}} =$ rational number + rational number = rational number

Alternatively, $\sqrt{5}^{100} + \frac{1}{\sqrt{5}^{100}} = 5^{50} + \frac{1}{5^{50}} = \frac{5^{100}+1}{5^{50}} = \frac{p}{q} =$ Rational number

A dilemma—Is 0.9999999.....equal to 1?

Obviously, when we write this down mathematically, 0.9999999..... $= 0.\overline{9} = \frac{9}{9} = 1$

Although a more logical question arises that any number of times we write 9 after the decimal as in 0.99999....., this should not be equal to 1.

An explanation to this is given in the following manner:

Do not look at 0.99999.....as if we are approaching towards 1, see this number as if we are moving away from 1 and look at the distance between 1 and this number.

So, distance between 1 and 0.9 = 0.1

Distance between 1 and 0.99 = 0.01

Distance between 1 and 0.999 = 0.001

Distance between 1 and 0.9999 = 0.0001

We can see that the distance is very slowly getting smaller, and is tending towards zero. So after writing a number of 9's as in 0.999999....., distance will become equal to zero, and number = 1.

Integers

Integers can be either positive or negative or zero. Integers are also classified as odd or even integers.

Even Integers Any integer that can be written in the format of 2N is an even integer, where N is an integer. In other words, any number that is divisible by 2 is an even integer. For example, 2, 4, 100 etc.

Is (–10) an even integer → Yes (–10) is an even integer.
Is zero an even integer → Yes, 0 is an even integer.

Odd Integers Any integer that can be written in the format of 2N + 1 is an odd integer, where N is an integer. For example, 1, 5, 101, –23 are odd integers.

Odd ± Odd = Even

Even ± Even = Even

Odd ± Even = Odd

Even ± Odd = Odd

$(Odd)^{Even}$ = Odd

$(Even)^{Odd}$ = Even

Odd × Even = Even

Odd × Odd = Odd

Even × Even = Even

Well, the idea is not asking you to mug it up. Rather, develop an understanding and whenever required, you should be able to verify it on your own.

Worked-out Example 4 Let x, y and z be distinct integers x and y are odd and positive, and z is even and positive. Which one of the following statements cannot be true? **{CAT 2001}**

(a) $y(x–z)^2$ is even
(b) $y^2(x–z)$ is odd
(c) $y(x–z)$ is odd
(d) $z(x–y)^2$ is even

☝ ***Solution*** Answer can be verified by assuming the values x = 1, y = 3, z = 2. Let us verify options:

(a) (x–z) = Odd. So $y(x–z)^2$ is also odd. Hence option a is not true.
(b) Since y and (x–z) both are odd, hence $y^2(x–z)$ is odd. So option b is true.
(c) Since y and (x–z) both are odd, hence $y(x–z)$ is odd. So option c is true.
(d) Since z and (x–y) both are even, hence $z(x–y)^2$ is even. So option d is true.
Hence answer = option a.

Worked-out Example 5 1st 100 natural numbers are written on a black board. Two persons *A* and *B* are playing a game of putting '+' and '–' sign one by one between any two consecutive integers out of these 100 natural numbers. Both *A* and *B* are free to put any sign (+ or –) anywhere provided there is no sign placed already between the two natural numbers. At the end, when the signs are put between all such two consecutive natural numbers, result is calculated. If the result is even, then A wins and if the result is odd, then B wins. Who will win?

☝ ***Solution*** It can be seen that there are 50 odd and 50 even numbers.

So, 100 consecutive natural numbers = set of 50 odd and 50 even numbers

Whatever sign we put between two odd numbers, resultant of 50 odd numbers = Even, and similarly, whatever sign we put between two even numbers, resultant of 50 even numbers = Even.

Hence the net result is Even + Even or Even – Even. In any case, net result = Even.

Hence A will win.

We can also see here that + sign or – sign does not matter here. Irrespective of the signs put by A or B, A is always going to win.

Worked-out Example 6 When 98 is added to a perfect square, another perfect square is obtained. How many such pairs of perfect squares exist?

☝ ***Solution*** $A^2 = B^2 + 98$

$\Rightarrow A^2 - B^2 = 98$

$\Rightarrow (A - B)(A + B) = 98$

Different possibilities for A and B are:

A	B
Even	Even
Odd	Odd
Even	Odd
Odd	Even

Consequently, different possibilities for A^2 and B^2 are:

A^2	B^2	$A^2 - B^2$	Remarks
Even	Even	Even	May be possible
Odd	Odd	Even	May be possible
Even	Odd	Odd	Not possible (as the difference = 98)
Odd	Even	Odd	Not possible (as the difference = 98)

If both A and B are even, then both A^2 and B^2 are going to be multiples of 4 (even^2). Hence $A^2 - B^2 = 4K \neq 98 \Rightarrow$ So, A = Even and B = Even are not possible.

Last possibility for this calculation to be true is A and B both are odd. Let us verify that:

If A and B both are odd then (A – B) and (A + B) both will be even. Hence product of (A – B) (A + B) = Even × Even = $4K \neq 98 \Rightarrow$ So, A = Odd and B = Odd are not possible.

Hence we conclude now that no such set exists.

Whole Numbers

When 0 is added to the set of natural numbers, we obtain whole numbers. So whole numbers are 0, 1, 2, 3, 4, 5,.....upto infinity. Whole numbers are also known as Non-negative integers.

Lowest whole number = 0

Natural Number

Natural numbers are counting numbers: 1, 2, 3, 4, 5.....upto infinity. Natural numbers are also known as positive integers.

Lowest natural number = 1 = 1st natural number. Since the difference between any two consecutive natural number = 1, we say that 10th natural number from starting = 10, or 22nd natural number from starting = 22 and so on.

Equation format of natural number $\rightarrow a_n = a_{n-1} + 1$, where, $a_1 = 1$ and a_1 is the lowest natural number.

Function format of natural number $\rightarrow f(n) = f(n-1) + 1$, where $f(1) = 1$ and $f(1)$ is the lowest natural number.

Natural numbers, further, can be categorized as either prime number or composite number except 1. [1 is neither prime number nor composite number]. We will see this on next pages.

Formula/generalization related to Natural numbers:

1. Sum of 1st N natural numbers $= \frac{N(N+1)}{2}$. Understand that this formula is applicable only if we are adding the numbers from 1, then 2, and so on.
2. Sum of the natural numbers from 1 – 10 = 55
   Sum of the natural numbers from 11 – 20 = 155
   Sum of the natural numbers from 21 – 30 = 255, and so on.

Worked-out Example 7 Find the sum of 1st 100 consecutive natural numbers.

☝ ***Solution*** N = 100. Using the formula given above, sum of 1st 100 natural numbers = 100 × 101/2 = 5050.

Worked-out Example 8 A child was asked to add first few natural numbers (i.e., 1 + 2 + 3 + ...) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was **{CAT 2002}**

(a) less than 10 (b) 10 (c) 15 (d) more than 15

☝ ***Solution*** Let us first try to see that summation till what number gives us a value close to 575.

Sum of the natural numbers from 1 – 10 = 55

Sum of the natural numbers from 11 – 20 = 155

Sum of the natural numbers from 21 – 30 = 255

Net summation from 1 – 30 = 55 + 155 + 255 = 465, so we are required to add some more numbers.

Next numbers = 31, summation from 1 – 31 = 496

Next numbers = 32, summation from 1 – 31 = 528

Next numbers = 33, summation from 1 – 31 = 561

Next numbers = 34, summation from 1 – 31 = 595.

Hence, child has missed out 595 – 575 = 20.

(Look at the end of this chapter under the section "CAT questions" to see a different method of solving the same question.)

Worked-out Example 9 A child tore off one leaf from a book having page number from 1 to N. Summation of the remaining number of pages = 1010. Which are the page numbers on that leaf torn off from this book?

☝ ***Solution*** We are required to find out the 1st number larger than 1010 in the summation 1 to N.

Going through hit and trial, summation from 1 to 40 = 820. So we are required to add more numbers to it.

820 + 41 = 861. 861 + 42 = 903. 903 + 43 = 946. 946 + 44 = 990. 990 + 45 = 1035.

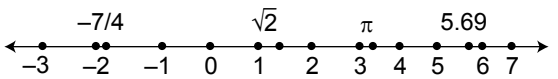
Ideally the summation should have been = 1035, but it is 1010. So its 25 less than the actual summation that it should have been.

So the page numbers missing from the book = page numbers 12 and page numbers 13. (it will be on one leaf only).

Real Number Line

Real number line (in short also known as number line) consists of the union of the rational and irrational numbers.

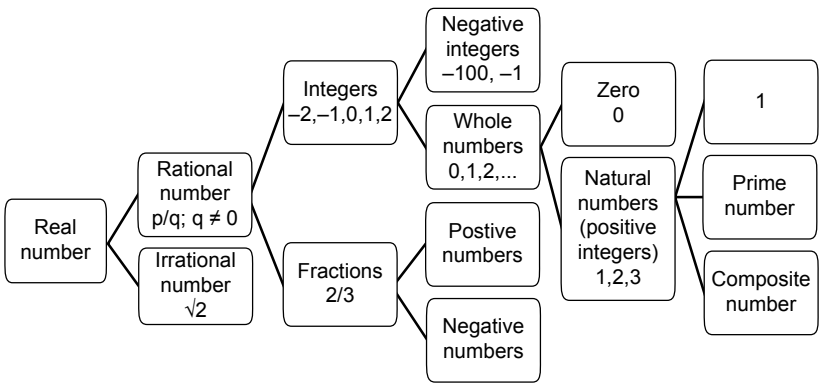
Every real number can be associated with a single point on the **real number line.**



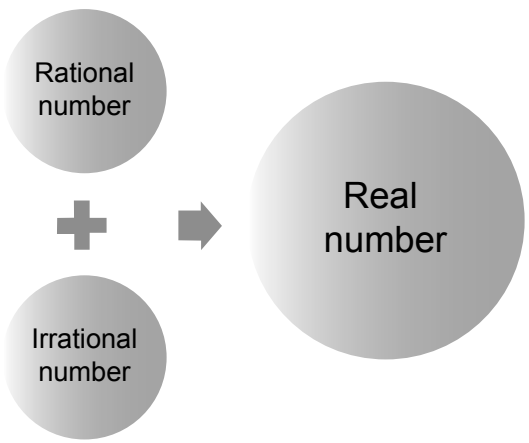
- On the left hand side, it extends from—infinity (–) to + infinity (+) on the right hand side.
- Numbers on the right hand side are always larger than the numbers on the left hand side.
- When we do addition operation, we move on to the right hand side of numbers line.
  - For example: when we add 4 units to 3 (= 3 + 4), we are required to move 4 units right hand side to 3, landing up at 7.
- When we do subtraction operation, we move on to the left hand side of numbers line.

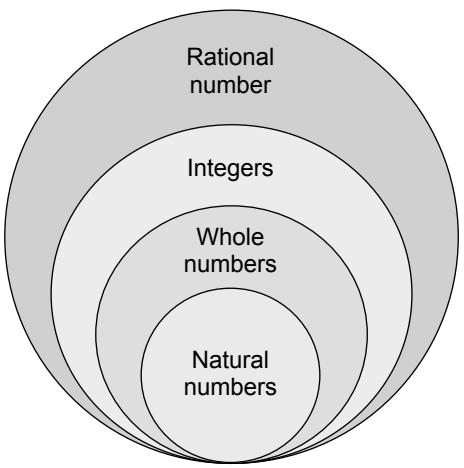
- For example: when we subtract 3 units from 4 (= 4 – 3), we are required to move 3 units left hand side to 4, landing up at 1.

SUMMARY OF THE WHOLE DISCUSSION



We can present the above graphics using sets theory too:





Worked-out Example 10 The sum of four consecutive two-digit odd numbers, when divided by 10, becomes a perfect square. Which of the following can be one of these four numbers? **{CAT 2006}**

1. 21 **2.** 25 **3.** 41 **4.** 67 **5.** 73

☝ ***Solution*** Maximum sum of any four consecutive two-digit odd numbers = 389. Since question is asking for a number divisible by 10, sum can be 360, 250, 160 or 90 and hence the number will be an odd number close to 90, 62, 40 or 22.

Now we will go through the options.

As the numbers are consecutive, for any of the options, sum should be approximately four times of the value given in the option. Going by this logic, 5th option can be easily eliminated as 4×73 is not close to any of the probable numbers. Doing a bit of hit and trial will ravel that $(43 + 41 + 39 + 37) = 160$, and hence 41 is the right answer. Look at the end of this chapter under CAT questions for 2nd method of solving this question.

Now we will see the types of natural numbers:

All the natural numbers greater than 1 has at least two factors—and on the basis of number of factors we can categorize natural numbers greater than 1 either as Prime number or Composite number.

Prime Number

Any number that has exactly two distinct factors is known as prime number. In other words, any number which is divisible by 1 and itself only is a prime number. For example, 2 is a prime number, because 2 is divisible by 1 and 2 only (two distinct factors only). Similarly, 103 is a prime number because it is divisible by 1 and 103 only. 2 is the only even prime number, and all the other prime numbers are odd.

In that way prime numbers are building blocks of numbers i.e., prime numbers are used to build the other numbers—known as composite numbers.

Worked-out Example 11 A, B, C, D and E are five prime numbers, not necessarily consecutive. Sum of these five prime numbers = 264. It is also given that $A < B < C < D < E$. What is the value of A^5?

☝ ***Solution*** You should not try to solve this question by trying to find out the actual values of prime numbers. Rather think of the logic behind the question.

Summation of five prime numbers = 264 = Even number

We know that except 2, all the other prime numbers are odd. Since summation of five prime numbers is even, hence four of these prime numbers are odd and one is even.

Since A is the lowest prime number, hence A = Even prime number = 2

So, $A^5 = 2^5 = 32$.

Properties of Prime Numbers

(i) There are infinite prime numbers.
(ii) There are infinite set of prime numbers such that difference between them = 2 (represented as N and N + 2, where N is a prime number). For example, 3 and 5, 5 and 7, 11 and 13 etc. Set of these two prime numbers are known as "Twin Primes."
(iii) There is exactly one set of prime number triplet N, N + 2, N + 4, where N is a prime number (3, 5, 7).
(iv) All the natural numbers greater than 1 has atleast one prime factor.
(v) If a number P has a prime factor N < P, then it has another prime divisor $M \leq \sqrt{P}$.
(vi) All the prime numbers greater than 3 are of the format $6N \pm 1$. It means that all the prime numbers will give either a remainder of +1 or –1 when divided by 6. Understand that this is only a sufficient condition and not necessary one. In other words, if a number is a prime number, it will be of the format $6N \pm 1$, but it does not mean that if any number is of the format $6N \pm 1$, then its going to be a prime number. For example, 25 = 6N + 1, but 25 is not a prime number.

Test of Primality—Using conditions v and vi, we will now learn the methods to check if a number N is a prime number:

Step 1—1st divide the number by 6, if number does not give remainder = either + 1 or –1, then its not a prime number. If in step 1, number gives either + 1 or –1 as the remainder, then we will go through the next steps.

Step 2—Find out the approximate value of square root of N.

Step 3—Starting from 2 to $\sqrt{N}$, check one by one if any of these numbers divide N.

Step 4—If N gets divided by any one of 2 to $\sqrt{N}$ (other than 1), then N is not prime, otherwise prime.

Worked-out Example 12 Check if N = 142 is a prime number?

☝ ***Solution*** 2 is the only even prime number. Hence, 142 is not a prime number.

Worked-out Example 13 Check if N = 143 is a prime number?

Step 1—Remainder obtained when 143 is divided by 6 = –1. Since remainder obtained = –1, hence, we will now go to next steps.
Since ,143 is divisible by 11, hence, 143 is not prime.

Step 2—$\sqrt{143} \cong 12$

Step 3—Starting from 1 to 12, we will try dividing 143 by all the numbers.

Outcome when 143 is divided by numbers from 2 to 12:

2	Not divisible
3	Not divisible
4	Not divisible
5	Not divisible
6	Not divisible
7	Not divisible
8	Not divisible
9	Not divisible
10	Not divisible
11	**Divisible**
12	Not divisible

Note Wilson theorem also gives the test for the primality of a natural number. See "Remainder" chapter to know more about this.

Worked-out Example 14 What is the number of natural numbers n in the range of $2 < n < 20$ such that (n–1)! is not divisible by n?

☝ ***Solution*** Take some initial values of n and try to find a pattern.

$N = 3 \Rightarrow (n-1)! = 2!$ Which is not divisible by 3.

$N = 4 \Rightarrow (n-1)! = 3!$ Which is not divisible by 4.

$N = 5 \Rightarrow (n-1)! = 4!$ Which is not divisible by 5.

$N = 6 \Rightarrow (n-1)! = 5!$ Which is divisible by 6.

$N = 7 \Rightarrow (n-1)! = 6!$ Which is not divisible by 7.

Now we can generalize that for N = Prime numbers, (n–1)! is not divisible by n.

Hence, values of n (greater than 6) for which (n–1)! is not divisible by n = Prime values of n = 7, 11, 13, 17, 19.

Total values = 3, 4, 5, 7, 11, 13, 17, 19 = 8 numbers

Composite Number

A composite number is a positive integer > 1 which is not prime number. In other words, composite numbers have factors other than 1 and itself, and hence, composite numbers will have atleast 3 factors or divisors.

Lowest composite number = 4.

List of 1st few composite factors:

Number	4	6	8	9	10	12	14	15
Prime Factorization	2^2	$2^1 \times 3^1$	2^3	3^2	$2^1 \times 5^1$	$2^2 \times 3^1$	$2^1 \times 7^1$	$3^1 \times 5^1$
Number of factors	3	4	4	3	4	6	4	4

Properties of Composite Number

(i) A composite number has atleast three factors.
(ii) All the natural numbers other than prime numbers and 1 are composite numbers.

Euler wrote in 1770 – Mathematicians have tried in vain to discover some order in the sequence of prime numbers but we have every reason to believe that there are some mysteries which the human mind will never penetrate.

Methods of Counting

Generally, we encounter two types of counting problems – 1st one related to digits counting and 2nd one related to numbers counting. Let us see this one by one with the help of questions.

Digit Counting In these types of questions, some numbers will be given and occurrence of a particular digit will be asked.

Worked-out Example 15 How many times the digits of a computer keyboard will be required to be pressed in typing 1st 100 natural numbers?

☝ ***Solution*** Following observations are required to be made:

1. Numbers are going to be of 1 digit, 2 digits, 3 digits.
2. Number of numbers of 1 digit/2 digits/3 digits are going to be different.

   Number of natural numbers of 1 digit = 9 $\Rightarrow$ Number of digits = $9 \times 1 = 9$

   Number of natural numbers of 2 digits = 90 $\Rightarrow$ Number of digits = $90 \times 2 = 180$

   Number of natural numbers of 3 digits = 1 $\Rightarrow$ Number of digits = $1 \times 3 = 3$

   Hence, total number of digits from 1 – 100 = 9 + 180 + 3 = 192

   Observation 1—There are 9 single digit natural number (1–9)

   Observation 2—There are 90 2-digits natural number (10–99)

   Observation 3—There are 900 3-digits natural number (100–999)

   It can be generalized now that number of four digit natural numbers = 9000 and so on.

Worked-out Example 16 How many times does the digit 6 appear when writing from 6 to 400?

☝ ***Solution*** One method of solving this is obviously through the actual counting method – 6, 16, 26, 36, 46, 56, 60, 61,……and so on.

Best method is to understand that how actually numbers occur or numbers are made in succession;

Every digit from 1 to 9 appears ten times at units place, and ten times at tens place.

For example, let us count number of times digit '4' appears in 1st 100 natural numbers.

At the units place – **0**4, **1**4, **2**4, **3**4, **4**4, **5**4, **6**4, **7**4, **8**4, **9**4 – 10 time (Given in bold)

At the tens place – 4**0**, 4**1**, 4**2**, 4**3**, 4**4**, 4**5**, 4**6**, 4**7**, 4**8**, 4**9** – 10 time (Given in bold)

It can also be seen that digits have actually interchanged their positions.

Observation 1—All the digits from 1 to 9 appear 10 times at units place and ten times at tens place. In other words, when we count any 100 consecutive natural numbers, i.e., it may not be from 1 to 100, even if it is from 23–122, number of times any digit from 1 to 9 will appear a total of 20 times–10 times at the unit place and 10 times at the tens place. We are required to count the occurrence at the 100's place separately.

Observation 2—All the digits from 1–9 appear to be a total of 300 times from 1 to 999 (including both the limits):

	Unit place	**Tens place**	
1 – 100	10	10	
101 – 200	10	10	
201 – 300	10	10	
301 – 400	10	10	
401 – 500	10	10	
501 – 600	10	10	
601 – 700	10	10	
701 – 800	10	10	
801 – 900	10	10	
901 – 999	10	10	
Sum	100	100	200

Number of times any of the digits (1 – 9) occurs at units place = 100

Number of times any of the digits (1 – 9) occurs at tens place = 100

Number of times any of the digits (1 – 9) occurs at hundreds place = 100

Number of times any of the digits (1 – 9) occurs from 1 to 999 = 100 + 100 + 100 = 300

Number Counting

Out of 1st N consecutive natural numbers,

(i) Every 2nd number is divisible by 2.

So if we take 1st 100 natural numbers, 50 are going to be divisible by 2 and remaining 50 are not divisible by 2.

Out of 1st 99 natural numbers, since 1st number is not divisible by 2, 49 numbers are going to be divisible by 2 and 50 numbers are not divisible by 2.

(ii) Every 3rd number is divisible by 3.

So, if we take 1st 100 natural numbers, $\frac{100}{3}$ = 33 are going to be divisible by 3 and remaining 67 are not divisible by 3.

However, if we have to find out number of numbers divisible by 3 in any 100 natural numbers, we cannot find out exact number which are divisible by 3. Key is to know if the 1st number or the last number is divisible or not?

Worked-out Example 17 Difference between two natural numbers A and B = 101. How many natural number between A and B (excluding both A and B) will be divisible by 3?

☝ ***Solution*** Let us take the values:

A = 102, B = 1. Excluding both 1 and 102, numbers which are divisible by 3 = 3, 6, 9, 12, ..., 99 = 33 numbers.

A = 103, B = 2. Excluding both 2 and 103, numbers which are divisible by 3 = 3, 6, 9, 12, ..., 99, 102 = 34 numbers.

So, it may be 33 or 34 numbers.

(iv) Every 4th number is divisible by 4.
So, if we take 1st 100 natural numbers, $\frac{100}{4}$ = 25 are going to be divisible by 4 and remaining 75 are not divisible by 4.

(v) Now we generalize this. Out of 1st N consecutive natural numbers, every mth number is divisible by m.

Out of any N consecutive natural numbers

(i) Out of any 2 consecutive natural numbers, exactly one number is divisible by 2.
(ii) Out of any 3 consecutive natural numbers, exactly one number is divisible by 3.
(iii) Out of any 4 consecutive natural numbers, exactly one number is divisible by 4.
(iv) Out of any 5 consecutive natural numbers, exactly one number is divisible by 5.

Worked-out Example 18

(a) Out of any 10 consecutive natural numbers, exactly 5 natural numbers are divisible by 2. (True/False)

(b) Out of any 11 consecutive natural numbers, exactly 4 natural numbers are divisible by 3. (True/False)

(c) Out of any 100 consecutive natural numbers, exactly 1 natural number is divisible by 100. (True/False)

(d) Out of any 101 consecutive natural numbers, exactly 1 natural number is divisible by 100. (True/False)

(e) Out of any 99 consecutive natural numbers, exactly 1 natural number is divisible by 100. (True/False)

☝ ***Solution***

(a) True. $\frac{10}{5} = 2$

(b) It may be 3 or 4 numbers depending upon which is the 1st number. For example if we start from 3, then there are 4 numbers divisible by 3. However if we start from 4, then there are going to be three numbers only.

(c) True. $\frac{100}{100} = 1$

(d) False. There may be one or two numbers divisible by 100.

(e) False. There may be zero or one number divisible by 100.

If N is a natural number, number of natural numbers in the range:

$A \le N \le B = (B - A) + 1$ [Both ends included]

$A \le N < B = (B - A)$ [Only one of the ends included]

$A < N \le B = (B - A)$ [Only one of the ends included]

$A < N < B = (B - A) - 1$ [None of the ends included]

For example, number of natural numbers in the following range is as given below:

Both the ends included	• $100 \le N \le 250$ • $(250 - 100) + 1 = 151$
Only one end included	• $100 \le N < 250$ • $250 - 100 = 150$
Only one of the ends included	• $100 < N \le 250$ • $250 - 100 = 150$
None of the ends included	• $100 < N < 250$ • $(250 - 100) - 1 = 149$

Alternatively, we can generate a pattern to count the number of numbers. This can be seen through the following example.

Worked-out Example 19 How many natural numbers from 200 to 500 (including both the limits) will be divisible by 3?

☝ ***Solution***

Method 1 Pattern method

1st number = 201 and next numbers are 204, 207,....., 498.

$201 = 67 \times 3$, ..., $498 = 166 \times 3$

So we have to count till 166th multiple of 3 leaving 1st 66 multiple.

Hence number of numbers = 166 – 66 = 100.

Method 2

1st number = 201 and next numbers are 204, 207,....., 498.

So, number of numbers $= \frac{498-201}{3}+1 = \frac{297}{3}+1 = 99 + 1 = 100$

Some General Properties

1. **If we reverse any two digit natural number, the difference between the original and resulting number will be equal to 9 times the difference between tens place digit and units place digit.**
   To understand this more precisely, let us see it in detail.
   If xy is a two digit number, with x = Tens place digit, and y = Units place digit, then xy – yx = 9 (x–y)
   For example, 91 – 19 = 9(9–1) = 72
   74 – 47 = 9 (7–4) = 27; 85 – 58 = 9 (8–5) = 27 and so on.
   It is actually very simple to see that why this happens.
   xy = 10x + y and yx = 10y + x
   So, xy – yx = 10x + y – (10y + x) = 9 (x–y)
   Similarly, if we add the numbers, let us see what happens then:
   xy + yx = 10x + y + 10y + x = 11 (x + y) $\Rightarrow$ 11 times sum of the digits at unit and tens place.

Worked-out Example 20 When a two digit number is reversed, another two digit number is obtained. Difference between original number and resulting number = 48. How many such two digit natural numbers are possible?

☝ ***Solution*** As we have seen that difference between original number and resulting number should be a multiple of 9, and 48 is not a multiple of 9, hence, no such number exists.

Worked-out Example 21 When a two digit number is reversed, another two digit number is obtained. Difference between original number and resulting number = 45. How many such two digit natural numbers are possible?

☝ ***Solution*** Taking a cue from the above discussions, 9 (x–y) = 45

$\Rightarrow$ x – y = 5 $\Rightarrow$ Sets of (x, y) = (5, 0), (6, 1), (7, 2), (8, 3), (9, 4).

However, we cannot take set (5, 0) as interchanging this will lead to '05' which is not a two digit number. Hence, number of sets possible = 4.

Worked-out Example 22 Digits of a two digit number are interchanged. Sum of new number obtained and original number is a perfect square. If tens place digit is larger than units place digit, how many such two digit numbers exist?

☝ ***Solution*** Assume xy is the two digit number. After interchanging the digits, new number obtained = yx.

According to the question, xy + yx = 10x + y + 10y + x = 11 (x + y)

11 (x + y) is perfect square ⇒ Since, 11 is a prime number, (x + y) should be equal to 11. Now we will see different possibilities so that x + y = 11.

Set of (x, y) = (9, 2), (8, 3), (7, 4), (6, 5) [Since tens place digit is larger than units place digit].

Hence, 4 sets are possible.

2. **If we reverse any three digit natural number, the difference between the original and resulting number will be equal to 99 times the difference between hundreds place digit and units place digit.**
   To understand this more precisely, let us see it in detail.
   If xyz is a three digit number, with x = Units place digit, and y = Tens place digit, z = Units digit then, xyz – zyx = 100x + 10y + z – (100z + 10y + x) = 99 (x–z)

Worked-out Example 23 The digits of a three-digit number A are written in the reverse order to form another three-digit number B. If B>A and B – A is perfectly divisible by 7, then which of the following is necessarily true? **{CAT 2005}**

(a) 100<A<299 (b) 106<A<305 (c) 112<A<311 (d) 118<A<317

☝ ***Solution*** Assume A = xyz = 100x + 10y + z and B = zyx = 100z + 10y + x.

B – A = 99 (z – x) = Divisible by 7

⇒ (z – x) should be divisible by 7.

It is also given that B – A > 0, or, B – A = 99 (z – x) > 0. Since, 99 is +ve, (z – x) should also be positive. Hence, z > x.

We can infer that only possible value for (z – x) = 7 ⇒ z = 9, x = 2 and z = 8 and x = 1.

So, A should be of the format 1y8 or 2y9 ъ.

Minimum value of A = 108 and maximum value of A = 299.

Only one option accommodates both of these values. Hence, option (b) is the answer.

[See "Divisibility Rules" chapter to see another approach to solve this question].

3. **Digital Sum of a number**
   When we keep on adding the digits of a number until a single digit is obtained is known as the digital sum of the number.
   For example, Digital sum of 458.
   458 = 4 + 5 + 8 = 17
   Further, we add digits of 17 = 1 + 7 = 8
   Hence, 8 is the digital sum of 458.

Worked-out Example 24 Find the digital sum of 28^{100}.

☝ ***Solution*** Vedic maths tells us that instead of multiplying the numbers and then calculating the digital sum, we can add the number 1st and then multiply the numbers.

Digital sum of $28 = 2 + 8 = 1 + 0 = 1$

Digital sum of 28^{100} = Digital sum of $1^{100} = 1$

Worked out Example 25 Find the digital sum of 34×56.

☝ ***Solution*** We are not required to find the product and then find the digital sum of the final product.

We can 1st find the digital sum of the individual number, and then multiply the numbers to find final digital sum.

Digital sum of $34 = 3 + 4 = 7$

Digital sum of $56 = 5 + 6 = 1 + 1 = 2$

Digital sum of 34×56 = Digital sum of 7×2 = Digital sum of $14 = 5$

[It can be verified that $34 \times 56 = 1904$, and digital sum of $1904 = 1 + 9 + 0 + 4 = 5$]

4. **Perfect Number**

A perfect number is a natural number for which when we add up all of the factors of the number less than that number, we get that number.

For example,

(i) Number 6

Factors of 6 less than 6 = 1, 2, 3 $\Rightarrow$ Sum of factors = $1 + 2 + 3 = 6$

Since sum of factors of 6 = Number 6, hence, 6 is a perfect number.

(ii) Number 28

Factors of 28 less than 28 = 1, 2, 4, 7, 14 $\Rightarrow$ Sum of factors = $1 + 2 + 4 + 7 + 14 = 28$

(iii) Number 496

Summation of factors of 496 less than 496 = $1 + 2 + 4 + 8 + 16 + 31 + 62 + 124 + 248$

At this stage, we can observe a pattern—There are powers of 2 from 1 up to a certain number, and then a prime number that is equal to 1 less than twice the last power of two.

In case of 6, $2 \times 2 - 1 = 3$

In case of 28, $4 \times 2 - 1 = 7$

In case of 496, $16 \times 2 - 1 = 3$

The rest of the factors are each twice of that prime number in a sequence. For example, in case of 496, $31 \times 2 = 62$. Next term is $62 \times 2 = 124$. Next term is $124 \times 2 = 248$.

This derives a formula for perfect number,

Perfect Number = $2^{n-1} \times (2^n - 1)$, where n is any natural number.

Note Till now, nobody has been able to find any odd perfect number, but we do not know if any odd perfect number exists or not? It may be only a distant rare possibility that an odd perfect number will exist. Till now there is no conclusive proof that there will not be any odd perfect number. So, till now all the perfect numbers that have been found are even numbers.

5. **Properties of perfect square**

(i) Unit digit of a perfect square can be—0 (even number of zeroes), 1, 4, 5, 6, 9. So, any perfect square cannot end with odd number of zeroes, 2, 3, 8.

(ii) Digital sum of any perfect square can be = 1, 4, 9.

Digital sum of only $0^2 = 0$. None of the other numbers can have digital sum = 0.

(iii) $88^2 = 7744$ is the only number of the format $(aa)^2 = bbcc$

(iv) $38^2 = 1444$ gives us the largest number of times a non-zero number can be there at the end.

Maximum number of times any non-zero number can be there at the end = 3 (4 occurs thrice at the end).

(v) Difference between perfect squares of any two consecutive natural numbers is an odd number equal to 2n + 1, where, n is the 1st number out of the two numbers taken.

So, $26^2 - 25^2 = 2 \times 25 + 1$ = An odd number

$100^2 - 99^2 = 2 \times 99 + 1$ = An odd number

This gives us an important method to find out $(n + 1)^2$ if n^2 is known.

For example, square of 35 = 1225, hence, square of 36 = $1225 + 2 \times 35 + 1 = 1225 + 71 = 1296$.

(vi) For any non-negative value of x

$x^2 > x$ for $x > 1$. For example $8^2 > 8$.

$x^2 < x$ for $x < 1$. For example $0.8^2 < 0.8$

$x^2 = x$ for $x = 0$ and 1. This goes like $1^2 = 1$ and $0^2 = 0$

Worked-out Example 26 How many perfect squares can be formed by using all the digits 1, 2 and 3, each of the digits given four times?

☝ ***Solution*** We have 1111, 2222, 3333. Any number formed by using all these digits will have digital sum = 1 + 1 + 1 + 1 + 2 + 2 + 2 + 2 + 3 + 3 + 3 + 3 = 2 + 4 = 6. We have already seen that digital sum of any perfect square cannot be equal to 6, hence, no perfect square can be made.

SIMPLIFICATION

When a series of calculation or operation signs like '+', '–', '×', '÷' are given, a precedence of operation is to be followed.

For example, 2 + 2 ÷ 2. If we do addition first and then division, then this question becomes (2 + 2) ÷ 2 = 4 ÷ 2 = 2

If we do division first and then addition, then this question becomes 2 + (2 ÷ 2) = 2 + 1 = 3

We know this precedence popularly as—VBODMAS.

V -------- Vinculum
B -------- Brackets–in the order (),{},[]

O -------- Of
D -------- Division
M -------- Multiplication
A -------- Addition
S -------- Subtraction

Types of Brackets—there are three standard types of brackets: Parentheses ( ), Square brackets [ ], Curly brackets { }.

BODMAS

B brackets
O of
DM divide/multiply
AS add/subtract

After we have done Vinculum calculation,

(i) 1st we do the bracket calculation.
(ii) Then we do the "of" calculation.
(iii) Next, we do division or multiplication. Division and multiply can be done interchangeably.
(iv) Finally we do addition or subtraction. Addition and subtraction can be done interchangeably.

Worked-out Example 27 Do the following simplifications.

(a) $5 + 7 \times 6$

(b) $90 - 8 \times 7$

(c) $90 \div 10 - 5 \times 1$

(d) $65 - 3 \times 4 + 3 \times 2$

(e) $40 \div (2 + 2 \times (10 - 6))$

Solution

(a) 1st we will do the multiplication and then addition
$5 + 7 \times 6 = 5 + 42 = 47$

(b) 1st we will do the multiplication and then subtraction.
$90 - 8 \times 7 = 90 - 56 = 34$

(c) This question involves three operations.
1st we will do division, then multiplication (or vice versa) and then subtraction.
$90 \div 10 - 5 \times 1 = 9 - 5 = 4$

(d) $65 - 12 + 6 = 59$

(e) $40 \div (2 + 2 \times (10 - 6)) = 40 \div (2 + 2 \times 4) = 40 \div (2 + 8) = 40 \div 10 = 4$

TEST YOUR LEARNING 1

1. Do the following simplification:

1. $((3 + 5) \times 3 - 20) \times 2 =$

2. $6 \times ((2 + 3) \times 2 - 4) =$

3. $(5 - 3) \times 6 + (20 - 2) \div 6 =$

4. $30 \div (((3 + 3) \times 2 + 3) \div 3) =$

PRACTICE EXERCISE 1

1. Find the value of missing figure (?) in the following question:

$$7\frac{1}{2}-\left[2\frac{1}{4}\div\left\{1\frac{1}{4}-?\left(1\frac{1}{2}-\frac{1}{3}-\frac{1}{6}\right)\right\}\right]=3$$

1. $\frac{1}{4}$ **2.** $\frac{3}{4}$ **3.** $\frac{4}{3}$ **4.** 1

5. None of these

2. The simplification of $\frac{0.8\times0.8\times0.8-0.5\times0.5\times0.5}{0.8\times0.8+0.8\times0.5+0.5\times0.5}$ gives

1. 0.8 **2.** 0.24 **3.** 0.3 **4.** 0.13

5. 0, 23

3. If we multiply a fraction by itself and divide the product by its reciprocal, the fraction thus obtained is $18\frac{26}{27}$. The fraction is

1. $\frac{8}{27}$ **2.** $2\frac{2}{3}$ **3.** $1\frac{1}{3}$ **4.** 1

5. None of these

4. Between two fractions $\frac{1}{2}$ and $\frac{1}{8}$, how many fractions are there in all?

1. 3 **2.** 0 **3.** 8 **4.** 16

5. Infinite

5. The value of $\left(2-\frac{1}{3}\right)\left(2-\frac{3}{5}\right)\left(2-\frac{5}{7}\right)...\left(2-\frac{999}{1001}\right)$ is

1. $\frac{1003}{3}$ **2.** $\frac{1003}{1001}$ **3.** $\frac{1}{1001}$ **4.** 1

5. None of these

6. If the fractions $\frac{9}{13},\frac{2}{3},\frac{8}{11},\frac{5}{7}$ are arranged in ascending order, then the correct sequence is

1. $\frac{9}{13},\frac{2}{3},\frac{8}{11},\frac{5}{7}$ **2.** $\frac{2}{3},\frac{9}{13},\frac{5}{11},\frac{8}{7}$ **3.** $\frac{2}{3},\frac{8}{11},\frac{5}{7},\frac{9}{13}$ **4.** $\frac{5}{7},\frac{8}{11},\frac{2}{3},\frac{9}{13}$

7. $\dfrac{(0.6)^0-(0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1}\left(\frac{3}{2}\right)^3-\left(\frac{1}{3}\right)^{-1}}$

1. $-\frac{3}{2}$ **2.** $-\frac{1}{2}$ **3.** $\frac{2}{3}$ **4.** $\frac{3}{2}$

5. 1

8. Arrange the following in ascending order of values:

$$\frac{87}{83},\left(\frac{87}{83}\right)^2,\frac{87}{89},\left(\frac{87}{89}\right)^2$$

1. $\frac{87}{89}, \frac{87}{89}, \left(\frac{87}{89}\right)^2, \left(\frac{87}{83}\right)^2$ **2.** $\frac{87}{83}, \frac{87}{89}, \left(\frac{87}{83}\right)^2, \left(\frac{87}{89}\right)^2$

3. $\left(\frac{87}{89}\right)^2, \frac{87}{89}, \frac{87}{83}, \left(\frac{87}{83}\right)^2$ **4.** $\left(\frac{87}{89}\right)^2, \frac{87}{83}, \frac{87}{89}\left(\frac{87}{89}\right)^2$

9. Which of the following is the lowest?

1. $\frac{1}{4^2}+\frac{1}{5^2}+\frac{1}{2^2}$ **2.** $\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{6^2}$ **3.** $\frac{1}{3^2}+\frac{1}{3^2}+\frac{1}{5^2}$ **4.** $\frac{1}{7^2}+\frac{1}{2^2}+\frac{1}{2^2}$

10. Find the natural number from the given options, which when added to the numerator and denominator of $\frac{2}{3}$, will result in a fraction nearest to $\frac{13}{15}$.

1. 6 **2.** 5 **3.** 4 **4.** 3

5. 8

11. If p is a positive fraction less than 1, then

(i) $\frac{2}{p} - p$ is a positive number (ii) p^2 is less than p

(iii) $\frac{1}{p}$ is a positive integer (iv) $\frac{1}{p}$ is less than 1

How many of the statements are true?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. 0

12. $2^8 \times 3^6$ is divisible by

1. $2^7 \times 3^7$ **2.** $2^6 \times 3^5$ **3.** $2^4 \times 3^7$ **4.** $2^5 \times 3^8$

5. None of these

13. The expression, $\frac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$ simplifies to

1. (x –1) **2.** (x – 2) **3.** (x – 7) **4.** 1/(x – 7)

5. None of these

14. Consider the following statements:

(a) If $a^x = b$, $b^y = c$, $c^z = a$, then xyz = 0.

(b) If $p = a^x$, $q = a^y$, $(p^y q^x)^z = a^z$, then xyz = 1.

(c) If $x^a = y^b = z^c$ and ab + bc + ca = 0, then xyz = 1.

Out of these statements:

1. (a) and (b) are correct. **2.** (b) and (c) are correct

3. Only (a) is correct. **4.** (a) and c are correct.

5. none is correct

15. If r = –2, then $r^4 + 2r^3 + 3r^2 + r$ =?

1. –8 **2.** –4 **3.** 0 **4.** 10

5. 2

16. If $\frac{x^4}{x^9} = 5^{-5}$, then x is equal to

1. 5 **2.** 4 **3.** 9 **4.** 6

5. None of these

17. If $a * b = \frac{ab}{a+b}$, *find* $3*(3*-1)$

1. –3 **2.** –1.5 **3.** 2/3 **4.** 3 **5.** –1

18. Which of the given numbers is the greatest?

1. $6\sqrt[3]{5}$ **2.** $8\sqrt[3]{2}$ **3.** $2\sqrt[3]{130}$ **4.** $\sqrt[3]{900}$

PRACTICE EXERCISE 2

1. For how many prime numbers N, N and N + 2 will be prime numbers?

1. 1 **2.** 2 **3.** More than 1 but finite

4. Infinite

2. For how many prime numbers P, P + 10 and P + 14 will be prime numbers?

1. 1 **2.** 2 **3.** More than 1 but finite

4. Infinite

3. How many three digit numbers are divisible by 3 but not divisible by 2?

1. 300 **2.** 450 **3.** 150 **4.** None of these

4. How many three digit numbers are divisible by 2 but not divisible by 3?

1. 300 **2.** 450 **3.** 150 **4.** None of these

5. What is the sum of all the natural number between 200 and 400 that are divisible by 11?

1. 5423 **2.** 5434 **3.** 5445 **4.** 5456

6. A number is said to be 'right prime' if despite dropping the left most digits successively, number continues to be a prime number. For example 223 is a right prime because despite dropping 2 from left most part, we obtain 23 as the prime number. Next even after dropping 2 from left, 3 is prime.

How many two digit numbers are 'right prime'?

7. N = 1! + 2! + 3! + 4! + ……+K! where K is a natural number less than 1000. For how many different values of K, will N be a perfect square?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. More than 3

8. P is a prime number greater than 37. Then the largest number that will always divide $(P - 1)\times(P + 1)$ is

1. 32 **2.** 16 **3.** 8 **4.** 4

5. 24

Directions for questions (9–18): Read the information given below and solve the questions based on it.

Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4—if question cannot be answered even by using both the statements.

9. Is x an even no?

A. $4x + 3y$ is an even no.

B. $3x + 4y$ is an even no.

10. What is a two digits number?

A. The number obtained by interchanging the digits is smaller than the original number by 63.

B. Sum of the digits is 11.

11. X, Y, and Z are integers, is X an odd number?

A. An odd number is obtained when X is divided by 5.

B. (X + Y) is an odd number.

12. N is a natural number. Is N a perfect square?

A. When N is divided by 100, remainder obtained is 2.

B. Total number of factors of N is odd.

1. 1 **2.** 3 **3.** 2 **4.** 4

5. 5

13. Which of a, b, c and d is/are odd?

A. ad + bc is odd.

B. ac + bd is odd.

14. What is the value of X?

A. X and Y are unequal even integers, less than 10, and X/Y is an odd integers.

B. X and Y are even integers, each less than 10, and product of X and Y is 12.

15. If m and n are consecutive positive integers, is $m > n$?

A. $m - 1$ and $n + 1$ are consecutive positive integers.

B. m is an even integer.

16. Is Q a positive number?

A. $4Q + 24 > 0$ B. $4Q - 24 < 0$

17. Is $X + Y + Z > A + B$?

A. $X > A + B$ B. $Y > A + B$

18. A, B, C, D, E and F are six integers such that $E < F$, $B > A$, $A < D < B$. C is the greatest integer. Is A the smallest integer?

A. $E + B < A + D$ B. $D < F$

19. On being asked about the his age, Siddharth replied that if his age was increased by 6 years, the square root of the resultant number obtained is equal to his age decreased by 6 years. What is Siddharth's age?

1. 9 **2.** 10 **3.** 19 **4.** 30

5. 3

20. How many natural numbers up to 200 are divisible by 7?

1. 28 **2.** 29 **3.** 27 **4.** 100

5. None of these

21. How many natural numbers up to 500 are divisible by 23?

1. 23 **2.** 27 **3.** 21 **4.** 19

5. None of these

22. How many natural numbers up to 200 are divisible 2 and 3 both?

1. 35 **2.** 33 **3.** 29 **4.** 27

5. None of these

23. How many natural numbers between 100 and 300 are divisible by 11?

1. 11 **2.** 10 **3.** 12 **4.** 18

5. 13

24. When a certain natural number is multiplied by 7, the product obtained consists only of fives. Find the least value of such a natural number.

1. 78465 **2.** 79365 **3.** 755 **4.** 7965

5. None of these

25. How many natural numbers between 150 and 500 are divisible by 2, 3 and 7 together?

1. 9 **2.** 8 **3.** 10 **4.** 11

5. 12

26. The sum of two numbers is 100 and their difference is 37. The difference of their square is

1. 37 **2.** 370 **3.** 463 **4.** 3700

5. More than one value possible

27. In an examination, a student answered 15 questions correctly and secured 40 marks. If there were two types of questions (2 marks and 4 marks questions), how many questions of 2 marks did he answer correctly?

1. 5 **2.** 10 **3.** 20 **4.** 40

5. None of these

28. A lady went to the market with a few one-rupee notes and a few 20-paise coins. When she returned, she had as many one-rupee notes as she originally had 20-paise coins and vice versa. She actually came back with about one-third of what she had gone with. How much did she spend in the market?

1. Rs 14.40 **2.** Rs 14.70 **3.** Rs 15.50 **4.** Rs 17.40

5. None of these

29. A student was asked to divide a number by 17/8. Instead, he actually multiplied it by 17/8 and hence got 225 more than the expected answer. What was the expected answer?

1. 126 **2.** 136 **3.** 64 **4.** 133

5. 85

30. There are two examination halls, P and Q. If 10 students are sent from P to Q, then the number of students in each hall is the same. If 20 students are sent from Q to P, then the number of students in P is double of that in Q. The number of students in halls P and Q respectively are

1. 100, 80 **2.** 80, 60 **3.** 70, 50 **4.** 60, 40

5. 70, 40

31. At the first stop on his route, a driver unloaded 2/5 of the packages in his van. After he unloaded another three packages at his next stop, 1/2 of the original number of packages remained. How many packages were in the van before the first unloading?

1. 25 **2.** 10 **3.** 30 **4.** 36

5. None of these

32. Let a; b; c; d be positive integers. Then the product (a–b) (a–c) (a–d) (b–c) (b–d) (c–d) is definitely divisible by

(i) 2 (ii) 3 (iii) 8 (iv) 12

1. (i), (ii) and (iv) only **2.** (i) and (iii) only

3. (i) only **4.** (ii) only

5. (i), (ii), (iii) and (iv) only

33. Let a and b be natural numbers and such that ab + 1 be divisible by 24. Then a + b is divisible by

(i) 2 (ii) 3 (iii) 8 (iv) 12

1. (i), (ii) and (iv) only **2.** (i) and (iii) only

3. (i) only **4.** (ii) only

5. (i), (ii), (iii) and (iv) only

34. What is the value of $\frac{1}{3-\sqrt{2}}+\frac{1}{3+\sqrt{2}}$?

1. $\frac{5}{7}$ **2.** $\frac{6}{7}$ **3.** 1 **4.** $\frac{8}{7}$

5. $\frac{4}{7}$

35. m and n are natural numbers with m × n = 160, 000. If neither m nor n is a multiple of 10, then what is the value of m + n?

1. 784 **2.** 824 **3.** 881 **4.** 924

5. 961

Directions for questions (36–40): Convert the following numbers into fraction form.

36. 3.891

37. 3.891891891891.....

38. 3.891919191.....

39. 3.89111111.....

40. $\sqrt{3.891}$

PRACTICE EXERCISE 3

1. Let A be a set with 8 elements. Find the maximum number of distinct 3-element subsets of A such that the intersection of any two of them is not a 2-element set.

1. 28 **2.** 56 **3.** 112 **4.** 14

5. 168

2. Given are:

(i) a and b are natural numbers

(ii) $\frac{5}{a}+\frac{1}{b}=\frac{1}{16}$

If a is an odd integer less than 100, how many pairs of values of a and b will satisfy the given conditions?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. 5

3. When 96 is added to a perfect square, another perfect square is obtained. How many such pairs of perfect squares exist?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

4. When 101 is added to a perfect square, another perfect square is obtained. How many such pairs of perfect squares exist?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

5. How many three digit natural numbers are there which will have even digits even number of times (count 0 time also)?

1. 300 **2.** 350 **3.** 400 **4.** 450

5. None of these

6. If $a_1 = 1$ and $a_{n+1} - 3a_n + 2 = 4n$ for every positive integer n, then a_{100} equals

1. $3^{99} - 200$ **2.** $3^{99} + 200$ **3.** $3^{100} - 200$ **4.** $3^{100} + 200$

5. None of these

7. Let S be a set of positive integers such that every element n of S satisfies the conditions:

(i) $1000 \le n < 1200$

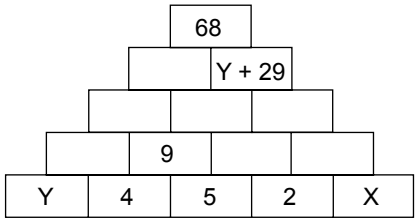
(ii) Every digit in n is odd

Then how many elements of S are divisible by 3?

1. 9 **2.** 10 **3.** 11 **4.** 12

5. 13

8. In the figure, number in any cell is obtained by adding two numbers in the cells directly below it. For example, 9 in the second row is obtained by adding the two numbers 4 and 5 directly below it. The value of X – Y is



1. 2 **2.** 3 **3.** 4 **4.** 5
5. 6

Directions for questions (9–10): Go through the information below and solve the questions based on it.

A, B, C, D, E and F are six positive integers such that,

$B + C + D + E = 4A$

$C + F = 3A$

$C + D + E = 2F$

$F = 2D$

$E + F = 2C + 1$

A is a prime number between 12 and 20.

Answer the following questions.

9. The value of F is

1. 14 **2.** 16 **3.** 20 **4.** 24
5. 28

10. Which of the following must be true?

1. D is the lowest integer and D = 14

2. C is the greatest integer and C = 23

3. B is the lowest integer and B = 12

4. F is the greatest integer and F = 24

5. A is the lowest integer and A = 13

11. What is the sum of all the natural numbers between 200 and 400 which are divisible by 11?

1. 5423 **2.** 5424 **3.** 5435 **4.** 5445
5. 5465

12. How many natural numbers P are there such that P, P + 8, P + 22 are all prime numbers?

1. 0 **2.** 1 **3.** 2
4. More than 2 but finite **5.** Infinte

Directions for questions (13–14): Read the passage given below and solves the questions based on it.

(i) A, B, C, D and E are positive integers.
(ii) A + B + C is odd and D × E is even.

13. How many of the above numbers, at the most, could be odd?

1. 5 **2.** 2 **3.** 3 **4.** 4

5. Cannot be determined

14. How many of the above numbers at least, are odd?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. Cannot be determined

15. The positive integers A, B, A – B and A + B are all prime numbers. The sum of these four prime numbers is a

1. Multiple of 5 **2.** Prime **3.** Even **4.** Multiples of 7
5. Multiple of 3

16. A and B are two prime numbers such that their sum A + B and their difference A – B each is prime. How many set of values of A and B are possible?

1. None **2.** 1 **3.** 2

4. More than 2 but finite **5.** None of these

17. When a two digit number is reversed, another two digit number is obtained. Difference between original number and resulting number = 63. How many such two digit natural numbers are possible?

1. 1 **2.** 2 **3.** 3 **4.** 5

5. 6

18. Four digits of the number 29138576 are omitted so that the result is as large as possible. The largest omitted digit is

1. 9 **2.** 8 **3.** 7 **4.** 6

5. 5

Directions for questions (19–20): A statement is followed by three conclusions. Select the answer from the following options.

Mark 1. Using the given statement, only conclusion I can be derived.

Mark 2. Using the given statement, only conclusion II can be derived.

Mark 3. Using the given statement, only conclusion III can be derived.

Mark 4. Using the given statement, all conclusions can be derived.

Mark 5. Using the given statement, none of the three conclusions I, II and III can be derived.

19. An operation "#" is defined by $a \# b = 1 - \frac{b}{a}$

Conclusion I. $(2 \# 1) \# (4 \# 3) = -1$

Conclusion II. $(3 \# 1) \# (4 \# 2) = -2$

Conclusion III. $(2 \# 3) \# (1 \# 3) = 0$

20. A, B, C and D are whole numbers such that,

$A + B + C = 118$

B + C + D = 156

C + D + A = 166

D + A + B = 178

Conclusion I. A is the smallest number and A = 21.

Conclusion II. D is the largest number and D = 88.

Conclusion III. B is the largest number and B = 56.

21. Rohan is typing natural numbers on a computer. How many times is he required to press the keys of the keyboard in order to type the first 200 natural numbers?

1. 493 **2.** 492 **3.** 495 **4.** 489

5. None of these

Directions for questions (22–25): In each of the questions, a statement is given. State whether True or False.

22. $5^{12} + 2^{10}$ is a composite number.

23. For any natural number value of n, (n + 1) (n + 30) is always even.

24. $\frac{5^{125}-1}{5^{25}-1}$ is a composite number.

25. Difference between the squares of any two consecutive odd numbers is a multiple of 8.

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING 1

1. **1.** 8 **2.** 36 **3.** 15 **4.** 6

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	**3**	**2**	**5**	**1**	**2**	**1**	**3**	**3**	**2**
11.	12.	13.	14.	15.	16.	17.	18.		
2	**2**	**2**	**2**	**4**	**1**	**1**	**1**		

1. Ans. 2

Let missing figure = x

$\frac{15}{2}-\left[\frac{9}{4}\div\left\{\frac{5}{4}-x\right\}\right]=3,$ or, $\frac{15}{2}-3=\frac{9/4}{5/4-x}$

$\frac{9}{2}=\frac{9}{5-4x}$,

or, $5 - 4x = 2$

$\Rightarrow x=\frac{3}{4}$

2. Ans. 3

We know that, $\frac{a^3-b^3}{a^2+ab+b^2}=a-b$

So the given expression = 0.8 – 0.5 = 0.3.

3. Ans. 2

Let the fraction be $\frac{a}{b}$. Then,

$\left[\frac{a}{b}\times\frac{a}{b}\right]\div\frac{b}{a}=18\frac{26}{27}=\frac{512}{27}$ or, $\left(\frac{a}{b}\right)^3=\left(\frac{8}{3}\right)^3$

So, $\frac{a}{b}=\frac{8}{3}=2\frac{2}{3}$

4. Ans. 5, There are infinite fractions between any two fractions.

5. Ans. 1

Given product $=\frac{5}{3}\times\frac{7}{3}\times\frac{9}{7}.....\times\frac{1003}{1001}=\frac{1003}{3}.$

6. Ans. 2

$\frac{9}{13}=0.692,\frac{2}{3}=0.666$

$\frac{8}{11}=0.727,\frac{5}{7}=0.714.$

7. Ans. 1

$$\frac{1-\left(\frac{1}{10}\right)^{-1}}{\left(\frac{2^3}{3}\right)\left(\frac{3}{2}\right)^3+(-3)}=\frac{1-10}{\frac{2^3}{3}\cdot\frac{3^3}{2^3}-3}$$

$$=\frac{-9}{9-3}=-\frac{3}{2}.$$

8. Ans. 3

9. Ans. 3

10. Ans. 2

Let $\frac{2+K}{3+K}=\frac{13}{15}\Rightarrow 2K=9\Rightarrow K=4.5\approx 5.$

11. Ans. 2

12. Ans. 2

13. Ans. 2

Given expression $=\frac{(x-1)(x-2)(x-7)(x-2)}{(x-7)(x-2)(x-1)}=x-2$

14. Ans. 2

(b) and (c) are correct.

15. Ans. 4

$r^4 + 2r^3 + 3r^2 + r = 16 + 2 \times (-8) + 3 \times 4 - 2 = 10.$

16. Ans. 1

17. Ans. 1

Given expression $=3*\left(\frac{3\times(-1)}{3+(-1)}\right)=3*\left(\frac{-3}{2}\right)$

$$=\frac{3\times\left(\frac{-3}{2}\right)}{3+\left(\frac{-3}{2}\right)}=\frac{\frac{-9}{2}}{\frac{3}{2}}=-3$$

18. Ans. 1

$(6\times 3\sqrt{5})^3=216\times=1080$

$(8\times 3\sqrt{2})^3=512\times 2=1024$

$(2\times 3\sqrt{130})^3=8\times 130=1040$

$(3\sqrt{900}^3)=900.$

PRACTICE EXERCISE 2

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	**1**	**3**	**1**	**3**	**5**	**3**	**5**	**4**	**3**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
1	**2**	**4**	**4**	**1**	**4**	**4**	**4**	**2**	**1**
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
3	**2**	**4**	**2**	**1**	**4**	**2**	**1**	**2**	**1**
31.	32.	33.	34.	35.					
3	**1**	**5**	**2**	**3**					

1. It can be observed that there are actually infinite pairs of prime numbers having difference of 2.

2. Some of the unit digit of P can be easily eliminated. For example, P = 1 (P + 4 will become divisible by 5), 2, 4, 5, 6, 8.

So the possible unit digits of P are = 3, 7, 9.

For P = 3, P + 10 and P + 14 = 13, 17. For all other prime values of P, one of the two (P + 10 and P + 14) will be divisible by 3.

3. Three digit numbers divisible by 3 = 300. Half of these will be divisible by 2, and other half not divisible. So, 3-digit numbers divisible by 3 but not divisible by 2 = 300 – 150 = 150.

4. Three digit numbers divisible by 2 = 450. Out of these 1/3rd of the numbers will be divisible by 3, and remaining numbers will not be divisible by 3. So, 3-digit numbers divisible by 2 but not divisible by 3 = 450 – 150 = 300.

7. Unit digit of 1! + 2! + 3! + 4! = 3, and unit digit of all the numbers beyond this like 1! + 2! + 3! + 4! + 5! etc., = 3 (since unit digit of 5! = unit digit of 6! = 0). We know that unit digit of a perfect square cannot be equal to 3, hence, any number equal to or beyond 1! + 2! + 3! + 4! cannot be perfect square. Let us check for the starting numbers,

1! = 1 = perfect square

1! + 2! = 3 = Not a perfect square

1! + 2! + 3! = 9 = Perfect square. Hence, only two values of K are possible for perfect square.

8. Since, P is a prime number greater than 37, it has to be odd. So both (P – 1) and (P + 1) are even numbers, that also consecutive even numbers—product of these two consecutive even numbers will be divisible by 8 always (one of these two will be a multiple of 4, and another of 2). It may or may not be divisible by 16 or 32.

Next Prime numbers are of the format 6N ± 1. If P = 6N + 1, then, P – 1 = 6N and if P = 6N –1, then, P + 1 = 6N. So we conclude that (P – 1)×(P + 1) will be a multiple of 3 also (we are not writing it a multiple of 6 since we have already extracted the powers of 2 earlier).

Hence, the largest number that will always divide (P – 1)×(P + 1) = 8 × 3 = 24.

9. Using statement A alone, if 4x + 3y is an even number then both 4x and 3y should be either odd or both should be even. Both 4x and 3y cannot be odd(as 4x is even) and hence both of 4x and 3y are even.

Now even if 4x is even, then we cannot conclude that x is even or odd.

Using statement B alone, if 3x + 4y is an even number then both 3x and 4y should be either odd or both should be even.

If each of 3x and 4y is even, then x can take values like x = 4/3, which is neither odd nor even.

If each of 3x and 4y is odd, then 3x is odd. But using this we cannot conclude that x is also odd. X may be a fraction also like x = 1/3 or even it can be an odd number.

Answer—

10. Using statement A alone gives that the difference between the digits of the number is 7. This gives the possibility of many numbers like 81, 92 etc.

Using statement B alone also gives the possibility of many numbers.

Using both the statements together, we get a unique number.

Answer—

11. Using statement A alone, we get that the quotient obtained when x is divided by an odd number is odd. Hence, x has to be odd.

Using statement B alone, one of x and y will be odd and other one will be even. So we cannot determine which of x or y is even?

Answer—

12. Using statement A, if the remainder obtained when N is divided by 100 is 2, then the unit digit of N is 2. Hence, N cannot be a perfect square.

Using statement B, since the total number of factors of N is odd, hence N is a perfect square. [Total number of factors of any perfect square is odd and vice-versa is also true.]

Hence, the answer is—option

13. Option 4 is the answer.

14. Option 4 (Do not forget to take the negative values of X and Y)

15. This question can be solved by using statement A alone but not by using statement B alone. Hence, option 1 is the answer.

16. A. $4Q + 24 > 0$

Or, $4Q > -24$, or, $Q > -6$

Hence, Q can be negative or positive or zero as well.

B. $4Q - 24 < 0$

Or, $4Q < 24$, or, $Q < 6$

Hence, Q can be negative or positive or zero as well.

Answer—4

17. Using statement A or statement B alone will not give the unique answer as Y and/or Z may have negative values. Even if we use both the statements together, since we do not know the sign of Z (which may be negative), we cannot answer this question.

Hence, answer is option 4.

20. Ans. 1.

Quotient obtained when 200 is divided by 7 is 28.

21. Ans. 3.

Quotient obtained when 500 is divided by 23 is 21.

22. Ans. 2.

Quotient obtained when 200 is divided by the L.C.M. of 2 and 3, i.e., 6 is 33.

23. Ans. 4.

Quotients obtained when 300 and 100 are divided by 11 are 27 and 9. Hence, between 300 and 100, there are 27 –9 i.e., 18 numbers.

24. Divide 555555 by 7 (None of 5, 55, 555, 5555, 55555 is divisible by 7).

$7 \mid 555555$

$\quad\quad\quad\quad\quad\quad$

$\quad\quad\quad 79365$

So the least number is 79365.

25. A number divisible by 2, 3 and 7 is divisible by their L.C.M. i.e., 42. Up to 100, there are 2 numbers divisible by 42. Up to 500, there are 11 numbers divisible by 42.

So, between 100 and 500, number of numbers divisible by 42 = 11 – 2 = 9.

26. Let the numbers be a and b.

Given is a + b = 100 and a – b = 37

So, $a^2 - b^2 = (a + b)(a - b) = 100 \times 37 = 3700$

27. Suppose number of 2-mark questions = x

Suppose number of 4-mark questions = y

x + y = 15

2 x + y = 40 $\Rightarrow$ x = 10, y = 5

28. Originally, the lady had, say, x One-rupee notes and y 20-paise coins,
On returning, she had y one-rupee notes and x 20-paise coins.

The balance was Rs $\frac{1}{3}\left(x+\frac{y}{5}\right)$

$$\Rightarrow \left(x+\frac{y}{5}\right)-\left(y+\frac{x}{5}\right)=\frac{1}{3}\left(x+\frac{y}{5}\right)$$

$$\Rightarrow \frac{x}{y}=\frac{13}{7}$$

If we take x = 13 and y = 17, then originally she had Rs 14.40 with her.

$\Rightarrow$ On returning, she had Rs 9.60. It means that the lady spent Rs 4.80 or a multiple of Rs 4.80 i.e., Rs 14.40.

29. Assume x is the number.

$x \times \frac{17}{8} - \frac{x}{17/8} = 225 \Rightarrow 225x = 136 \times 225$. So x = 136.

30. Go through options. x = 100, y = 80.

31. Suppose there were x packages originally.

$$\frac{x}{2}-\frac{2x}{5}=3\Rightarrow\frac{5x-4x}{10}=3\Rightarrow\frac{x}{10}=3\ \ x=30$$

33. a + b will be divisible by 24. Hence, all the four numbers will divide a + b.

34. $$\frac{1\times\left(3+\sqrt{2}\right)}{\left(3-\sqrt{2}\right)\left(3+\sqrt{2}\right)}+\frac{1\times\left(3-\sqrt{2}\right)}{\left(3+\sqrt{2}\right)\left(3-\sqrt{2}\right)}=\frac{\left(3+\sqrt{2}\right)+\left(3-\sqrt{2}\right)}{9-2}=\frac{6}{7}$$

36. $\frac{3891}{1000}$

37. $3.891891891891.....=3\frac{891}{999}=\frac{3888}{999}$

38. $P = 3.891919191.....\Rightarrow 10\,P = 38.9191..... = 38\frac{91}{99}=\frac{3853}{99}$

Hence, $P = \frac{3853}{990}$

39. $P = 3.891111.....\Rightarrow 100\,P = 389.111..... = 389\frac{1}{9}=\frac{3502}{9}$

Hence, $P=\frac{3502}{900}$

40. It cannot be converted since its irrational number.

PRACTICE EXERCISE 3

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	**2**	**5**	**1**	**4**	**3**	**1**	**3**	**5**	**3**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
4	**1**	**4**	**1**	**2**	**2**	**2**	**5**	**5**	**2**
21.	22.	23.	24.	25.					
2	**True**	**True**	**True**	**True**					

1. There are going to be ${}^{8}C_{3}$ sets.
An example with 8 sets is:
123; 145; 167; 246; 278; 348; 357; 568

3. $A^2 = B^2 + 96 \Rightarrow A^2 - B^2 = 96 \Rightarrow (A - B)(A + B) = 96$

Both of A and B should be even numbers for this calculation to hold good.
$(A - B)(A + B) = 48 \times 2$ or $(A - B)(A + B) = 24 \times 4$ or $(A - B)(A + B) = 16 \times 6$ or $(A - B)(A + B) = 12 \times 8$

These four combinations will give four different sets.

$A + B = 48$, and $A - B = 2 \Rightarrow A = 25$, $B = 23$ and so on.

4. $A^2 = B^2 + 101 \Rightarrow A^2 - B^2 = 101 \Rightarrow (A - B)(A + B) = 101$
Since, 101 is a prime number, hence, $A + B = 101$, and $A - B = 1$
$\Rightarrow A = 51$ and $B = 50$ is the only solution.

5. Half of the times it will be even and half of the times it will be odd. Hence, $900/2 = 450$.
Alternatively, we can use permutation and combination to solve this question;
We have four types of numbers:
EEO = $4 \times 5 \times 5 = 100$
EOE = $4 \times 5 \times 5 = 100$
OEE = $5 \times 5 \times 5 = 125$
OOO = $5 \times 5 \times 5 = 125$
Total = 450.

6. $a_1 = 1, \quad a_{n+1} - 3a_n + 2 = 4n$
$A_{n+1} = 3a_n + 4n - 2$
When, $n = 2$ then, $a^2 = 3 + 4 - 2 = 5$
When $n = 3$ then, $a_3 = 3 \times 5 + 4 \times 2 - 2 = 21$
From the options, we get an idea that a_n can be expressed in a combination of some power of3 & some multiple of 100.

(1) $3^{99} - 200$; tells us that a_n could be: $3^{n-1} - 2 \times n$; but it does not fit a_1 or a_2 or a_3.

(2) $3^{99} + 200$; tells us that a_n could be : $3^{n-1} + 2 \times n$; again not valid for a_1, a_2 etc.

(4) $3^{100} + 200$; tells $3^n + 2n$: again not valid.

So, (3) is the correct answer.

7. The 100th and 1000th position value will be only 1.
Now, the possibility of unit and tens digits are (1, 3), (1, 9), (3, 1), (5, 5), (3, 7), (7, 3), (7, 9), (9, 1), (9, 7). Hence, 9 values.

9–10

$B + C + D + E = 4A$ ----------(i)
$C + F = 3A$ ----------(ii)
$C + D + E = 2F$ ----------(iii)
$F = 2D$ ----------(iv)
$E + F = 2C + 1$ ----------(v)
From (iii) and (iv), we get, $C + E = 3D$ ----------(vi)
From (iv) and (v) we get,
$E = 2C - 2D + 1$ ----------(vii)
$\Rightarrow 3C - 2D + 1 = 3D \Rightarrow 3C + 1 = 5D$ ----------(viii)
From (iv) we get, $3C + 1 = 5F/2 \Rightarrow (6C + 2)/5 = F$ ----------(ix)
From (ix) and (ii) we get, $11C + 2 = 15A$ ----------(x)
Possible values of A = 13 or 17 or 19 and all the numbers are natural numbers.
So, using (x), we get integer value for C only when A is 17 $\Rightarrow$ A = 17 and C = 23
Using (ix), we get, F = 28 and using (viii), we get D = 14
From (vii), we get E = 19 and using (i), we get B = 12

9. Option 5.

10. Option 3.

13. A, B and C would all be odd. Either D or E could also be odd. (One of these, namely, either E or D, must be even). Hence, four numbers, at the most, could be odd.

14. We are trying to arrive at the minimum possible value of odd numbers. Only one number among A, B and C need be odd. Both D and E could be even. (That is, neither D, nor E, need to be odd). Hence, atleast one number is odd.

A = 5, B = 2 is the only solution.

For any other value of A and B, either A – B or A + B will be divisible by 3. Even in this case A – B = 3 which is divisible by 3, but coincidentally this is divisible by 3 *only.*

17. Given is $9(x - y) = 63 \Rightarrow x - y = 7$

Sets of (x, y) = (7, 0), (8, 1), (9, 2).

However, we cannot take set (7, 0) as interchanging this will lead to '07' which is not a two digit number. Hence, number of sets possible = 2.

19. Option 5. None can be derived.

20. Option 2. Only conclusion II can be derived.

21.

		Number of digits
Total Number of Single Digit Numbers	9	9
Total Number of Double Digit Numbers	90	180
Total Number of Triple Digit Numbers	101	303
		492

22. True. Hint – calculating the unit digit or last two digits will not answer this question. Unit digit = 9. Calculating the unit digit will not do as it does not tell us if N is prime. Last two digits = 25 + 24 = 49. This also does not answer the question.

23. True. One of (n + 1) or (n + 30) has to be even, for all the values of n. Hence even.

24. True

25. True. Write down any two odd numbers in the format of (a + b) (a – b). One of these numbers will be divisible by 4, and other by 2. Hence, always divisible by 8.

CAT QUESTIONS FROM THIS CONCEPT

CAT 2008

1. The integers 1, 2, ..., 40 are written on a blackboard. The following operation is then repeated 39 times: In each repetition, any two numbers, say a and b, currently on the blackboard are erased and a new number $a + b - 1$ is written. What will be the number left on the board at the end?

1. 820 **2.** 821 **3.** 781 **4.** 819

5. 780

2. Three consecutive positive integers are raised to the first, second and third powers respectively and then added. The sum so obtained is a perfect square whose square root equals the total of the three original integers. Which of the following best describes the minimum, say m, of these three integers?

1. $1 \leq m \leq 3$ **2.** $4 \leq m \leq 6$ **3.** $7 \leq m \leq 9$ **4.** $10 \leq m \leq 12$

5. $13 \leq m \leq 15$

CAT 2007

3. A confused bank teller transposed the rupees and paise when he cashed a cheque for Shailaja, giving her rupees instead of paise instead of rupees. After buying a toffee for 50 paise, Shailaja noticed that she was left with exactly three times as much amount as on the cheque. Which of the following is a valid statement about the cheque amount?

1. Over Rupees 22 but less than Rupees 23

2. Over Rupees 18 but less than Rupees 19

3. Over Rupees 4 but less than Rupees 5

4. Over Rupees 13 but less than Rupees 14

5. Over Rupees 7 but less than Rupees 8

4. Consider four digit numbers for which the first two digits are equal and also equal. How many such numbers are perfect squares?

1. 4 **2.** 0 **3.** 1 **4.** 3

5. 2

CAT 2006

5. If $x = -0.5$, then which of the following has the smallest value?

1. $2^{1/x}$ **2.** $1/x$ **3.** $1/x^2$ **4.** 2^x

5. $1/\sqrt{-x}$

6. The sum of four consecutive two-digit odd numbers, when divided by 10, becomes a perfect square. Which of the following can possibly be one of these four numbers?

1. 21 **2.** 25 **3.** 41 **4.** 67

5. 73

7. The number of employees in Obelix Menhir Co. is a prime number and is less than 300. The ratio of the number of employees who are graduates and above, to that of employees who are not, can possibly be:

1. 101 : 88 **2.** 87 : 100 **3.** 110 : 111 **4.** 85 : 98

5. 97 : 84

8. When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed?

1. 5 **2.** 6 **3.** 7 **4.** 8

5. 10

CAT 2005

9. For a positive integer n, let p_n, denote the product of the digits of n and S_n, denote the sum of the digits of n. The number of integers between 10 and 1000 for which $p_n + s_n = n$ is

1. 81 **2.** 16 **3.** 18 **4.** 9

CAT 2004

10. Suppose n is an integer such that the sum of digits on n is 2, and $10^{10} < n < 10^{11}$. The number of different values of n is

1. 11 **2.** 10 **3.** 9 **4.** 8

CAT 2003

Directions for questions (11–13): Go through the passage given below and solve the questions based on it.

The seven basic symbols in a certain numeral system and their respective values are as follows:

I = 1, V = 5, X = 10, L = 50, C = 100, D = 500 and M = 1000

In general, the symbols in the numeral system are read from left to right, starting with the symbol representing the largest value; the same symbol cannot occur continuously more than three times; the value of the numeral is the sum of the values of the symbols. For example, XXVII = 10 + 10 + 5 + I + 1 = 27. An exception to the left-to-right reading occurs when a symbol is followed immediately by a symbol of greater value; then the smaller value is subtracted from the larger.

For example, XLVI (50 — 10) + 5 + 1 = 46.

11. The value of the numeral MDCCLXXXVII is

1. 1687 **2.** 1787 **3.** 1887 **4.** 1987

12. The value of the numeral MCMXC IX is

1. 1999 **2.** 1899 **3.** 1989 **4.** 1889

13. Which of the following represents the numeral for 1995?

I. MCMLXXV

II. MCMXCV

III. MVD

IV. MVM

1. Only I and II

2. Only Ill and IV

3. Only II and IV

4. Only IV

14. An intelligence agency forms a code of two distinct digits selected from 0, 1, 2, ..., 9 such that first digit of the code is non-zero. The code, handwritten on a slip, can however potentially create confusion when read upside down—for example, the code 91 may appear as 16. How many codes are there for which no such confusion can arise?

1. 80 **2.** 78 **3.** 71 **4.** 69

15. If x and y are integers, then the equation, $5x + 19y = 64$ has

1. no solution for $x < 300$ and $y < 0$

2. no solution for $x > 250$ and $y > -100$

3. a solution for $250 < x < 300$

4. a solution for $-59 < y < -56$

16. If a, a + 2 and a + 4 are prime numbers, then the number of possible solutions for a is

1. one **2.** two **3.** Third

4. more than three

CAT 2002

17. Let x and y be positive integers such that x is prime and y is composite. Then,

1. y – x cannot be an even integer

2. xy cannot be an even integer

3. $\frac{(x+y)}{x}$ cannot be an even integer

4. None of these

18. Number S is obtained by squaring the sum of digits of a two-digit number D. If difference between S and D is 27, then the two-digit number D is

1. 24 **2.** 54 **3.** 34 **4.** 45

19. The owner of a local jewellery store hired three watchmen to guard his diamonds, but a thief still got in and stole some diamonds. On the way out, the thief met each watchman, one at a time. To each he gave $\frac{1}{2}$ of the diamonds he had then, and 2 more besides. He escaped with one diamond. How many did he steal originally?

1. 40 **2.** 36 **3.** 25 **4.** None of these

20. A rich merchant had collected many gold coins. He did not want anybody to know about him. One day, his wife asked, ' How many gold coins do we have?" After a brief pause, he replied, "Well! if I divide the coins into two unequal numbers, then 48 times the difference between the two numbers equals the difference between the squares of the two numbers." The wife looked puzzled. Can you help the merchant's wife by finding out how many gold coins the merchant has?

1. 96 **2.** 53 **3.** 43 **4.** None of these

CAT 2001

21. A child was asked to add first few natural numbers (i.e., 1 + 2 + 3 + ...) so long his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was

1. less than 10 **2.** 10 **3.** 15 **4.** more than 15

22. Let x, y and z be distinct integers x and y are odd and positive, and z is even and positive. Which one of the following statements cannot be true?

1. $y(x-z)^2$ is even **2.** $y^2(x-z)$ is odd

3. $y(x-z)$ is odd **4.** $z(x-y)^2$ is even

23. Of 128 boxes of oranges each box contains at least 120 and at most 144 oranges. The number of boxes containing the same number of oranges is at least:

1. 5 **2.** 103 **3.** 6

4. Cannot be determined

24. In a four-digit number the sum of the first 2 digits is equal to that of the last 2 digits. The sum of the first and last digits is equal to the third digit. Finally, the sum of the second and fourth digits is twice the sum of the other 2 digits. What is the third digit of the number?

1. 5 **2.** 8 **3.** 1 **4.** 4

25. Anil had to do a multiplication. Instead of taking 35 as one of the multipliers, he took 53. As a result, the product went up by 540. What is the new product?

1. 1050 **2.** 540 **3.** 1440 **4.** 1590

26. Three friends, returning from a movie, stopped to eat at a restaurant. After dinner, they paid their bill and noticed a bowl of mints at the front counter. Sima took one-third of the mints, but returned four because she had a monetary pang of guilt. Farida then took one-fourth of what was left but returned three for similar reason. Eden then took half of the remainder but threw two back into the bowl. The bowl had only 17 mints left when the raid was over. How many mints were originally in the bowl?

1. 38 **2.** 31 **3.** 41 **4.** None of these

27. All the page numbers from book are added, beginning at page 1. However, one page number was added twice by mistake. The sum obtained was 1000. Which page number was added twice?

1. 44 **2.** 45 **3.** 10 **4.** 12

28. Atul is given 158 in one-rupee denominations. He has been asked to allocate them into a number of bags such that any amount required between Re 1 and Rs 158 can be given by handing out a certain number of bags without opening them. What is the minimum number of bags required?

1. 11 **2.** 12 **3.** 13 **4.** None of these

29. In some code, letters a, b, c, d and e represent numbers 2, 4, 5, 6 and 10. We just do not know which letter represents which number. Consider the following relationships:

I. a + c = e, II. b – d = d and III. e + a = b

Which of the following statements is true?

1. b = 4, d = 2 **2.** a = 4, e = 6 **3.** b = 6, e = 2 **4.** a = 4, c = 6

CAT 2000

30. A change-making machine contains one-rupee, two-rupee and five-rupee coins. The total number of coins is 300. The amount is Rs 960. If the numbers of one-rupee coins and two-rupee coins are interchanged, the value comes down by Rs 40. the total number of five-rupee coins is

1. 100 **2.** 140 **3.** 60 **4.** 150

31. Let D be a recurring decimal of the form D = 0. a_1a_2 a_1a_2 a_1a_2... where digits a_1 and a_2 lie between 0 and 9. Further, at most one of them is zero. Which of the following numbers necessarily produces an integer, when multiplied by D?

1. 18 **2.** 108 **3.** 198 **4.** 288

32. Let x, y and z be distinct integers, that are odd and positive. Which one of the following statements cannot be true?

1. xyz^2 is odd

2. $(x - y)^2 z$ is even

3. $(x + y - z)(x + y)$ is even

4. $(x - y)(y + z)(x + y - z)$ is odd

CAT 1999

33. Each of the numbers $x_1, x_2, \ldots, x_n$, $n \geq 4$, is equal to 1 or –1. Suppose, $x_1x_2x_3x_4 + x_1x_2x_3x_4x_5 + x_3x_4x_5x_6 + \ldots + x_{n-3}x_{n-2}x_{n-1}x_nx_1 + x_{n-1}x_nx_1x_2 + x_nx_1x_2x_3 = 0$, then

1. n is even **2.** n is odd **3.** n is an odd multiple of 3

4. n is prime

34. The number of positive integer valued pairs (x, y), satisfying 4x – 17y = 1 and x < 1000 is:

1. 59 **2.** 57 **3.** 55 **4.** 58

35. Let a, b, c be distinct digits. Consider a two digit number 'ab' and a three digit number 'ccb', both defined under the usual decimal number system. If $(ab)^2 = ccb$ and ccb > 300 then the value of b is

1. 1 **2.** 0 **3.** 5 **4.** 6

36. If n = 1 + x, where, x is the product of four consecutive positive integers, then which of the following is/are true?

A. n is odd B. n is prime C. n is a perfect square

1. A and C only **2.** A and B only **3.** A only **4.** None of these

37. If n^2 = 123456787654321, what is n?

1. 12344321 **2.** 1235789 **3.** 11111111 **4.** 1111111

Directions for questions (38–40): Read the passage below and solve the questions based on it.

There are fifty integers a1, a2,..., a50, not all of them necessarily different. Let the greatest integer of these fifty integers be referred to as G, and the smallest integer be referred to as L. The integers a1 through a24 form sequence S1, and the rest form sequence S2. Each member of S1 is less than or equal to each member of S2.

38. All values in S1 are changed in sign, while those in S2 remain unchanged. Which of the following statements is true?

1. Every member of S1 is greater than or equal to every member of S2.

2. G is in S1.

3. If all numbers originally in S1 and S2 had the same sign, then after the change of sign, the largest number of S1 and S2 is in S1.

4. None of the above

39. Elements of S1 are in ascending order, and those of S2 are in descending order. a24 and a25 are interchanged. Then, which of the following statements is true?

1. S1 continues to be in ascending order.

2. S2 continues to be in descending order.

3. S1 continues to be in ascending order and S2 in descending order.

4. None of the above

40. Every element of S1 is made greater than or equal to every element of S2 by adding to each element of S1 an integer x. Then x cannot be less than:

1. 210

2. The smallest value of S2

3. The largest value of S2

4. (G–L)

SOLUTIONS AND EXPLANATIONS

1. Let *A* and *B* are any two numbers between 1 and 40, which are removed and (A + B – 1) is written in the first turn. When this is repeated 39 times, sum of the numbers at the end = (1 + 2 + 3 + 4 + ...+ 40) – 1 × 39 = 781.

2. Assume that the integers are : n – 1, n and n + 1

Given that $(n-1)+n^2+(n+1)^3$ is perfect square and $(n-1)+n^2+(n+1)^3 = (n-1)+n+(n+1) \Rightarrow n = 0, 1, 4$

Now, n should be positive and minimum.

So, n≠ 1 or 0 (if n = 1, then n – 1 will not be positive)

Hence, n = 4, n – 1 = 3

So, $1 \leq m \leq 3$

3. Let the original amount be Rs x and y paise. Then interchanged amount = Rs y and x paise. From the given condition, 3 (100 x + y) = 100 y + x – 50

$\Rightarrow 300x + 3y = 10y + x - 50$

$\Rightarrow 299x = 97y - 50$

$\Rightarrow y = \dfrac{299x+50}{97}$

Considering the options, only for x = 18, we get an integral value for y i.e., y = 56. Hence, [2]

4. Let the four-digit number be, 1000a + 100a + 10b + b = 1100a + 11b

This number will be a perfect square if, $1100a + 11b = k^2$; where k is an integer

$\Rightarrow 11(100a+b) = k^2$

$\Rightarrow 100a+b = \dfrac{k^2}{11}$

Therefore, k should be a multiple of 11 such that 100a + b is a 3-digit number k = 44, 55, 66, …99. Corresponding values of 100a + b will be 176, 275, 396, 539, 704, 891.

Therefore, a, b < 9 only '704' satisfies this.

Therefore, $1100a + 11b = 7700 + 44 = 7744 = 88^2$

Hence, [3].

Alternatively, we can solve this question by using elimination and fundas of number system.

Unit digit of a perfect square can be – 0, 1, 4, 5, 6, 9

Only these digits will get repeated at the last two places of the said square in this question.

So, the last two digits can only be – 00, 11, 44, 55, 66, 99

A perfect square cannot have 11, 55, 66, 99 as its last two digits.

(Reason – A perfect square, when divided by 4, should give either 0 or 1 as the remainder).

So, now we are left with = 00 and 44.

Perfect square lies in between 1000 and 9999 (both including). If the last two digits are 00, then the unit digit should be '0' in the number. Then it is

not possible to have other two digits (at the thousands and hundreds place) same.

So the only option left, if possible = 44

Taking a cue from $12^2 = 144$, $88^2 = 7744$

This is the only number satisfying the condition given in the question.

Note

1. The same question with a different format was asked in CAT 99. The question was—Given $(aa)^2 = bbcc$, where a, b and c are non-zero digits. Find the value of b?
2. $38^2 = 1444$ is another such interesting number. "444" occurring at the end of this square is the maximum number of times a non-zero digit is occurring at the end of any perfect square.

5. Going through options,

Obviously options 1 and 4 are not possible (Since any power +ve or –ve raised to any positive number will be positive).

Options 3 and 5 will be positive. Only option 2 is negative.

6. Odd digits are 1, 3, 5, 7 and 9. If sum of four consecutive odd numbers is divisible by 10, then the digits should be 7, 9, 1 and 3.

Now using the option 3 – the numbers are 37, 39, 41 and 43.

Hence, option 3 is the answer.

7. The sum of the numerator and denominator of the ratio should be a prime number.

Note that the number of employees is less than 300.

Consider options.

(1) 101 : 88

101 + 88 = 189

$189 \times 2 > 300 \Rightarrow$ Number of employees who are graduates and above = 101 and the others are 88 in number.

189 is not a prime number. Option 1 is eliminated.

(2) 87 : 100

87 + 100 = 187, $187 \times 2 > 300$

187 is not a prime number.

(3) 110 : 111

110 + 111 = 221, $221 \times 2 > 300$

221 is not a prime number.

(4) 85 : 98

85 + 98 = 183, $183 \times 2 > 300$

183 is not a prime number.

Hence, option 5 is the answer.

8. We know that $xy - yx = 9(x - y)$, where, xy is a two digit number and $xy \geq yx$.

So, $x - y = 2$

The possible pairs of (x, y) is (3, 1) (4, 2) (5, 3) (6, 4), (7, 5) (8, 6) (9, 7)

But we want the number other than 13 so, there are 6 possible numbers are there i.e., 24, 35, 46, 57, 68, 79.

So total possible numbers are 6.

Hence, option 2 is the answer.

9. (4) $10 < n < 1000$

Let n be the two digit number

$n = 10a + b \Rightarrow p_n = ab$, $s_n = a + b$

Then, $ab + a + b = 10a + b$

$\Rightarrow ab = 9a \Rightarrow b = 9$

There are 9 such numbers 19, 29, 33, ….99

Then, Let n be three digit number

$\Rightarrow n = 100a + 10b + c \Rightarrow P_n = abc$, $s_n = a + b + c$

Then, $ab + a + b + c = 100a + 10b + c$

$\Rightarrow abc = 99a + 9b$

$\Rightarrow bc = 99 + 9\ \frac{b}{a}$

But the minimum value for bc = 81 (when both b & c are 9 i.e.,)

And RHS is more than 99. Hence, no such number is possible.

Hence, option 4.

10. (1) We have,

(1) $10^{10} < n < 10^{11}$

(2) Sum of the digits for 'n = 2 clearly

(n)min = 10000000001 (1 followed by 9 zeros and finally 1)

Obviously, we can form 10 such numbers by shifting '1' by one place from right to left again and again. Again, there is another possibility for n

n = 20000000000

So finally, number of different values for n = 10 + 1 = 11 answer.

11. (2) MDCCLXXXVII = 1000 + 500 + 100 + 100 + 50 + 10 + 10 + 10 + 5 + 1 + 1 = 1787

12. (1) MCMXCIX = 1000 + (1000 – 100) + (100 – 10) + (10 – 1)

= 1000 + 900 + 90 + 9 = 1999

13. (3) (I) MCMLXXV = 1000 + (1000—100) + 50 + 10 + 10 + 5 = 1975

(II) MCMXCV = 1000 + (1000—100) + (100—10) + 5 = 1995

(III) MVD = 1000 + (500 —5) = 1495

(IV) MVM = 1000 + (1000—5) = 1995

Therefore, the answer is (II) and (IV), i.e., option 3.

14. (3) Total codes which can be formed = 9 x 9 = 81. (Distinct digit codes)

The digits which can confuse are 1, 6, 8, 9 and from these digits we can form the codes = $4 \times 3 = 12$

Out of these 12 codes two numbers 69 and 96 will not create confusion.

Therefore, (12 – 2) = 10 codes will create a confusion.

Therefore, total codes without confusion = 81 – 10 = 71.

15. (3) $5x + 19y = 64$

We see that if $y = 1$, we get an integer solution for $x = 9$, now if y changes (increases or decreases by 5x will change decrease or increase) by 19. Looking at the options, if $x = 256$, we get, $y = 64$. Using these values we see options 1, 2 and 4 are eliminated and also that there exists a solution for $250 < x \leq 300$.

16. (1) As any prime number greater than 3 can be expressed in the form $6n \pm 1$ minimum difference between three consecutive prime numbers will be 2 and 4. The values that satisfy the given conditions are only 3, 5 and 7 i.e., only one set is possible.

17. (4) Let us solve this question through elimination

$9 - 7 = 2$ is even, therefore, option 1 not possible.

$2 \times 9 = 18$ is even, therefore, option 3 is not possible.

$\frac{3+9}{3} = \frac{12}{3} = 4$ is even, therefore, option 3 is not possible.

The correct option is 4.

18. (2) Go through the options.

Choice (2) $54\ S = (5 + 4)2 = 81$

$\Rightarrow D - S = 81 - 54 = 27$. Hence, the number = 54

19. (2) Since thief escaped with 1 diamond,

Before 3rd watchman he had $(1 + 2) \times 2 = 6$

Before 2nd watchman he had $(6 + 2) \times 2 = 16$

Before 1st watchman he had $(16 + 2) \times 2 = 36$

Alternative method

Go through the options.

20. (4) Let the number of gold coins = x + y

$48(x - y) = x^2 - y^2$

$48(x - y) = (x - y)(x + y) \Rightarrow x + y = 48$

Hence the correct choice would be none of these. Option d.

21. (4) $575 = -x$

$1150 = n^2 + n - 2x$

$n(n + 1) \geq 1150$

$n + n \geq 1150$

The smallest value for it is $n = 34$.

For $n = 34$

$40 = 2x \Rightarrow x = 20$

22. Go through the options.

Odd × Odd = Odd

Odd × Even = Even

Even × Even = Even

Option a is the answer.

23. There are 25 different possibilities if there are 26 boxes, at least 2 boxes con-

tain the same number of oranges (i.e., even if each of the 25 boxes contain a different number of oranges, the 26th must contain one of these numbers).
Similarly, if there are 51 boxes at least 3 boxes contain the same number of oranges.
Hence, at least 6 boxes have same of oranges for 128 boxes. Hence, option c is the answer.

24. Let the four-digit number be abcd.

$a + b = c + d$ .....(i)

$b + d = 2(a + c)$ .....(ii)

$a + d = c$ .....(iii)

From (i) and (iii), $b = 2d$ .....(iv)

We also obtain $3b = 4c + d$ .....(v)

Using (iv) and (v), $3(2d) = 4c + d \Rightarrow 5d + 4c = 0$

$C = \frac{5}{4}d \Rightarrow$ Now d can be 4 or 8.

But if $d = 8$, then $c = 10$ not possible. So $d = 4$ which gives $c = 5$.

25. Let the number be x.
Increase in product = $53x - 35x = 18x$
$= 18x = 540\ x = 30$
Hence, new product = $53 \times 30 = 1590$. Hence, option d is the answer.

26. Sima was the first person to pick and she picks up $\frac{1}{3}$ of the mint. Going through the options, none of the options is a multiple of 3. Hence, option d is the answer.

27. Let the total number of pages in the book be n.

Let page number x be repeated. Then, $\sum_{i=1}^{n} + x = 1000$

$$\frac{n(n+1)}{2} + x = 1000$$

Thus, $\frac{n(n+1)}{2} \leq 1000$ gives $n = 44$

Since, $\frac{n(n+1)}{2} = 990$ (for $n = 44$)

Hence, $x = 10$.

28. Number of one-rupee coins = 158. Possible arrangements of coins are listed as 1, 2, 4, 8, 16, 32, 64 and 31.
Number of arrangements = 8.
So the least number of bags required = 8.
Also Understand that there are going to be a large number of different sets of 8 bags only, and this particular set is not the only one. One more set other than what we have seen here is—1, 2, 4, 8, 16, 32, 63, 32.

29. From II, b = 2d

Hence, b = 10, d = 5 or b = 4, d = 2

From III, e + a = 10 or e + a = 4

From I, a + c = e or e – a = c

From III and and I, we get, 2e = 10 + c or 2e = 4 + c

$$e = 5 + \frac{c}{2} \quad \text{...(i)}$$

$$\text{Or } e = 2 + \frac{c}{2} \quad \text{...(ii)}$$

From (i), we can take c = 2, 4, 6, 10.

For c = 2, e = 6

C = 4, e = 7 (Not possible)

C = 6, c = 8 (Not possible)

C = 10, e = 10 (Not possible since both c and e cannot be 10)

From (ii), we have c = 2, 4, 6, 10.

For c = 2, e = 3 (Not possible)

C = 4, e = 4 (Not possible)

C = 6, e = 5 (Possible)

C = 10, e = 7 (Not possible)

Considering the possibility from B that c = 6 and e = 5 means e + a = 4, a =–1 (not possible)

Hence, only possibility is b = 10, d = 5, c = 2, e = 6. e + a = 10, a = 4

30. Let the number of five-rupee, two-rupee and one-rupee coins be x, y and z respectively.

$$\begin{aligned} x + y + z &= 300 \\ 5x + 2y + z &= 960 \\ \underline{5x + y + 2z} &= \underline{920} \\ y - z &= 40 \end{aligned}$$

And, x + 2y = 340

Use the answer choices now.

If x = 140, y = 100 and z = 60, this satisfies the given conditions.

31. $99 \times D = a_1a_2$. Hence, $D = \frac{a_1a_2}{99}$. So D must be multiplied by 198 as 198 is a multiple of 99. Hence, c is the answer.

32. Take any three odd and positive numbers and check this out. Hence d is the answer.

33. Each term has to be either 1 or –1. Hence, if the sum of n such terms is 0, then n is even. Hence, a is the answer.

34. The difference between two integers will be 1, only if one is even and the other one is odd. 4x will always be even, so 17y has to be odd and hence, y has to be odd.

Moreover, the number 17 y should be such a number that is 1 less than a multiple of 4. In other words, we have to find all such multiples of 17, which are 1 less than a multiple of 4, the first such multiple of 17 goes on increasing, the difference between it and its closest higher multiple of 4 is in the following pattern, 0, 3, 2, 1, e.g., 52–51 = 1.

68–68 = 0. 88–85 = 3, 104–102 = 2, 120–119 = 1, 136–136 = 0

So the multiples of 17 that we are interested in are 3, 17, 11, 15,

Now since, $x < 1000$, $4x < 4000$. The multiple of 17 closest and less than 4000 is 3995 (17×235). And incidentally, 3996 is a multiple of 4, i.e., the difference is 4.

This means that in order to find the answer, we need to find the number of terms in the AP formed by 3, 7, 11, 15… 235, where, $a = 3$, $d = 4$. Since, we know that $Tn = a + (n-1)\,d$, so $235 = 3 + (n-1) \times 4$. Hence, $n = 59$. So option 1 is the answer.

35. $(ab)^2 = ccb$, the greatest possible value of 'ab' to be 31, Since $31^2 = 961$ so $18 < ab < 31$. So, the possible value of ab which satisfies $(ab)^2 = ccb$ is 21, so $21^2 = 441$, $a = 2$, $b = 1$, $c = 4$. Option 1.

37. Consider the pattern: $1^2 = 1$, $11^2 = 121$, $111^2 = 12321$, $1111^2 = 1234321$ and so on.

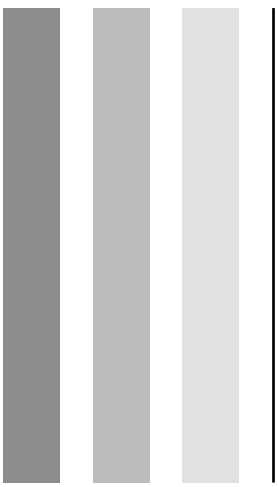
Hence, the square root = 11111111. Option 3.

38. None of the statements is true. Option 4.

39. S1 remains the same, but S2 changes. Option 3.

40. x must be equal to the greatest difference in the value of numbers of S1 and S2. Option 4.

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Chapter 2

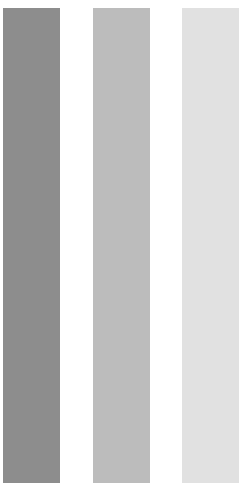
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DEMYSTIFYING NUMBER SYSTEM

LCM AND HCF

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- What is LCM?—Different ways of looking at it
- What is HCF?
- How to connect HCF to LCM?
- Some generalizations and their applications
- Types of questions asked from this chapter in the CAT

LCM

LCM stands for Least Common Multiple. In other words, LCM is the lowest number divisible by all the numbers taken. This is also the 1st number that comes common in the list of multiples (or table) of all the numbers taken. To understand this better, let us see the following example:

Worked-out Example 1 LCM of 10, 20 and 25

1st few multiples of 10 = 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, ...

1st few multiples of 20 = 20, 40, 60, 80, 100, 120.....

1st few multiples of 25 = 25, 50, 75, 100, 125.....

1st number common in the list of multiples of 10, 20 and 25 = 100. Hence, 100 is the LCM of 10, 20 and 25.

Meaning of LCM

Taking a cue from above example, we can also say that 100 is the lowest number divisible by 10, 20 and 25. Giving a second thought, is 100 actually the lowest number divisible by 10, 20 and 25?

Some numbers that are less than 100 and divisible by 10, 20 and 25 are = 0, –100, –200 etc. Despite these numbers being lower than 100, why these numbers are not the LCM of taken numbers?

Reason—0 is not a multiple of 10 or 20 or 25. At the same time, –100 or –200 are also not the multiples of 10 or 20 or 25. Since these are not the multiples, hence not the LCM.

Worked-out Example 2 LCM of 10, –20 and –25

LCM is defined only for positive numbers. In other words, LCM is defined only for positive fractions and natural numbers—positive numbers.

In other words, LCM for negative numbers as well as zero is not defined. Hence, no LCM of 10, –20 and –25 is possible.

Process to Find out LCM

(i) Standard method—Standard method of finding LCM is through factorization of numbers. Let us understand this 3-step process:

Step 1—Factorize the numbers.

Step 2—Collect all the distinct factors.

Step 3—Raise each factor to its maximum available power and multiply these.

Finding LCM of 10, 20 and 25

Step 1:

$10 = 2^1 \times 5^1$

$20 = 2^2 \times 5^1$

$25 = 5^2$

Step 2:

Distinct factors = 2, 5

Step 3:

Highest power of $2 = 2^2$

Highest power of $5 = 5^2$

Hence, LCM $= 2^2 \times 5^2 = 100$

(ii) Straight Line method—Straight Line method of finding LCM involves taking numbers in succession one by one and then observing which factors are to be multiplied.

Let us understand this through an example:

Worked-out Example 3 Find LCM of 24, 30, 52, 76.

☝ ***Solution*** Let us start with one number, and slowly keep moving to the next numbers.

We are here starting with 24.

Take next number 30, and think what number should be multiplied to 24 so that 30 divides 24 × N (where N is the number multiplied to 24). Obviously, it is going to be the factors of 30 which are not present in 24. Now, $30 = 2^1 \times 3^1 \times 5^1$ and $24 = 2^3 \times 3$. Factors of 30 which are missing in $24 = 5^1$. Hence, LCM of 24 and 30 = 24 × 5

[Alternatively, if we start with 30, and then we will be required to think that which number should be multiplied to 30 so that (30 × K) is divisible by each of 30 and 24. Now, $30 = 2^1 \times 3^1 \times 5^1$ and $24 = 2^3 \times 3$. Factors of 24 that is not present in $30 = 2^2$. Hence, LCM of 30 and 24 = 30 × 4]

Next number is $52 = 13^1 \times 2^2$. 2^2 is already present in the LCM (that we have calculated till now) of 24 and 30, hence we will multiply it by 13^1 (which is not present in 24 × 5). So, LCM of 24, 30 and 52 = 24 × 5 × 13.

Next number is $76 = 19^1 \times 2^2$. It can be seen that 2^2 is already present in the LCM (that we have calculated till now) of 24, 30 and 52. Hence, we will multiply this by 19^1.

Hence, LCM of 24, 30, 52, 76 = 24 × 5 × 13 × 19 = 29640

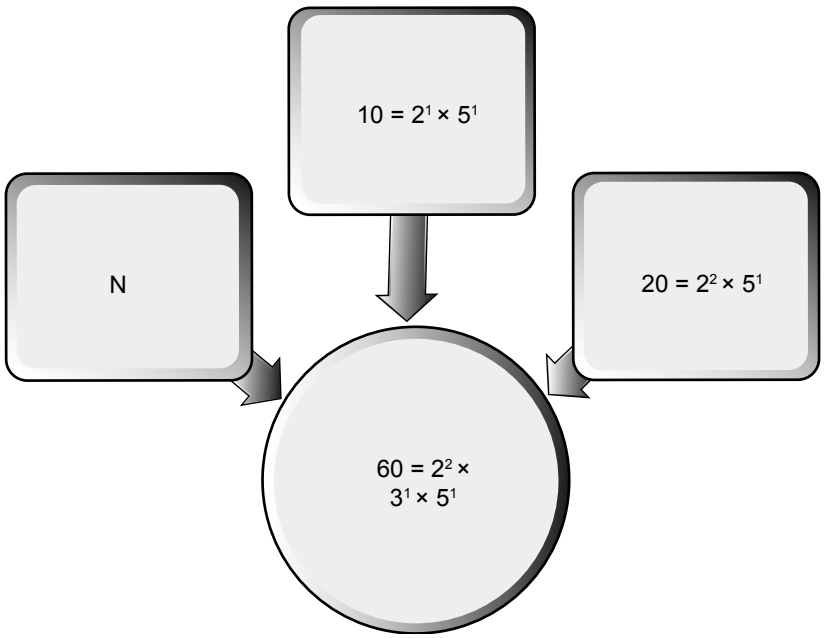
Straight line approach can also be shown by the following expression:

LCM (A, B, C) = LCM (LCM (A, B), C) = LCM (A, LCM (B, C)) = LCM (B, LCM (A, C))

Similarly, LCM (A, B, C, D) = LCM (LCM (A, B), LCM (C, D)) and so on.

Worked-out Example 4 LCM of three natural numbers 10, 20 and N = 60. How many values of N are possible?

☝ ***Solution*** We have already seen that to generate the LCM we multiply the prime numbers with the highest available power. So let us start with factorizing the numbers:



$2^2 \times 5^1$ is already present in 20, however, 3 is not present in either 10 or 20. So we can conclude that 3^1 has to come from N. This is the minimum value of N = 3. Secondly, we can also say that N may contain powers of 2 and 5 as long as maximum power of 2 = 2 and maximum power of 5 = 1 (as in $2^2 \times 5^1$)

So total different values of N = $(3^1 \times 2^0 \times 5^0)$, $(3^1 \times 2^1 \times 5^0)$, $(3^1 \times 2^2 \times 5^0)$, $(3^1 \times 2^0 \times 5^1)$, $(3^1 \times 2^1 \times 5^1)$, $(3^1 \times 2^2 \times 5^1)$ = 3, 6, 12, 15, 30, 60 = 6 values.

Worked-out Example 5 LCM of two natural numbers = 200. How many set of values of two natural numbers are possible?

☝ ***Solution*** $200 = 2^3 \times 5^2$

Let us assume natural numbers = A and B.

We understand that the only prime numbers present in each of A and B can be 2 or 5 only. It also means that each of A and B cannot contain any prime number other than 2 or 5.

Since there are two prime numbers there are two cases possible—either both the prime numbers are coming from the same number or else one is coming from A and 2nd one is coming from B.

Let us formulate the cases:

A	B
$2^3 \times 5^2$	$2^{0-3} \times 5^{0-2}$

Since B can take 4 × 3 = 12 different values for one value of A, there are 12 different sets possible.

Next, let us assume that 2^3 is coming out from A and 5^2 is coming out from B.

A	B
2^3	5^2

Further, A can have 5^0 or 5^1 besides 2^3. Similarly, B can have 2^0, 2^1, 2^2 besides 5^2.

Hence, A can have either $2^3 \times 5^0$ or $2^3 \times 5^1$; B can have either $5^2 \times 2^0$ or $5^2 \times 2^1$ or $5^2 \times 2^2$.

So A can have 2 values, and B can have 3 values. So total number of sets = 2 × 3 = 6

Hence total number of sets = 12 + 6 = 18

In general, for $N = P^x \times Q^y$ (where P and Q are prime numbers and N is a natural number), total sets of two natural numbers which have their LCM as $N = [(x + 1)(y + 1) + xy]$. Understand that this would give us un-ordered pairs of sets.

In this formula, (x + 1) (y + 1) is same as number of factors of a natural number. Its all because if one number A is taken as LCM, then other number B can be any factor of A, such that their LCM will be equal to the number A.

TEST YOUR LEARNING—LCM

1. Find the LCM of
- (i) 36, 46, 56, 76
- (ii) 22, 32, 42, 52
- (iii) 24, 38, 56, 78, 82
- (iv) 12, 44, 78, 105
- (v) 25, 40, 52, 88, 13

2. LCM of 1st 100 natural numbers is N. What is the LCM of 1st 105 natural numbers?

3. What is the ratio of the LCM of first 20 natural numbers and first 10 natural numbers?
- **1.** 11 × 13 × 17 × 19
- **2.** 2 × 11 × 13 × 17 × 19
- **3.** 11 × 13 × 17
- **4.** 2 × 11 × 13 × 17 × 19
- **5.** None of these

4. If the ratio of the LCM of first N natural numbers and first 10 natural numbers is 11 × 13 × 17 × 19 × 2, then what is the maximum possible value of N?

5. What is the ratio of the LCM of 1st 75 natural numbers to the LCM of 1st 50 natural numbers?

HCF

HCF stands for Highest Common Factor. It is also known as GCD (Greatest Common Divisor). This is the largest number that can divide the given set of numbers.

LCM AND HCF

A number is a common factor of two or more than two numbers if it divides all the numbers.

For example, let us see the factors of some of the numbers:

Factors of 20 = 1, 2, 4, 5, 10, 20

Factors of 40 = 1, 2, 4, 5, 8, 10, 20, 40

Factors of 45 = 1, 3, 5, 9, 15, 45

Common factors of 20, 40 and 45 = 1 and 5 (these two factors will divide 20, 40 and 45).

The Highest Common Factor of the numbers is, as the name suggests, the largest factor common across all the given numbers.

In the above example, 5 is the Highest Common Factor.

In other words, in the above example, 5 is the largest number which will divide each of 20, 40 and 45.

Properties of HCF

1. HCF (a, a) = a
2. HCF (a, b) = HCF (b, a)
3. If K = HCF (a, b) then HCF (a/k, b/k) = 1 (where all the digits are non zero).
4. If a > b > 0, and a = bq + r, then, HCF (a, b) = HCF (b, r).
   Let us understand this with the help of an example:
   Compute HCF (803, 154).
   HCF (803, 154) = HCF (154, 33) since 803 = 154 · 5 + 33
   HCF (154, 33) = HCF (33, 22) since 154 = 33 · 4 + 22
   HCF (33, 22) = HCF (22, 11) since 33 = 22 · 1 + 11
   HCF (22, 11) = 11 since 22 = 11 · 2 + 0
   Hence, HCF (803, 154) = 11.
   This is also known as Euclidean Algorithm for finding HCF.
5. If HCF of (a, b) = N (where, a and b are natural numbers and a > b), then, HCF of [b, (a + b)] = HCF of [b, (a – b)] = N
   For example, HCF of (28, 20) = 4
   Hence, HCF of [20, (28 + 20)] = HCF of [20, (28 – 20)] = 4
6. For all a, b, n (non-zero values), HCF (na, nb) = n HCF (a, b). Same rule is applicable for LCM too.
   LCM (na, nb) = n LCM (a, b)
   For example, HCF (15, 25) = 5, hence HCF (15 × 10, 25 × 10) = 5 × 10 = 50
   Similarly, LCM (15, 25) = 25 × 3, hence LCM (15 × 10, 25 × 10) = 25 × 3 × 10 = 750

Summarizing LCM and HCF

It is very essential to understand the mechanism of find out LCM and HCF. We can simply understand the mechanism to find out lowest common multiple and highest common factor through this example.

Worked-out Example 6 Find out LCM and HCF of 16, 12, 24.

Solution

Number	Multiples	Factors
16	16, 32, 48, 64, 80, 96, 112, 128,…….	1, 2, 4, 8, 16
12	12, 24, 36, 48, 60, 72, 84, 96, 108,….	1, 2, 3, 4, 6, 12
24	24, 48, 72, 96, 120, 144, 168, 192,….	1, 2, 3, 4, 6, 8, 12, 24
	Common multiple	Common factor
	48	1, 2, 4…..
	Least common multiple	Highest common factor
	48	4

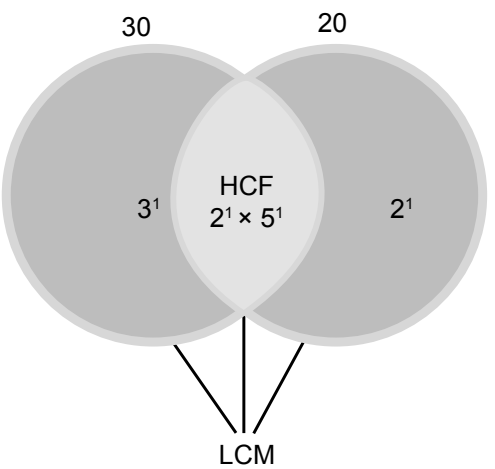
Finally, analogy between Union of the sets and Intersection can be understood with the help of LCM and HCF of sets. Consider this for the following example:

Finding LCM and HCF of 20 and 30.

$20 = 2^2 \times 5^1$

$30 = 2^1 \times 3^1 \times 5^1$

LCM = $2^2 \times 3^1 \times 5^1$ and HCF = $2^1 \times 5^1$



It can be seen that HCF is the intersection of the sets, and LCM is the union of the sets.

Worked-out Example 7 How many different HCF is/are possible for the numbers N and N + 10 (N is a natural number)?

☝ ***Solution*** Assuming different values of N here may not serve the purpose, as it will keep on giving us different values of HCFs.

For example, if we take N = 1, then the HCF (1, 11) = 1

Similarly, if we take N = 10, then the HCF (10, 20) = 10

Similarly, if we take N = 12, then the HCF (12, 22) = 2

So, the point is that we do not get exact number of different HCF by assuming values. Despite we might get some of the HCFs repeating after taking different values of N, its difficult to ensure that we would get only those values of HCF.

Hence by assuming the values, probably it is difficult to count exact number of different HCF.

So, instead of going through the method of plugging in the values, we will go through the process that can be scaled up in any case.

Let us use difference method of finding HCF.

We say that the HCF of N and (N + 10) = HCF of N and (N + 10 – N) = HCF of N and 10

Now, it can be seen that possible values of HCF can be the factors of 10.

So, total number of HCF = Number of factors of 10 = 4 (Namely 1, 2, 5, 10)

Worked-out Example 8 How many different HCF is/are possible for the numbers N, N + 10, N + 20, N + 25, N + 26 (N is a natural number)?

☝ ***Solution*** By assuming different values of N, we can see that HCF = 1. However, this does not answer the natural query—If this is going to happen always or not? To answer the question, we should either get unique value or get conclusive evidence that answer is not unique. None of these happens in this case.

So we need to explore that if HCF is always = 1 or not? Besides, we also need to find out even if HCF is always = 1, why this happens?

Why HCF = 1?

We can see that last two numbers N + 25 and N + 26 are consecutive natural numbers. And HCF of any two consecutive natural numbers = 1 (two consecutive natural numbers are always co-prime).

Now we can conclude that since (N + 25) and (N + 26) are having only one factor common between them (=1), so despite N, N + 10, N + 20 may different values of HCF not equal to 1, all these numbers N, N + 10, N + 20, N + 25, N + 26 will have only one factor common among them = 1.

Alternatively, we can see the same using difference method of finding HCF too.

HCF of (N + 25), (N + 26) = HCF of (N + 25) and [(N + 26) – (N + 25)] = HCF of (N + 25), 1 = 1

Hence, HCF of all the numbers = 1

Worked-out Example 9 What is the HCF of $(x^2 - 5x + 6)$ and $(x^2 - 7x + 12)$?

☝ ***Solution*** $(x^2 - 5x + 6) = (x - 2)(x - 3)$

$(x^2 - 7x + 12) = (x - 3)(x - 4)$

HCF of $[(x - 2)(x - 3)]$ and $[(x - 3)(x - 4)]$ cannot be determined as HCF of $(x - 2)$ and $(x - 4)$ depends upon the value of x.

It can be seen through the values taken too:

For $x = 5$, $(x^2 - 5x + 6) = 6$ and $(x^2 - 7x + 12) = 2$, Hence HCF = 2 $\Rightarrow$ In this case HCF = $(x - 3)$

For $x = 6$, $(x^2 - 5x + 6) = 12$ and $(x^2 - 7x + 12) = 6$, Hence HCF = 6 $\Rightarrow$ In this case HCF $\neq (x - 3)$

Hence, unique value of HCF cannot be determined.

TEST YOUR LEARNING—HCF

1. What is the HCF of 12, 15, 18, 21?

2. What is the HCF of 120, 150, 180, 210?

3. What is the HCF of 72, 47, 78, 46?

4. What is the HCF of 403, 465, 496?

5. What is the HCF of x^2, x^4, x^3?

6. What is the HCF of $(x^{24} - y^{24})$, $(x^{12} - y^{12})$, $(x^4 - y^4)$?

7. What is the HCF of $(x^{20} - y^{20})$, $(x^{16} - y^{16})$, $(x^{14} - y^{14})$?

8. What is the HCF of $(x^{36} - y^{36})$, $(x^{10} - y^{10})$, $(x^4 - y^4)$, $(x^2 - y^2)$, $(x - y)$?

9. What is the HCF of $(x - 6)(x - 8)$ and $(x - 8)(x - 10)$?

Application of LCM/HCF

Case 1—Time and work

In the questions involving individual efficiency, we use the concept of LCM to assume the total units of work instead of going through the actual method of solving the question.

Worked-out Example 10 Tatto can do a work in 10 days and Tappo can do same work in 12 days. How many days will it take if both of them start working together?

☝ ***Solution***

Normal Method:

Work done by Tatto in one day = 1/10 of the total work

Work done by Tappo in one day = 1/12 of the total work

Total work done by Tatto and Tappo in one day $= \left[\frac{1}{10} + \frac{1}{12}\right] = \frac{11}{60}$ of the work

Hence, total time taken by Tatto and Tappo to finish the work $= \frac{60}{11} = 5\frac{5}{11}$ days

LCM Method:

Let us assume total work = LCM of (10, 12) units = 60 units. Now since 60 units of work is being done by Tatto in 10 days, so Tatto is doing 6 units of work per day and similarly, Tappo is doing 5 units of work per day. Hence, they are doing 11 units of work in one day together.

So, time taken to finish the work by Tatto and Tappo $= \frac{\text{Total work}}{\text{Work done in one day}}$

$= \frac{60}{11} = 5\frac{5}{11}$ days.

Case 2—Time speed and distance: Circular Motion

Worked-out Example 11 Speed of A is 15 m/s and speed of B is 20 m/s. They are running around a circular track of length 1000 m in the same direction. Find after how much time they will meet for the first time at the starting point if they start running at the same time.

☝ ***Solution*** Time taken by A in taking one circle = 66.66 sec

Time taken by B in taking one circle = 50 sec

LCM (66.66, 50) = 200 sec

Case 3—Number system: Tolling the bell

Worked-out Example 12 There are two bells in a temple. Both the bells toll at a regular interval of 66.66 sec and 50 sec respectively. After how much time will they toll together for the first time?

☝ ***Solution*** Time taken by 1st bell to toll = 66.66 sec

Time taken by 2nd bell to toll = 50 sec

LCM (66.66, 50) = 200 sec

It can be observed here that Mathematical interpretation of both the questions 11 and 12 are same, just the language has been changed.

Case 4—Number system: Number of rows

Worked-out Example 13 There are 24 peaches, 36 apricots and 60 bananas and they have to be arranged in several rows in such a way that every row contains same number of fruits of one type. What is the minimum number of rows required for this to happen?

☝ ***Solution*** We can put one fruit in one row, and in (24 + 36 + 60) 120 rows, we can arrange all the fruits. Or, we may put two fruits in one row and can arrange all the fruits in 60 rows. But for the rows to be minimum, number of fruits should be maximum in one row.

HCF of 24, 36, 60 = 12, so 12 fruits should be there in one row.

Hence number of rows = 10

Case 5—Number system: Finding remainder

Worked-out Example 14 There is a number which when divided by 4 and 5 gives 3 as the remainder. What is the lowest three digit number which satisfy this condition?

☝ ***Solution*** Let us assume that there is no remainder. So, number has to be a multiple of LCM of 4 and 5. Now, LCM (4, 5) = 20

But there is a remainder of 3 when divided by 4 and 5. So, the number will be of the form (20N + 3).

Hence, numbers are 23, 43, 63, 83, 103 and so on….

So, 103 is the answer.

Standard Formula

1. LCM × HCF = Product of two numbers.
   This formula can be applied only in case of two numbers. However if the numbers are relatively prime to each other (i.e., HCF of numbers = 1), then this formula can be applied for any number of numbers.
2. LCM of fractions = LCM of numerator of all the fractions/HCF of denominator of fractions

   If the fractions are $\frac{x}{y}, \frac{a}{b}$ and $\frac{m}{n}$,

   Then LCM of fractions $= \frac{\text{LCM of numerators}}{\text{HCF of denominators}} = \frac{\text{LCM of } (x, a, m)}{\text{HCF of } (y, b, n)}$

   LCM of decimals can be calculated by converting decimals into fractions first, and then using above method.
3. HCF of fractions = HCF of numerator of all the fractions/LCM of denominator of fractions.

   If the fractions are $\frac{x}{y}, \frac{a}{b}$ and $\frac{m}{n}$,

   Then HCF of fractions $= \frac{\text{HCF of numerators}}{\text{LCM of denominators}} = \frac{\text{HCF of } (x, a, m)}{\text{LCM of } (y, b, n)}$

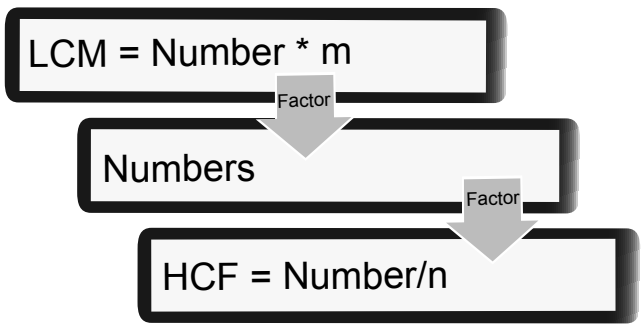
   HCF of decimals can be calculated by converting decimals into fractions first, and then using above method.
4. For three natural numbers a, b and c, following relations hold good:

   (a) $$\frac{\text{LCM } (a, b, c)^2}{\text{LCM } (a, b) \times \text{LCM } (b, c) \times \text{LCM } (a, c)} = \frac{\text{HCF } (a, b, c)^2}{\text{HCF } (a, b) \times \text{HCF } (b, c) \times \text{HCF } (a, c)}$$

   (b) $$\text{HCF } (a, b, c) = \frac{a \times b \times c \times \text{LCM } (a, b, c)}{\text{LCM } (a, b) \times \text{LCM } (b, c) \times \text{LCM } (c, a)}$$

(c) $\text{LCM}(a, b, c) = \dfrac{a \times b \times c \times \text{HCF}(a, b, c)}{\text{HCF}(a, b) \times \text{HCF}(b, c) \times \text{HCF}(c, a)}$

5. HCF of (sum of two numbers and their LCM) = HCF of numbers.
6. For $N = a^p \times b^q \times c^r$……, where a, b and c are prime numbers and N is factorized.
   Number of sets of two numbers such that their LCM is N
   $= \dfrac{(2 \times p + 1)(2 \times q + 1)(2 \times r + 1) - 1}{2} + 1$
   This will give un-ordered pairs of sets.
7. LCM for a given set of numbers has to be greater than or equal to the HCF of the same set of numbers.
   It can be further understood through the following discussion:



Whatever numbers do we take; LCM is a multiple of these numbers. So LCM ≥ Numbers.

At the same time, HCF is a factor of Numbers or Numbers taken are multiple of HCF.

Hence, Number ≥ HCF

Combining these two, we get LCM ≥ Numbers ≥ HCF

Corollary 1—LCM can be equal to HCF only when all the numbers taken are equal. HCF obtained in this case is also going to be largest possible HCF.

Corollary 2—It also gives that HCF has to be a factor of LCM.

Worked-out Example 15 Product of two natural numbers = 100, and their HCF = 20. What is their LCM?

☝ ***Solution*** We know LCM × HCF = Product of two numbers

$\Rightarrow$ LCM × 20 = 100 $\Rightarrow$ LCM = 5

Think—Is this possible?

Worked-out Example 16 Product of two natural numbers = 144, and their HCF = 18. What is their LCM?

☝ ***Solution*** We know LCM × HCF = Product of two numbers

$\Rightarrow$ LCM × 18 = 144 $\Rightarrow$ LCM = 8

Think—Is this possible?

As we have discussed above, LCM cannot be less than HCF. Hence, we conclude that if product of two natural numbers = 144, their HCF cannot be equal to 18, and in turn their LCM cannot be equal to 8.

Reason it out:

However, a larger question arises here is that since LCM cannot be equal to 8, can HCF be equal to 18? Obviously the answer is NO because if HCF = 18 becomes true, then LCM = 8 will also become true. So, it means that if product of two natural numbers = 144, then their HCF cannot be 18. This leads to another question—that if 18 is not a possible value of HCF, then what values can be taken by HCF? Or, primarily, why 18 cannot be HCF?

Let us reason it out (although i will suggest the users/reader of this book to ponder about this on their own and try to find the reason)

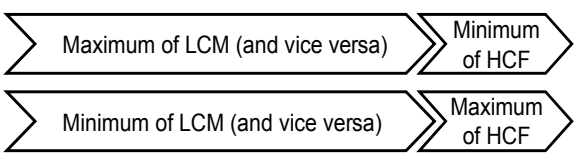
Understand that if 18 is the HCF of two natural numbers, then each of the two numbers are divisible by 18 (or 18 is a factor of each of the numbers). So, the format of the numbers are 18a, 18b such that HCF of (a, b) = 1. So, the product of two numbers = 18a × 18b = 324 ab ⇒ this number has to be larger than or equal to 324 (since a and b will be natural numbers). Hence we conclude that HCF = 18 is not possible.

Maxima and Minima in Case of LCM/HCF

If product of two numbers is given, and none of LCM or HCF is given, then this gives rise to the case of maxima and minima.

Primarily the formula that we are going to use is—LCM × HCF = Product of two numbers. Although this formula only provides the basic framework, and to solve these questions we would be required to visualize the situation.

Going by the formula LCM × HCF = Product of two numbers, we can say that, since RHS is constant, LHS will be inversely proportional to HCF (subject to the values being natural numbers).



Worked-out Example 17 Product of two natural numbers = 144. What is the (a) largest possible (b) smallest possible HCF of these two natural numbers?

☝ ***Solution*** Let us 1st factorize $144 = 12 \times 12 = (2^2 \times 3) \times (2^2 \times 3)$

Largest possible HCF occurs when LCM = HCF ⇒ when LCM = HCF, numbers are equal.

We already know that—Product of two natural numbers = LCM × HCF.

Since numbers have to be equal, each of the numbers = 12, and Largest possible HCF = 12.

(b) Smallest possible HCF, obviously, has to be equal to 1. (possible set of numbers = 144, 1)

Worked-out Example 18 Product of two natural numbers = 144. How many different values of LCM are possible for these two natural numbers?

☝ ***Solution*** we have already seen in the above question that largest possible value of HCF = 12. And consequently smallest possible value of LCM = 12.

Let us see the different values of HCF and corresponding values of LCM:

HCF = 12	HCF = 6	HCF = 4	HCF = 3	HCF = 2	HCF = 1
LCM = 12	LCM = 24	LCM = 36	LCM = 48	LCM = 72	LCM = 144

So, total different values of LCM = 6.

Worked-out Example 19 Product of two natural numbers A and B = 590 and their HCF = 59. How many set of values of A and B are possible?

☝ ***Solution*** Since HCF of the numbers = 59, so we can write

A = 59 x and B = 59 y

Using LCM × HCF = Product of two numbers

59 x × 59 y = 59 × 590 $\Rightarrow$ xy = 10

So sets of x, y possible are = (1, 10), (10, 1) and (2, 5), (5, 2)

Hence, 4 sets of numbers are possible = (59, 590), (590, 59), (118, 295), (295, 118).

Worked-out Example 20 If the LCM of two positive integers (x, y), where x < y, is 60 times the HCF, which of the following is not a possible value of $\frac{y}{x}$?

1. 3.75 **2.** 6.66 **3.** 2.4 **4.** 60 **5.** None of these

☝ ***Solution*** LCM × HCF = Product of x and y

Let us convert all the values in terms of HCF.

x = a × HCF and y = b × HCF.

$\Rightarrow$ 60 HCF × HCF = a HCF × b HCF $\Rightarrow$ ab = 60

Since x < y, hence a < b, and HCF (a, b) = 1.

Possible values of a and b are (in order) – (1, 60), (3, 20), (4, 15), (5, 12), so all the 1st four options are possible.

PRACTICE EXERCISE 1

1. Find the number of pairs of two natural numbers having product = 3600 and HCF = 30.

1. 0 **2.** 1 **3.** 2 **4.** 5

5. 6

2. Four bells toll at regular interval of 24 sec, 36 sec, 40 sec and 48 sec respectively. How many times they next toll together in 1 hour (not counting T = 0 situation)?

1. 10 **2.** 4 **3.** 5 **4.** 6

5. 12

3. Three runners are running on a circular track of length 1000 m. Speed of runners are 10 m/s, 20 m/s and 40 m/s respectively. All of them are starting from the same point at the same time in the same direction. After how much time will they meet at the starting point for the 1st time after they have started?

1. 100 sec **2.** 50 sec **3.** 200 sec **4.** 250 sec

5. None of these

4. Two runners are running on a circular track of length 1000 m. Speed of runners are 20 m/s and 50 m/s. Both of them are starting from the same point at the same time in the same direction. After how much time will they meet at a point diametrically opposite to their starting point for the 1st time after they have started?

1. 100 sec **2.** 50 sec **3.** 200 sec **4.** 250 sec

5. None of these

5. What is the least number of soldiers in a regiment, such that they stand in rows of 18, 15 and 25 and also form a solid square?

1. 900 **2.** 225 **3.** 1600 **4.** 900

5. None of these

6. Given that sum of two numbers is ab and their difference is $\frac{1}{7}$ of their sum. Find their HCF.

1. $a + b$ **2.** $\frac{(a-b)}{ab}$ **3.** $\frac{12}{7}$ ab **4.** ab

5. $\frac{6}{7}$ ab

Directions for questions (7–9): Go through the passage given below and solve the questions based on it.

Ankit Goel is having 60 apples, 156 oranges and 204 guavas. He distributes the fruits equally among a group of kids such that each kid gets maximum number of fruits, and all the kids get exactly one type of fruit.

7. Find the number of fruits that each child will get.

1. 12 **2.** 24 **3.** 18 **4.** 6

5. 35

8. Find the number of children.

1. 12 **2.** 24 **3.** 18 **4.** 6

5. 35

9. Radii of the front wheel and the rear wheel are 14 cm and 21 cm respectively. A blue mark is put on the point of contact of each of the wheel and the ground when the bike is stationary. Now the bike starts moving. Find after what distance will the 2 blue marks touch the ground simultaneously again.

1. 42 cm **2.** 294 cm **3.** 84 cm **4.** 264 cm

5. 1056 cm

Directions for questions (10–13): Read the information below and solve the questions based on it.

Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4 as the answer—if question cannot be answered even by using both the statements together.

10. Find the value of X.

A. The LCM of X and 18 is 36.

B. The HCF of X and 18 is 2.

11. What is the HCF of natural numbers A and B?

A. LCM of the natural numbers A and B is 101.

B. A and B are distinct natural numbers.

12. What is the HCF of two natural numbers M and N?

A. M – N = 5

B. M + N = 5

13. LCM of 1st N natural number = K. What is the LCM of 1st (N + 10) natural numbers in terms of K?

A. LCM (N + 1, N + 2, ……N + 10) = M

B. LCM (1, 2, ……10) = P

14. LCM of 1st N natural number = LCM of 1st 10 natural number × 11 × 13. Which of the following is the maximum possible value of N?

1. 13 **2.** 14 **3.** 15 **4.** 16

5. 17

15. P is a natural number (3 < P < 500). How many P's are there so that HCF of P and 100 is 1?

1. 200 **2.** 198 **3.** 150 **4.** 197

5. 251

16. LCM of 12^{24}, 16^{18} and N is 24^{24}. Number of all the possible values of N = S. What is the value of S?

1. 25 **2.** 1800 **3.** 1825 **4.** 1600

5. 125

17. What is the side of the largest square slabs which can be paved on the floor of a room 5 m 44 cm long and 3 m 74 cm wide?

1. 56 **2.** 42 **3.** 38 **4.** 34

5. 48

18. What is the least number of square tiles required to pave the ceiling of a room 15 m 17 cm long and 9 m 2 cm broad?

1. 656 **2.** 738 **3.** 814 **4.** 902

5. 478

19. LCM of two natural numbers is 4800 and their HCF is 160. If one of the numbers is 480, then the second number is

1. 16 **2.** 16000 **3.** 160 **4.** 1600

5. 800

20. HCF of 3240, 3600 and a third number N is 36 and their LCM is $2^4 \times 3^5 \times 5^2 \times 7^2$. Find N.

1. $2^4 \times 5^3 \times 7^2$ **2.** $2^2 \times 3^5$ **3.** $3^5 \times 7^2$ **4.** $2^5 \times 5^2 \times 7^2$

5. None of these

21. An electronic device makes a beep after every 60 sec. Another device makes a beep after every 62 sec. They beeped together at 10 a.m. They time when they will next make a beep together at the earliest is

1. 10.30 a.m. **2.** 10.31 a.m. **3.** 10.59 a.m. **4.** 11 a.m.

5. 11:01 a.m.

22. The sum and difference of the LCM and the HCF of two numbers are 312 and 264, respectively. Find the value of the numbers if their sum is 168.

1. 96, 72 **2.** 76, 92 **3.** 108, 66 **4.** 88, 80

5. None of these

23. The HCF of two expressions p and q is 1. Their LCM is

1. (p + q) **2.** (p – q) **3.** p q **4.** 1

5. $\frac{1}{pq}$

24. Determine the HCF of $25ab^2c^3$, $80ab^3c^2$ and $12a^2b^2c$.

1. $20a^2b^2c^3$ **2.** $10ab^2c$ **3.** $80ab^2c$ **4.** $10ab^2c^2$

5. $20ab^2c^2$

25. (y + k) is the HCF of $(y^2 + ay + b)$ and $(y^2 + cy + d)$. What is the value of y?

1. $\frac{b+d}{a+c}$ **2.** $\frac{a+b}{c+d}$ **3.** $\frac{a-b}{c-d}$ **4.** $\frac{b-d}{a-c}$

5. None of these

PRACTICE EXERCISE 2

1. It is a fact that every natural number can be LCM for a set of numbers. For example, consider 24.

LCM of 8, 3 = 24

LCM of 12, 24 = 24

LCM of 2, 8, 24 = 24 and so on. In this case, we have atleast three sets of distinct numbers [1st set (8, 3), 2nd set (12, 24), 3rd set (2, 8, 24)] for which LCM = 24.

How many natural numbers from 1 to 100 can be the LCM of two or more than two distinct natural numbers in only one way (consider x, y = y, x $\Rightarrow$ interchangeability of numbers is not allowed)?

1. 0 **2.** 1 **3.** 24 **4.** 25

5. 26

2. LCM of two distinct natural numbers = 263. How many different value/s of HCF is/are possible for these two distinct natural numbers?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

3. It is known that n! Is always divisible by the LCM of 1st n natural numbers. For how many values of n, LCM of 1st n natural numbers will also be divisible by n!?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

4. For how many values of n, LCM of 1st n natural numbers will be half of n!?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

5. $$\frac{n!}{\text{LCM of 1st n consecutive natural number}}$$

$$= \frac{(n+1)!}{\text{LCM of 1st}(n+1)\text{ consecutive natural numbers}} = K(K > 1)$$

How many values of n are possible such that $n \leq 40$?

1. 10 **2.** 11 **3.** 12 **4.** 31

5. 41

6. LCM of two natural numbers A and B = 300. How many set of (A and B) are possible?

1. 74 **2.** 38 **3.** 75 **4.** 37

5. None of these

7. LCM of three natural numbers = 150. How many set of three natural numbers are possible?

1. 112 **2.** 48 **3.** 178 **4.** 142

5. None of these

8. Which of the following is true for N = LCM of [($^{20}C_0$), ($^{20}C_1$), ($^{20}C_2$), ..., ($^{20}C_{20}$)]?

1. N will be divisible by 7

2. N will be divisible by 3^2

3. N will be divisible by 2^2

4. N will be divisible by 5^2

5. More than one statements are true.

9. If p is a prime, find the possible values of HCF (N, N + p).

1. 1 **2.** 2 **3.** 3

4. Depends upon the value of N **5.** Depends upon the value of P

10. Given that, $X_1 = 5$, $X_2 = 25$ and X_{n+2} = GCD ($X_{n+1,}$ X_n) + X_n.
What is the LCM of (X_{19} and X_{20})?

1. 1840 **2.** 2560 **3.** 2160 **4.** 2440

5. 2040

11. LCM of three consecutive natural numbers (a, b, c) = N. Which of the following is equal to
[LCM (a, b) × LCM (a, c) × LCM (b, c)]?

1. 2N **2.** N^2 **3.** N^3

4. Will not have a unique value **5.** Cannot be determined

12. LCM of three consecutive natural numbers (a, b, c) = N. How many different value/s of HCF (a, b, c) is/are possible?

1. 1 **2.** 2 **3.** 3

4. More than one but finite **5.** Depends upon the value of N

13. LCM of three consecutive natural numbers (a, b, c) = 60. Which of the following is equal to [HCF (a, b) × HCF (a, c) × HCF(b, c)]?

1. 1 **2.** 2 **3.** 3

4. More than one but finite **5.** Depends upon the value of N

14. LCM of three natural numbers (a, b, c) = 60. How many different value/s of HCF (a, b, c) is/are possible?

1. 1 **2.** 6 **3.** 12 **4.** 18

5. None of these

Directions for questions (15–16): Read the directions given below and solve the questions based on it.

There are three natural number a, b and c.

HCF (a, b, c) = 10

LCM (a, b, c) = 100.

15. What is the minimum product of (a × b × c)?

1. 1000 **2.** 10000 **3.** 20000 **4.** 40000

5. None of these

16. How many different values of $(a \times b \times c)$ are possible?

1. 4 **2.** 5 **3.** 6 **4.** 7

5. None of these

17. HCF of [(N – 1)! and N] > 1. Consider the statements given below:

(i) N is a natural number

(ii) N is a composite number

(iii) N is a prime number

(iv) N is a rational number

Choose the best option:

1. Exactly one of the statements is definitely true.

2. At least one of the statements is definitely true.

3. Exactly two of the statements are definitely true.

4. At max two of the statements are definitely true.

5. All are definitely true.

18. LCM of 1st 20 natural number i.e., LCM (1 – 20) = LCM (1 – N), where N is a natural number ≠ 20. What is the difference between the maximum and minimum value of N?

1. 5 **2.** 1 **3.** 2 **4.** 3

5. 4

19. What is the unit digit of the LCM of $(3^{2003} - 1)$ and $(3^{2003} + 1)$?

1. 8 **2.** 2 **3.** 4 **4.** 6

5. 0

20. What is the HCF of [(1111111.....written 105 times), (11111.....104 times)]?

1. 1 **2.** 11 **3.** 101

4. (11...written 11 times) **5.** None of these

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING—LCM

1.

(i) $36 \times 23 \times 7 \times 2 \times 19$

(ii) $22 \times 16 \times 21 \times 13$

(iii) $24 \times 19 \times 7 \times 2 \times 13 \times 41$

(iv) $12 \times 11 \times 13 \times 7 \times 5$

(v) $25 \times 8 \times 13 \times 11$

2. We will consider all the natural numbers from 101 to 105 one by one.

LCM of 1st 101 natural numbers = LCM of 1st 100 natural numbers with 101 = LCM (N, 101) = N × 101

LCM of 1st 102 natural numbers = LCM of 1st 101 natural numbers with 102 = LCM (101 N, 102) = 101 N

LCM of 1st 103 natural numbers = LCM of 1st 102 natural numbers with 103 = LCM (101 N, 103) = 101 N × 103

LCM of 1st 104 natural numbers = LCM of 1st 103 natural numbers with 104 = LCM (101 × 103 N, 102) = 101 N × 103

LCM of 1st 105 natural numbers = LCM of 1st 104 natural numbers with 102 = LCM (101 N, 102) = 101 N × 103 = 10403 N

3. To solve this question, we just have to understand that which factors (of numbers exceeding 10) will be multiplied to the LCM of first 10 natural numbers.

So what numbers should be multiplied? Did you get—$11 \times 13 \times 17 \times 19$?

Observe once again—LCM of first 10 natural numbers will have 2^3 only, whereas LCM of first 20 natural numbers should have 2^4 (because of presence of 16).

Hence ratio of (1st 20 natural numbers/LCM of 1st 10 natural numbers) = $11 \times 13 \times 17 \times 19 \times 2$.

4. In the above question, we have seen that N = 20 is one of the possible values. Now we just have to see that we do not multiply the existing LCM of first 20 natural numbers by any new prime number. So let us see the numbers greater than 20 one by one:

$21 = 7 \times 3$, we have already included both 7 and 3, so not needed to multiply with any new number.

$22 = 2 \times 11$, we have already included both 2 and 11, so not needed to multiply with any new number.

23 = We have not included 23 yet in the LCM of first 22 natural numbers, hence if we take 23 also, we will be required to multiply the LCM of first 22 natural number by 23, resulting into the ratio being = $11 \times 13 \times 17 \times 19 \times 2 \times 23$.

So the largest possible value of N = 22.

5. What is the ratio of the LCM of 1st 75 natural numbers to the LCM of 1st 50 natural numbers?

To answer this question, we just need to find the factors that will be required to be multiplied to the LCM of 1st 50 natural numbers. Let us assume LCM of 1st 50 natural numbers = N.

1st of all, all the prime numbers > 50 will be multiplied to N, and then all the prime factors that have cropped up above 50. Let us see this:

Numbers	51	52	53	54	55	56	57	58	59	60	61	62	63
LCM (1–50) will be multiplied by	1	1	53	1	1	1	1	1	59	1	61	1	1

Numbers	64	65	66	67	68	69	70	71	72	73	74	75
LCM (1–50) will be multiplied by	2	1	1	67	1	1	1	71	1	73	1	1

Hence ratio of LCM of 1st 75 natural numbers to the LCM of 1st 50 natural numbers = $53 \times 59 \times 61 \times 2 \times 67 \times 71 \times 73$.

TEST YOUR LEARNING—HCF

1. Step 1 goes like → $12 = 2^2 \times 3^1$; $15 = 3^1 \times 5^1$; $18 = 2^1 \times 3^2$; $21 = 3^1 \times 7^1$
Step 2 goes like → 3 (only prime factor that is common across all the numbers)
Step 3 goes like → 3^1 (minimum power of 3 = 1). Hence HCF = 3.

2. HCF = 30

3. HCF = 1

4. HCF = 31

5. HCF = x^2

6. HCF = $(x^4 - y^4)$

7. HCF = cannot be determined. Understand that in this case, HCF is dependent upon the values of x and y. For example, if y = 0, then HCF of $(x^{20} - y^{20})$, $(x^{16} - y^{16})$, $(x^{14} - y^{14})$ = HCF of (x^{20}), (x^{16}), $(x^{14}) = x^{14}$. And in case $y \neq 0$, then HCF of the given numbers is other than x^{14}.

8. HCF = (x – y)

9. Cannot be determined

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	**3**	**1**	**5**	**4**	**5**	**1**	**5**	**4**	**3**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
3	**1**	**4**	**3**	**2**	**3**	**4**	**3**	**4**	**5**
21.	22.	23.	24.	25.					
2	**1**	**3**	**2**	**4**					

1. Since HCF = 30, numbers can be assumed to be 30a and 30b.
Hence, $30a \times 30b = 3600 \Rightarrow ab = 4 \Rightarrow$ There is only one set possible a = 4, b = 1. [we cannot take a = b = 2, as in that case HCF will become = 60].
Hence, numbers are 30, 120. Answer = option 2.
Understand that we are considering 'un-ordered pairs'. Hence, (4, 1) and (1, 4) are the same sets.

2. Time taken to toll together = LCM of 24, 36, 40 and 48 = 720 seconds = 12 minutes
So they will toll together 60/12 = 5 times in 1 hour.
Answer = option 3.

3. Time taken by 1st runner to complete one round $= \frac{1000}{10} = 100$ sec

Time taken by 2nd runner to complete one round $= \frac{1000}{20} = 50$ sec

Time taken by 3rd runner to complete one round $= \frac{1000}{40} = 25$ sec

Hence time taken for all of them to meet at the starting point = LCM (100, 50, 25) = 100 sec
Understand that mathematical input in 2 and 3 remains same. Only thing that has got changed is Language.

4. Total number of meeting points = 5 – 2 = 3. We also understand that these three points are going to be equidistant from each other, and hence going to be 333.33 m, 666.66 m, 1000 m (starting point). Hence they will never meet at a point diametrically opposite to the starting point.

5. Number of soldiers should be a multiple of 18, 15 and 25 $\Rightarrow$ Number should be a multiple of LCM (18, 15, 25) = 450. Now Lowest multiple of 450 that is a perfect square = 900.

9. Circumference of the wheels are 88 cm and 132 cm. LCM (88, 132) = 264 cm

10. Using statement A alone, Different values that X can take = 4, 12, 36. Hence, statement A alone is not sufficient.
Using statement B alone, Different values that X can take = 2, 4, 8 etc. Hence, 2nd statement alone is not sufficient.
However, using both the statements together, we get a unique value of X = 4.
Hence, answer is option 3.

11. Using statement A only gives us two different sets of values of A and B—(101, 101) and (101, 1). So, unique value of HCF cannot be obtained.
Using the statement B only also does not give us a unique number.
However, using both the statements together gives us the values of A and B as—(101, 1). And HCF of these two numbers = 1.
Hence, the answer is—option 3, both the statements together are required.

12. Using statement A alone, M – N = 5 will give us two different values of HCF of M and N = 1 and 5. So, statement A alone is not sufficient.

If M + N = 5, then different set of values of M and N possible are—(1, 4) and (2, 3). In any case HCF = 1. Hence, statement B alone is sufficient.

Hence, option 1 is the answer.

13. Statement A could have answered this question on its own if the common factors between M and K would have been known. Since its not given, statement A alone cannot answer this question.

Statement B alone also does not answer this question. Even by using both the statements together, we cannot find the common factors.

Hence, option 4 is the answer.

14. LCM of 1st N natural number = LCM of 1st 10 natural number × 11 × 13 = K (assume)

Since, K contains 13 as a prime factor, N has to contain 13. However, this may not be the largest possible value of N. Let us verify it,

For N = 14, LCM (1 – 14) = LCM (1 – 13) × 1

For N = 15, LCM (1 – 15) = LCM (1 – 13) × 1 × 1

For N = 16, LCM (1 – 16) = LCM (1 – 13) × 2

$\Rightarrow$ N can take maximum value of 15.

15. HCF of P and 100 = 1 $\Rightarrow$ none of the factors of 100 is present in P and vice versa.

$100 = 2^2 \times 5^2$. So we have to find out such values of P that do not have factors = 2 or 5 or any multiple of 5. In other words, we have to find out the values of P which are neither divisible by 2 nor divisible by 5.

We have two methods now;

Method 1—Using set theory

Let us rewrite the range $3 < P < 500 = 4 \leq P \leq 499 = 496$ natural numbers (subtract 1, 2, 3, 500 namely four numbers from the 1st 500 natural numbers)

All these numbers are either divisible or not divisible by any of the two. To solve this question we are required to find out the numbers which are not divisible by 2 or 5. For this, through set theory we find out the numbers divisible by either of 2 or 5, and then will subtract it from total number of numbers.

$n(2 \cup 5) = n(2) + n(5) - n(2 \cap 5)$

where, $n(2 \cup 5)$ = number of numbers which are divisible by either 2 or 5.

$n(2)$ = Number of numbers which are divisible by 2.

$n(5)$ = Number of numbers which are divisible by 5.

$n(2 \cap 5)$ = number of numbers which are divisible by both 2 and 5 = divisible by 10.

$n(2) = 248$ (we have to eliminate 2 and 500 from the 1st 500 numbers = 250 – 2).

$n(5) = 99$

$n(2 \cap 5) = n(10) = 49$

$n(2 \cup 5) = n(2) + n(5) - n(2 \cap 5) = 248 + 99 - 49 = 298 \Rightarrow$ so there are 248 numbers which are divisible either by 2 or 5.

Hence numbers which are not divisible = 496 – 298 = 198

Method 2—Using divisibility

In the range $4 \leq P \leq 499 = 496$ natural numbers, half of the numbers will be divisible by 2 and other half will not be divisible by 2.

Hence, number of numbers not divisible by 2 = 248
Out of these 248 numbers, 50 numbers will be divisible by 5 (numbers like 5, 15, 25, 35,..... 495). Hence, number of numbers not divisible by 5 out of the number not divisible by 2 = 248 – 50 = 198.

16. LCM of 12^{24}, 16^{18} and N is 24^{24}
$12^{24} = 2^{48} \times 3^{24}$, $16^{18} = 2^{72}$, $24^{24} = 2^{72} \times 3^{24}$
It can be seen that even if N is not present, LCM of 12^{24}, 16^{18} is 24^{24}.
Hence N can take any value from $2^{0-72} \times 3^{0-24}$. Number of values possible = $73 \times 25 = 1825$

17. The side of the square slab is the HCF of 544 cm and 374 cm. HCF (544, 374) = 34.

18. Side of each tile = (HCF of 1517 and 902) cm = 41 cm
Hence number of tiles $= \dfrac{1517 \times 902}{41 \times 41} = 814$

19. Second number $= \dfrac{\text{HCF} \times \text{LCM}}{\text{First number}} = \dfrac{4800 \times 160}{480} = 1600.$

21. LCM of 60 and 62 seconds is 1860 sec = 31 min.
Hence, they will beep together at 10.31 a.m.

22. Assume x and y are the LCM and the HCF of the two numbers respectively.
Given is x + y = 312 and x – y = 264
$\Rightarrow x = \dfrac{312 + 264}{2} = 288$ and $y = \dfrac{312 - 264}{2} = 24$
Since, the HCF is 24, let the two numbers be 24p and 24q.
Then, 24p + 24q = 168 $\Rightarrow$ a + b = 7
LCM × HCF = product of the numbers
Or, 288 × 24 = 24a × 24b or ab = 12
Hence, $a - b = \sqrt{(a+b)^2 - 4ab} = \sqrt{49 - 48} = 1$
This gives a = 4 and b = 3
Numbers are 24a and 24b, i.e., 96 and 72.
Alternatively this question can be done with the help of options too.

23. LCM $= \dfrac{\text{Product of expressions}}{\text{HCF}} = \dfrac{pq}{1} = pq.$

24. $25ab^2c^3 = 5^2 \times a \times b^2 \times c^3$
$80ab^3c^2 = 5 \times 2^4 \times a \times b^3 \times c^2$
$12a^2b^2c = 3 \times 2^2 \times a^2 \times b^2 \times c$
Hence, required HCF = $5 \times 2 \times a \times b^2 \times c$

25. Since, y + k is the HCF, putting x = –k will make each of the expressions zero.
$\Rightarrow k^2 - ak + b = k^2 - ck + d = 0$
Or, $k = \dfrac{b - d}{a - c}$

PRACTICE EXERCISE 2

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	**2**	**4**	**3**	**2**	**2**	**5**	**3**	**2**	**2**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
2	**1**	**1**	**3**	**2**	**1**	**1**	**4**	**3**	**1**

1. This is possible only for prime numbers. There are 25 prime numbers from 1 to 100.

2. 263 is a prime number. So only possible set = (263, 1)

3. This will happen only if LCM of 1st n consecutive natural numbers = n!.
IT can be observed that this is possible only for n = 1, or n = 2 or n = 3. Hence 3 values.

4. In the last question, we saw that till n = 3, for every value of n, n! = LCM of 1st n consecutive natural numbers.
For n = 4, LCM of 1st n consecutive natural numbers = n!/2 [this is happening all because of the fact that 4 introduces 2^2 for the first time, and LCM of 1st 3 consecutive natural numbers contain only 2^1]
For n = 5 also same thing happens.
For n > 5, it can be seen that LCM of 1st n consecutive natural numbers will multiply n! By 6 and then 7 and so on...however LCM will not be multiplied in the similar fashion.

5. Let us consider right hand side ratio. Ratio will not change with respect to LHS ratio only if (n + 1)th number is a prime number, so that the multiplier on the numerator = multiplier on the denominator. So, the job is to find out the prime numbers less than 40.
Number of prime numbers = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37 = 12 Numbers.
But we will not consider n = 2, since for n = 2, both LHS = RHS.
Hence, total values of n = 11.

6. $300 = 5^2 \times 2^2 \times 3^1$
Sets possible = 38

8. For n being a natural number, following relation holds good:
$(n + 1) \times \text{LCM}\ [({}^nC_0), ({}^nC_1), ({}^nC_2), \ldots\ldots\ldots, ({}^nC_n)] = \text{LCM}\ ((1, 2, 3, \ldots\ldots\ldots\ldots n, (n + 1))$
$\Rightarrow \text{LCM}\ [({}^nC_0), ({}^nC_1), ({}^nC_2), \ldots\ldots\ldots, ({}^nC_n)] = (\text{LCM}\ ((1, 2, 3 \ldots\ldots\ldots\ldots\ldots\ldots\ldots n, (n + 1)))/(n + 1)$
$\Rightarrow \text{LCM}\ [({}^{20}C_0), ({}^{20}C_1), ({}^{20}C_2), \ldots\ldots\ldots, ({}^{20}C_{20})]$

$$= \frac{\text{LCM}\ (1, 2, 3)\ldots\ldots\ldots\ldots\ldots\ldots\ldots 20, 21)}{21}$$

LCM (1, 2, 3,.....21) = $2^4 \times 3^2 \times 5 \times 7 \times$ some natural number
When LCM (1, 2, 3,.....21) is divided by 21, power of 3 = 1, and power of 7 = 0.
Hence option 3 is the answer.

9. Either 1 or P can be the HCF of N, N + P.

11. We know that:

$$\frac{\text{LCM } (a, b, c)^2}{\text{LCM } (a, b) \times \text{LCM } (b, c) \times \text{LCM } (a, c)} = \frac{\text{HCF } (a, b, c)^2}{\text{HCF } (a, b) \times \text{HCF } (b, c) \times \text{HCF } (a, c)}$$

Since HCF of any two or three consecutive numbers = 1, hence,

$$= \frac{\text{HCF } (a, b, c)^2}{\text{HCF } (a, b) \times \text{HCF } (b, c) \times \text{HCF } (a, c)} = 1.$$

Hence, LCM $(a, b, c)^2$ = LCM (a, b) × LCM (b, c) × LCM (a, c)

12. The only possible HCF for any two (or three or more) consecutive natural number = 1. Hence, answer = 1.

13. The only possible HCF for any two (or three or more) consecutive natural number = 1. Hence, answer = 1.

14. Different sets of LCM (a, b, c) = 60. It can be seen that if we fix two of the numbers at 60 each, then 3rd number can be taken as *any* factor of 60. Hence different value of HCF = total number of factors of 60 = (2 + 1) (1 + 1) (1 + 1) = 12.

Solutions for questions (15–16):

Since, HCF (a, b, c) = 10, we can assume a = 10x, b = 10y, c = 10z

So, LCM (10x, 10y, 10z) = 100 ⇒ LCM (x, y, z) = 10 and

HCF (10x, 10y, 10z) = 10 ⇒ HCF (x, y, z) = 1.

Obviously, largest possible value of x or y or z can be 10, and value of x or y or z can take only those values that are factors of 10 = 1 or 2 or 5 or 10.

Possible sets are:

x	y	z
1	1	10
1	2	10
1	5	10
1	10	10
2	5	10
2	5	1

These are the six different sets of (x, y, z).

15. Minimum possible value of (a × b × c) = 10 × 10 × 100 = 10000

16. Since there are six different set of (x, y, z), however, not all these values will give different product.

Total different value of x × y × z = 10, 20, 50, 100, hence, only four different values of (a × b × c) are possible.

17. HCF of [(N – 1)! and N] > 1 none of the factors of N and (N – 1)! are common or in other words, none of the factors of N are present in (N – 1)! (other than 1). This is possible only if N = Prime number. Hence, statement 1 is the only definitely true statement. So, answer is option (1).

18. Lowest possible value of N = 19, and maximum possible value of N = 22. Hence, difference between maximum and minimum value of N = 22 – 19 = 3.

19. $(3^{2003} - 1)$ and $(3^{2003} + 1)$ are two consecutive even natural numbers, hence, their HCF = 2.

We know that,

LCM × HCF = Product of natural numbers

Since we have to find out only the unit digit of LCM, we focus only upon the unit digit of the numbers too.

Unit digit of $(3^{2003} - 1) = 6$ and Unit digit of $(3^{2003} + 1) = 8$.

LCM × 2 =...6 × ....8 $\Rightarrow$ Unit digit of LCM = 2.

Note Its wrong to calculate the LCM of unit digit of the numbers (6, 8) = 24, and then saying that unit digit of the LCM will also be equal to 4.

20. HCF (111.....written n + 1 times, 111.....written n times) = 1

This can be proved by taking a small example:

HCF (1111, 11111) = HCF (1111, 10000) [Using difference method of finding HCF]

HCF (1111, 10000) = HCF (1, 10000) [Using Euclidean Algorithm of HCF] = 1

CAT QUESTIONS FROM THIS CONCEPT

CAT 2002

1. At a bookstore, 'MODERN BOOK STORE' is flashed using neon lights. The words as individually flashed at the intervals of $2\frac{1}{2}$ s, $4\frac{1}{4}$ s and $5\frac{1}{8}$ s respectively, and each word is put off after a second. The least time after which the full name of the bookstore can be read again is

1. 49.5 s **2.** 73.5 s **3.** 1744.5 s **4.** 855 s

CAT 2001

2. Three pieces of cakes of weight $4\frac{1}{2}$ lb, $6\frac{3}{4}$ lb and $7\frac{1}{5}$ lb respectively are to be divided into parts of equal weight. Each part must be as heavy as possible. If one such part is served to each guest, then what is the maximum number of guests that could be entertained?

1. 54 **2.** 72 **3.** 20

4. None of these

3. A red light flashes three times per minute and a green light flashes five times in 2 min at regular intervals. If both lights start flashing at the same time, how many times do they flash together in each hour?

1. 30 **2.** 24 **3.** 20 **4.** 60

SOLUTIONS AND EXPLANATIONS

1. 2. Because each word is lit for a second,

$$\text{LCM}\left(\frac{5}{2}+1, \frac{17}{4}+1, \frac{41}{8}+1\right) = \text{LCM}\left(\frac{7}{2}, \frac{21}{4}, \frac{49}{8}\right)$$

$$\frac{\text{LCM}\,(7, 21, 49)}{\text{HCM}\,(2, 4, 8)} = \frac{49\times 3}{2} = 73.5\ \text{s}$$. Hence option 2.

2. 4. $\text{HCF}\left(\frac{9}{2}, \frac{27}{4}, \frac{36}{5}\right) = \frac{\text{HCF}\,(9, 27, 36)}{\text{LCM}\,(2, 4, 5)} = \frac{9}{20}\ \text{lb}$

= weight of each piece total weight = 18.45 lb

Maximum number of guests $= \frac{18.45\times 20}{9} = 41$. Hence option 4.

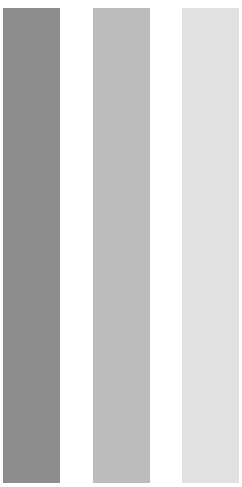
3. First light after 20 s.

Second light blinks after 24 s.

They blink together after LCM of 20 and 24 = 120 s = 2 min.

So, No. of times they blink together in one hour $= \frac{60}{2} = 30$

So, option 1.



Chapter 3

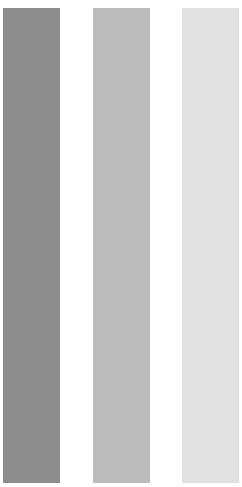
of

DEMYSTIFYING NUMBER SYSTEM

REMAINDER

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- What is Remainder?
- Properties of Remainder
- Methods of finding out remainder
- Theorems related to remainder—CRT, Fermat's etc.
- Types of questions asked from this chapter in the CAT

REMAINDER

A number is said to be divisible by another number if the quotient obtained is an integer and remainder obtained is zero. This statement provides two necessary conditions for a number to be divisible by other:

1. Quotient obtained should be an integer. This can be a +ve integer, or a –ve integer or zero.
2. Remainder should be equal to zero.

So when I distribute 30 apples among 6 children, each of the children get equal integral number of apples (5 in this case—this is known as quotient), as well as I am left with zero apple (this is known as remainder). Going through the same logic, if i distribute zero apple among 6 children, each of the children still get equal integral number of apples (0 in this case) and I am left with zero apple (this is known as remainder). And so we say that $\frac{0}{6} = 0$.

Consider another case: I have now 33 apples, and I want to distribute these apples among 6 kids. I may divide it in two ways:

(i) I first distribute 5 apples each, and then distribute the remaining 3 apples by cutting in halves. So quotient obtained = 5.5
(ii) If I distribute 5 apples to each of the kids, I am left with 3 apples. This is the case when we obtain remainder. Quotient = 5, remainder = 3.

Basic Framework of Remainder: Different Cases

(i) When N is a number divisible by 7, it can be written as, 7 K = N, where K is the quotient.
(ii) When N is divided by 7, remainder obtained is 3 $\Rightarrow$ it can be written as, 7 K + 3 = N, where K is the quotient.
(iii) When N is divided by 7, remainder obtained = 3 is equivalent of saying remainder obtained is (–4), when divided by 7. It can be understood that when N is divided by 7, remainder obtained is 3 $\Rightarrow$ N is 3 more than a multiple of 7 $\Rightarrow$ So N is 4 short of another multiple of 7. So remainder obtained = –4.
(iv) When divided by 8, different remainders obtained can be = 0, 1, 2, 3, 4, 5, 6, 7 (8 different remainders)

Similarly, when divided by 5, different remainders obtained can be = 0, 1, 2, 3, 4 (5 different remainders)

Properties of Remainder

1. Remainders are Additive

It means that remainder can be obtained by breaking down a number in two or more than two parts, and then finally calculating the resultant.

For example,

Worked-out Example 1 What is the remainder obtained when 40 is divided by 7?

☝ ***Solution*** Obviously the answer is 5.

$40 = 5 \times 7 + 5$ [Dividend = Quotient × Divisor + Remainder]

Using the property of remainder being additive, we can break down 40 in two or more than two parts, find the individual remainder and finally add up those individual remainder to get the final remainder.

$$\frac{40}{7}=\frac{35+5}{7}=\frac{36+4}{7}=\frac{30+10}{7}=\frac{20+12+8}{7}$$

It can be seen in the above example that in each of the cases remainder obtained is going to be same.

$\frac{35+5}{7}=\frac{35}{7}+\frac{5}{7}$ (Remainder obtained individually = 0 + 5 = 5)

We could have even written down 40 as one number being subtracted from other too.

$$\frac{40}{7}=\frac{50-10}{7}=\frac{49-9}{7}$$

Remainder obtained from $\frac{50-10}{7}=\frac{50}{7}-\frac{10}{7}$

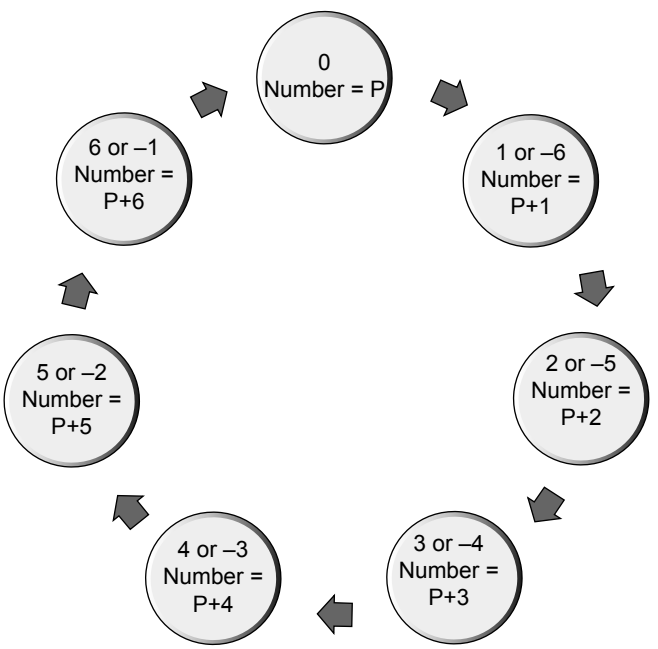
Remainder obtained from $\frac{50}{7}=1$ and remainder obtained from $\frac{10}{7}=3$

So net remainder = 1 – 3 = –2 = 5

Concept enhancer 1 Remainder obtained = –2 when divided by 7 means that this number is 2 less than a multiple of 7 ⇒ Hence this number is 5 more than the previous multiple of 7.

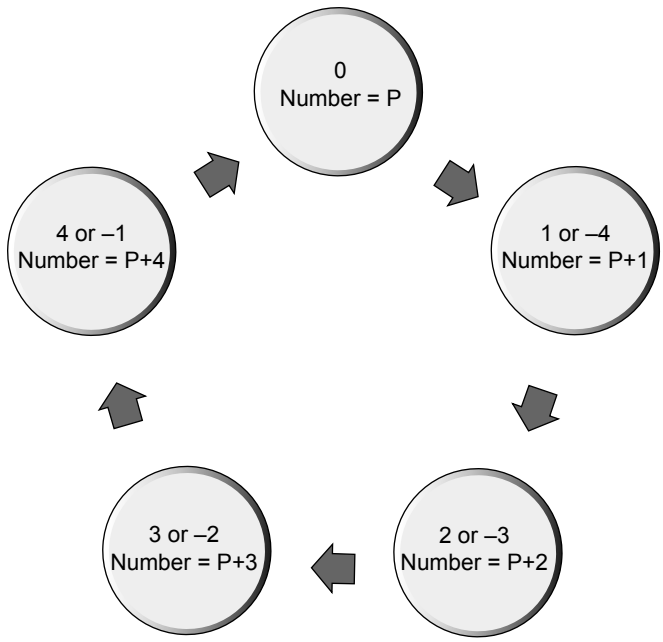
Following is the expression of what we just discussed, 7 K – ***2*** = 7 (K – 1) + ***5***
It can be further understood through the following graphics:

Graphics 1—When divided by 7



It shows that when divided by 7, P is divisible. So when P+1 will be divided by 7, remainder obtained will be either 1 or –6. Similarly, when P+2 is divided by 7, remainder obtained will be 2 or –5. And so on.

Graphics 2—When divided by 5



It shows that when divided by 5, P is divisible. So when P+1 will be divided by 5, remainder obtained will be either 1 or –4. Similarly, when P+2 is divided by 7, remainder obtained will be 2 or –3. And so on.

2. Remainders are Multiplicative

It means that remainder can be obtained by breaking down a number in two or more than two parts, and then finally calculating the resultant.

For example,

Worked-out Example 2 What is the remainder obtained when (10 × 20) is divided by 7?

☝ ***Solution*** Going by Actual method of finding remainder, remainder obtained = 200/7 = 4

Since remainders are multiplicative, we can find out the individual remainder, and then multiply these to obtain final remainder.

Remainder obtained when 10 is divided by 7 = 3

Remainder obtained when 20 is divided by 7 = 6

Remainder obtained $\frac{10\times20}{7}$ = Remainder obtained $\frac{3\times6}{7}$ = Remainder obtained $\frac{18}{7}=4$

Let us understand that why this happens:

$10 = 7p + R_1$ and $20 = 7q + R_2$ where R_1 and R_2 are the respective remainder obtained when 10 and 20 are divided by 7. p and q are respective quotients.

$10 \times 20 = (7p + R_1)(7q + R_2) = (7p \times 7q) + (7p \times R_2) + (7q \times R_1) + (R_1 \times R_2)$

----------(1)

When (1) is divided by 7, each of the terms $(7p \times 7q) + (7p \times R_2) + (7q \times R_1)$ is divisible by 7. So the net remainder is obtained from the last term $(R_1 \times R_2)$. This is exactly what we have done in the last example.

Worked-out Example 3 What is the remainder when $(1421 \times 1423 \times 1425)$ is divided by 12? **{Based on CAT 2000}**

☝ ***Solution*** Remainder of 1421/12 = 5

Remainder of 1423/12 = 7

Remainder of 1425/12 = 9

Remainder $(1421 \times 1423 \times 1425)/12$ = Remainder $(5 \times 7 \times 9)/12$ = Remainder $(5 \times 63)/12$ = Remainder $(5 \times 3)/12 = 3$

Worked-out Example 4 What is the remainder when 4^{100} is divided by 7?

☝ ***Solution*** Remainder obtained from $\frac{4^{100}}{7}=\frac{16^{50}}{7}$ = Remainder obtained from $\frac{2^{50}}{7}=\frac{32^{10}}{7}$ = Remainder obtained

Ideally, when we are using this method to find out the remainder, we should be constantly on a lookout for an expression giving us remainder either +1 or –1. Although it is not essential that we always get remainder +1 or –1 for every expression. We will discuss this once again while discussing the pattern method of finding remainder.

Let us see that how the same question becomes quite easier now:

Remainder obtained from $\frac{4^{100}}{7}$ = Remainder obtained from $\frac{[4^3]^{33}\times4^1}{7}$ = Remainder obtained from = Remainder obtained from $\frac{1^{33}\times4^1}{7}$ = Remainder obtained from $\frac{4}{7}=4$

3. Its wrong to calculate the Remainder by reducing the fraction to its lower terms. For example, Look at the following fractions:

   (a) ½ (b) 10/20 (c) 100/200

   Despite the fact that all the fractions taken here are same, remainder obtained in each of the cases are going to be different,

Remainder obtained in (a) ½ = 1
Remainder obtained in (b) 10/20 = 10
Remainder obtained in (a) 100/200 = 100

Worked-out Example 5 $\dfrac{1!+2!+3!+4!+5!+\ldots\ldots\ldots\ldots\ldots 1000!}{2^3\times 3\times 5}$

1. 3 2. 1 3. 5 4. 4 5. None of these

☝ ***Solution*** Denominator = $2^3 \times 3^1 \times 5 = 120 = 5!$

So the remainder will come only from 1! + 2! + 3! + 4!. Hence, net remainder 1! + 2! + 3! + 4! = 33 (since numerator is less than denominator, numerator itself will be the remainder).

Summarizing the discussion regarding the property of remainder:

Let remainder obtained when 'a' is divided by n = b and remainder obtained when 'c' is divided by n = d.

Then remainder obtained for LHS and RHS will be same.

(i) $\dfrac{a+c}{n}=\dfrac{b+d}{n}$

(ii) $\dfrac{a-c}{n}=\dfrac{b-d}{n}$

(iii) $\dfrac{ac}{n}=\dfrac{bd}{n}$

Let us now see some standard formats of remainder questions

Worked-out Example 6–9 FORMAT 1

A number when divided by 7 and 9 gives remainder 1 and 1 respectively.

6. Which is the lowest such natural number?

☝ ***Solution*** Lowest such natural number = 1, and not 64.

7. What is the remainder when lowest such natural number is divided by 11?

☝ ***Solution*** When 1 is divided by 11, remainder obtained = 1.

8. For every ..... natural numbers, there is exactly one such number which satisfies the property given above. (Fill in the blanks)

☝ ***Solution*** For every 63 numbers (LCM of 9 and 7), there will be exactly one such number satisfying this property.

9. Only natural numbers can satisfy the above given statement. (True/False)

☝ ***Solution*** False. There may be negative numbers satisfying this statement.



Concept enhancer 2 Understand that the family of numbers satisfying this statement is, 1 ± 63P, where P is an integer. Some of the numbers satisfying this statement are, 1, 64, –62 etc.

Worked-out Example 10 FORMAT 2

A number when divided by 7 and 6 gives remainder 1 and 4 respectively.

10. Which is the lowest such natural number?

Solution Let us assume N is that natural number.

N is divided by 7, remainder obtained = 1 $\Rightarrow$ N = 7 x + 1

N is divided by 6, remainder obtained = 4 $\Rightarrow$ N = 6 y + 4

N = 7 x + 1 = 6 y + 4 $\Rightarrow$ 7 x – 6y = 3

Now we are required to insert the values of x and y to satisfy the equation given above.

We obtain x = 3 and y = 3, and the number obtained = N = 7 × 3 + 1 = 6 × 3 + 1 = 22

Alternatively, it can be seen through making some combinations directly:

Lowest natural number N divided by 7, remainder obtained = 1 is N = 1, however, when N is divided by 6, remainder obtained = 1 and not 4. Hence, 1 is not that number.

Next Natural number N divided by 7, remainder obtained = 1 is N = 8, however, when 8 is divided by 6, remainder obtained = 2 and not 4. Hence, 8 is not that number.

Next natural number N divided by 7, remainder obtained = 1 is N = 15, however, when 15 is divided by 6, remainder obtained = 3 and not 4. Hence, 15 is not that number.

Next natural number N divided by 7, remainder obtained = 1 is N = 22, and when 22 is divided by 6, remainder obtained = 4. Hence, 22 is that number.



Concept enhancer 3 Although a larger question arises that till which number we should keep on checking, and which number should be that limit till which if we do not get any number satisfying the given statements, we should be in a position to conclude that we are not going to get any such number on the whole number line?

Answer to the above question is—Till the LCM of the divisors. In this case, it is LCM of (7 and 6) = 42.

So if we do not get any such number till 42, we can conclude that no such number exists that will give remainder 1 and 4 when divided by 7 and 6 respectively. In other words, if at all numbers satisfying this statement exists, one and exactly one such natural number will be obtained from 1 to 42, exactly one such number from 43 to 84, and so on for every next 42 numbers.

It can also be inferred that for every 42 numbers, not necessarily starting from 1, will have exactly one such natural number satisfying this statement. For example, from 100 to 141, there is exactly one number that will satisfy the statement given in this question.

Worked-out Example 11–13 FORMAT 3

11. What is the remainder when this number is divided by 11? (use data from previous question)

Solution It can be seen that there are more than one natural numbers satisfying the given statement, 22, 64, 106 etc.

Remainder obtained when 22 is divided by 11 = 0

Remainder obtained when 64 is divided by 11 = 9

Remainder obtained when 106 is divided by 11 = 7

So, more than one remainders will be obtained.

12. For every ..... natural numbers, there is exactly one such number which satisfies the property given above. (Fill in the blanks)

☝ ***Solution*** Using the Concept enhancer 3, for every 42 natural numbers, there is exactly one such number which satisfies the statements given in the question. Although the term natural number can be replaced by integers, and this statement will be true even in that case.

13. Only natural numbers can satisfy the above given statement. (True/False)

☝ ***Solution*** False. Using the answer to 8, even negative integers will satisfy. Understand that the family of numbers satisfying this statement is, 22 ± 42P, where P is an integer.

Worked-out Example 14 FORMAT 4

A number when divided by 6 and 9 gives remainder 4 and 6 respectively. Which is the lowest such natural number?

☝ ***Solution*** Let us go by the discussion that we had in the above question. If we are going to get a number satisfying the conditions given in this question, we should obtain this number in the first 18 (18 = LCM of 9 and 6) natural numbers. It is quite evident that no such number exist satisfying the above statements. Hence, no such number is possible.

Alternatively, let us assume that N is the natural number.

So, $N = 6x + 4 = 9y + 6$

$\Rightarrow 6x - 9y = 6 - 4 \Rightarrow 3(2x - 3y) = 2$

We can see that LHS of this equation is divisible by 3, whereas, RHS is not. Hence, if a number divided by 9 gives remainder 6, then the same number cannot give remainder 4 when divided by 6 and vice-versa.

Worked-out Example 15 A number when divided by 6 and 9 gives remainder x and y respectively. How many different values can $|x - y|$ take?

☝ ***Solution*** We can solve this question through taking different values. Or else we can use following to solve this question:

Concept enhancer 4 Difference between the remainders = $|x - y|$ has to be divisible by the HCF of the divisors i.e., in this case = HCF (9,6) = 3

Hence different values of $|x - y| = 0, 3 \Rightarrow$ So, only two values are possible.

Worked-out Example16 What is the remainder when 55^{100} is divided by 110?

☝ ***Solution*** HCF of 55 and 100 = 55, so the remainder obtained has to be divisible by 55. So the remainders possible are 0 or 55.

Now, 55^{100} is an odd number and 110 is an even number, so remainder obtained can never be equal to zero (or in other words, since an odd number can never be divisible by an even number, 55^{100} can never be divisible by 110).

Hence, 55 is the only possible option left out for the remainder. So, 55 is the remainder.

Worked-out Example 17–18 FORMAT 5

An integer when divided by 17 and 9 gives remainder 14 and 6 respectively.

☝ ***Solution*** Remainder obtained = 14 when divided by 17 is same as remainder being (–3) when divided by 17.

Remainder obtained = 6 when divided by 9 is same as remainder being (–3) when divided by 9.

17. Which is the lowest such integer?

☝ ***Solution*** Integer is going to be of the format—(–3 ± 153K, where K is any integer). Hence, the lowest such integer is going to be (–∞).

18. Which is the lowest such positive integer?

☝ ***Solution*** Taking a cue from the above solution, lowest positive integer is going to be (–3 + 153×1) = 150.

Summarizing the above discussion:

	Problem states that…..	Solution
1	Find the Greatest Number that will exactly divide a, b and c.	Required number = HCF of a, b and c (greatest common divisor)
2	Find the Greatest number that will divide x, y and z leaving remainders a, b and c respectively.	Required number (greatest common divisor) = HCF of (x–a), (y–b) and (z–c)
3	Find the least number which is exactly divisible by a, b and c.	Required number = LCM of a, b and c
4	Find the least number which when divided by x, y and z leaves the remainders a, b and c respectively, and (x–a) = (y–b) = (z–c) = N.	Required number = LCM of (x, y and z) – N
5	Find the least number which when divided by x, y and z leaves the same remainder 'r' each case.	Required number = (LCM of x, y and z) + r

[This table presents just a part of the above discussion, and does not present the exhaustive list that we discussed above.]

TEST YOUR LEARNING 1

(You may solve some of these questions using options. However, I have removed the options, because I want you to develop the equation method of solving which may be the only method of solving a particular question like Question 4 as given below)

1. A natural number divided by 9 gives remainder 4, and when divided by 7 gives remainder 2. What will be remainder obtained when lowest such natural number is divided by 11?
2. A natural number divided by 15 gives remainder 3, and when divided by 6 gives remainder 4. What will be remainder obtained when lowest such natural number is divided by 13?
3. What is the smallest five-digit number which when divided by 7, 11 and 21 leaves a remainder of 3 in each case?
4. What is the sum of digits of the least multiple of 13 which when divided by 6, 8 and 12 leave 5, 7 and 11 as the remainder respectively?
5. Find the lowest four digit natural number which when divided by 5 leaves remainder 1 but when divided by 6 leaves remainder 2.
6. How many three digit natural numbers are there which when divided by 6 gives remainder 2, but when divided by 7 gives remainder 1?
7. A natural number divided by 19 gives remainder 7, and when divided by 17 gives remainder 5. What will be remainder obtained when lowest such natural number is divided by 13?
8. A natural number divided by 7 gives remainder 4, when divided by 17 gives remainder 4, and when divided by 9 gives remainder 6. Which is the lowest such natural number?
9. A natural number N follows the following property:
   (i) Remainder obtained when divided by 4 = 3
   (ii) Remainder obtained when divided by 13 = 4
   (iii) Remainder obtained when divided by 23 = 9
   (iv) Remainder obtained when divided by 15 = 12

   How many different values of N are possible?
10. In the above question, how many natural numbers less than 10000 will satisfy the properties given above?

Fermat's Remainder Theorem

Let P be a prime number and N be a number, not divisible by P (in other words N and P are co-prime). Then, remainder obtained when N^{P-1} is divided by P is 1.

Remainder obtained when $\frac{N^{P-1}}{P} = 1$ provided HCF (N, P) = 1, and P is a prime number.

Using Fermat's theorem we can say that:

Remainder of $\frac{6^6}{7}$ = Remainder of $\frac{5^6}{7}$ = Remainder of $\frac{4^6}{7}$ = Remainder of $\frac{3^6}{7}$ = Remainder of $\frac{2^6}{7}$ = Remainder of $\frac{1^6}{7}$ = 1

Worked-out Example 19 What is the remainder when 2^{100} is divided by 101?

Solution Since it satisfies the Fermat's theorem format, remainder = 1

Worked-out Example 20 What is the remainder when 7^{82} is divided by 11?

Solution Using Fermat's, remainder obtained when 7^{10} is divided by 11 = 1. Hence, remainder obtained when 7^{80} is divided by 11 = 1.

So remainder obtained $\frac{7^{82}}{11}$ = Remainder obtained $\frac{7^{80}\times 7^2}{11}$ = Remainder obtained $\frac{1\times 7^2}{11}$ = remainder obtained when 49 is divided by 11 = 5.

Wilson's Remainder Theorem

Statement If P is a prime number, then (p–1)! + 1 is divisible by P.

In other words, remainder obtained when (p–1)! is divided by p (p = prime number) = –1.

For example, Remainder obtained when 4! Is divided by 5, remainder obtained = –1

Remainder obtained when 6! Is divided by 7, remainder obtained = –1

Remainder obtained when 12! Is divided by 13, remainder obtained = –1

Corollary to Wilson's Theorem: If p is a composite number, then (p–1)! is divisible by P (except for p = 4).

Worked-out Example 21 FORMAT 6—Pattern/Cycle method of finding Remainder

Remainder follows a pattern specific to the example or question given. Like when 4^N is divided by 7, there is going to be a pattern of remainders. And once we have obtained this pattern of remainder, we can find out the remainder for any higher power.

Reason why pattern exists can be attributed to the fact that when a series of powers are divided by 7 (or any other number) (as in case of 4^N divided by 7), total different remainder possible = 7 (namely 0,1,2,3,4,5,6). Hence, even if the value of N = 8, there has be atleast one 7^K having same remainder as 7^M. And once even a single remainder has got repeated, which has to happen, the whole pattern can be generated, and used further.

Let us now see the method to find out pattern of remainder:

4^1 divided by 7 → Remainder = 4

4^2 divided by 7 → Remainder = 2

4^3 divided by 7 → Remainder = 1

4^4 divided by 7 → Remainder when $4^3 \times 4^1$ is divided by 7 → Since 4^3 gives remainder = 1 when divided by 7, net remainder when 4^4 divided by 7 = Remainder when 4^1 is divided by 7 = 4

Summarizing the above discussion:

Remainder obtained when 4^N is divided by 7	4^1	4^2	4^3	4^4	4^5	4^6	4^7	4^8	4^9
	4	2	1	4	2	1	4	2	1

Conclusion 1 It can be seen that once we have got remainder 1, we have got the pattern of remainder. Power of number that gives remainder 1 (in this case 4^3), will give us the length of pattern. In this case length of pattern = 3.

Conclusion 2 Pattern will be obtained when any of the remainders is getting repeated—it may be 1 or any other number. In other words, it is not necessary that unless we have got remainder as 1, we will not get pattern.

Understand this with the help of following example:

Finding pattern of 4^N is divided by 6:

Remainder obtained when 4^N is divided by 6	4^1	4^2
	4	4

In this case remainder is getting repeated when 4^2 is divided by 6 = 4. So length of pattern = 1.

Hence, all the next powers of 4^{3+} will have the same remainder = Remainder obtained when 4^1 or 4^2 is divided by 6 = 4.

Worked-out Example 22 What is the remainder when $x = 32^{32^{32}}$ is divided by (a) 2, (b) 3, (c) 4, (d) 5, (e) 6, (f) 7, (g) 8, (h) 9.

(a) It can be clearly seen that x is divisible by 2. Hence, remainder obtained = 0

(b) Remainder obtained when 32 is divided by 3 = –1

Hence remainder obtained from $\frac{32^{32^{32}}}{3}$ = Remainder obtained from $\frac{-1(32^{32})}{3}$ = Remainder obtained from $\frac{1}{3} = 1$

(c) Remainder obtained when 32 is divided by 4 = 0

Hence, remainder = 0

(d) Remainder obtained when 32 is divided by 5 = 2

Hence, remainder obtained from, $\frac{32^{32^{32}}}{5}$ = Remainder obtained from $\frac{2^{(32^{32})}}{5}$ = Remainder obtained from $\frac{2^{4n}}{5}$ = Remainder obtained

(e) Remainder obtained when 32 is divided by 6 = 2

Hence, remainder obtained from, $\frac{32^{32^{32}}}{6}$ = Remainder obtained from $\frac{2^{(32^{32})}}{6}$

Let us find the cycle when 2^n is divided by 6:

Remainder obtained when 2^n is	2^1	2^2	2^3	2^4	2^5
Divided by 6	2	4	2	4	2

It can be seen that cycle of remainder = 2, since remainder obtained from $2^1 = 2^3 = 2^5$ and so on.

For all the odd powers of 2, remainder obtained = 2 and for all the even powers, remainder obtained = 4.

Hence, Remainder obtained from $\frac{32^{(32^{32})}}{6} = \frac{2^{\text{even power}}}{6} = 4$

(f) Remainder obtained when 32 is divided by 7 = 4

Hence, remainder obtained from, $\frac{32^{32^{32}}}{7}$ = Remainder obtained from $\frac{4^{(32^{32})}}{7}$

Let us find the cycle when 4^n is divided by 7:

Remainder obtained when 4^n is	4^1	4^2	4^3	4^4	4^5
Divided by 7	4	2	1	4	2

It can be seen that cycle of remainder = 3. We can see that remainder obtained when 4^{3n} is divided by 7 = 1, remainder obtained when 4^{3n+1} is divided by 7 = 4, remainder obtained when 4^{3n+2} is divided by 7 = 2.

So we need to check that 32^{32} is 3n or 3n+1 or 3n+2?

Remainder obtained when 32^{32} is divided by 3 = Remainder obtained when $(-1)^{32}$ is divided by 3 = 1.

Hence, 32^{32} is 3n+1.

So final remainder = remainder obtained when 4^{3n+1} is divided by 7 = 4.

(g) It can be seen that 32 is divisible by 8. Hence, x is also divisible by 8. So remainder obtained = 0.

(h) Remainder obtained when 32 is divided by 9 = 5.

Hence, remainder obtained from, $\frac{32^{32^{32}}}{9}$ = Remainder obtained from $\frac{5^{(32^{32})}}{9}$

Let us find the cycle when 5^n is divided by 9:

Remainder obtained when 5^n is	5^1	5^2	5^3	5^4	5^5	5^6	5^7
Divided by 9	5	7	8	4	2	1	5

It can be seen that cycle of remainder = 6. We can see that remainder obtained when 5^{6n} is divided by 9 = 1, remainder obtained when 5^{6n+1} is divided by 9 = 5, remainder obtained when 5^{6n+2} is divided by 9 = 7 and so on.

So we need to check that 32^{32} is 6n or 6n+1 or 6n+2 or so on?

Remainder obtained when 32^{32} is divided by 6 = Remainder obtained when $(2)^{32}$ is divided by 6 = 4

Hence, 32^{32} is 6n+4.

So final remainder = remainder obtained when 5^{6n+4} is divided by 9 = 4

Worked-out Example 23 What is the remainder when 5^3 is divided by 3^3?

☝ ***Solution*** First of all, remainder obtained when $\frac{5^3}{3^3} \neq$ Remainder obtained when $\left[\frac{5}{3}\right]^{\wedge}3$. Remainder obtained when $\frac{5^3}{3^3} =$ remainder obtained when 125 is divided by 27 = 17.

Worked-out Example 24 What is the remainder when 5^{15} is divided by 3^3?

☝ ***Solution*** Remainder obtained when $\frac{5^{15}}{3^3} =$ Remainder obtained when $\frac{[5^3]^5}{27} =$ Remainder obtained when $\frac{17^5}{3^3} =$ Remainder obtained when $\frac{289 \times 289 \times 17}{3^3} =$ Remainder obtained when $\frac{(-8)(-8) \times 17}{27}$ = Remainder obtained when $\frac{64 \times 17}{27}$ = Remainder obtained when 170 is divided by 27 = 8.

Relationship between Remainder and Factors in Case of a Polynomial

In case of polynomials, we are not required to find the remainder by actual division. By using factor theorem, we can find out the remainder without actual division. Although we should know the method of synthetic division which is atleast better and less time taking than actual long division method.

Synthetic Division Method

Synthetic division is a shortcut method for dividing a polynomial by a linear factor of the form (x – k).

To write the dividend we need to take care that we write the powers of variable (starting with the highest one) in descending powers and insert 0's for any missing terms. For example, if we divided $(x^4 - 3x + 6)$, the polynomial $(x^4 - 3x + 6)$, starts with degree 4, then the next degree is 1 and finally zero. So the missing degrees are 3 and 2. So if we were to divide it, we would write the co-efficient as follows:

Co-efficient of x^4	Co-efficient of x^3	Co-efficient of x^2	Co-efficient of x^1	Co-efficient of x^0
1	0	0	–3	6

Let us understand the modus operandi with the following example:

Worked-out Example 25 What is the remainder when the polynomial $p(x)= x^4 - 3x^2 + 1$ is divided by $x - 2$?

Step 1: Start with writing 2 (after putting $x - 2 = 0$, value obtained of $x = 2$) on the left hand side and co-efficient on the right hand side.

	x^4	x^3	x^2	x^1	x^0
2	1	0	−3	0	1

Step 2: Write co-efficient of x^4 at the bottom. These values written below 2nd line is the result that we will obtain at the end.

2	1	0	−3	0	1
	↓				
	1				

Step 3: Multiply 1 (inside circle) by 2 (that we have written on the left hand side).

2	1	0	−3	0	1
		2			
	1				

Step 4: Add the value above 1st line (0) and the value between 1st and 2nd line (2) and write the result below 2nd line.

2	1	0	−3	0	1
		+ 2			
		↓			
	1	2			

Step 5: Next multiply this 2 written below 2nd line with 2 (written on the left hand side).

2	1	0	−3	0	1
		2	4		
	1	2			

Step 6: Add the value above 1st line (–3) and the value between 1st and 2nd line (4) and write the result below 2nd line.

2	1	0	−3	0	1
		2	4		
	1	2	1		

Step 7: Next multiply this 1 written below 2nd line with 2 (written on the left hand side).

2	1	0	–3	0	1
		2	4	2	
	1	2	1		

Step 8: Add the value above 1st line (0) and the value between 1st and 2nd line (2) and write the result below 2nd line.

2	1	0	–3	0	1
		2	4	2	
	1	2	1	2	

Step 9: Next multiply this 1 written below 2nd line with 2 (written on the left hand side).

2	1	0	–3	0	1
		2	4	2	4
	1	2	1	2	

Step 10: Add the value above 1st line (1) and the value between 1st and 2nd line (4) and write the result below 2nd line.

2	1	0	–3	0	1
		2	4	2	4
	1	2	1	2	5

Remainder obtained is equal to the number on the bottom right hand side = 5.

Alternatively same question can be done by using factor theorem too.

Factor Theorem

If g(x) divides f(x), we say that f(x) is divisible by g(x) or g(x) is a factor of f(x).

Factor theorem gives us a method to determine whether a polynomial g(x) is a factor of a polynomial f(x) or not without actual division.

Let f(x) be a polynomial of degree greater than or equal to 1 and 'a' be a real number such that f(a) = 0, then (x – a) is a factor of f(x), Conversely, if (x – a) is a factor of f(x) then f(a) = 0.

Note

1. (x + a) is a factor of a polynomial f(x) if f(–a) = 0.
2. (ax – b) is a factor of a polynomial f(x) if f(b/a) = 0.
3. ax + b is a factor of polynomial f(x) if f(–b/a) = 0.
4. (x – a) (x – b) is a factor of a polynomial f(x) if f(a) = 0 and f(b) = 0.

Worked-out Example 26 Remainder obtained when polynomial $p(x)= x^4-3x^2+1$ is divided by $x-2$.

☝ ***Solution*** Remainder obtained when polynomial $p(x)= x^4-3x^2+1$ is divided by $x-2 \Rightarrow p(2) = 2^4-3\times2^2+1 = 5$.

Worked-out Example 27 Find the remainder when $p(y) = y^3 + y^2 + 2y + 3$ is divided by $y+2$.

☝ ***Solution*** $P(y) = K(y+2)$ + remainder $\Rightarrow$ It we put $y = -2$ in $p(y)$, we will obtain the remainder.

Compute $p(-2)$ = Value of $p(y)$ when y is replaced by -2.

We get $P(-2)=(-2)^3 + (2)^2 + 2(-2) + 3 = -8 + 4 - 4 + 3 = -5$

It follows from this example that the remainder obtained when $p(x)$ is divided by $(x-a)$ is equal to $p(a)$ i.e., the value of $p(x)$ at $x = a$.

Worked-out Example 28 Find out if $(x-3)$ is a factor of the polynomial $x^3-3x^2+4x-12$?

☝ ***Solution*** Let $f(x) = x^3-3x^2+4x-12$ be the given polynomial. By factor theorem, $(x-a)$ is a factor of a polynomial $f(x)$ if $f(a) = 0$. Therefore, in order to prove that $(x-3)$ is a factor of $f(x)$, it is sufficient to show that $f(3) = 0$.

Now, $f(x) = x^3-3x^2+4x-12$

$\Rightarrow f(3) = 3^3-3\times3^2+4\times3-12 = 27-27+12-12=0$

Hence, $(x-3)$ is a factor of $f(x)$.

Worked-out Example 29 Find out if $(x-1)$ is a factor of (i) $\left(x^{10}-1\right)$ and (ii) $\left(x^{11}-1\right)$.

☝ ***Solution*** Let $f(x) = x^{10}-1$ and $g(x)=x^{11}-1$.

To find out that $(x-1)$ is a factor of both $f(x)$ and $g(x)$, it is sufficient to show that $f(1) = 0$ and $g(1) = 0$.

Now, $f(x) = x^{10}-1$ and $g(x) = x^{11}-1$

$\Rightarrow f(1) = 1^{10}-1 = 0$ and $g(1)=1^{11}-1 = 0$

$\Rightarrow (x-1)$ is a factor of both $f(x)$ and $g(x)$

Note

1. If a polynomial $p(x)$ is divided by $(x+a)$, the remainder is the value of $p(x)$ at $x = -a$, i.e., $p(-a)$.
2. If a polynomial $p(x)$ is divided by $(ax-b)$, the remainder is the value of $p(x)$ at $x = b/a$, i.e., $P(b/a)$.
3. If a polynomial $p(x)$ is divided by $(ax + b)$, the remainder is the value of $p(x)$ at $x = -b/a$, i.e., $p(-b/a)$.
4. If a polynomial $p(x)$ is divided by $(b - ax)$, the remainder is equal to the value of $p(x)$ at $x = b/a$, i.e., $p(b/a)$.

Worked-out Example 30 Determine the remainder when the polynomial $p(x)= x^4 – 3x^2 + 1$ is divided by x–1.

☝ ***Solution*** We understand that the required remainder is equal to p(1).

Now, $p(x) = x^4 – 3x^2 + 1$

$\Rightarrow p(1) = (1)^4 – 3 \times 1^2 + 1 = 1 – 3 + 1 = –1$

Hence, remainder obtained = p(1) = –1.

SOME GENERALIZATIONS

1. (i) $(a^n + b^n)$ is divisible by (a + b), if n is odd.

(ii) Extension of the above formula—$(a^n + b^n + c^n)$ is divisible by (a + b + c), if n is odd and a, b and c are in Arithmetic Progression.

(iii) $(a^n + b^n + c^n + d^n)$ is divisible by (a + b + c + d), if n is odd and a, b, c and d are in Arithmetic Progression.

Similarly, this formula can be extended for any number of terms provided that the terms as in a, b, c, d etc, are in AP, and power n = Odd.

(iv) $(a^n – b^n)$ is divisible by (a + b), if n is even.

(v) $(a^n – b^n)$ is divisible by (a – b), if n is any natural number.

(vi) $(a^n – b^n)$ is divisible by $(a^k – b^k)$, if k is a factor of n.

Worked-out Example 31 Which of the following will definitely divide $x^{32} – y^{32}$ (for all the values of x and y)? Answer in Yes or No.

(a) $x^2 – y^2$ (b) $x^4 – y^4$ (c) $x^6 – y^6$ (d) $x^8 – y^8$ (e) $x^3 – y^3$ (f) $x^5 – y^5$

☝ ***Solution*** As we have already seen above, all the powers of x and y that are factors of 32 will divide $x^{32} – y^{32}$.

It can be seen in the following way:

$x^{32} – y^{32} = (x^{16} – y^{16}) (x^{16} + y^{16})$ $[a^2 – b^2 = (a+b) (a–b)]$

Now, $(x^{16} – y^{16}) = (x^8 – y^8) (x^8 + y^8) \Rightarrow x^8 – y^8 = (x^4 – y^4) (x^4 + y^4) \Rightarrow x^4 – y^4 = (x^2 – y^2) (x^2 + y^2)$

(a) Yes. Since 2 is a factor of 32, hence $x^{32} – y^{32}$ will be divisible by $x^2 – y^2$.

(b) Yes. Since 4 is a factor of 32, hence $x^{32} – y^{32}$ will be divisible by $x^4 – y^4$.

(c) No. Since 6 is not a factor of 32, hence $x^{32} – y^{32}$ will not be divisible by $x^6 – y^6$. Please understand that this question is set in the background of "all the values of x and y". And in no way, this example says that "$(x^{32} – y^{32})$ will never be divisible by $(x^6 – y^6)$". For example, if y = 0 and $x \neq 0$, then, obviously, $(x^{32} – y^{32}) = x^{32}$ will become divisible by $(x^6 – y^6) = x^6$.

(d) Yes.

(e) No. As stated in (c).

(f) No. As stated in (c).

Worked-out Example 32 What is the remainder when $(15^{23} + 23^{23})$ is divided by 19? **{CAT 2004, 2 marks}**

☝ ***Solution*** It can be observed that $(15^{23} + 23^{23})$ is divisible by 38, so it will be divisible by 19 also. Hence, remainder = 0.

Alternatively, this problem can be done either by cyclicity method or theorem method.

Worked-out Example 33 What is the remainder when $(16^3 + 17^3 + 18^3 + 19^3)$ is divided by 70? **{CAT 2005, 1 mark}**

☝ ***Solution***

Method 1

Let us solve this question in parts:

$(16^3+19^3)+(17^3+18^3)=35\,(\text{odd number})+35\,(\text{odd number})=35\,(\text{odd number}+\text{odd number})$ = 35 × even number = 70 K.

[(16^3+19^3) is an odd number. Since odd × odd = Odd. 35 is an odd number, hence, (16^3+19^3) = 35 (odd number)]

Hence, $(16^3+17^3+18^3+19^3)$ = 70 K ⇒ Remainder obtained = 0.

Method 2

16, 17, 18 and 19 are in AP, and power is odd. Sum of (16 +17+18+19) = 70. Hence, $(16^3+17^3+18^3+19^3)$ will be divisible by 70.

2. $\frac{(A+1)^N}{A}$ will always give 1 as the remainder.(For all natural values of A and N)

Worked-out Example 34 What is the remainder when 9^{100} is divided by 8?

☝ ***Solution*** For A = 8, it satisfies the above condition. So, remainder = 1

Alternatively, we can apply either of cyclicity or theorem method to find the remainder.(Do this yourself).

3. $\frac{A^N}{A+1}$ When N is even, remainder is 1 and when N is odd, remainder is A itself.

Worked-out Example 35 What is the remainder when 2^{10} is divided by 3?

☝ ***Solution*** Since here N is even, so remainder = 1

Relationship between Remainder and Factors in General

Statement

1. If N is divisible by Q, then all the multiples of N will be divisible by Q and so remainder obtained in each of the cases = 0.
   Example—Remainder obtained when 70 is divided by 35 = 0. Hence, remainder obtained when any multiple of 70 (140, 210 etc.) is divided by 35 = 0.
2. If N is divisible by Q, then all the factors of Q will also divide N and so remainder obtained in each of the cases = 0. However, when multiples of Q are dividing N, we may not get the exact remainder, rather we get a range of remainders.

Example

Remainder obtained when 100 is divided by N	N = 70	N = 70 × 2 = 140	N = 70 × 3 = 210	N = 70 × 4 = 280
Remainder obtained	30	100	100	100

Remainders obtained = 30 and 100.

Worked-out Example 36 A number N gives a remainder of 7 when divided by D and a remainder of 20 when divided by 2D. What will be remainder obtained when 2N is divided by D?

☝ ***Solution*** N = DQ + 7 (when Q = Quotient, and D = Divisor) ----------(1)

N = 2D×K + 20 (where K = Quotient and 2D = Divisor) ----------(2)

Putting the value of N from (2) into (1):

We can say that when N = 2D×K + 20 is divided by D, remainder obtained = 7.

$$\frac{2DK+20}{D}=\frac{2DK}{D}+\frac{20}{D}$$

Remainder obtained from $\frac{2DK}{D}=0$

Hence, remainder obtained from $\frac{20}{D}=7 \Rightarrow D=13$

Now, 2N = 2 (DQ + 7) = 4DQ + 14.

Remainder obtained from $\frac{2N}{D}$ = Remainder obtained from $\frac{14}{13}=1$.

Hence, 1 is the remainder.

Some Special Cases

1. Using Binomial Theorem

Binomial expansion of $(a+b)^n = {}^nC_0a^n + {}^nC_1a^{n-1}b + {}^nC_2a^{n-2}b^2$

$+ {}^nC_3a^{n-3}b^3 + ... + {}^nC_nb^n$

It can be seen that power of a is decreasing from n to 0 and power of x is increasing from 0 to n.

Worked-out Example 37 What is the remainder when $(5^{50} + 7^{50})$ is divided by (a) 6, (b) 36, (c) 72?

☝ ***Solution***

(a) $(5^{50} + 7^{50})$ is divisible by 12. Hence, also divisible by 6. So, when $(5^{50} + 7^{50})$ is divided by 6, remainder obtained = 0.

(b) $(5^{50} + 7^{50}) = (6 - 1)^{50} + (6 + 1)^{50}$

Now we will expand each of these two terms binomially one by one:

$(6+1)^{50} = ({}^{50}C_0 \times 8^{50} \times 1^0) + ({}^{50}C_1 \times 8^{49} \times 1^1) + ({}^{50}C_2 \times 8^{48} \times 1^2) + \ldots\ldots\ldots\ldots + ({}^{50}C_{48} \times 8^2 \times 1^{48}) + ({}^{50}C_{49} \times 8^1 \times 1^{49}) + ({}^{50}C_{50} \times 8^0 \times 1^{50})$

Understand that when $(6+1)^{50}$ is divided by 36 $(=6^2)$, except last two terms of the above expansion, all the terms will be divisible. Hence, the remainder will come only from $({}^{50}C_{49} \times 8^1 \times 1^{49}) + ({}^{50}C_{50} \times 8^0 \times 1^{50})$

Similarly, $(6-1)^{50} = ({}^{50}C_0 \times 8^{50} \times 1^0) - ({}^{50}C_1 \times 8^{49} \times 1^1) + ({}^{50}C_2 \times 8^{48} \times 1^2) - \ldots\ldots\ldots\ldots + ({}^{50}C_{48} \times 8^2 \times 1^{48}) - ({}^{50}C_{49} \times 8^1 \times 1^{49}) + ({}^{50}C_{50} \times 8^0 \times 1^{50})$.

In this case too, the remainder will come only from $[-({}^{50}C_{49} \times 8^1 \times 1^{49}) + ({}^{50}C_{50} \times 8^0 \times 1^{50})]$

So the net remainder obtained from $(5^{50} + 7^{50})$ is same as remainder obtained from $({}^{50}C_{49} \times 8^1 \times 1^{49}) + ({}^{50}C_{50} \times 8^0 \times 1^{50}) - ({}^{50}C_{49} \times 8^1 \times 1^{49}) + ({}^{50}C_{50} \times 8^0 \times 1^{50}) =$

2. Chinese Remainder Theorem

Around AD 100, Chinese mathematician Sun-Tsu solved the following problem:

Worked-out Example 38 There is a positive number that gives remainder 2 when divided by 3, remainder 3 when divided by 5, and remainder 2 when divided by 7. Which is the lowest such number?

Mathematically this is written as:

$x = 2 \pmod 3$

$x = 3 \pmod 5$

$x = 2 \pmod 7$

A number which when divided by 3 gives remainder 2 is of the format $N = 3K + 2$

Now, we will try to find out the lowest value of K such that N gives remainder 3 when divided by 5 = 2. Hence, lowest such natural number = 8. Next number will be obtained when we add 15 [=[LCM (3,5)] to 8 = 23.

And hence the family of numbers satisfying $x = 2 \pmod 3$ and $x = 3 \pmod 5$ $\Rightarrow N = 15P + 8$.

Now, we are required to find a value of P such that $15P + 8$ gives remainder 2 when divided by 7.

Let us check it for various values of P:

P	P = 1	P = 2	P = 3	P = 4	P = 5	P = 6	P = 7
Remainder obtained when divided by 7	2	3	4	5	6	0	1

Obviously for P = 1, all the three conditions are satisfied. Hence, the lowest such number = 23.

Next number will be obtained when we add LCM (3, 5, 7) to 23. Hence, family of numbers satisfying following conditions:

$x = 2 \pmod 3 \Rightarrow$ Remainder obtained is 2 when divided by 3

$x = 3 \pmod 5 \Rightarrow$ Remainder obtained is 3 when divided by 5

$x = 2 \pmod 7 \Rightarrow$ Remainder obtained is 2 when divided by 7

$N = 105\ K + 23$.

This example illustrates Chinese Remainder Theorem.

3. Finding Unit Digit or Tens Digit

Remainder obtained when a number is divided by 10 = Unit digit of the number. For example, when we divide 156 by 10, remainder obtained = 6, which is the unit digit of 156.

Remainder obtained when a number is divided by 100 = Last two digits of the number. For example, when we divide 1564 by 10, remainder obtained = 64, which are the last two digits of 1564.

Remainder obtained when a number is divided by 1000 = Last three digits of the number. For example, when we divide 15643 by 1000, remainder obtained = 643, which are the last three digits of 643.

We will discuss more of unit digit/tens digit in the chapter cyclicity.

4. Euler's Theorem of Remainder

Let f(n) be the number of integers less than n and co-prime with n, then the remainder obtained when $m^{f(n)}$ is divided by n = 1, where m and n are co-prime to each other.

Consider the following example:

f(8) = Number of integers less than 8 and co-prime with 8 = 4 (namely 1, 3, 5, 7).

[To obtain the value of "Number of integers less than n and co-prime with n" directly, we use Euler' Totient Function—See "Factor" chapter to know more about this]

So, Remainder obtained when $\frac{5^{f(8)}}{8}$ = Remainder obtained when $\frac{5^4}{8} = 1$

Or, Remainder obtained when $\frac{7^{f(8)}}{8}$ = Remainder obtained when $\frac{7^4}{8} = 1$

Or, Remainder obtained when $\frac{9^{f(8)}}{8}$ = Remainder obtained when $\frac{9^4}{8} = 1$

Or, Remainder obtained when $\frac{11^{f(8)}}{8}$ = Remainder obtained when $\frac{11^4}{8} = 1$

Or, Remainder obtained when $\frac{13^{f(8)}}{8}$ = Remainder obtained when $\frac{13^4}{8} = 1$

Euler' theorem can be used to find out the length of cycle in finding remainder using "pattern method."

5. When a Digit Repeats Itself n Times

These questions will involve one or more than one digit being repeated n times, and question will seek the remainder obtained when this number is divided by another number.

Some facts:

(i) Any digit written three times side by side will be divisible by 3 and 37.

For example, 888 is divisible by 3 and 37.

Obviously, from here we can conclude that a digit written six times, or 9 times or 3n times will also be divisible by 3 and 37.

For example, 888888 is divisible by 3 and 37.

(ii) Any number of the format 'abcabc' will be divisible by 7, 11 and 13.

(iii) If P is a prime number > 5, then any digit written (N–1) times will be divisible by P, where N is the recurring decimal of P. [Recurring decimal is the number of times after which the digits get repeated after decimal. For example, to calculate recurring decimal of 7, we write 1/7 = 0.142857142857……. (repeats after 6 digits).]

For example, 888888 (8 written 6 times) is divisible by 7.

Or, 222222222222 (2 written 12 times) is divisible by 13.

Worked-out Example 39 What is the remainder when 77777……upto 56 digits is divided by 19?

☝ ***Solution*** Since 19 is a prime number, it will divide any digit written 18 times side by side.

Hence, 77777……..54 times will be divisible by 19. So net remainder will be obtained when 77 (last two digits) is divided by 19 = 1.

Worked-out Example 40 What is the remainder when 7 + 77 + 777 + 7777 + ……(till 100 terms) is divided by 8?

☝ ***Solution*** You might have seen this expression 7 + 77 + 777 + 7777 + …… (till 100 terms) in algebra, where we first find out the sum till n terms, and then proceed to solve whatever is asked.

However, we are not required to use algebra to solve this question. Ideally in these scenarios, one is required to obtain a pattern of remainders.

P	P = 7	P = 77	P = 777	P = 7777	P = 77777	P = 777777
Remainder obtained when divided by 8	7	5	1	1	1	1

So, except for the 1st two terms, all the other terms will give remainder = 1.

Hence, remainder obtained = (7 + 5 + 1 × 98) is divided by 8 = Remainder obtained = (7 + 5 + 1 × 98) is divided by 8 = 6

Concept of Successive Division

Suppose we say that N is any number which is divided successively by 3 and 5, then what we mean to say is—At first, we divide N by 3 and then the quotient obtained from this division is divided by 5.

Example Let us see the case when 50 is divided by 5 and 3 successively.

50 divided by 5 gives 10 as the quotient. Now, we will divide 10 by 3. It gives finally a quotient of 3 and remainder of 1. So, we say that when 50 is divided by 5 and 3 successively, remainders obtained are 0 and 1 respectively.

Worked-out Example 41 When a number N is divided successively by 3 and 5, remainder obtained are 1 and 2 respectively. What is the remainder when N is divided by 15?

☝ ***Solution*** It can be seen that we are required to calculate it from back-end.

Family of numbers which when divided by 5 gives remainder 2 = 5 P + 2

So, N = 3 (5 P + 2) + 1 = 15 P + 7

Now, when N is divided by 15, remainder = 7

PRACTICE EXERCISE 1

1. A and B are two natural numbers, each divisible by 5. Which of the following is not necessarily true?

1. A + B is divisibly by 5.

2. A + B is divisible by 10.

3. A – B is divisible by 5.

4. $A^2 - B^2$ is divisible by 5.

5. All of these are definitely true.

Directions for questions (2–20): Find the remainder in each of the questions when numerator is divided by denominator:

2. $90^{90}/13$

1. 2 **2.** 1 **3.** 3 **4.** 4

5. None of these

3. $3^{202}/101$

1. 8 **2.** 11 **3.** 9 **4.** 7

5. None of these

4. $10^{200}/8$

1. 0 **2.** 1 **3.** 3 **4.** 4

5. None of these

5. $(10^{10}+10^{100}+10^{1000})/7$

1. 5 **2.** 4 **3.** 2 **4.** 1

5. None of these

6. $\dfrac{7^{11^{22^{33^{44}}}}}{5}$

1. 1 **2.** 2 **3.** 3 **4.** 4

5. None of these

7. $\dfrac{1!+2!+3!+4!+5!+\ldots\ldots\ldots\ldots..100!}{2^3\times 3^1}$

1. 9 **2.** 10 **3.** 12 **4.** 8

5. None of these

8. $\dfrac{1!^{1}+2!^{2}+3!^{3}+4!^{4}+5!^{5}+\ldots\ldots\ldots\ldots\ldots 100!^{100}}{2^3\times 5^1}$

1. 5 **2.** 37 **3.** 15 **4.** 24

5. None of these

9. $\dfrac{1!^{1!}!+2!^{2!}!+3!^{3!}!+4!^{4!}!+5!^{5!}\text{----------}+1000!^{1000!}}{24}$

1. 21 **2.** 5 **3.** 7 **4.** 19

5. None of these

10. How many prime numbers are there which when divided by another prime number, give a quotient which is same as the remainder?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. More than 2

11. When a natural number is divided by 4 and 7, it gives remainder 3 and 2 respectively. What is the remainder obtained when the same natural number is divided by 11?

1. 3 **2.** 2 **3.** 1 **4.** 7

5. Cannot be determined

12. When a natural number M is divided by 4 and 7, it gives remainder 3 and 2 respectively. Even when another natural number N is divided by 4 and 7, it also gives remainder 3 and 2 respectively. It is also given that there is no other number in between M and N exhibiting these properties. P is a natural number that gives the same remainder after dividing each of M and N. How many values of P are possible?

1. 2 **2.** 3 **3.** 6 **4.** 7

5. None of these

13. When each of 7179 and 9699 are divided by another natural number N, remainder obtained is same. How many values of N are possible?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. None of these

14. What is the remainder when 7 + 77 + 777 + 7777 + ……(till 100 terms) is divided by 8?

1. 0 **2.** 2 **3.** 4 **4.** 6

5. None of these

15. Find the remainder when $1^{1995} + 2^{1995} + \ldots\ldots\ldots\ldots\ldots + 1996^{1995}$ is divided by 1997.

1. 3 **2.** 0 **3.** 1 **4.** 7 **5.** 1996

16. x, y, z are three integers. What is the remainder when $[xyz\,(x^3 - y^3)\,(y^3 - z^3)\,(z^3 - x^3)]$ is divided by 7?

1. 1 **2.** 0 **3.** 3 **4.** 6

5. Cannot be determined

17. P is a prime number greater than 3. What is the remainder when p^2 is divided by 12?

1. 1 **2.** 0 **3.** 3 **4.** 6

5. Cannot be determined

18. A natural number N is divisible by 43. Same natural number N when divided by 3, 4 and 5 leaves remainder 1 in each case. What is the least value of N?

1. 241 **2.** 301 **3.** 361 **4.** 441

5. None of these

19. The least number which must be added to 10964 to make it exactly divisible by 7 is:

1. 1 **2.** 3 **3.** 5 **4.** 6 **5.** 2

20. $(7^{119} + 1)$ is divisible by:

1. 2 **2.** 4 **3.** 6 **4.** 8

5. All but option 3

Directions for questions (21–40): Go through the directions given below and solve the questions based on it.

Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4 as the answer—if question cannot be answered even by using both the statements.

21. If N is an odd natural number, what is the remainder when p is divided by 4?

Statement 1—When N is divided by 8, the remainder is 5.

Statement 2—N is the sum of the cubes of two positive integers.

22. What is the remainder when N is divided by M?

Statement 1—When N is divided by P, remainder obtained = 10

Statement 2—P is a factor of M.

23. What is the remainder when natural number P is divided by 6?

Statement 1—P is a multiple of 3.

Statement 2—P is a multiple of 4.

24. I have to pack some apples into some boxes. If I pack 3, or 4 or 5 or 6 apples in a box, I am left with one apple in each of the cases. However, if I pack 7 apples in a box, I am left with none. What is the minimum number of apples that I have?

1. 301 **2.** 601 **3.** 61 **4.** 436

5. None of these

25. Remainder obtained in how many of the following cases would be same?

(i) 10^6 divided by 7

(ii) 10^{16} divided by 17

(iii) 10^{18} divided by 19

1. 0 **2.** 1 **3.** 2 **4.** 3

26. For what values of x is $25^x + 1$ divisible by 13?

1. All real values of x **2.** Odd natural values of x

3. Even values of x **4.** All the integral values of x

27. $(17^{21} + 19^{21})$ is not divisible by

1. 36 **2.** 8 **3.** 9 **4.** 18

28. When x is divided by 6, remainder obtained is 3. Find the remainder when $x^4 + x^3 + x^2 + x + 1$ is divided by 6.

1. 3 **2.** 4 **3.** 1 **4.** 5

29. If $y^2 = a^2 \dfrac{(n+1)}{(n-1)}$; and n is a natural number > 1.

$F(y) = \left(\dfrac{y}{y-a}\right)^2 + \left(\dfrac{y}{y+a}\right)^2$ is always divisible by

1. 2 **2.** 4 **3.** 5 **4.** 6

5. 3

30. Consider the function $g(x) = x^4 + 6x^3 + 11x^2 + 6x$ where x is an integer. G(x) is always divisible by

1. 24 **2.** 28 **3.** 21 **4.** 14

5. None of these

31. In a question of division, the divisor is ten times the quotient and five times the remainder. What is the dividend, if the remainder is 46?

1. 5366 **2.** 5436 **3.** 5336 **4.** 5436

5. None of these

32. Which one of the following is the least number of four digits divisible by 71?

1. 1016 **2.** 1065 **3.** 1094 **4.** 1056

5. 1003

33. A natural number when divided by 2, 3, 4, 5 and 6 leaves 1, 2, 3, 4 and 5 as remainders, respectively, but when divided by 7 leaves no remainder. What is such least number?

1. 210 **2.** 119 **3.** 126 **4.** 154

5. 59

34. What is the greatest number of five digits which when divided by 4, 6, 10 and 15 leaves the same remainder 3 in each case?

1. 99993 **2.** 99063 **3.** 90093 **4.** 99963

5. None of these

35. If (x –a) is the HCF of $x^2 - x - 6$ and $x^2 + 3x - 18$, find the value of a.

1. 3 **2.** 6 **3.** 9 **4.** 12

5. None of these

36. Find the remainder obtained when $[y^{106} - y^{105} + (y+1)^3]$ is divided by $(y^2 - 1)$.

1. 0 **2.** 1 **3.** y + 3 **4.** 3y + 5

5. None of these

37. Given that $(x^2 - 4)$ is a factor of $2x^3 + px^2 + qx + 12$, where p and q are constants, find the value of p and q.

1. –3, 8 **2.** 3, 8 **3.** 3, –8 **4.** –3, –8

5. More than one value possible

38. How many natural numbers greater than 1000 but less than 3000 exist which when divided by 5 leaves remainder 1 but when divided by 6 leaves remainder 2?

1. 66 **2.** 67 **3.** 33 **4.** 34

5. 68

39. $x^{3n}+y^{3n}$ is divisible by x + y if

1. N is any integer ≥ 0 **2.** N is an odd positive integer

3. N is an even positive integer **4.** N is a rational number

5. N is any real number

40. What is the remainder when 46! is divided by 47?

1. 1 **2.** 46 **3.** 34 **4.** 11

5. None of these

PRACTICE EXERCISE 2

1. What is the remainder when x is divided by 100, where, x = greatest integer value of $[\sqrt{21} + \sqrt{20}]^{50}$?

1. 301 **2.** 601 **3.** 61 **4.** 201

5. 1

2. What is the remainder when $[22^5 + 23^5 + 24^5 + ...+87^5 + 88^5]$ is divided by 110?

1. 0 **2.** 60 **3.** 110 **4.** 55

5. None of these

3. For how many set of natural number values of x and y, $xy^2 + y + 7$ will divide $x^2y + x + y$?

1. 0 **2.** 1 **3.** 2 **4.** 5

5. None of these

4. If p is a prime and n is an integer such that p is divisible by $(4n^2 + 1)$, what is the remainder obtained when p is divided by 4?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. Cannot be determined

Directions for questions (5–6): Read the information below and solve the questions based on it.

P is an odd prime number and x_i is defined in the following way:

For i = 1, $x_i = 1/P$

For i = 2, $x_i = 2/P$ and so on.....

5. What is the remainder when $\sum_{i=1}^{p-1} x_i$ is divided by P (P>20)?

1. 1 **2.** 0 **3.** P–3 **4.** P–1

5. Depends upon the values of p

6. If $\sum_{i=1}^{n} x_t$ is divisible by P. What is the minimum value of n if P = 17?

1. 33 **2.** 16 **3.** 17 **4.** 34

5. None of these

7. N is a natural number. For how many values of N, $\frac{2^N + 1}{N^2}$ will be an integer?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. More than 3

8. N is a natural number exhibiting following property:

(i) N is a multiple of 8.

(ii) N is less than 2000.

(iii) N leaves the remainders 2, 4 and 5 on being successively divided by 5, 6 and 7.

How many values of N are possible?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. None of these

Directions for questions (9–11): Read the passage below and solve the questions based on it:

$(N!)_{10} = (x)_{20}$, where, N and x are natural numbers and x is on base 20. Also given are:

(i) Sum of digits of $x = (95)_{10}$

(ii) Sum of digits at odd places of x = Sum of digits at even places of x

9. What is the remainder when x is divided by $(y)_{20} = (19)_{10}$?

1. 0 **2.** 1 **3.** 10 **4.** 9

5. Cannot be determined

10. What is the remainder when x is divided by $(399)_{20}$?

1. 0 **2.** 1 **3.** 10 **4.** 9

5. Cannot be determined

11. What is the remainder when x is divided by $(20)_{10}$?

1. 0 **2.** 1 **3.** 10 **4.** 9

5. Cannot be determined

Directions for question (12): Read the direction given below and solve the questions based on it.

Following method of presenting the digits is given:

$(11)_{10}$ = N2

$(10)_{10}$ = N1M1

$(101)_{10}$ and $(110)_{10}$ are presented as N2M1.

As it can be seen above, 1 is presented by N and 0 is presented by M. Further, number of times the digits occur is written with the help of numerals. As obvious, digit 0 cannot come as the left most digit (like 0101 is not possible) as this will be same as 101.

So what matters is how many times 1 and 0 are coming and not the order of their occurrence. Total number of 1 coming is written after N and total number of 0 is coming is written after M.

So, N3M2 can be written in following possible ways:

11100, or, 10101, or 11001, or 10010 and so on.

12. When P =MxNy (a natural number), where x and y can take only whole number values, is divided by Q (a natural number). For which of the following values of Q, remainder obtained can never be equal to 0 for any value of x and y?

1. 37 **2.** 213 **3.** 412 **4.** 589

5. None of these

13. Given that HCF (Q, 10) = 1. P = Mx, where x can take only whole number values, is divided by Q (a natural number). For which of the following values of Q, remainder obtained can never be equal to 0?

1. 107 **2.** 99 **3.** 19 **4.** 23

5. None of these

14. For all the values of x and y, which of the following will not divide the natural number $N = (x^{36} - y^{36})$?

1. $x - y$ **2.** $x^3 - y^3$ **3.** $X^{12} - y^{12}$ **4.** $X^{24} - y^{24}$

5. $X^{18} - y^{18}$

15. The first term of a sequence (a_n) is a positive integer. The subsequent terms of the sequence are defined by the rule,

$$a_n = \begin{cases} \frac{1}{2}a_{n-1}, & \text{if } a_{n-1} \text{ is even.} \\ 3a_{n-1}+1, & \text{if } a_{n-1} \text{ is odd.} \end{cases}$$

For all the values of (a_n), this sequence must contain a number which is divisible by ....(fill in the blanks)?

1. 3 **2.** 4 **3.** 5 **4.** 6

5. 7

16. Remainder obtained when (n–1)! Is divided by n = –1. Which of the following is true for n?

1. N is a composite number.

2. N is a prime number.

3. N is a perfect square.

4. N cannot be a prime number.

5. N is a cube.

17. P C Sorkar, the famous magician, is famous for his card tricks, *inter alia*. In one of his shows, he presents the following trick:

He has one hundred cards numbered 1 to 100. He puts them into three boxes, a red one, a white one and a blue one in such a way that each box contains at least one card.

Now, he requests the audience to select two of the three boxes, chooses one card from each of the boxes and announces the sum of the numbers on the chosen cards. Given this sum, P C Sorkar identifies the box from which no card has been chosen.

Total number of ways in which he can put all the cards into the three boxes so that this trick always Works = N. Which of the following is the true for N?

(Two ways are considered different if at least one card is put into a different box.)

1. 0 **2.** $1 \le N \le 9$ **3.** $10 \le N \le 19$ **4.** $20 \le N \le 29$

5. $30 \le N \le 39$

18. It is given that $M = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + ... + \frac{1}{23} = \frac{n}{23!}$, where n is a natural number.

What is the remainder when n is divided by 13?

1. 0 **2.** 3 **3.** 6 **4.** 7 **5.** 11

19. A large number of apples = $2009^{2009^{2009}}$ is distributed among few groups. Every group gets 42 apples, except for one group X. What is the number of apples that group X gets?

1. 0 **2.** 14 **3.** 21 **4.** 28

5. 35

20. P is a prime number > 71. N = (P + 2) (P – 1) (P + 1) (P – 2). Which of the following will definitely divide N for all the values of P?

(i) 120 (ii) 240 (iii) 360 (iv) 480

1. (i) and (ii) only **2.** (i) and (iii) only

3. (ii) and (iv) only **4.** (i), (ii) and (iii) only

5. None of these

21. Find the sum of all positive two-digit integers x such that $(x + 2)(x^2 + 7x + 13)$ is divisible by 9.

1. 565 **2.** 675 **3.** 845 **4.** 135

5. None of these

22. P is a natural number ≥ 3. How many different remainders would be obtained when, $p^{p^{p^p}} - p^{p^{p^p}}$ is divided by 1989?

1. 5 **2.** 6 **3.** 4 **4.** 3

5. None of these

23. N = 8888.....88 times.

M = 7777777.....written N times.

What is the remainder when M is divided by 11?

1. 7 **2.** 1 **3.** 0 **4.** 3

5. 5

24. In the above question, what is the remainder when M is divided by 13?

1. 4 **2.** 1 **3.** 0 **4.** 6

5. 12

25. Consider the function $f(y) = y^2 + 8y - 857$. Y takes only such values for which f(y) is perfect square. What is the remainder when $\sum f(y)$ is divided by 100?

1. 16 **2.** 36 **3.** 68 **4.** 36

5. 52

26. Consider the function $f(x) = x^3 + 2x$, where, x is a natural number.

How many of the following statement/s is/are true?

Statement 1—f(x) is divisible by 3 for all odd integers x.

Statement 2—f(x) is divisible by 3 for all x.

Statement 3—f(x) is divisible by 6 for all even integers x.

1. 0 **2.** 1 **3.** 2 **4.** 3

5. Cannot be determined

27. Consider the function $f(x) = 2^{2x} - 1$, where, x is a natural number.

How many of the following statement/s is/are true?

Statement 1—f(x) is divisible by (x–1) for x > 1.

Statement 2—f(x) is divisible by x for all x.

Statement 3—f(x) is divisible by 3 for all x.

1. 0 **2.** 1 **3.** 2 **4.** 3

5. Cannot be determined

28. What is the remainder when $(33^{13} + 36^{13} + 39^{13} + 42^{13} + 45^{13} + 48^{13} + 51^{13})$ is divided by 147?

1. 0 **2.** 2 **3.** 49 **4.** 6

5. 42

29. A milkman has 3 jars containing 57 ℓ, 129 ℓ and 177 ℓ of pure milk respectively. A measuring can leave the same amount of milk unmeasured in each jar after a different number of exact measurements of milk in each jar. What is the volume of largest such can?

1. 12 ℓ **2.** 24 ℓ **3.** 16 ℓ **4.** 48 ℓ

5. None of these

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING 1

1. 3. Lowest number is LCM(9, 7)–5 = 63 – 5 = 58.

2. Not possible.

3. 10167

4. 8

5. 1016

6. 21. Lowest is (42 × 3 + 8) and largest = (42 × 23 + 8)

7. 12.

8. 123

9. Infinite

10. 1. Only number possible = 147

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	**1**	**3**	**1**	**1**	**2**	**1**	**2**	**2**	**2**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
5	**3**	**5**	**4**	**2**	**2**	**2**	**2**	**3**	**5**
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
1	**4**	**3**	**1**	**4**	**2**	**2**	**3**	**1**	**1**
31.	32.	33.	34.	35.	36.	37.	38.	39.	40.
3	**2**	**2**	**4**	**1**	**4**	**4**	**2**	**1**	**2**

1. If A and B are divisible by 5, then (A+B) is also divisible by 5. It may or may not be divisible by 10. For example, for A = 10, B = 15, A+B is not divisible by 10. Hence, option 2 is not necessarily true.

2. Remainder obtained from $90^{90}/13$ = Remainder obtained from $(-1)^{90}/13 = 1$

3. Using Fermat's theorem, $(3^{100}/101)$ will give remainder = 1. Hence, $(3^{200}/101)$ will give remainder = 1. So, remainder obtained when $(3^{202}/101) = 9$.

4. Remainder obtained when $10^3/8 = 0$. Hence, all the higher powers (3+) of 10 will also be divisible by 8.

5. Remainder obtained when $10^{10}/7$ = Remainder obtained when $3^{10}/7$

Using Fermat's theorem, $(3^6/7)$ will give remainder = 1. Hence, remainder obtained when $3^{10}/7$ = remainder obtained when $(3^6 \times 3^4)/7$ = remainder obtained when $3^4/7$ = remainder obtained when $81/7 = 4$.

Similarly, remainder obtained when $10^{100}/7$ = remainder obtained when $3^{100}/7$ = remainder obtained when $3^{96} \times 3^4/7 = 4$.

Similarly, when 10^{1000} is divide by 7, remainder = 4.

Hence, net remainder = (4+4+4)/7 = 5.

6. Remainder obtained $\frac{7^{11^{22^{33^{44}}}}}{5}$ = Remainder obtained $\frac{2^{11^{22^{33^{44}}}}}{5}$

Now, let us 1st find out the pattern of remainder obtained when powers of 2 is divided by 5.

Remainder obtained when 2^n is	2^1	2^2	2^3	2^4
Divided by 5	2	4	3	1

Hence, length of the cycle = 4. Now we are required to check what is the remainder obtained when $11^{22^{33^{44}}}$ is divided by 4.

Remainder obtained when $11^{22^{33^{44}}}$ is divided by 4 = Remainder obtained when $(-1)1^{22^{33^{44}}}$ (= 1) is divided by 4 = 1.

Remainder obtained is $\frac{2^{11^{22^{33^{44}}}}}{5}$

= Remainder obtained when $2^{(4n+1)}$ is divided by 5. = Remainder obtained when $2^{4n} \times 2^1$ is divided by 5 = 2 (remainder obtained when 2^{4n} is divided by 5 =1).

7. Denominator = $2^3 \times 3^1 = 24 = 4!$

So the remainder will come only from 1! + 2! + 3!. Hence, net remainder 1! + 2! + 3! = 9 (since numerator is less than denominator, numerator itself will be the remainder).

8. Denominator = $2^3 \times 5 = 40$

Net remainder will come from the 1st four terms $1!^1 + 2!^2 + 3!^3 + 4!^4 = 1 + 4 + 216 + 576 \times 576$

Net remainder = 37.

9. Denominator = 24 = 4!

Net remainder will come from the 1st three terms only = $1!^{1!} + 2!^{2!} + 3!^{3!} = 1 + 4 + 6^6$

Net remainder = 5

11. Format of the number satisfying the above conditions = 28 K + 23. For different values of K, remainder obtained when divided by 11 would be different.

For K = 1, number = 51 $\Rightarrow$ Remainder obtained = 7

For K = 2, number = 79 $\Rightarrow$ Remainder obtained = 2. Hence, cannot be determined uniquely.

12. Taking a cue from the above question, format of the number satisfying the conditions (When a natural number M is divided by 4 and 7, it gives remainder 3 and 2 respectively) = 28 K + 23

Let us assume that M = 28K + 23, and N = 28 R + 23

Since in all the cases, remainder obtained from 23 will be same, remainder obtained from 28K or 28R should also be same. So the only possible values of P are the factors of 28 = 1, 2, 4, 7, 14, 28 = 6 different values.

13. $7179 = NQ_1 + R$ and $9699 = NQ_2 + R$, where, Q_1 and Q_2 are the respective quotients when 7179 and 9699 are divided by N.

Or, $9699 - 7179 = 2520 = N(Q_2 - Q_1)$.

$2520 = 5 \times 2^3 \times 3^2 \times 7 \Rightarrow$ Number of factors $= (1+1)(3+1)(2+1)(1+1) = 2 \times 4 \times 3 \times 2 = 48$

Now, one of the factors of 2520 will be divisor and another part will be the value of $(Q_2 - Q_1)$.

15. Write the equidistant terms from right and left hand side together.

$(1^{1995} + 1996^{1995}) + (2^{1995} + 1995^{1995}) + (3^{1995} + 1994^{1995}) + \ldots\ldots\ldots\ldots + (997^{1995} + 998^{1995}) = 1^{1995} + 2^{1995} + \ldots\ldots\ldots\ldots\ldots + 1996^{1995}$.

Each of the terms in bracket like $(1^{1995} + 1996^{1995})$ or $(2^{1995} + 1995^{1995})$ etc. Is divisible by 1997. Hence, net remainder = 0.

16. Different remainder possible when a number N is divided by 7 = 0, 1, 2, 3, 4, 5, 6.

Different remainder possible when a number N^3 is divided by 7 = 0, 1, 6.

If there are three numbers x, y and z, there are different possibilities for the remainders:

Either remainder obtained from one of them = 0 or atleast two of x^3, y^3 or z^3 will have same remainder. (if remainder obtained $\neq 0$, then atleast two out of three numbers x^3, y^3 or z^3 will certainly have same remainder from a list of possible remainders 1 and 6.)

In any case remainder obtained when $[xyz(x^3 - y^3)(y^3 - z^3)(z^3 - x^3)]$ is divided by 7 = 0.

17. $P = (6N \pm 1)^2 = 36N^2 + 12N + 1$

Hence, remainder obtained = 1.

18. Go through the options.

19. Remainder obtained when 10964 is divided by 7 = 2, hence, 5 should be added.

20. $(7^{119} + 1) = (7^{119} + 1^{119})$ = Divisible by 8. Hence, divisible by 2 and 4 too.

Also remainder obtained when $(7^{119} + 1)$ is divided by 3 = 2, hence, not divisible by 6.

21. Using statement 1—N = 8K + 5, hence, when N is divided by 4, remainder = 1.

Using statement 2—$N = a^3 + b^3 =$ Odd $\Rightarrow$, Exactly one of a and b will be even and other will be odd.

Reminder obtained when $Even^3$ when divided by 4 = 0

Reminder obtained when Odd^3 when divided by 4 = 1 or 3

So using statement 2 only, we cannot answer the question uniquely.

Hence, option 1 is the answer.

22. Obviously individual statement itself is not sufficient. Even both the statements together will not get the exact unique value of remainder.

Hence, option 4 is the answer.

23. Using statement 1, P = 3, 6, 9 etc. So unique value of remainder cannot be calculated.

Statement 2—P = 4, 8, 12, etc. So, unique value of remainder cannot be calculated.

Using both the statements together, P is a multiple of 12, so P = 12, 24, 36, 48 etc. In all these cases, remainder obtained = 0. Hence, option 3 is the answer.

24. LCM of 3,4,5,6 = 60. So the number should be of the format 60K + 1. 1st number of the format 60K + 1 which is divisible by 7 = 301.

25. Using Fermat's theorem, remainder obtained in each of the given cases = 1. Hence, answer is option 4.

26. Remainder obtained when $25^x + 1$ is divided by 13 = $(-1)^x + 1$.

For number to be divisible, remainder = 0. So, $(-1)^x + 1 = 0 \Rightarrow$ x should be odd.

27. $(17^{21} + 19^{21})$ is divisible by 36 $\Rightarrow$ hence, also divisible by factors of 36.

28. Assume any value of x, and solve the question.

30. $g(x) = x^4 + 6x^3 + 11x^2 + 6x = x\,(x+1)\,(x+2)\,(x+3) \Rightarrow$ product of four consecutive integers, which is always divisible by 4!.

31. Remainder = 46 $\Rightarrow$ Divisor = 5 × 46 = 230

Given that 10 × quotient = 5 × remainder

Remainder = 2 × quotient $\Rightarrow$ Quotient = 23

Dividend = 23 × 230 + 46 = 5336

32. 71) 1000 (14

$\underline{-71}$

290

$\underline{-284}$

6

So, least number = 1000 – 6 + 71 = 1065

33. L.C.M. of 2, 3, 4, 5, 6 is 60.

Hence, the family of numbers will be 60K – 1. Numbers of this family are 59, 119 etc. Out of these 119 is divisible by 7.

34. L.C.M. of 4, 6, 10, 15 = 60

Greatest number of five digits which is divisible by 60 = 99960.

Hence, required number = 99960 + 3 = 99963

35. Let $p(x) = x^2 - x - 6$, $q(x) = x^2 + 3x - 18$

Since (x – a) is the HCF of p(x) and q (x), (x – a) is a factor of p (x) and q (x) both.

$\Rightarrow$ p(a) = 0 and q (a) = 0 $\Rightarrow$ p (a) = q (a)

$\Rightarrow a^2 - a - 6 = a^2 + 3a - 18$

$\Rightarrow 4a = 12 \Rightarrow a = 3$.

37. Go through options.

38. Lowest such natural number = 26.

Next number will be obtained when LCM of (5, 6) is added to 26 = 56

Family of numbers satisfying the property = 30 K + 26

Lowest four digit number = 990 + 26 = 1016

Largest number less than 3000 = 2970 + 26 = 2996

Hence, number of numbers satisfying the property = $\frac{2996-1016}{30}+1=67$

39. Whatever is the value of n, where n is any whole number, power of both x and y are odd and equal. Hence, divisible by all the whole number values of n.

40. Using Wilson theorem, (46! + 1) is divisible by 47. Hence, when 46! Is divided by 47, then remainder obtained = –1 = 46.

PRACTICE EXERCISE 2

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
5	**4**	**5**	**2**	**2**	**2**	**3**	**3**	**1**	**1**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
5	**5**	**5**	**4**	**2**	**2**	**3**	**4**	**5**	**2**
21.	22.	23.	24.	25.	26.	27.	28.	29.	
1	**5**	**3**	**5**	**3**	**4**	**2**	**1**	**2**	

1. Last four digits of x = 0001

Hence, remainder = 1

2. Observe the numbers given once. There may be some link between the numerator and the denominator 110, which is 'atleast somewhat' unusual number in the questions of remainder.

We can see that if we take the equidistant terms from both the ends like ($22^5 + 88^5$), or ($23^5 + 87^5$) and so on, these terms are going to be divisible by (22 + 88 = 23 + 87 = ... =110).

Next, we see that from 22 to 88 (including both the limits) we have 67 terms, hence, 33 pairs of numbers (as given in above bracket), and one number left out. All those pairs of numbers will give remainder zero each. So the net remainder will be obtained by the number that has been left behind. Now we are required to find out that number.

Further, we also see that the number left behind will be the middle number in the series 22, 23, 24, ..., 87, 88. Since, there are 33 pairs, so 34th number from 22 will be the middle number = 55. Alternatively, middle number is going to be the average of this series (since average is always the middle number of any Arithmetic Progression series).

Average of the given series 22,23,24,.....,87,88 = $\frac{22+88}{2}=\frac{23+87}{2}=55.$

So, the remainder [$22^5 + 23^5 + 24^5 + \ldots\ldots\ldots\ldots + 87^5 + 88^5$] is divided by 110 = Remainder obtained when 55^5 is divided by 110 = 55. (See Worked-out Example 16)

Solutions for Questions (5–6):

5. Understand that for every odd prime number P,

$$\frac{1}{P}+\frac{2}{P}+\ldots\ldots\ldots\ldots+\frac{P-1}{P}=0$$

This question is asking us exactly the same thing expressed in the format of summation sign (Σ). So, what we are supposed to do is to take any value of P > 20 as per the question, and write the above expression. Remainder obtained finally = 0.

Please understand that this question mentions P > 20 only to increase the difficulty level. Had I given just P instead of P > 20 here, you might have done this question by assuming any value of P = 3 or 5 etc. This I did not want to happen because in that case probably you might not have thought so longer to obtain the relation that is given above. And once you go through the actual process or by visualizing the situation, you can see that summation of the remainder obtained from the 1st number and last number = summation of the remainder obtained from the 2nd number and second last number = summation of the remainder obtained from equidistant terms from both the ends = 0. Since there are an even number of such pairs going to be formed, net remainder = 0

6. Using the discussion in 5, we can say now that its going to happen for n = P – 1 = 16

7. N = 1 and N = 3 are the only two solutions.

8. Let us first find out the format of N:

When divided by 7, remainder obtained = 5 ⇒ So format = (7 K + 5).

When divided by 6, remainder obtained = 4 ⇒ So format = 6 (7 K + 5) + 4.

When divided by 5, remainder obtained = 2 ⇒ So format = 5[6 (7 K + 5) + 4] + 2 = 210 K + 172.

Since, N is a multiple of 8, we are required to choose the values of K in such a way that [210 K + 172] is divisible by 8. We can see that 172/8 gives 4 as the remainder. So, 210K should also give remainder 4.

This is possible for K = 2 ⇒ N = 592 and for K = 6 ⇒ N = 1432. Less than 2000, only these two values of N will satisfy.

Solutions for Questions (9–11):

Consider the divisibility rule of 9 on base 10 = Sum of digits should be divisible by 9.

Generalizing the above rule, divisibility rule of (N – 1) on base N = Sum of digits of the number should be divisible by (N–1).

Similarly, divisibility rule of (N + 1) on base N = Divisibility rule of 11 on base 10 = Difference in the sum of digits at odd places and sum of digits at even places should be divisible by 11.

For example, each of $(83)_{12}$, $(74)_{12}$, $(56)_{12}$, $(65)_{12}$ is divisible by $11_{10}=1A_{12}$

9. Since sum of digits = 95, hence, divisible by $(y)_{20} = (19)_{10}$

10. 399 = 21 × 19. This number is divisible by 21 also since sum of digits at the odd places = sum of digits at even place. So it is divisible and remainder obtained = 0

11. This is a simple question. Sum of digits = 95 can be obtained in many ways—having different unit digits—0, 1, 2, 3 etc. And in each of these cases, remainder obtained when divided by 20 is going to be different. Hence, answer is option 5.

12. All the natural numbers will have atleast one multiple made up of only 1's and/or 0's. Hence answer is None of these.

13. If the HCF (Q, 10) = 1, then all such Q will have atleast multiple made up of only 1's.

16. Wilson's theorem provides both necessary and sufficient condition. So if remainder = –1 when (n–1)! Is divided by n, n has to be a prime number.

17. Purpose of putting this question in remainder chapter—This question uses the concept of remainder. Understand that he would be able to do this only if he knows what remainder can he obtain by taking two cards from two particular boxes. And in turn what remainder he does not obtain if he does not take a card from that third particular box.

Consider a small case—there are only three cards numbered 1, 2 and 3, put in three boxes of different color (as given in this question) as per the following arrangement:

Red box—Card Number 1; White box—Card Number 2; Blue box—Card Number 3

If now summation is said to be 3, remainder obtained = 0, hence, card from blue box has not been picked up.

Now, if we scale this situation up, let us assume that there are 6 cards numbered 1 – 6, kept in three boxes. To be able to identify that card from which box has not been picked up, he would put cards in the following way,

Red box—Card Number 4; White box—Card Number 5; Blue box—Card Number 6, and so on cards should be put.

Total number of ways this can be done = 12. Hence option 3. $\mathbf{10 \le N \le 19}$ is the answer.

18. $n = 23!\ (1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+.....+\frac{1}{23})$. When n is divided by 13, all but only one term $(=\frac{23!}{13})$ will not be divisible by 13. Hence remainder will be obtained only from $=\frac{23!}{13}$.

Remainder obtained when n is divided by 13 = 7.

19. This question can be reframed as—what is the remainder when $2009^{2009^{2009}}$ is divided by 42?

Remainder obtained when $2009^{2009^{2009}}$ is divided by 42 = Remainder obtained when $35^{2009^{2009}}$ is divided by 42

To obtain the remainder, we take 7 common from both numerator and denominator.

$$\frac{35^{2009^{2009}}}{42} = \frac{7}{7} \times \frac{5^{\text{odd number}} \times 7^{\text{even number}}}{6}$$

Remainder obtained when $\frac{5^{\text{odd number}}}{6} = -1 = 5$

Remainder obtained when $\frac{7^{\text{even number}}}{6} = 1$

Hence, net remainder = $7 \times 5 \times 1 = 35$

20. P is a prime number > 71, hence P is an odd number.

(P–1) and (P+1) are two consecutive even natural numbers, hence, one of these two numbers will be divisible by 4 and other by 2.

Also P will be of the format 6K ± 1, hence, one of (P–1) or (P+1) = 6K [if 6K+1, then P–1 = 6K and if 6K–1, then P+1 = 6K]. Since we have already considered these two number to be even numbers, hence (P–1)(P+1) will be divisible by 24 (=$2 \times 4 \times 3$). ----------(1)

Next P will be of the format 6K ± 1, so one of (P+2) or (P–2) will be a multiple of 3. ----------(2)

Also, since P is an odd prime number, so unit digit of P = 1 or 3 or 7 or 9.

If unit digit = 3 or 7, then one of (P–2) or (P+2) will be a multiple of 5. Or

If unit digit = 1 or 9, then one of (P–2) or (P+2) will be a multiple of 5

----------(3)

Combining (1), (2) and (3), N is a multiple of $24 \times 3 \times 5 = 360$.

Hence, N is definitely divisible by 120 and 360. It may or may not be divisible by 240.

So, option 2 is the answer.

21. Values of x for which expression is divisible by 9 will be obtained by taking (x+2) and $(x^2 + 7x + 13)$ individually.

It can be seen that (x + 2) is divisible by 9 for x = 16, 25, 34, ...97. Summation of these numbers = 565.

Now, $(x^2 + 7x + 13) = (x+3)^2 + (x + 4)$ Þ this is never divisible by 9 for any two digit numbered value of x.

Hence, summation of all such two digit x = 565.

22. $p^{p^{p^p}} - p^{p^p}$ is divisible by 1989 for all P is natural number ≥ 3. Hence there is only one remainder. Hence, option 5 is the answer.

23. N = Even number

M = 7 written even number of times

It can be seen that when 7 is written even number of times, remainder obtained when divided by 11 = 0; and when 7 is written odd number of times, remainder obtained = 7.

Hence, in this case, remainder obtained = 0

24. We can explore that any digit written 6 times is divisible by 13. For example, 111111, or 222222 etc., are divisible by 13. We can extend this further by saying that any digit written 6K times (12 times, or 18 times, or 24 times etc.) will also be divisible by 13. Hence, when 8 is written 8888.....84 times, its divisible by 13.

Let us now consider M.

For N = 8, M = 77777777 (7 written 8 times) $\Rightarrow$ Remainder will be obtained from last two 7 only $\Rightarrow$ Remainder obtained when 77 is divided by 13 = 12.

For N = 88, M = 77.....88 times (7 written 88 times) $\Rightarrow$ Remainder will be obtained from last four 7 only $\Rightarrow$ Remainder obtained when 7777 is divided by 13 = 3.

For N = 888, M = 7777.....888 times (7 written 888 times) $\Rightarrow$ Remainder obtained will be equal to 0 since 888 is divisible by 6 and we have already discussed that any digit written 6K times will be divisible by 13.

So we can conclude that even for N = 888888 (8 written 6 times), M is divisible by 13. Extending this, for N = 88888.....87 times, M is divisible by 13.

So net remainder will be obtained when 777.....8 times is divided by 13 = 12.

25. $y^2 + 8y - 857 = k^2 \Rightarrow y^2 + 8y + 16 = k^2 + 873 \Rightarrow (y+4)^2 = k^2 + 873$. Now we will take all the variables on one side keeping 873 only on right hand side. Next we obtain $873 = 3^2 \times 97$.

F(y) can take only following values as perfect square = 436^2, 144^2, 44^2

When, $436^2 + 144^2 + 44^2$ is divided by 100, remainder obtained = 68

26. Answer can be verified by taking some values.

27. Answer can be verified by taking some values.

28. We can see that the terms are in AP, and power is odd.

Hence, $(33^{13} + 36^{13} + 39^{13} + 42^{13} + 45^{13} + 48^{13} + 51^{13})$ is divisible by (33 + 36 + 39 + 42 + 45 + 48 + 51) = 294. If a number is divisible by 294, then it is going to be divisible by all the factors of 294 also, and 147 is a factor of 294. Hence, remainder obtained = 0.

CAT QUESTIONS FROM THIS CONCEPT

CAT 2008

Suppose, the seed of any positive integer n is defined as follows:

Seed(n) = n, if n < 10 = seed(s(n)), otherwise, where, s(n) indicates the sum of digits of n. For example, seed(7) = 7, seed(248) = seed(2 + 4 + 8) = seed(14) = seed(1 + 4) = seed(5) = 5 etc.

1. How many positive integers n, such that n < 500, will have seed(n) = 9?

1. 39 **2.** 72 **3.** 81 **4.** 108

5. 55

CAT 2005

2. If $x = (16^3+17^{13}+18^3+19^3)$, then x divided by 70 leaves a remainder of

1. 0 **2.** 1 **3.** 69 **4.** 35

3. Let n! = 1 x 2 x 3 x... x n for Integer n ≥ 1. lf p = 1! + (2x2!) + (3 x 3!) + ... + (l0 x 10!), then p+2 when divided by 11! leaves a remainder of

1. 10 **2.** 0 **3.** 7 **4.** 1

CAT 2004

4. The remainder, when $(15^{23} + 23^{23})$ is divided by 19, is

1. 4 **2.** 15 **3.** 0 **4.** 18

CAT 2003

5. What is the remainder when 4^{96} is divided by 6?

1. 0 **2.** 2 **3.** 3 **4.** 4

CAT 2002

6. When 2^{256} is divided by 17, the remainder would be

1. 1 **2.** 16 **3.** 14 **4.** None of these

7. After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?

1. 80 **2.** 75 **3.** 41 **4.** 53

8. $7^{6n} - 6^{6n}$, where n is an integer > 0, is divisible by

1. 13 **2.** 127 **3.** 559 **4.** All of these

CAT 2001

9. Let b be a positive integer and $a = b^2-b$. If $b \geq 4$, then $a^2 - 2a$ is divisible by

1. 15 **2.** 20 **3.** 24 **4.** All of these

CAT 2000

10. Let N = 1421 × 1423 × 1425. What is the reminder when N is divided by 12?

1. 0 **2.** 9 **3.** 3 **4.** 6

11. The integers 34041 and 32506, when divided by a three-digits integer n, leave the same remainder. What is the value of n?

1. 289 **2.** 367 **3.** 453 **4.** 307

CAT 1999

12. The remainder when 7^{84} is divided by 342 is

1. 0 **2.** 1 **3.** 49 **4.** 341

SOLUTIONS AND EXPLANATIONS

1. This question is asking for the number of numbers less than 500 and divisible by 9 = 500/9 = 55. Hence, option 5 is the answer.

2. 1. $\frac{a^n + b^n + c^n + ...}{a + b + c + ...}$ is divisible for all the odd values of n.

Hence option 1.

Alternatively, $(16^3+17^{13}+18^3+19^3) = (16^3+19^3) + (17^{13}+18^3) = 35$ (odd number) + 35 (odd number) = 35 (odd + odd) = 35 × even number = 70 K (where K is a natural number).

When, 70 K is divided by 70, remainder obtained = 0.

3. 4. If p = 1! = 1

Then, p + 2 = 3 when divided by 2! Remainder will be 1.

If p = 1! + 2 × 2! = 5

Then, p + 2 = 7 when divided by 3! Remainder is still 1.

Hence, p = 1! + (2 × 2!) + (3 × 3!)+ … + (10 × 10!) when divided by 11! Leaves remainder 1.

Hence, the remainder is 1.

4. 3. $15^{23} = (19 - 4)^{23} - 19x + (-4)^{23}$ where x is a natural number.

$23^{23} - (19 + 4)^{23} = 19y + (4)^{23}$ where y is a natural number.

$15^{23} + 23^{23} - 19 (x + y) + 4^{23} + (-4)^{23}$

$= 19 (x + y)$

5. 4. Remainder when $\frac{4^{96}}{6}$

Let's find the pattern of remainders when power of 4 is divided by 6.

$\frac{4^1}{6} = 4$ $\quad$ $\frac{4^2}{6} = 4$

$\frac{4^3}{6} = 4$ $\quad$ $\frac{4^4}{6} = 4$

Hence, any power of 4 when divided by 6 leaves a remainder of 4.

6. 1. $(2^4)^{64} = (17-1)^{64} = 17n+(-1)^{64} = 17n + 1$ Hence, remainder = 1

7. 4. $3 (4 (7x+4) + 1) + 2 = 84x + 53$

Therefore, remainder is 53.

8. 4. $7^{6n} - 6^{6n}$

put n = 1

$7^6 - 6^6 = (7^3 - 6^3) (7^3 + 6^3)$

This is a multiple of $7^3 + 6^3 = 559$ and 7+6 = 13.

9. 4. $a = b^2 - b$, $b \geq 4$

$a^2 - 2a = (b^2 - b)^2 - 2 (b^2 - b)$

$= (b^2 - b) (b^2 - b - 2)$

Using different values to b $\geq$ 4 and we find that it is divisible by 15, 20, 24. Hence, all of these is the answer. Option 4 is the right answer.

10. 3. N = 1421 × 1423 × 1425. When divided by 12, it shall look like $\frac{\left[(1416+5)\times(1416+7)\times(1416+9)\right]}{12}$.

Now the remainder will be governed by the term 5×7×9. Which when divided by 12 leaves the remainder 3. Hence, option 3 is the answer.

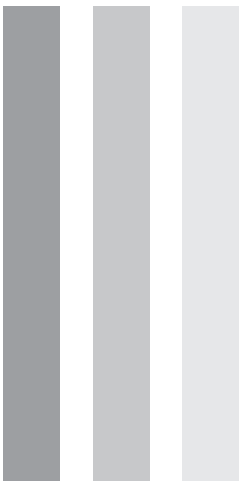
11. 4. Let r be the remainder. Then 34041 – r and 32506 – r are perfectly divisible by n. Hence, their difference should also be divisible by the same. (34041–r) – (32506 – r) = 1535, which is divisible by only 307. Hence, option 4 is the answer.

12. 2. $7^{84} = (7^3)^{28} = 343^{28}$

When, 343^{28} is divided by 342, remainder obtained = 1.

Hence, option 2 is the answer.

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Chapter 4

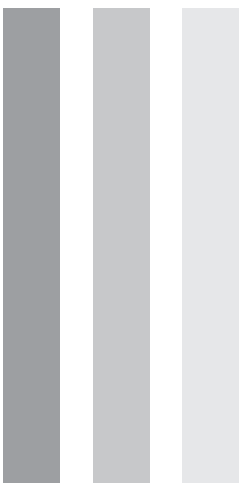
of

DEMYSTIFYING NUMBER SYSTEM

BASE SYSTEM

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- How numbers are constructed on different bases?
- Inter-conversion of numbers on bases
- Algebraic calculations—Addition, subtraction and multiplication
- Types of questions asked from this chapter in the CAT

BASE SYSTEM

Number of digits or symbols used in a particular system of writing the numbers is known as base. For example, in decimal system, we use 10 digits (0–9) to write the numbers, and hence base = 10. Similarly, in binary system, we use 2 digits (0–1) to write the numbers, hence, base = 2.

Let us 1st understand how we generate counting numbers:

In our decimal system of writing the numbers, we use 10 digits (0-9). So we start writing the numbers from 0, then 1, 2, 3, 4, 5, 6, 7, 8, 9.

In this system, largest number of single digit = 9. If we have to form a number bigger than this number, we are required to take resort to two-digit numbers starting from 10. So we write 11, 12, 13, 14, 15, 16, 17, 18, 19. After this we once again repeat the cycle with taking 2 at the tens place digit—20, 21, 22, 23, 24, 25, 26, 27, 28, 29. After this we take 3 at the tens place and repeat the cycle—30, 31, 32, ...and so on. Once we have reached at the largest number of two digits = 99, we move on to three digits number starting with 100, a number of three digits. Then we write 101, 102, 103,....., 110, ...199, 200, 201, 202,.....299, 300, 301, ...and so on till the largest number of three digits 999. So, basically number formation is nothing but repeating the cycle, and once we are through with the cycle in single digit, we move on to two digits, then three digits, and so on.

Consider now a system of writing the number that has only 8 digits (0-7). To camouflage this with some reality let us assume that educational institutions have mandated that digit 8 and 9 will not be used for writing the numbers.

To generate the counting numbers now, we start writing the numbers from 0, then 1, 2, 3, 4, 5, 6, 7. Once we have written down 7, we do not have any single digit to take our counting forward. So to form a number larger than 7, we are required to take resort to two digit numbers here only. And which is the lowest two digit natural number? Obviously, it is 10. Then we move it forward to 11, then 12, 13, 14, 15, 16, 17. We do not have '18' or '19' in this system, since we do not have digits 8 and 9, or, 8 and 9 are not allowed to be used. So, from 17, we will directly jump to 20, then 21, 22, 23, 24, 25, 26, 27, 30.....50, 60, 70, 71, 72, 73, 74, 75, 76, 77. Obviously we do not have any 78 or 79 or 80 etc., in this system, as reasoned above. To make a number larger than 77, we will now move on to 3-digits number starting with 100.

Similarly, had there been only 6-digits (0-5) to write down the numbers, numbers would have been formed like this; 0, 1, 2, 3, 4, 5, 10, 11, 12, 13, 14, 15, 20, 21, 22, 23, 24, 25, 30, ...54, 55, 100, 101, ...554, 555, 1000.. and so on.

Alternatively process of number generation can be seen with the help of the example of speedometer attached to vehicles also:

Initially, when the vehicle has just been rolled out, speedometer shows = 0000. 1st of all, rightmost digit place rotates from 0 to 9, and once this has become 9, 2nd rightmost digit becomes 1 and rightmost digit shows = 0. Similarly rightmost

digit keeps on moving from 0 to 9, and for every such rotation, 2nd rightmost digit keeps on moving 1 unit ahead.

So starting from 000, it moves on to 010, then 020, then 090, then 099, then 100 and so on.

Consider a different case now—that there is another vehicle B having speedometer with only 7 digits from 0 to 6, vehicle A having speedometer with 0 to 9 digits, vehicle C with only 4 digits from 0 to 3.

Actual distance covered and data shown by both the speedometers is tabulated below:

Actual distance covered	Speedometer A (0-9)	Speedometer B (0-6)	Speedometer C (0-3)
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	10
5	5	5	11
6	6	6	12
7	7	10	13
8	8	11	20
9	9	12	21
10	10	13	22
11	11	14	23

Some Facts

1. System of writing the numbers always starts from the value 0. So, if it is given that there are 8 digits given in a system, the digits will be 0, 1, 2, 3, 4, 5, 6, and 7 given in ascending order. To represent a number on base N, we write N as the subscript with the number as the base. For example, 200 on base 3 will be written as 200_3 or $(200)_3$ (or) 200 on base 12 will be written as 200_{12} or $(200)_{12}$.
2. Lowest possible system of writing the numbers = Binary using two digits (0 and 1). In other words, there cannot be a system of writing the numbers of only one digit i.e., 0 only. Reason for this can be attributed to the fact that any number of zero put side by side = 0 only. (0= 00= 000 = 0000). So we will not be able to generate the counting numbers greater than 0.
3. A system of writing the numbers may have any number of digits—suitably accommodated by the numerals or symbols to represent that number. For example, a system of writing the number having 100 digits will have 100 numerals or 100 symbols.

Whole of the above discussion can be summed up in the following table:

$(0\text{-}9)_{10}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$(0\text{-}F)_{16}$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11
$(0\text{-}8)_{9}$	0	1	2	3	4	5	6	7	8	10	11	12	13	14	15	16	17	18
$(0\text{-}7)_{8}$	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17	20	21
$(0\text{-}6)_{7}$	0	1	2	3	4	5	6	10	11	12	13	14	15	16	20	21	22	23
$(0\text{-}5)_{6}$	0	1	2	3	4	5	10	11	12	13	14	15	20	21	22	23	24	25
$(0\text{-}3)_{4}$	0	1	2	3	10	11	12	13	20	21	22	23	30	31	32	33	100	101
$(0\text{-}2)_{3}$	0	1	2	10	11	12	20	21	22	100	101	102	110	111	112	120	121	122

Some of the conclusions that we can draw from the above table:

Conclusion 1 Numeral representation of a number on base N increases when we go to a system of writing the numbers having fewer digits than that of base N.

For example, $(17)_{10} = (11)_{16} = (21)_8 = (122)_3$

Above example illustrates that:

When digits used = 10, numeral presentation of number = 17.

When digits used = 16, which is more than 10, numeral presentation of number = 11, which is less than 17 (numeral presentation on base 10).

When digits used = 8, which is less than 10 as well as 16, numeral presentation of number = 21, which is more than both 17 and 11.

When digits used = 3 which is less than each of 8, 10, 16, numeral presentation of number on base 3 = 122, which is more than each of 21, 17 and 11.

However, understand that actual value of all these numbers $(17)_{10} = (11)_{16} = (21)_8 = (122)_3$ are equal.

Conclusion 2 As seen above, there is no difference in the digits of one system to the other system.

For example, $2_3 = 2_4 = 2_5 = 2_6 = 2_7 = 2_8 = 2_9 = 2_{10} = 2_{100}$

It is all because of the fact that when we start counting, digit 2 occurs in all the systems as the 3rd number after 0 and 1.

Conclusion 3 $(10)_N = (N)_{10} \Rightarrow (10)_3 = (3)_{10}$ or $(10)_5 = (5)_{10}$

This can be further extended to any power of LHS and RHS.

$[(10)_N]^2 = [(N)_{10}]^2 \Rightarrow [(10)_5]^2 = [(5)_{10}]^2$

$[(10)_N]^3 = [(N)_{10}]^3$

Worked-out Example 1 Largest symbol of a system of writing the number having 100 digits = Z. What is the decimal system (having 10 digits from 0-9) equivalent of Z?

☝ ***Solution*** Decimal system (having 10 digits from 0-9) equivalent of Z = Value of Largest possible symbol on base 100 = Starting from 0, 100th number = 99. Hence, $Z_{100} = 99_{10}$. So 99 is the answer.

Worked-out Example 2 Given is $2_N \times 4_N = 8_N$, where N is number of digits in that the system of writing. For how many values of N is the given calculation true?

☝ ***Solution*** 1st of all, it can be seen that this calculation holds true for N = 10. Giving this calculation a deeper thought, for this calculation to hold true we should have the following digit:

0	1	2	3	4	5	6	7	8

Essentially, $2 \times 4 = 8$ is nothing but 2 added four times starting from 0 (2 + 2 + 2 + 2). So, even if the digits as given above are present, $2_N \times 4_N = 8_N$ holds true. In this case value of N = 9.

Let us see a different case:

0	1	2	3	4	5	6	7	8	9

Does this calculation hold true if we have one more digit = 9? System above given is decimal system. Obviously, N = 10 is also a probable value.

Even $2_N \times 4_N = 8_N$ is true for below given system of writing the numbers too:

0	1	2	3	4	5	6	7	8	9	A

In this case N = 11.

Now we can summarize our result – $N \geq 9$ is the set of values of N. Hence, there are infinite values of N for which $2_N \times 4_N = 8_N$ will hold true.

Worked-out Example 3 Given is $4_N \times 5_N = 20_N$, where N is number of digits in that the system of writing. For how many values of N is the given calculation true?

☝ ***Solution*** Let us find out the answer to this question through contradiction:

N = 10 is one of the solutions. Let us assume that there is atleast one more system of M digits that will satisfy this calculation.

So, $20_{10} = 20_M$. It is possible only if 10 = M. So we can conclude that only value of N possible = 10.

(this example can be seen with the case of an speedometer attached to vehicles—if in covering the same distance, two vehicles are showing the same number as the distance, then both the speedometers are having the same number of digits.

Worked-out Example 4 What is the difference between the largest 5 digit hexadecimal number and smallest four digit hexadecimal number?

☝ ***Solution*** Largest 5 digit hexadecimal number = FFFFF and smallest four digit hexadecimal number = 1000.

$$\begin{array}{r} \text{FFFFF} \\ \text{-1000} \\ \hline \text{FEFFF} \end{array}$$

F – 1 = E, and F – 0 = F. When we subtract 1 from F, we would obtain a digit less than F. [As in the case of 9 – 1 = 8.]

Inter-conversion of Numbers from One Base to other Base

While discussing the method to generate the numbers, we observed that how numbers go from 0–9, next to 10–19 and the cycle keeps on repeating for every

10 numbers. Had there been only 7 digit, length of cycle would have been equal to 7 i.e., starting from 0–6, next numbers would have 10–16, then 20–26 and so on. We will keep this in our mind while learning the conversion of numbers from one base to other base.

For the sake of easing whole **conversion exercise,** we would distribute it in two types:

1. $(\text{Base})_{10}$ to any other base and vice-versa.
2. $(\text{Base})_x$ to $(\text{Base})_y$ and vice-versa; none of x and y being equal to 10 but x and y will be given.
   (i) $(\text{Base})_{10}$ to any other base and vice-versa
   (ii) Base 10 to base N

1. $(\text{Base})_{10}$ to any other base and vise-versa

Method 1

Let us see in case of $(74)_{10}$.

$(74)_{10} = 7 \times 10^1 + 4 \times 10^0$, since the base is 10.

Now, if we have to convert this number on base 9, then we will try to write it in terms of powers of 9.

$(74)_{10} = 8 \times 9^1 + 2 \times 9^0 = (82)_9$

$(74)_{10} = 1 \times 8^2 + 1 \times 8^1 + 2 \times 8^0 = (112)_8$

$(74)_{10} = 1 \times 7^2 + 3 \times 7^1 + 4 \times 7^0 = (134)_7$

$(74)_{10} = 2 \times 6^2 + 0 \times 6^1 + 2 \times 6^0 = (202)_6$

While converting the numbers from decimal system to any other system of writing the numbers, we need to take care of the following two rules:

(i) Take maximum possible power of the base and then keep writing rest of the number with the help of lesser power of base (as illustrated in above example).
(ii) Once we have used $(\text{base})^n$; where n is the maximum power, we will be required to write the co-efficient of all the powers of base from 0 to (n–1) as in the case of $(74)_{10} = (202)_6$.

Example 1 Now, suppose we have to convert $(356)_7$ in the base of 10.

$(356)_7 = 3 \times 7^2 + 5 \times 7^1 + 6 \times 7^0 = (188)_{10}$

Example 2 Convert 143_{10} on base 8.

Step 1—Maximum power of the desired base 8 that can be taken out from $143 = 8^2$. 8^2 will come 2 times $= 8^2 \times 2 = 128$.

Since maximum power of 8 = 2, this number is going to have 3 digits, and we are required to write down the co-efficient of 8^2, 8^1 and 8^0.

$143_{10} = 2 \times 8^2 + _ \times 8^1 + _ \times 8^0$

Now we are required to fill the two blanks in RHS of above expression. We have already written 128 out of 143, and we are required to write 15 (143 – 128) with the help of [_ × 8^1 + _ × 8^0].

It can be clearly seen that → one time 8^1 can be accommodated and 7 times 8^0 can be accommodated.

Hence, $143_{10} = 217_8$.

Method 2

Divide the number by the resultant base and write the quotient and remainder in the following way:

Converting $(74)_{10}$ to the base of $(\ )_9$:

Base	74	Remainder
9	8	2

Quotient

So, $(74)_{10} = (82)_9$

Converting $(74)_{10}$ to the base of $(\ )_8$:

Base	74	Remainder
8	9	2
8	1	1

Quotient

So, $(74)_{10} = (112)_8$

Converting $(74)_{10}$ to the base of $(\ )_7$:

Base	74	Remainder
7	10	4
7	1	3
	1	

So, $(74)_{10} = (134)_7$

Converting $(74)_{10}$ to the base of $(\ )_6$:

Base	74	Remainder
6	12	2
6	2	0

Quotient

So, $(74)_{10} = (202)_6$

TEST YOUR LEARNING 1

1. Convert the number from one base to other base (using the above given method):

	Number	From base	To base
1.	123	10	7
2.	250	10	9
3.	252	10	6
4.	460	10	5

[Go online—to check your base conversion calculations, you may refer to the following website (java enabled pc required)]

[http://pages.swcp.com/~spsvs/resume/BaseConversion/BaseConversion.html]

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2. Converting any number on base N to base 10

To go to base N from base 10, we were required to divide the number by base N. To go back to base 10 from base N, we are now required to use the multiplication by N.

Method 1

Converting $(120)_N$ to base 10:

We write $(120)_N = 1 \times N^2 + 2\ N^1 + 0 \times N^0$. Idea behind writing this is simple and can be understood best by taking an example on base 10.

$120_{10} = 1 \times 10^2 + 2 \times 10^1 + 0 \times 10^0$ [120 represents – there is 1 hundred, 2 tens and 0 units.]

Hence, $(120)_3 = 1 \times 3^2 + 2 \times 3^1 + 0 \times 3^0 = 9 + 6 + 1 = (16)_{10}$

$(120)_8 = 1 \times 8^2 + 2 \times 8^1 + 0 \times 8^0 = 64 + 16 + 1 = (81)_{10}$

$(120)_{12} = 1 \times 12^2 + 2 \times 12^1 + 0 \times 12^0 = 144 + 24 + 1 = (169)_{10}$

$(1357)_9 = 1 \times 9^3 + 3 \times 9^2 + 5 \times 9^1 + 7 \times 9^0 = 729 + 243 + 45 + 7 = (1024)_{10}$

Method 2

Understand this method through the following mechanism:

$(ABC)_N = ((A \times N + B)N + C)N$

$(120)_3$

Step 1—Multiply the left most digit with the base = $1 \times 3 = 3$

Step 2—Add 2 (2nd digit from left) to 3 (obtained from step 1) = $2 + 3 = 5$

Step 3—5×3(base) = 15

Step 4—Add 0 (last digit from left) to 15 (obtained from step 3) = $0 + 15 = 15_{10}$.

Summarizing the whole discussion $(120)_3 = [\{(1 \times 3)+2\}\times 3]+0 = 15_{10}$

$(120)_8$

Step 1—Multiply the left most digit with the base = $1 \times 8 = 8$

Step 2—Add 2 (2nd digit from left) to 8 (obtained from step 1) = $2 + 8 = 10$

Step 3—10×8(base) = 80

Step 4—Add 0 (last digit from left) to 80 (obtained from step 3) = $0 + 80 = 80_{10}$.

Summarizing the whole discussion $(120)_8 = [\{(1 \times 8)+2\}\times 8]+0 = 80_{10}$

$(234)_7$

Step 1—Multiply the left most digit with the base = $2 \times 7 = 14$.

Step 2—Add 3 (2nd digit from left) to 14 (obtained from step 1) = $3 + 14 = 17$.

Step 3—17×7(base) = 119

Step 4—Add 4 (last digit from left) to 119 (obtained from step 3) = $4 + 119 = 123_{10}$.

Summarizing the whole discussion $(234)_7 = [\{(2 \times 7)+3\}\times 7]+4 = 123_{10}$

$(2542)_6 = [\{(2 \times 6 + 5) \times 6\} + 4] \times 6 + 2 = 638_{10}$

We have reached now a stage to solve the elementary questions based upon base conversion. Let us start with Worked-out Example 3 again, that we have done earlier.

Worked-out Example 5 Given is $4_N \times 5_N = 20_N$, where N is number of digits in that system of writing. For how many values of N is the given calculation true?

☝ ***Solution*** Standard process of solving these questions involve writing both the Right Hand side and Left hand side on one common base—preferably on base 10.

$4_N = 4_{10}$ and $5_N = 5_{10}$

Hence, LHS = $4_N \times 5_N = 4_{10} \times 5_{10} = 20_{10}$

Now convert RHS on base 10. $20_N = 2 \times N + 0 \times N^0 = 2N$

LHS = RHS $\Rightarrow 20 = 2N \Rightarrow N = 10$. And this is the only value of N.

Worked-out Example 6 Given is $4_N \times 5_N = 21_N$, where N is number of digits in that system of writing. For how many values of N is the given calculation true?

☝ ***Solution*** Going by the above solution, LHS = 20

RHS = $2 \times N + 1 \times N^0 = 2N + 1$

LHS = RHS $\Rightarrow 2N + 1 = 20$

This does not give the natural number value of N. Hence, this calculation is not possible on any base.

Worked-out Example 7 Given is $(12N)_4 = (124)_N$. [In LHS, N is the unit digit and in RHS, N is the base]. How many values of N is/are possible?

☝ ***Solution*** Let us convert each of LHS and RHS on base 10.

$(12N)_4 = 1 \times 4^2 + 2 \times 4^1 + N \times 4^0 = 24 + N$

$(124)_N = 1 \times N^2 + 2 \times N^1 + 4 \times N^0 = N^2 + 2N + 4$

LHS = RHS $\Rightarrow N^2 + 2N + 4 = 24 + N \Rightarrow N^2 + N - 20 = 0$

$\Rightarrow (N + 5)(N - 4) = 0 \Rightarrow N = 4$ or -5

Obviously, $N = -5$ cannot be the solution. Hence, $N = 4$ is the only solution?

Is it possible to have $N = 4$ in this case? If $N = 4$, then none of LHS and RHS will be possible because on base 4 there cannot be a digit like '4'. [largest digit on base 4 = 3].

Now we conclude that N cannot take any value. So number of values of N possible = 0.

Alternatively, this question can be done logically too.

LHS = $(12N)_4 \Rightarrow$ on base 4, largest digit possible = 3, so largest possible value of N = 3.

RHS = $(124)_N \Rightarrow$ Presence of digit '4' is possible only for $N \geq 5$. So minimum possible value of N = 5.

LHS states that maximum value of N = 3, and RHS states that minimum possible value of N = 5. So there is no common value between LHS and RHS.

Hence, number of value of N possible = 0.

TEST YOUR LEARNING 2

1. Convert the number from one base to other base:

	Number	From base	To base
1.	123	8	10
2.	250	6	10
3.	352	10	6
4.	460	10	4
5.	368	10	8
6.	909	10	100
7.	640	8	10

2. $(\text{Base})_x$ to $(\text{Base})_y$ and vice-versa; none of x and y being equal to 10 [x and y are given]

We will see this in two parts:

1. Conversion from one base to other base where one base is not the power of other base (like base 3 to base 4 or base 6 to base 2).
2. Conversion from one base to other base where one base is the power of other base (like base 2 to base 4 or base 8 to base 2).

Let us start with **(a) Conversion from one base to other base where one base is not the power of other base (like base 3 to base 4 or base 6 to base 2).**

We have the method of base conversion defined through base 10. Although we can convert any base to any other base directly too, but that needs more of

mental calculation. So we will see 1st the method of conversion from one base to other base through base 10, and then the second method – without base 10.

Convert $(345)_8$ to $(\ )_9$.

We will do this problem with the help of creating a bridge of base 10 between base 8 and base 7.

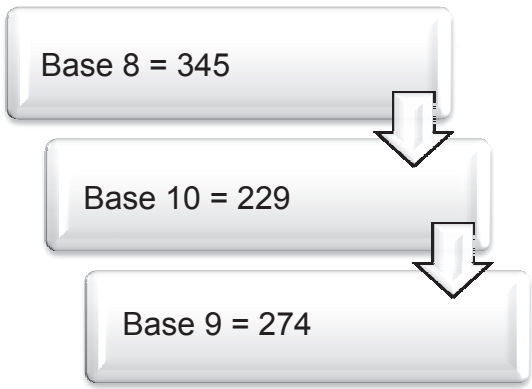
Step 1—Convert $(345)_8$ into base 10.

$$345 = 3 \times 8^2 + 4 \times 8^1 + 5 \times 8^0 = (229)_{10}$$

Step 2—Now convert this number in base 10 into base 9.

$$(229)_{10} = 2\times 9^2 + 7 \times 9^1 + 4 \times 9^0 = (274)_9$$

Whole process can be summarized with the help of following graphics:



Now recall Method 2 of converting the numbers from base N to base 10.

$(ABC)_N = ((A \times N + B)N + C)N$. This is what we have seen on previous pages. We will now use this method to convert the number from one base to other base directly without going through base 10.

Following is the way we convert $(345)_8$ on base 10:

$(345)_8 = [\{(3 \times 8) + 4\}\times 8]+5 = 229_{10}$

Let us look at it step wise, and make analogy between the conversion on base 10 and conversion on base 9:

345	on base 10	on base 9	Remarks
Step 1	$3 \times 8 = 24$	$3 \times 8 = 26$	is equal to 24 of base 10 converted on base 9
Step 2	$24 + 4 = 28$	$26 + 4 = 31$	is equal to 28 of base 10 converted on base 9
Step 3	$28 \times 8 = 224$	$31 \times 8 = 268$	is equal to 224 of base 10 converted on base 9
Step 4	$224 + 5 = 229$	$268 + 5 = 274$	is equal to 229 of base 10 converted on base 9

Once you have understood above table, let us see this:

345_8	on base 9
Step 1	$3 \times 8 = 26$
Step 2	$26 + 4 = 31$
Step 3	$31 \times 8 = 268$
Step 4	$268 + 5 = 274$

This is the same table given without numbers "on base 9" and "Remarks".

Are you able to understand this now? If yes, you have learnt this method of converting the numbers from one base to other base directly without going through base 10.

See one more example:

243 on base 6	on base 8
Step 1	$2 \times 6 = 14$
Step 2	$14 + 4 = 20$
Step 3	$20 \times 6 = 140$
Step 4	$140 + 3 = 143$

I hope now you must have understood this method. Use this newly acquired method to solve the following questions. *USE this method only.* You may be required to use digits larger than 9 to solve these questions – A = 10, B = 11, C = 12, and so on Z = 35.

TEST YOUR LEARNING 3

	Number	From base	To base
1.	123	8	5
2.	250	6	12
3.	352	9	15

(b) Conversion from one base to other base where one base is the power of other base (like base 2 to base 4 or base 8 to base 2).

We can do this conversion through base 10, or else we can go directly from one base to other as long as one base is the power of other base.

Convert a number from base n to base $n^2 \Rightarrow$ power of the resultant base will decide that how many digits are to be coupled. In this case power of the resultant base = 2, hence, two digits will be taken together from base 2.

(i) **Convert $(1011100101)_2$ to $(\ )_4$ system:**

Step 1—$(1011100101)_2$ will be rewritten as $(10)_2\,(11)_2\,(10)_2\,(01)_2\,(01)_2$

Step 2—$(10)_2\,(11)_2\,(10)_2\,(01)_2\,(01)_2$ – these numbers will be convert-

ed on base 4 now

Step 3—$(10)_2 = 2$; $(11)_2 = 3$; $(10)_2 = 2$; $(01)_2 = 1$; $(01)_2 = 1$.

Hence, final number $(101110010)_2 = (23211)_4$

Remarks Understand that $2_3 = 2_{10}$; $3_3 = 3_{10}$ and so on. So even if step 2 would have been "these numbers will be converted on base 10 now", final answer would have been same.

(ii) **Convert $(1011100101)_2$ to octal ( )$_8$ system:**

Step 1—$(1011100101)_2$ will be re-written as $(001)_2(011)_2(100)_2(101)_2$.

Step 2—$(001)_2(011)_2(100)_2(101)_2$ – these numbers will be converted on base 8, now;

Step 3—$(001)_2 = 1$; $(011)_2 = 3$; $(100)_2 = 4$; $(101)_2 = 5$

Hence, final number = $(1011100101)_2 = (1345)_8$

(iii) **Convert $(13121232)_4$ to hexadecimal ( )$_{16}$ system:**

Initial base = 4 and Final base = $16 = 4^2$. So two digits will be clubbed.

Step 1—$(13121232)_4$ will be re-written as $(13)_4(12)_4(12)_4(32)_4$

Step 2—$(13)_4(12)_4(12)_4(32)_4$ – these numbers will be converted on base 16 now

Step 3—$(13)_4 = 7$; $(12)_4 = 6$; $(12)_4 = 6$; $(32)_4 = E$ (=14)

Hence, final number = $(13121232)_4 = (766E)_{16}$

(iv) **Convert $(123012)_4$ to base 8:**

Step 1—We will 1st convert $(123012)_4$ to base 2 = $(1\ 10\ 11\ 00\ 01\ 10)_2$.

Step 2—We will now convert $(11011000110)_2$ to base 8.

$(11011000110)_2 = (011\ 011\ 000\ 110)_2$

Step 3—$(011)_2 = 3$; $(011)_2 = 3$; $(000)_2 = 0$; $(110)_2 = 6$.

Hence, $(123012)_4 = (3306)_8$

Alternatively, we could have 1st converted $(123012)_4$ on base 10, and from there we could have converted the number back to base 8.

$(123012)_4 = (1734)_{10} = (3306)_8$

(v) **Convert $(3458)_9$ to ( )$_3$:**

Step 1—$(3458)_9$ write each of the digits with the help of two digits using two digits

Step 2—$(3458)_9 = (10\ 11\ 12\ 22)_3$.

(vi) **Convert $(32132)_4$ to ( )$_{16}$:**

Since, $4^2 = 16$, hence, we will couple two digits from right together.

$03 = 3$; $(21)_4 = 9$; $(32)_4 = E$

Hence, $(32132)_4$ to $(39E)_{16}$

Decimal calculations with base system

So far we have seen the calculations involving natural numbers only. Lets work now with decimals.

Converting decimal system numbers to any other system

(i) Suppose (12.725) is a number in decimal system which is required to be converted into octal system (8 digits)

We will first convert 12 into octal system.

$(12)_{10} = (14)_8$

Now to convert $(0.725)_{10}$ into $(\quad)_8$, we will apply following method:

$0.725 \times 8 = \mathbf{5}.8$ Take out integral part from here.

$0.8 \times 8 = \mathbf{6}.4$ Take out integral part from here.

$0.4 \times 8 = \mathbf{3}.2$ Take out integral part from here.

$0.2 \times 8 = \mathbf{1}.6$ Take out integral part from here.

$0.6 \times 8 = \mathbf{4}.8$ Take out integral part from here.

And keep doing this till the moment we get decimal part as zero i.e., the product should be an integer.

$(0.725)_{10} = (0.56314\ldots\ldots)_8$

So, $(12.725) = (14.56314\ldots\ldots)_8$

And so on, it will keep on going like this.

(ii) Convert $(10.125)_{10}$ to $(\,)_8$:

$(10)_{10} = (12)_8$

$(0.125)_{10} = 0.125 \times 8 = 1.0$

Hence, $(10.125)_{10} = (12.1)_8$

Converting any other system numbers to decimal system

Now suppose if $(15.453)_7$ is to be converted into decimal system, then the process is as follows:

We will first convert $(15)_7$ into decimal system.

$(15)_7 = 1 \times 7^1 + 5 \times 7^0 = (12)_{10}$

Now, $(0.453)_7 = 4 \times 7^{-1} + 5 \times 7^{-2} + 3 \times 7^{-3}$

$= 0.682215743.....$

So, $(15.453)_7 = (12.682215743.....)_{10}$

TEST YOUR LEARNING 4

1. Convert the number from one base to other base (using the above given method):

	Number	From base	To base
1.	12.571	10	7
2.	25.214	8	10
3.	25.212	10	6
4.	46.31	11	10

Algebraic Calculations on Different Base System

We will discuss here Addition, Subtraction and Multiplication on base other than 10.

Before that, we should understand that any such calculation is possible only on same base. In other words, two numbers on two different bases cannot be added or subtracted or multiplied to each other without bringing both the numbers on any common base. For example, 346_8 cannot be added to 643_7. Only way this addition can be done is by bringing both these numbers on the same base—either we convert 346_8 to the base 7 or 643_7 to the base 8 or both of these numbers to any common base.

Addition

Process of addition on any base system is exactly the same as that on decimal system.

Consider the addition on binary system:

0 + 0 = 0 with carry forward 0

0 + 1 = 1 with carry forward 0

1 + 1 = 0 with carry forward 1 and so on.

Let us understand more of addition through some simple examples:

527_{10}
$+ 436_{10}$
$= 963_{10}$

On base 10

1st we add 7 and 6 = 13. 3 is written as unit digit and 1 is carries forward to the tens place. Next is 2 + 3 = 5 and 1 is added to it carried forward from unit place = 5 + 1 = 6. Finally, 5 + 4 = 9.

Hence, summation = 9 6 3.

Let us do the same summation on base 9:

527_9
$+ 436_9$
1064

Method 1 Convert all the numbers on decimal system, do the subtraction and convert the result back to the asked base.

$527_9 = 430_{10}$

$436_9 = 357_{10}$

Hence, $527_9 + 436_9 = 430_{10} + 357_{10} = 787_{10}$

Now we will convert 787_{10} on base 9 $\Rightarrow 787_{10} = 1064_9$

Method 2 Without going through the decimal system, working with the given system.

7 + 6 = 13 on base 10. We will convert this on base 9 = 14. This 4 will go the unit place and 1 will be carried forward to the tens place.

At tens place = 2 + 3 + 1 (carried forward from unit place) = 6

At hundreds place = 5 + 4 = 9 on base 10. As there is no digit like 9 on base 9, we convert this on base 9 = 10.

Hence, result = 1064

Now, Let us do the same summation on base 8 directly without going through base 10:

527_8
$+ 436_8$
1165_8

7 + 6 = 13 on base 10. We will convert this on base 8 = 15. This 5 will go the unit place and 1 will be carried forward to the tens place.

At tens place = 2 + 3 + 1 (carried forward from unit place) = 6.

At hundreds place = 5 + 4 = 9 on base 10. As there is no digit like 9 on base 8, we convert this on base 8 = 11.

Hence, result = 11 6 5

Subtraction

Once again, Process of subtraction on any base system is exactly the same as that on decimal system.

647_{10}
$- 328_{10}$

At the unit place, it is not possible to subtract 8 from 7, hence, we borrow from the tens place. Since, the place value of any digit at the tens place = 10, hence, the number that will be borrowed will be always 10 on the base of 10.

Now it becomes:

63	17_{10}
– 32	8_{10}
	9_{10}

Now tens and hundreds place will can be calculated simply = 63 – 32 = 31.

So resultant = 319_{10}

Let us do the same subtraction on base 9:

647_9
$-\ 328_9$

Method 1 Convert all the numbers on decimal system, do the subtraction and convert the result back to the asked base.

$647_9 = 529_{10}$

$328_9 = 269_{10}$

Hence, $647_9 - 328_9 = 529_{10} - 269_{10} = 260_{10} = 318_9$

Now we will convert 260_{10} on base 9 $\Rightarrow 260_{10} = 318_9$

Method 2 Without going through the decimal system, working with the given system.

At the unit place, it is not possible to subtract 8 from 7, hence, we borrow from the tens place. Since the place value of any digit at the tens place = 9, hence, the number that will be borrowed from the tens place will be always 9 on the base of 9.

It can also be understood as follows: $647_9 = 6 \times 9^2 + 4 \times 9^1 + 7 \times 9^0 \Rightarrow$ At the tens place, value of the numbers = 4×9^1. So if we borrow 1, 1×9^1 will be borrowed.

Now it becomes:

63_9	16_{10}
-32_9	8_{10}
	8_{10}

Since, $8_{10} = 8_9$, we can write the unit digit = 8 without doing any further conversion.

Next $63_9 - 32_9 = 31_9$

Hence, resultant = 318_9

Multiplication

As seen above in case of subtraction and addition, in multiplication too we have two methods of doing the calculation—Method 1 will be through base 10, and Method 2 will be doing the calculation through the given base. Let us see this one by one using an example:

(i) 46×53 On base 10

		4	6
		× 5	3
+	1	3	8
2	3	0	
2	4	3	8

(ii) 46×53 On base 9

Method 1 Convert all the numbers on decimal system, do the subtraction and convert the result back to the asked base.

$46_9 = 42_{10}$

$53_9 = 48_{10}$

So $46_9 \times 53_9 = 42_{10} \times 48_{10} = 2016_{10}$

Now we will convert 2016_{10} on base 9.

$2016_{10} = 2680_9$

Method 2 Without going through the decimal system, working with the given system,

	2(c.f)	
	4	6
	×	3
1	5	0

1st line of multiplication on base 9. (C.f. = Carried forward)

		3(c.f.)	
		4	6
		5	
2	5	3	

2nd line of multiplication on base 9. (C.f = Carried forward)

Now we will do the final addition:

+	1	5	0
2	5	3	
2	6	8	0

So, $46_9 \times 53_9 = 2680_9$

Some Standard Facts

1. Some standard system of writing
   Decimal system
   Digits used—0,1,2,3,4,5,6,7,8,9 $\Rightarrow$ Total digits used = 10 digits
   Hexa-decimal system
   Digits used—0,1,2,3,4,5,6,7,8,9, A,B,C,D,E,F $\Rightarrow$ Total digits used = 16
   Octal system
   Digits used—0,1,2,3,4,5,6,7 $\Rightarrow$Total digits used = 8
   Binary system
   Digits used—0,1 $\Rightarrow$ Total Digits used = 2
2. If a number is a prime number on base 10, then it will be prime on all the bases it is converted upon. Similarly, if a number is a composite number on base 10, then it will be composite number on all the bases it is converted upon.
   5_{10} is a prime number = 101_2 is a prime number = 12_3 is a prime number and so on.

PRACTICE EXERCISE 1

Directions for questions (1–5): Mark your response as true or false.

1. Any amount of stamp that is greater than or equal to Rs 8 can be formed by infinite supply of Rs 3 and Rs 5 stamps. (True/False)

2. An accountant interchanges the digits of a two-digit number xy, where x is the tens place digit and y is the unit digit, while making an entry in his ledger and ends up with an error in his accounts. If the base 10 equivalent of the error is 48, more than one values of xy are possible. (True/False)

3. On any base system of writing the numbers, if a number is having unit digit = 0, then the number is divisible by 5. (True/False)

4. It is possible only for decimal system of writing the numbers that if a number is having unit digit = 0, then the number is divisible by 5. (True/False)

5. Largest possible digit of a system of writing the numbers having 7 digits is 7. (True/False)

Directions for questions (6–20): Solve the questions and mark the answer from the options given.

6. How many two digit natural numbers on base 10 when converted on base 5 will have unit digit = 1?

1. 18 **2.** 19 **3.** 20 **4.** 21

5. None of these

Directions for questions (7–9): Answer the questions on the basis of the tables given below. Two binary operations $\oplus$ and @ are defined over the set {a, e, f, g, h} as per the following tables:

$\oplus$	a	e	f	g	h
a	a	e	f	g	h
e	e	f	g	h	a
f	f	g	h	a	e
g	g	h	a	e	f
h	h	a	e	f	g

*	a	e	f	g	h
a	a	a	a	a	a
e	a	e	f	g	h
f	a	f	h	e	g
g	a	g	e	h	f
h	a	h	g	f	e

Thus, according to the first table, $f \oplus g = a$, while according to the second table $g * h = f$, and so on. Also, let $f^2 = f * f$, $g^3 = g*g*g$, and so on.

7. What is the smallest positive integer n such that $g^n = e$?

1. 4 **2.** 5 **3.** 2 **4.** 3

5. 1

8. Upon simplification, $f \oplus [f * \{f \oplus (f * f)\}]$ equals

1. e **2.** f **3.** g **4.** h

5. None of these

9. Upon simplification, $\{a^{10} * (f^{10} \oplus g^g)\} \oplus e^8$ equals

1. e **2.** f **3.** g **4.** h

5. None of these

Directions for questions(10–13): Read the passage below and solve the questions based on it.

Our decimal system has 10 digits to write the numbers—0 to 9. A new system of writing the numbers is to be devised by inserting a new symbol $ between 8 and 9 such that $ – 8 = 9 – $. All the other digits of decimal system are to be kept at the same place. So this new system has digits from 0 – 9 to write the number.

10. What is value of the number 6 of this new system when converted on base 10?

1. 6 **2.** 5 **3.** 7 **4.** 3

5. Cannot be determined

11. What is value of the number 65 of this new system when converted on base 10?

1. 65 **2.** 71 **3.** 70 **4.** 72

5. None of these

12. What is value of the number 98 of this new system when converted on base 10?

1. 107 **2.** 117 **3.** 108 **4.** 118

5. None of these

13. What is value of the number 9$ of this new system when converted on base 10?

1. 107 **2.** 98 **3.** 108 **4.** 119

5. None of these

Directions for questions (14–19): Do the following calculations on binary system.

14. $1000_2 + 1101_2 + 1111_2 =$

1. 100100 **2.** 111100 **3.** 101010 **4.** None of these

15. $111_2 + 101_2 + 011_2 =$

1. 1011 **2.** 1111 **3.** 1101 **4.** None of these

16. $10001_2 - 1111_2 =$

1. 101 **2.** 11 **3.** 10 **4.** None of these

17. $11111_2 - 10001_2 =$

1. 1010 **2.** 1111 **3.** 1110 **4.** None of these

18. Multiply 101 by 11

1. 1111 **2.** 1011 **3.** 1110 **4.** 11011

5. None of these

19. Multiply 11001 by 101

1. 1111101 **2.** 1110101 **3.** 1011101 **4.** 1100110

5. None of these

20. How many of the following numbers written on binary system are divisible by 2_{10}?

(i) 110101 (ii) 110010 (iii) 11111 (iv) 100010

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

21. How many natural numbers N are there such that the base 2001 representation of n! consists entirely of 1's?

1. 1 **2.** 2 **3.** 101 **4.** 1001

5. None of these

22. Following is a calculation on a particular base (other than 10). Identify the base.

$$\begin{array}{r} 2\,3\,_\,5\,_ \\ +\ 1\,_\,6\,4\,2 \\ \hline 4\,2\,4\,2\,3 \end{array}$$

1. 11 **2.** 9 **3.** 7 **4.** 8

5. 6

23. abcd is a four digit no in base 7 such that 2(abcd) = bcda. (a,b ≠ 0)
Find the value of a.

1. 1 **2.** 2 **3.** 3 **4.** 4

5. cannot be determined

26. $(100!)_N$ has 24 zeroes at its end. How many values of N are possible?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. None of these

25. How many natural numbers are there when you count from 11_6 to 400_6?

1. 137 **2.** 138 **3.** 136 **4.** 390

5. None of these

26. How many times does the digit 6 appear when we count from 6_8 to 400_8?

1. 37 **2.** 38 **3.** 79 **4.** 78

5. None of these

27. How many prime number factors will be there for $N = 420_6$?

1. 2 **2.** 3 **3.** 4 **4.** 5

5. Cannot be determined

28. What is the total number of factors of $N = 720_8$?

1. 5 **2.** 10 **3.** 15 **4.** 78

5. None of these

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING 1

1. 234

Base	Quotient	Remainder
7	123	4
	17	3
	2	

2. 307

Base	Quotient	Remainder
9	250	7
	27	0
	3	

3. 1100

Base	Quotient	Remainder
6	252	0
	42	0
	7	1
	1	

4. 3320

Base	Quotient	Remainder
5	460	0
	92	2
	18	3
	3	

TEST YOUR LEARNING 2

1. 83 **2.** 102 **3.** 1344 **4.** 13030
5. 560 **6.** 99 **7.** 416

TEST YOUR LEARNING 3

1. 313 **2.** 86 **3.** 14B

TEST YOUR LEARNING 4

1. 15.3665 **2.** 21.273 **3.** 41.1134 **4.** 50.28

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
True	**True**	**False**	**False**	**False**	**2**	**1**	**4**	**1**	**1**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
2	**4**	**4**	**1**	**2**	**3**	**3**	**1**	**1**	**3**
21.	22.	23.	24.	25.	26.	27.	28.		
2	**3**	**5**	**2**	**5**	**5**	**2**	**2**		

1. True

2. True

Decimal system equivalent of $(xy)_n - (yx)_n = nx + y - ny - x = (x - y)(n - 1) = 48$

Let us now see the different possible values of (x – y) and (n –1):

(x – y)	(n –1)		
48	1	Not possible	
24	1	Not possible	
16	3	Not possible	
12	4	Not possible	
8	6	Not possible	On base n = 7, difference x – y cannot be 8
6	8	Possible	71, 82 $\Rightarrow$ Two values are possible.

3. False. This is not possible for so many system of writing the numbers like N = 2, 3, 4 etc.

4. False. It is possible for N = 5, 10, 15 etc system of writing the numbers.

5. False. If there are 7 digits, then the digits are going to be 0 – 6. Hence given statement is false.

6. Numbers are going to be 11, 16, 21, ..., 96.

Hence, number of numbers = 19.

7. 1.

$g^2 = g * g = h$

$g^3 = g^2 * g = h * g = f$

$g^4 = g^3 * g = f * g = e$

$\therefore$ n = 4.

8. 4.

$f \oplus [f * \{f \oplus (f * F)\}] = f \oplus [f * \{f \oplus h\}] = f \oplus [f * e] = f \oplus [f] = h$

9. 1.

10. 1. As we have already discussed digits are having the same value on all the systems.

Hence, 6 of this new system containing 11 digits = 6 of decimal system

11. 2. This new system is having 11 digits.

$65_{11} = 6 \times 11^1 + 5 \times 11^0 = 71_{10}$

12. 4. 9 of this new system corresponds to 10 of decimal system.

$98_{11} = 10 \times 11^1 + 8 \times 11^0 = 118_{10}$

13. 4. 9 of this new system corresponds to 10 of decimal system and $ of this new system is equal to 9 of decimal system.

$9\$_{11} = 10 \times 11^1 + 9 \times 11^0 = 119_{10}$

14. 1.

$$\begin{array}{r} 1000 \\ 1101 \\ \underline{1111} \\ 100100 \end{array}$$

15. 2.

$$\begin{array}{r} 111 \\ 101 \\ \underline{011} \\ 1111 \end{array}$$

16. 3.

17. 3.

18. 1.

$$\begin{array}{r} 101 \\ \underline{11} \\ 101 \\ \underline{101} \\ 1111 \end{array}$$

19. 1.

$$\begin{array}{r} 11001 \\ \underline{101} \\ 11001 \\ 00000 \\ \underline{11001} \\ 1111101 \end{array}$$

20. 3. The idea is to observe the unit digit of the number—if unit digit of the number = 0, then it will be divisible by 2_{10}. Hence 2 numbers.

21. 1. It is possible only for N = 0, and N = 1.

25. 5. It is possible for N = 5, 9, 10, 15, 20 ... so, obviously more than 4. Hence option 5.

27. 2. $(420)_6 = (156)_{10} \Rightarrow$ 156 has 3 prime factors.

28. Number of factors of 720_8 = Number of factors of 464_{10}

$464 = 29 \times 2^4 \Rightarrow$ Number of factors = $2 \times 4 = 10$

CAT QUESTIONS FROM THIS CONCEPT

CAT 2003

1. A positive whole number M less than 100 is represented in base 2 notation, base 3 notation, and base 5 notation. It is found that in all three cases the last digit is 1, while in exactly two out of the three cases the leading digit is 1. Then M equals

1. 31 **2.** 63 **3.** 75 **4.** 91

5. None of these

CAT 2001

2. In a number system the product of 44 and 11 is 3414. The number 3111 of this system, when converted to the decimal number system, becomes

1. 406 **2.** 1086 **3.** 213 **4.** 691

5. None of these

CAT 2000

3. Convert the number 1982 from base 10 to base 12. The result is

1. 1182 **2.** 1912 **3.** 1192 **4.** 1292

5. None of these

SOLUTIONS AND EXPLANATIONS

1. 4. Since the last digit in base 2, 3 and 5 is 1, the number should be such that on dividing by either 2, 3 or 5 we should get a remainder 1. The smallest such number is 31. The next set of numbers are 61, 91. Among these only 31 and 91 are a part of the answer choices.

Among these, $(31)_{10} = (11111)_2 = (1011)_3 = (111)_5$ Thus, all three forms have leading digit 1.

Hence, the answer is 91.

2. 2. The product of 44 and 11 is 484. If base is x, then 3411

$= 3x^2 + 4x^2 + 1x^1 + 4 \times x^0 = 484$

$= 3x^3 + 4x^2 + x = 480$

This equation is satisfied only when x=5. So base is 5.

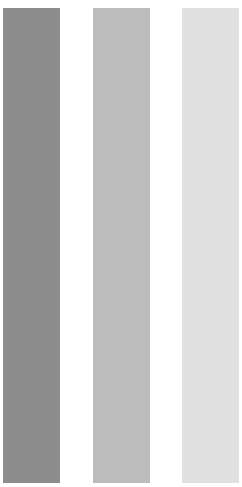
In decimal system, the number 3111 can be written,

$3 \times 5^3 + 1 \times 5^2 + 1 \times 5^1 + 1 \times 5^0 = 406$.

3. 3.

12	1982	
12	165	2
12	13	9
	1	1

The answer is 1192. Hence, option 3 is the answer.



Chapter 5

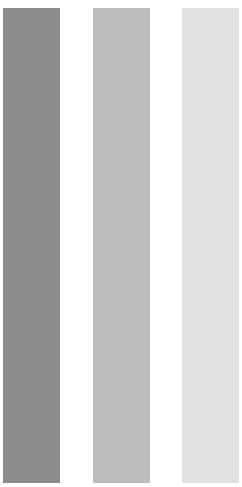
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DEMYSTIFYING NUMBER SYSTEM

EXPONENTS

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- Indices/exponents and its rules
- Surds and its rules
- Exponents and methods to find out exponent of one number into another
- Types of questions asked from this chapter in the CAT

EXPONENTS

Exponents or power is a way to show that how many times a number is multiplied to itself. For example, when we write y^2, we mean to say that y is multiplied to itself.

y^3 = y is multiplied to itself twice more, or a total of three times = $y \times y \times y$.

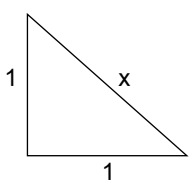
In the example given above, y is known as base, and 3 is known as exponent or power.

Rules of Exponents

1. $a^m \times a^n = a^{m+n}$
2. $a^m/a^n = a^{m-n}$
3. $(a^m)^n = a^{mn}$
4. $a^m \cdot b^m = (ab)^m$
5. $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$
6. $a^0 = 1$
7. $a^{-b} = \frac{1}{a^b}$
8. $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
9. $a^{b^c} = a^{[b^c]}$
   For example, $2^{3^4} = 2^{81}$ (and not 2^{12}).
   By convention, we always calculate powers from 'top to bottom'. So if we are given with $2^{3^{4^5}}$, first we will calculate 4^5 (=1024), then 3^{1024} and finally $2^{3^{1024}}$.
10. $a^x = a^y, \Rightarrow x = y$ [a is not equal to –1 or 0 or 1].

SURDS

Discussion on surds has been occurring for a long time, to be more precise, since the development of the Pythagorean Theorem. Look at the right-angled triangle with perpendicular = base = 1.



Using Pythagorean theorem, $x^2 = 1 + 1 = 2 \Rightarrow x = \sqrt{2}$

We have already seen that,

- A number that can be expressed as a fraction of integers (denominator $x \neq 0$) is called a **rational number**. Examples of rational numbers are $\frac{5}{7}, -\frac{3}{8}, 4$.
- A number which cannot be expressed as a fraction of two integers is called an irrational number. Examples of irrational numbers are π, $\sqrt{2}$, $\sqrt[3]{11}$ etc.

Now we will define **surd**.
An irrational number involving a root is called a surd. Examples of surds are $4\sqrt[3]{5}$, $\sqrt{3}$ etc.

Rules of Surds

1. $a^{\frac{1}{b}} = \sqrt[b]{a}$
2. Two basic laws of surds are, for all positive real numbers a and b:
   $\sqrt{a}.\sqrt{b} = \sqrt{ab}$
   $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
3. Conjugate surds are specially related surds such as, $\sqrt{a}+\sqrt{b}, \sqrt{a}-\sqrt{b}$
   Their product is a rational number, as,
   $(\sqrt{a}-\sqrt{b})(\sqrt{a}+\sqrt{b}) = (\sqrt{a})^2 - (\sqrt{b})^2 = a - b$
   This idea of conjugate surds is used for rationalizing the denominator.
4. Surds are further categorized as Pure Surds or Mixed Surds. A pure surd is a number without any visible natural number component added to it. For example, $\sqrt[4]{48}$ is a pure surd. A mixed surd is a number which has a visible natural number component attached to it. For example, $2\sqrt[4]{3}$ is a mixed surd.

Rationalizing the Denominator

Rationalization of the denominator is done in a fraction such that there are no surds in the denominator. This is done by multiplying the numerator and denominator by the conjugate surd of the denominator.

TEST YOUR LEARNING 1

1. Simplify the following:

1. -5^3

2. $-3^3 - (-3)^2 + (-2)^2$

3. $x^{-2}.\ x^{3}$
4. $\dfrac{x^{3}.x^{\frac{1}{2}}}{x^{4}}$
5. $x^{-1} \times 2x^{1/2}$
6. $\sqrt[3]{8a^{2}b} \times a^{\frac{1}{3}} \times b^{\frac{5}{3}}$

2. Find the conjugate surd of:

1. $\sqrt{2} + \sqrt{3}$
2. $\sqrt{19} - \sqrt{13}$
3. $\sqrt{2a} - \sqrt{3b}$

3. Simplify the following fractions:

1. $\dfrac{\sqrt{2}-\sqrt{5}}{\sqrt{2}+\sqrt{5}}$
2. $\dfrac{1}{\sqrt{2a}+\sqrt{b}}$

4. Solve for y:

1. $5^{y} = 1/625$
2. $16^{y} = 32$
3. $\dfrac{x^{-1}+y^{-1}}{x+y} - \dfrac{x^{-1}-y^{-1}}{x-y}$

Finding the Exponent of a Number in Other Number

Normally, the questions asked will be having, either direct or indirect connotation, one number as numerator and other number as denominator and exponent of denominator dividing the numerator will be required to calculate.

We will see these questions under two broad headings:

(i) Linear Format questions, and
(ii) Factorial format questions

Linear Format Question

Let us understand this format of questions with an example:

Worked-out Example 1 What is the maximum power of 5 that will divide $(25 \times 30 \times 35 \times 40 \times 42 \times 45)$?

☝ ***Solution*** To answer this question, we are required to count the number of 5's present in $25 \times 30 \times 35 \times 40 \times 42 \times 45$.

$25 = 5^{2}$; $30 = 5^{1} \times K$ (we are interested in counting only the number of 5's, and it does not matter that it is multiplied to which number. So we write K instead of 6); $35 = 5^{1} \times L$; $40 = 5^{1} \times M$; $42 = 5^{0} \times N$; $45 = 5^{1} \times P$.

Hence, total number of 5's present in $25 \times 30 \times 35 \times 40 \times 42 \times 45 = 5^2 \times 5^1 \times 5^1 \times 5^1 \times 5^0 \times 5^1 \times Q = 5^6 \times Q$.

So, maximum power of 5 that will divide $(25 \times 30 \times 35 \times 40 \times 42 \times 45) = 6$.

Worked-out Example 2 What is the maximum power of 10 that will divide $(25 \times 30 \times 35 \times 40 \times 42 \times 45)$?

☝ ***Solution*** $10 = 2^1 \times 5^1$. To answer this question, we are required to count the number of sets of (2's and 5's) present in $25 \times 30 \times 35 \times 40 \times 42 \times 45$.

We have already counted that there are six 5's present. We are required to find out the number of 2's in $(25 \times 30 \times 35 \times 40 \times 42 \times 45)$.

25, 35 and 45 does not contain any 2.

$30 = 2^1 \times K$

$40 = 2^3 \times M$

$42 = 2^1 \times N$

So, $(25 \times 30 \times 35 \times 40 \times 42 \times 45) = 2^1 \times K \times 2^3 \times M \times 2^1 \times N = 2^5 \times Q$

Number of times $(25 \times 30 \times 35 \times 40 \times 42 \times 45)$ is divisible by 10 = Number of times $(25 \times 30 \times 35 \times 40 \times 42 \times 45)$ is divisible by 2^1 and 5^1 (or in other words, number of times set of one 2 and one 5 can be made).

Number of sets of 2 and 5 that can be made = minimum of power of 2 or 5 = min (5, 6) = 5

Why do we take the minimum of exponent of 2 and 5 while making sets of 2 and 5? It can be understood in the following way—Let us assume you have 5 left foot shoes, and 6 right foot shoes. Number of people who can wear these shoes simultaneously = 5 persons.

Worked-out Example 3 What is the maximum power of 10 that will divide $(25^{70} \times 30^{120} \times 35 \times 40 \times 42 \times 45)$?

☝ ***Solution*** $10 = 2^1 \times 5^1$. To answer this question, we are required to count the number of sets of (2's and 5's) present in $(25^{70} \times 30^{120} \times 35 \times 40 \times 42 \times 45)$.

Total number of 5's in $25^{70} = 70 \times 2 = 140$

Total number of 5's in $30^{120} = 120$

Total number of 5's in 35 = Total number of 5's in 40 = Total number of 5's in 45 = 1

Total number of 5's in 42 = 0

Total number of 5's in $(25^{70} \times 30^{120} \times 35 \times 40 \times 42 \times 45) = 140 + 120 + 3 + 0 = 263$

Now we will count the number of 2's.

Total number of 2's in 25^{70} = Total number of 2's in 35 = Total number of 2's in 45 = 0

Total number of 2's in $30^{120} = 120$

Total number of 2's in 42 = 1

Total number of 2's in 40 = $(2^3 =) 3$

Total number of 2's in $(25^{70} \times 30^{120} \times 35 \times 40 \times 42 \times 45) = 120 + 3 + 1 = 124$

Number of sets of 2 and 5 that can be made = minimum of power of 2 or 5 = min (124, 263) = 124

So, maximum power of 10 that will divide ($25^{70} \times 30^{120} \times 35 \times 40 \times 42 \times 45$) = 124

TEST YOUR LEARNING 2

1. Find the maximum number of times following numbers will be divided by 2.

1. 100 **2.** 128
3. 144 **4.** 180
5. 10^5 **6.** 78×42
7. 391×467 **8.** 22^{40}
9. 12! **10.** $8! \times 25$

2. Find the maximum number of times following numbers will be divided by 5.

1. 200 **2.** 125
3. 148 **4.** 190
5. 10^5 **6.** 78×42
7. $35^{19} \times 25^4$ **8.** 25^{40}
9. 12! **10.** $8! \times 25$
11. 25^{25^2}

Factorial Format Questions

Let us understand this format of questions with the help of an example:

Worked-out Example 4 What is the maximum power of 5 that will divide 10!?

☝ ***Solution*** $10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. It can be clearly seen that maximum power of 5 that can divide 10! = 2 (comes from 10 and 5).

Worked-out Example 5 What is the maximum power of 5 that will divide 12!?

☝ ***Solution*** $12! = 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. It can be clearly seen that maximum power of 5 that can divide 10! = 2 (comes from 10 and 5). We can also conclude the next exponent of 5 will increase only with 15!, then next with 20!.

Worked-out Example 6 What is the maximum power of 5 that will divide 40!?

☝ ***Solution*** Now we can do the counting without actually writing 40!. Number of 5's can be counted in the following numbers (in the order they appear) – 5 (1), 10(1), 15(1), 20(1), 25(2), 30(1), 35(1), 40(1) – Number in bracket indicates the number of times 5 will appear in that number.

So total number of times 5 will appear in 40! = 1 + 1 + 1 + 1 + 2 + 1 + 1 + 1 = 9.

Now we will define a formula that will help us in finding the exponent of any prime number in n! Directly:

Exponent of any prime number P in n! $= \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \ldots\ldots\ldots\ldots + \left[\frac{n}{p^x}\right]$,

where, $n \geq p^x$ and [.] denotes the greatest integer value i.e., we have to consider only the integral value.

Let us find out exponent of 5 in 1000! = $1000/5 + 1000/5^2 + 1000/5^3 + 1000/5^4$ = 200 + 40 + 8 + 1 = 249.

Worked-out Example 7 What is the highest power of 5 which can divide N = (22! + 17894!)?

☝ ***Solution*** Number of times this number is divisible by 5 is same as number of zeroes at the end of this number. Since 22! have 4 zeroes at its end, so N will also be having only four zeroes at its end. Hence, highest power of 5 which can divide N is 4.

Process to Find out the Exponent of any Composite Number in N!

We have got three different kinds of composite numbers:

1. Product of two or more than two prime numbers with unit power of all the prime numbers
   e.g., 15(5 × 3), 30(2 × 3 × 5) etc.
2. (Any prime number)^n, where, n >1
   e.g., 4(=2^2), 27(=3^3)
3. Product of two or more than two prime numbers with power of any one prime number more than 1.
   e.g., 12(2^2 × 3), 72(2^3 × 3^2) etc.

Let us find out the exponents of the above written composite numbers one by one:

Worked-out Example 8 Let us find out the exponent of 15 in 100!

15 is the product of two distinct prime numbers 5 and 3. So, to find out the exponents of 15, we need to find out the exponents of 5 and 3 individually.

So, we will apply the same formula of finding out the exponents for any prime number in both of these cases individually, and minimum of those two will be the answer.

$100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$

$100/3^x = [100/3] + [100/3^2] + [100/3^3] + [100/3^4] = 33 + 11 + 3 + 1 = 48$

Now, we say that 100! is divisible by 5^{24} and 100! is divisible by 3^{48}. To answer the question that 100! is divisible by what power of 15, we need to make sets of one 5 and one 3. Obviously, number of sets of 5 and 3 = Minimum of (exponents of 5, exponents of 3) = Minimum of (24, 48) = 24.

Hence 100 is divisible by 15^{24}.

Note We can take a note here that actually we are not required to calculate for each of the prime factors—3 and 5 as in this case. Knowing the fact that exponent of 5 will be always less than the exponent of 3, only calculating the exponent of 5 will do. Similarly, if we have to find out the exponent of 35 in 50!, this is going to be equal to the exponent of 7 in 50! (Because exponent of 7 is less than the exponent of 5 in 50!).

Worked-out Example 9 Let us find out the exponent of 25 in 100!

$25 = 5^2$

In this case, we will first find out the exponents of 5 and then divide it by 2 (actually the power) to find out the exponents of 25.

$100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$

So, the largest power of 25 that will divide 100 = largest power of 5^2 that will divide = 24/2 = 12

Worked-out Example 10 Let us find out the exponent of 45 in 100!

$45 = 3^2 \times 5^1$. We will find out the exponent of 3^2 and 5^1 in 100! one by one.

Exponent of 5 in 100! = $100/5^x = [100/5] + [100/5^2] = 20 + 4 = 24$

To find out the exponent of 3^2 in 100!, we will first find out the exponent of 3^1 in 100! and then divide that by 2 (i.e., the power of 3).

$100/3^x = [100/3] + [100/3^2] + [100/3^3] + [100/3^4] = 33 + 11 + 3 + 1 = 48$

So exponent of 3^2 in 100! = 48/2 = 24

Minimum of (exponent of 5, exponent of 3^2) will be the largest power of 45 that will divide 100! = Min (24, 24) = 24

TEST YOUR LEARNING 3

1. Arrange the following Steps to define the right approach to solve the given question. Some of the statements given may not have any significance in the right arrangement of the steps.

1. To find out the exponent of 16 in 200!

(i) Find out the exponent of 2 in 200!. This can be calculated directly.

(ii) Find out the exponent of 16 in 200!. This can be calculated directly.

(iii) Whatever value has been obtained, divide it by 4.

(iv) This is equal to the exponent of 16 in 200!.

2. To find out the exponent of 65 in 250!.

(i) Factorize 65

(ii) Find the exponent of 17 in 250!. This can be calculated directly.

(iii) Find the exponent of 13 in 250!. This can be calculated directly.

(iv) Find the exponent of 5 in 250!. This can be calculated directly.

(v) Minimum of exponent of (ii, iii as calculated above) will be the answer.

(vi) Minimum of exponent of (iii, iv as calculated above) will be the answer.

2. What is the exponent of 5 in

1. 100! **2.** 200! **3.** 250!

3. What is the exponent of 8 in

1. 50! **2.** 30! **3.** 25!

4. What is the exponent of 35 in

1. 70! **2.** 40! **3.** 50!

5. What is the exponent of 72 in

1. 60! **2.** 80!

Finding Number of Zeroes on Base other than 10

Sometimes question may be asked on base other than 10. For example, number of zeroes at the end of n! when converted on base 8. Let us understand this with the help of next example;

Worked-out Example 11 How many zeroes will be there at the end of $(10!)_{10}$ when converted on base 8?

☝ ***Solution*** It can be seen that $(8)_{10} = (10)_8$

If number is divided by 8^3 once, then there will be one zero at the end of the number when converted on base 8, if the number is divided by 8^3 twice, then there will be two zeroes at the end of the number when converted on base 8 and so on.

Understand this better:

Number 16_{10} converted on base 8 = 20_8

Number 32_{10} converted on base 8 = 40_8

Number 64_{10} converted on base 8 = 100_8

Number 128_{10} converted on base 8 = 200_8

Number 96_{10} converted on base 8 = 140_8

So number of zeroes will be there at the end of $(10!)_{10}$ when converted on base 8 = Number of exponents of 8 in $(10!)_{10}$

$$\frac{10!}{2^x} = \frac{10}{2} + \frac{10}{2^2} + \frac{10}{2^3} = 5 + 2 + 1 = 8$$

Hence, $\frac{10!}{8^x} = \frac{8}{3} = 2$

So number of zeroes at the end = 2

PRACTICE EXERCISE 1

1. How many zeroes will be there at the end of $(10!)_{10}$ when converted on

1. Base 7 **2.** Base 5 **3.** Base 35 **4.** Base 15

5. Base 10 **6.** Base 17 **7.** Base 25 **8.** Base 4

2. How many two digit natural numbers (base 10) when converted on base 3 will have three zeroes at the end?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. None of these

3. $(20!)_{10}$ is converted on each of the bases from base 2 to base N (N < 20) one by one. For how many different values of N, number of zeroes obtained at the end of this new number will be equal to zero?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. None of these

4. $(20!)_{10}$ is converted on each of the bases from base 2 to base N (N < 20) one by one. For how many different values of N, number of zeroes obtained at the end of this new number will be equal to one?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. None of these

5. $(20!)_{10}$ is converted on each of the bases from base 2 to base N one by one. Number of zeroes obtained at the end of this new number = 0. What is the lowest value of N?

1. 19 **2.** 21 **3.** 22 **4.** 23

5. 29

6. How many four digit natural numbers (on base 10) when converted on base 5 will have exactly 3 zeroes at its end?

1. 42 **2.** 54 **3.** 58 **4.** 64

5. None of these

7. (123!) is divided by $(25!)^x$. What is the maximum integral value of x?

1. 4 **2.** 5 **3.** 6 **4.** 7

5. None of these

8. $100! = K \times (10!)^x$, where x and k both are natural numbers. What is the maximum value of x?

1. 14 **2.** 15 **3.** 16 **4.** 24

5. None of these

9. $100! = K \times (N!)^x$ (where K, N and x are natural number)gives the maximum value of x = 9. How many values of N are possible?

1. 1 **2.** 2 **3.** 3 **4.** More than 3

5. Infinite

10. How many zeroes will be there at the end of the number: $11^{(5!)!} - 1$?

1. 0 **2.** 1 **3.** 11 **4.** 49

5. None of these

11. N = Product of 1st 100 multiples of 9. What is the maximum power of 25 that will exactly divide N?

1. 48 **2.** 24 **3.** 12 **4.** 26

5. None of these

12. Find the number of zeroes at the end of the expression: $(10! + 40! + 70! + 100!)^{10! + 40! + 70! + 100!}$?

1. 16 (10! + 40! + 70! + 100!)

2. 16 (10! + 40! + 70! + 100!)

3. 51 × (10! + 40! + 70! + 100!)

4. $51^{(10! + 40! + 70! + 100!)}$

5. None of these

13. N = Product of 1st 100 multiples of 5. What is the maximum power of 125 that will exactly divide N?

1. 124 **2.** 62 **3.** 41 **4.** 26

5. None of these

14. What is the maximum power of 5 that will divide $25^{25^{25}} \times 20^{25^{25}} \times 15^{25^{25}} \times 10^{25^{25}} \times 5^{25^{25}} = K \times 25$. Find the value of K.

1. 125 **2.** 6 × 125 **3.** 4 × 125 **4.** 2 × 625

5. None of these

15. What is the co-efficient of x^{18} in the expansion of $(x + x^2 + x^3 + ... + x^6)^4$?

1. 60 **2.** 70 **3.** 80 **4.** 90

5. 100

16. How many zeroes will be there at the end of 625^{10}!?

1. $\frac{5^{40}-1}{4}$ **2.** $\frac{5^{40}-1}{5}$ **3.** $\frac{5^{39}-1}{4}$ **4.** $\frac{5^{39}-1}{5}$

5. None of these

17. What is the number of zeroes at the end of the number 18! + 19!?

1. 3 **2.** 4 **3.** 5 **4.** 6

5. Cannot be determined

18. N is the number of consecutive zeroes at the end of the decimal representation of the product K = 1! × 2! × 3! × 4!..... 99! × 100! × 101!. Find the remainder when N is divided by 10.

1. 4 **2.** 2 **3.** 3 **4.** 8

5. 5

19. How many of the following will be equal?

(i) Exponent of 5 in 100!

(ii) Exponent of 35 in 100!

(iii) Exponent of 15 in 100!

(iv) Exponent of 25 in 100!

1. 0 **2.** 1 **3.** 2 **4.** 3

5. 4

20. A number n! is having 23 zeroes at its end. How many zeroes will be there at the end of (n + 10)!?

1. 23 **2.** 24 **3.** 25 **4.** 26

5. None of these

21. $\frac{2000!}{1000!} = n(1\times3\times5\times7\ldots\ldots\ldots\times1997\times1999)$

What is the value of n in the above calculation?

1. 2^{100} **2.** 2^{1000} **3.** $2^{100}\times3^{100}$ **4.** 3^{100}

5. None of these

22. 85! ends with exactly 20 consecutive zeroes. Even if 85! is converted to base N, N being any natural number, it will have same number of zeroes at its end. What can be the maximum value of N?

1. 80 **2.** 160 **3.** 240 **4.** 200

5. None of these

23. What is the coefficient of $x^2\times y^2\times z^2$ in the expansion of $(x + y + z)^6$?

1. 10 **2.** 20 **3.** 40 **4.** 90

5. 120

24. What is the coefficient of (x^3y^2) in $(x + y)^5$?

1. 4 **2.** 6 **3.** 8 **4.** 10

5. None of these

25. How many zeroes will be there at the end of the expression N = 10 × 20 × 30 ……. × 1000?

1. 1280 **2.** 1300 **3.** 1320 **4.** 124

5. None of these

26. How many zeroes will be there at the end of the expression N = 7 × 14 × 21 × …….. × 777?

1. 24 **2.** 25 **3.** 26 **4.** 27

5. 28

27. N! is having x zeroes at its end, and N + 1! Is having x + 2 zeroes at its end. How many values of N will be there given that N is a natural number < 130?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. 5

28. How many terms are there in the expansion of $(a + b)^{50}$?

1. 49 **2.** 50 **3.** 51 **4.** 52

5. 101

29. How many terms are there in the expansion of $(a + b + c)^{10}$?

1. 11 **2.** 22 **3.** 44 **4.** 66

5. 99

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING 1

1.

1. $(-5)(-5)(-5) = -125$

2. $= -(3)(3)(3) - (-3)(-3) + (-2)(-2)$

$= -27 - 9 + 4 = -32$

3. x^1

4. $x^{-1/2}$

5. $\frac{2}{\sqrt{x}}$

6. $2ab^2$

2.

1. $\sqrt{2} - \sqrt{3}$

2. $\sqrt{19} + \sqrt{13}$

3. $\sqrt{2a} + \sqrt{3b}$

3.

1. $\frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}+\sqrt{5}} = \frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}+\sqrt{5}} \cdot \frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}-\sqrt{5}} = \frac{2-2\sqrt{10}+5}{2-5} = \frac{2\sqrt{10}-7}{3}$

2. $\frac{1}{\sqrt{2a}+\sqrt{b}} = \frac{1}{\sqrt{2a}+\sqrt{b}} \cdot \frac{\sqrt{2a}-\sqrt{b}}{\sqrt{2a}-\sqrt{b}} = \frac{\sqrt{2a}-\sqrt{b}}{(\sqrt{2a}-\sqrt{b})(\sqrt{2a}+\sqrt{b})} = \frac{\sqrt{2a}-\sqrt{b}}{2a-b}$

4.

1. $5^y = 1/625 = 5^{-4} \Rightarrow y = -4$

2. $2^{4y} = 2^5 \Rightarrow 4y = 5 \Rightarrow y = 1.25$

3. $\frac{2}{xy}$

TEST YOUR LEARNING 2

1. **1.** 2 **2.** 7 **3.** 4 **4.** 2 **5.** 5

6. 2 **7.** 0 **8.** 40

(i) $12! = 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. We will now count the number of 2's in each of these numbers individually.

12! =	12	11	10	9	8	7	6	5	4	3	2	1
Number of 2's	2	0	1	0	3	0	1	0	2	0	1	0

So, total number of 2's = 2 + 1 + 3 + 1 + 2 + 1 = 10.

(j) $8! \times 25$

We can see that 25 does not contain any 2's. In this number, only 8! contains 2's. We will now count this one by one.

$8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$.

So, total number of 2's = 3 + 1 + 2 + 1 = 7

2. **1.** 2 **2.** 3 **3.** 0 **4.** 1 **5.** 5

6. 0 **7.** $35^{19} \times 25^4$

Number of times 5 is present in 35^{19} = 19 (each of 35 contain one 5, hence $35 \times 35 \times 35$ .....19 times will contain 19 5's).

Number of times 5 is present in $25^4 = 2 \times 4 = 8$ (each of 25 contain two 5's, hence $25 \times 25 \times 25 \times 25$ will contain $2 \times 4 = 8$ 5's).

So, total number of 5's present in $35^{19} \times 25^4 = 19 + 8 = 27$ (Understand it will not be 19×8)

(h) $25^{40} = 25 \times 25 \times 25 \times$.....40 times.

Total number of times 5 is present in $25^{40} = 2 \times 40 = 80$

(i) $12! = 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. We will now count the number of 5's in each of these numbers individually.

It can be clearly seen that 5 as a factor is present only in 10 and 5, and in each of these numbers 5 is present once.

Total number of times 5 is present in 12! = 1 + 1 = 2

(j) $8! \times 25$

Number of times 5 is present in $25 = 5 \times 5 = 2$

Number of times 5 is present in 8! = 1 [$8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \Rightarrow$ 5 is present as a factor in 5 only].

Total number of times 5 is present in $8! \times 25 = 1 + 2 = 3$

(k) $25^{25^2} = 25^{625} = 25 \times 25 \times 25 \times$ .....625 times.

One 25 contain 2 times 5.

So total number of 5's in $25^{25^2} = 2 \times 625 = 1250$

TEST YOUR LEARNING 3

1. **1.** Sequence of steps will be—(i – iii – iv.)

2. Sequence of steps will be—(i – iii – iv – vi)

2. **1.** Number of exponent of 5 in $100! = \frac{100}{5} + \frac{100}{5^2} = 20 + 4 = 24.$

2. Number of exponent of 5 in $200! = \frac{200}{5} + \frac{200}{5^2} + \frac{200}{5^3} = 40 + 8 + 1 = 49.$

3. Number of exponent of 5 in $250! = \frac{250}{5} + \frac{250}{5^2} + \frac{250}{5^3} = 50 + 10 + 2 = 62.$

3. To find out the exponent of 8, we need to find out the exponent of 2, and then divide that by 3 (as $2 \times 2 \times 2 = 8 \Rightarrow$ three times 2 will be equal to 8.)

1. Exponent of 2 in $50! = \frac{50}{2} + \frac{50}{2^2} + \frac{50}{2^3} + \frac{50}{2^4} + \frac{50}{2^5} = 25 + 12 + 6 + 3 + 1 = 47$

Number of times 50! Is divisible by 2 = 47.

Hence, number of times 50! is divisible by $2^3 = 47/3 = 15$.

2. Exponent of 2 in 30! $= \frac{30}{2}+\frac{30}{2^2}+\frac{30}{2^3}+\frac{30}{2^4} = 15 + 7 + 3 + 1 = 26.$

Number of times 50! is divisible by 2 = 26.

Hence, number of times 50! is divisible by $2^3 = 26/3 = 8$.

3. Exponent of 2 in 25! $= \frac{25}{2}+\frac{25}{2^2}+\frac{25}{2^3}+\frac{25}{2^4} = 12 + 6 + 3 + 1 = 22$

Number of times 50! is divisible by 2 = 22.

Hence, number of times 50! is divisible by $2^3 = 22/3 = 7$.

4. $35 = 5^1 \times 7^1$. For any given factorial, exponent of 7 will be less than or equal to the exponent of 5. So to make out the pairs of one exponent of 5 and one exponent of 7, deciding factor will be the exponents of 7 (which will be less than or equal to the exponent of 5).

1. Exponent of 7 in 70! $= \frac{70}{7}+\frac{70}{7^2} = 10 + 1 = 11.$

Hence, exponent of 35 in 70! = 11

2. Exponent of 7 in 40! $= \frac{40}{7}+\frac{40}{7^2} = 5 + 0 = 5.$

Hence, exponent of 35 in 40! = 5

3. Exponent of 7 in 50! $= \frac{50}{7}+\frac{50}{7^2} = 7 + 1 = 8.$

Hence, exponent of 35 in 50! = 8.

5. $72 = 2^3 \times 3^2$

1. Exponent of 2 in 60! $= \frac{60}{2}+\frac{60}{2^2}+\frac{60}{2^3}+\frac{60}{2^4}+\frac{60}{2^5} = 30 + 15 + 7 + 3 + 1 = 56.$

So, exponent of 2^3 in 60! = 56/3 = 18

Exponent of 3 in 60! $= \frac{60}{3}+\frac{60}{3^2}+\frac{60}{3^3} = 20 + 6 + 2 = 28.$

So, exponent of 3^2 in 60! = 28/2 = 14.

Minimum of (exponent of 2^3, exponent of 3^2) = Min (18, 14) = 14

2. Exponent of 2 in 80! $= \frac{80}{2}+\frac{80}{2^2}+\frac{80}{2^3}+\frac{80}{2^4}+\frac{80}{2^5}+\frac{80}{2^6} = 40 + 20 + 10 + 5 + 2 + 1 = 78.$

So, exponent of 2^3 in 80! = 78/3 = 26.

Exponent of 3 in 80! $= \frac{80}{3}+\frac{80}{3^2}+\frac{80}{3^3} = 26 + 8 + 2 = 36.$

So, exponent of 3^2 in 80! = 36/2 = 18.

Minimum of (exponent of 2^3, exponent of 3^2) = Min (26, 18) = 18.

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
See ahead.	**4**	**1**	**4**	**4**	**3**	**2**	**3**	**2**	**2**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
3	**5**	**3**	**5**	**3**	**1**	**2**	**4**	**3**	**5**
21.	22.	23.	24.	25.	26.	27.	28.	29.	
2	**5**	**4**	**4**	**4**	**3**	**4**	**3**	**4**	

1. It can be seen that:

1. $(7)_{10} = (10)_7$. It can be extended to any power of LHS and RHS too, as long as the power on both the sides remain same.

$[(7)_{10}]^2 = [(10)_7]^2$, or, $[(7)_{10}]^3 = [(10)_7]^3 \Rightarrow [(7)_{10}]^N = [(10)_7]^N$, where, N is a natural number.

To find the number of zeroes at the end of 10! converted on base 7, we can either go through first calculating 10! Then converting it on base 7, and then observing how many zeroes are there at the end. We may apply this method for 10!, however, had this been 50! Instead of 10!, we definitely would not have used this.

Since number of zeroes at the end on base 7 = Number of times exponents of 7 occur on base 10, we will calculate exponents of 7 on base 10 (which is quite easier to calculate).

Number of exponents of 7 in 10! = $\frac{10}{7} = 1$

2. Base 5

$(5)_{10} = (10)_5$

Now, we are required to find out the exponent of 5 in 10!.

Number of exponents of 5 in 10! = $\frac{10}{5} = 2$

3. Base 35

$(35)_{10} = (10)_{35}$

We are required to find out the pair of exponent of 7 and 5. Obviously, exponent of 7 will be less than the exponent of 5 in 10!.

Exponent of 35 in 10! = Exponent of 7 in 10! = $\frac{10}{7} = 1$

4. Base 15

$(15)_{10} = (10)_{15}$

We are required to find out the pair of exponent of 3 and 5. Obviously exponent of 5 will be less than the exponent of 3 in 10!.

Exponent of 15 in 10! = Exponent of 5 in 10! = $\frac{10}{5} = 2$

5. Base 10

To obtain 10 at the end, we need to find the exponents of $(5^1 \times 2^1)$. Since number of exponents of 5^1 will be less than the exponents of 2^1 in 10!, hence, we are required to find the exponents of 5.

$10!/5^x = 10/5 = 2$.

So, number of zeroes at the end = 2.

6. Base 17

$(17)_{10} = (10)_{17}$

Number of zeroes at the end when 10! is converted on base 17 = Exponent of 17 on $(10!)_{10}$ = 10/17 = 0.

Hence, number of zeroes at the end of $(10!)_{10}$ when converted on base 17 = 0.

7. Base 25

$(25)_{10} = (10)_{25}$

So, we are required to find the exponent of 25 on 10!.

$$\frac{10!}{5^x} = \frac{10}{5} = 2$$

Number of exponent of 5 on 10! = 2 $\Rightarrow$ Number of exponent of 5^2 on 10! = 2/2 = 1.

8. Base 4

Number of zeroes at the end of $(10!)_{10}$ when converted on base 4 = Number of exponents of 4 in $(10!)_{10}$.

$$\frac{10!}{2^x} = \frac{10}{2} + \frac{10}{2^2} + \frac{10}{2^3} = 5 + 2 + 1 = 8$$

Hence, $\frac{10!}{4^x} = \frac{8}{2} = 4$

So, number of zeroes at the end = 4.

2. On base 3, number of zeroes will depend upon the exponents of 3.

For example, if there is only one exponent of 3 in the decimal presentation of the number, then there will be exactly one zero at the end of this natural number when converted on base 3.

Consider,

$(6)_{10} = (20)_3$ [6 is divisible by $3^1 \Rightarrow$ Exactly one zero at the end of equivalent number on base 3]

$(9)_{10} = (100)_3$ [9 is divisible by $3^2 \Rightarrow$ Exactly two zeroes at the end of equivalent number on base 3]

$(12)_{10} = (110)_3$ [12 is divisible by $3^1 \Rightarrow$ Exactly one zero at the end of equivalent number on base 3]

$(27)_{10} = (1000)_3$ [27 is divisible by $3^3 \Rightarrow$Exactly three zeroes at the end of equivalent number on base 3]

To solve this question, we need to find out that how many two digit numbers are there divisible by 27? Obviously answer is 3, namely 27, 54, 81.

Out of these, 81 will have 4 zeroes, and 27 and 54 will have three zeroes.

3. All the numbers N from 2 to 19 will have atleast one exponent when N divides 20! $\Rightarrow$ Atleast one zero has to be there at the end of number.

Hence, none of the numbers will have number of zero obtained at the end = 0.

5. For number of zeroes to be equal to 0, 20! should not contain the exponents of the number leading to zero. For example, on base 22, $22_{10} = 10_{22.}$ So to

check the number of zeroes on base 22, we should be dividing 20! by 11 and 2. Exponent of 11 in 20! = 20/11 = 1 $\Rightarrow$ Number of zeroes at the end of $(20!)_{10}$ converted on base 11 = 1.

Lowest value of N such that number of zeroes at the end of $(20!)_{10}$ converted on base 11 = 0 will be obtained at N = 23. In other words, 20! will not have any exponent of 23 $\Rightarrow$ Number of zeroes obtained at the end of 20! when converted on base 23 = 0.

6. All the numbers whose decimal notation is divisible by 125 will have three zeroes at the end when converted on base 5.

It can be seen through the following sequence of events:

(i) If decimal notation is divisible by 5^1 and not divisible by 5^2, some of these numbers = 5, 10, 15, 20 etc.

$5_{10} = 10_5$

$10_{10} = 20_5$

$15_{10} = 30_5$

$145_{10} = 1040_5$

(ii) If decimal notation is divisible by 5^2 and not divisible by 5^3, some of these numbers = 25, 50, 75, 350 etc.

$25_{10} = 100_5$

$50_{10} = 200_5$

$75_{10} = 300_5$

$350_{10} = 2400_5$

Hence, to find out the four digit numbers having exactly three zeroes at the end, we need to find out the numbers which are divisible by 125 and not divisible by 625.

Series of the four digit numbers divisible by 125 = 1000, 1125, 1250,.....,9875.

Total number of numbers = $\frac{9875-1000}{125}+1 = 71 + 1 = 72$ [1 is added because both the ends are counted].

[Alternatively, to count the number of numbers, we can see that the series goes like, 125×8, 125×9,....., 125×79. So from the 1st 79 multiples of 125, we do not have to consider 1st 7 multiples of 125. Hence, total number of numbers = 79 – 7 = 72]

But some of these numbers will be divisible by 625 or even higher powers of 5, and hence, these numbers will have more than 3 zeroes. We will eliminate these numbers.

Series of the four digit numbers divisible by 625 = 1250, ..., 9375.

Total number of numbers = $\frac{9375-1250}{625}+1 = 13 + 1 = 14$

[Alternatively, to count the number of numbers, we can see that the series goes like, 625×2, 625×3,....., 625×15. So from the 1st 15 multiples of 625, we do not have to consider 1st multiple of 125. Hence, total number of numbers = 15 – 1 = 14].

So, total number of four digit numbers which will have exactly three zeroes when converted on base 5 = 72 – 14 = 58.

We can summarize the whole discussion as: Number of four digit numbers = n(125) – n (625).

7. 24! Contains 23 as the largest prime number in it. Besides, the lowest prime number contained in 24! = 2, and 2 appears 22 times in 24! (in other words, 24! Is divisible by 2^{22}).

Number of times 23 is going to be contained in 123! = 123/23 = 5.

Number of times 2 is going to be contained in 123! =

$$\frac{123}{2}+\frac{123}{2^2}+\frac{123}{2^3}+\ldots\ldots+\frac{123}{2^6} = 61 + 30 + 15 + 7 + 3 + 1 = 117.$$

Number of times 2^{22} is going to be contained in 123! $= \frac{117}{22} = 5.$

Min of (Number of times 23 occurs, number of times 2^{22} occurs) will be the answer = Min (5, 5) = 5.

8. We need to check for the largest prime number contained in 10! = 7.

$100!/7^x = 100/7 + 100/7^2 = 16$

9. It is possible only for N = 11, 12. Understand that we need to keep the denominator fixed at 11. This is possible only for 11! and 12! (for N = 11 and N = 12).

It is not possible for N = 22 or 33 or higher multiples of 11. Question is not asking for $100! = K \times (N)^x$, its asking for $100! = K \times (N!)^x$

For example, in case of N = 22, we need to check $100! = K \times (22!)^x \Rightarrow 100!$ will be divided by the largest possible prime number = 19.

10. Unit digit of the $11^{(5!)!} = 1$. Hence, unit digit of $11^{(5!)!} - 1 = 0$. This is the only zero at the end. (None of the powers of 11 ends with 01.)

11. $N = 9 \times 18 \times 27 \times 36 \times \ldots\ldots\ldots\ldots\ldots\times 891 \times 900.$

Or, $N = 9^{100} (1 \times 2 \times 3 \times \ldots\ldots\ldots\times 99 \times 100) = 9^{100} \times 5^{24} \times K$

When N is divided by 25, maximum power of 25 = 24 = 24/2 = 12

13. $N = 5 \times 10 \times 15 \times 20 \times \ldots\ldots\ldots\ldots\ldots\times 495 \times 500.$

Or, $N = 5^{100} (1 \times 2 \times 3 \times \ldots\ldots\ldots\times 99 \times 100) = 5^{100} \times 5^{24} \times K$

When N is divided by 125, maximum power of 125 = 100 + 24 = 124/3 = 41

16. $625^{10}! = (5^4)^{10}! = 5^{40}!$

Number of zeroes at the end of $5^{40}! = \frac{5^{40}}{5} + \frac{5^{40}}{5^2} + \frac{5^{40}}{5^3} + \ldots\ldots\ldots\frac{5^{40}}{5^{40}} = 5^{39} + 5^{38}$

$+ 5^{37} + \ldots\ldots\ldots + 1 = 1\left[\frac{5^{40}-1}{5-1}\right] = \frac{5^{40}-1}{4}$. Hence, option 1 is the answer.

17. 18! + 19! = 18! (1 + 19) = 18! × 20. 18! will have three zeroes at its end. When multiplied by 20, total number of zeroes = 4.

18. Number of zeroes at the end of K = 1148 = N. Remainder obtained when N is divided by 10 = 8

19. Exponent of 5 in 100! = Exponent of 15 in 100!. Hence, only two values are equal.

20. None of the numbers is having 23 zeroes at its end. It can be seen that 100! Will have 24 zeroes at its end, and number before that 95! will have 22 zeroes at its end.

95! = 22 zeroes, 96! = 22 zeroes, 97! = 22 zeroes, 98! = 22 zeroes, 99! = 22 zeroes, 100! = 24 zeroes.

23. This question is similar to—How many different six-letter "words" can be formed using two X's, two Y's, and two Z's.?

Number of words/Co-efficient = $\frac{6!}{2! \times 2! \times 2!} = 90$

24. Coefficient of (x^3y^2) in $(x + y)^5 = \frac{5!}{3! \times 2!} = 10$ (see the solution of Q23).

25. $N = 10 \times 20 \times 30$ ……$\times 1000 = (10 \times 1) \times (10 \times 2) \times (10 \times 3) \times$ .....$\times (10 \times 100) = 10100 \times 100!$

100! contains 24 zeroes at its end. 10^{100} will have 100 zeroes at its end.

Total number of zeroes at its end = 100 + 24 = 124.

26. $N = 7 \times 14 \times 21 \times$……..$\times 777 = 7^{111} (1 \times 2 \times 3 \times$ ……$\times 111) = 7^{111} \times 111!$ 7^{111} will not have any zero at its end, zeroes will come from 111!.

Number of zeroes at the end of 111! = $\frac{111}{5} + \frac{111}{5^2} = 22 + 4 = 26$

27. This is possible if N + 1th number is having 2 5's at its end.

N = 24, 49, 74, 99.

28. For any binomial $(a + b)^n$, No. of terms = (n + 1)

Hence No. of terms in $(a + b)^{50} = 51$.

29. Answer is ${}^{12}C_2 = 66$

CAT QUESTIONS FROM THIS CONCEPT

CAT 2006

1. Which among $2^{1/2}$, $3^{1/3}$, $4^{1/4}$, $6^{1/6}$ and $12^{1/12}$ is the largest?

1. $2^{1/2}$ **2.** $3^{1/3}$ **3.** $4^{1/4}$ **4.** $6^{1/6}$

5. $12^{1/12}$

CAT 2005

2. If $R = \dfrac{30^{65} - 29^{65}}{30^{64} + 29^{64}}$, then

1. $0 < R \leq 0.1$ **2.** $0.1 < R \leq 0.5$ **3.** $0.5 < R \leq 1.0$ **4.** $R > 1$

CAT 2000

3. Let S be set of prime numbers greater than or equal to 2 and less than 100. Multiply all elements of S, with how many consecutive zeros will the product end?

1. 1 **2.** 4 **3.** 5 **4.** 10

SOLUTIONS AND EXPLANATIONS

1. Solution – Take LCM of 2, 3, 4, 6, 12 = 12

$2^{1/2} = (2^6)^{1/12} = (64)^{1/12}$

$3^{1/3} = (3^4)^{1/12} = (81)^{1/12}$

$4^{1/4} = (4^3)^{1/12} = (64)^{1/12}$

$6^{1/6} = (6^2)^{1/12} = (36)^{1/12}$

$12^{1/12} = (12)^{1/12} = (12)^{1/12}$

So, option (2) is largest and is the answer.

2. 4. $$\frac{30^{65} - (30-1)^{65}}{30^{64} + (30-1)^{64}} = \frac{30^{65} - 30^{65}\left(1 - \frac{1}{30}\right)^{65}}{30^{64} - 30^{64}\left(1 - \frac{1}{30}\right)^{64}}$$

$$\text{Or } R = 30\left\{\frac{1 - \left(1 - \frac{1}{30}\right)^{65}}{1 + \left(1 - \frac{1}{30}\right)^{64}}\right\}$$

$$R = 30\left\{\frac{1 - (0.96)^{65}}{1 + (0.96)^{64}}\right\}$$

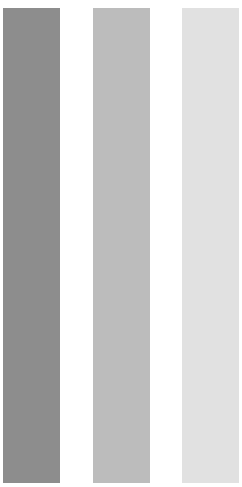
Numerator is only slightly less then 1 and denominator is only slightly more than 1. r is slightly less than 1. Hence, r > 1.

Alternatively, this question can also be done by induction, considering small powers 1st and then trying to see what happens when we increase the power and finally generalizing the whole situation.

So, had this been $R = \frac{30^2 - 29^2}{30^1 + 29^1} = 1$ or $R = \frac{30^3 - 29^3}{30^2 + 29^2} > 1$ (through actual calculations). So now we can generalize that R > 1. Hence, option 4 is the answer.

3. There is only one 5 and one 2 in the set of prime numbers. Hence, there would be only one zero at the end of the resultant product. Hence, option 1 is the answer.

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Chapter 6

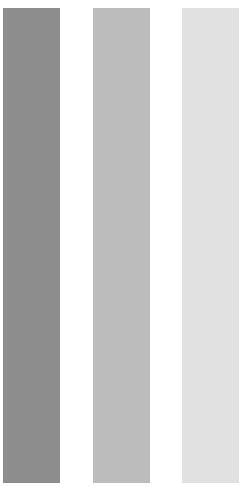
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DEMYSTIFYING NUMBER SYSTEM

FACTORS

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- Factors and multiples
- Some generalizations related to factors
- Number of factors, sum of factors
- Conditional questions on factors

FACTORS AND MULTIPLES

If one integer can be divided by another integer an exact number of times, then the first number is said to be a **multiple** of the second, and the second number is said to be a **factor** of the first.

For example, 48 is a **multiple** of 6 because 6 goes into 48 an exact number of times (8 times in this case). Similarly, 6 is a **factor** of 48. On the other hand, 48 is not a multiple of 5, because 5 does not go into 48 an exact number of times. So, 5 is not a factor of 48.

In other words, remainder obtained when one number A is divisible by B, then B is a said to be a factor of A. We also understand that:

(i) A divides B, quotient obtained = Q (or, AQ = B) implies that the quotient obtained in this case Q will also divide B, and the quotient in this case = A.

For example, when 20 is divided by 4, quotient obtained = 5. Hence, when 20 is divided by 5, then quotient obtained = 4. The essence of the whole discussion is the fact that factors will occur in pair.

Pairs of factors of 20 = (1, 20), (2, 10), (4, 5) = 3 pairs of distinct factors $\Rightarrow$ Total Number of factors = An Even number.

However, the discussion above is not valid for perfect squares. Let us understand that with the help of a perfect square:

Pair of factors of 36 = (1, 36), (2, 18), (3, 12), (4, 9), (6, 6)

We can see that first four pairs of factors are distinct, but (6, 6) is having only one distinct factor. Hence, number of factors of 36 = 9 = An odd number

We finally summarize this as:

All the perfect squares will have odd number of factors, whereas all the natural numbers other than perfect square will have even number of factors. Even the vice versa is true that if a number is having odd number of factors, then that number has to be a perfect square.

(ii) Natural number line can be categorized on the basis of number of factors.

Number of factors of 1st natural number i.e., 1 = 1

Number of factors of prime number = 2

Number of factors of all the composite numbers = Minimum 3

(iii) Perfect squares of prime numbers will have total number of factors = 3

Worked-out Example 1 There is a prison with 100 cells inside it. Cells are numbered from 1 to 100 and every cell is occupied by one prisoner only. One day jailer decides to release some of the prisoners and for this to happen, he defines an algorithm of 100 steps which follows:

Step 1—Reverse the position of all the cells which are divisible by 1.

Step 2—Reverse the position of all the cells which are divisible by 2.

Step 3—Reverse the position of all the cells which are divisible by 3.

……

…….

……..

Step 99—Reverse the position of all the cells which are divisible by 99.

Step 100—Reverse the position of all the cells which are divisible by 100.

Reversing the position of the cells means that if the cell is closed, open it and vice versa. Initially, all the cells are closed. After executing all these steps, prisoners of all the cells which remain open are released.

How many prisoners are released?

1. 25 2. 10 3. 5 4. 50 5. 75

☝ ***Solution*** Let us discuss the fate of any particular cell number as per the algorithm given:

Cell Number 45

Initially closed

After Step 1	After Step 2	After Step 3	After Step 4	After Step 5	After Step 6
Open	Open	Close	Close	Open	Open

After Step 7	After Step 8	After Step 9	After Step 15	After Step 16	After Step 45
Open	Open	Close	Open	Open	Close

It can be seen that for cell number 45, only those step numbers which are factors of 45 will have any impact of the position of cell number 45. These are going to be—Step 1, Step 3, Step 5, Step 9, Step 15, Step 45. Beyond step 45, none of the steps will have any impact on cell number 45.

It can be concluded that the moment 1st factor is obtained (in the form of Step 1), cell will be opened. And so on:

1st Factor	2nd Factor	3rd Factor	4th Factor	5th Factor	6th Factor	And So On
Open	Close	Open	Close	Open	Close	

We can see that when 1st or 3rd or 5th factor or any odd number of factor is obtained, cell gets opened. However, when 2nd or 4th or 6th or any even number of factor is obtained, cell will get closed.

Had it been the case that a number is having only odd number of factors, cell would have remained opened. This is possible only if the cell number is perfect square.

Hence, cell numbers which will remain open = 1, 4, 9, 16, 25, 36, 49, 64, 81, 100. These ten cells will remain open, and ten prisoners will be released.

Let us now learn the method to calculate the number of factors of a number. All the discussions given ahead are for natural numbers, unless stated otherwise.

Finding Total Number of Factors of a Number

We understand that if natural number N is divisible by natural number Q, then Q is a factor of N. We go a step ahead and say that if Q is not a factor of N, then N will not be divisible by Q.

Let us understand this with the help of an example:

Assume N = 150

Prime factorization of N = 150 = $2^1 \times 3^1 \times 5^2$

We can conclude the following:

(i) 150 is divisible by only three prime numbers = 2, 3, 5. It also means that 150 will not be divisible by any of the other prime numbers namely 7, 11, 13 etc.
(ii) Since largest power of 2^1 is contained in 150, largest power of 2 that will divide 150 = 1 $\Rightarrow$ So any number of the order 2^2 or 2^{2+} will not divide 150.
(iii) Since largest power of 3^1 is contained in 150, largest power of 3 that will divide 150 = 1 $\Rightarrow$ So any number of the order 3^2 or 3^{2+} will not divide 150.
(iv) Since largest power of 5^2 is contained in 150, largest power of 5 that will divide 150 = 2 $\Rightarrow$ So any number of the order 5^3 or 5^{3+} will not divide 150.
(v) Even the combination of two or more than two of $2^1, 3^1, 5^2$ are possible.

All this can be understood in the following way too:

$$\frac{(2^1 \times 3^1 \times 5^2)}{Q}$$

Since the numerator is having $2^1 \times 3^1 \times 5^2$, denominator can have only these powers only—either individually or a combination.

So we can see that the following powers of these prime numbers can be taken:

2^0 or 2^1, 3^0 or 3^1, 5^0 or 5^1 or 5^2 $\Rightarrow$ 2 can be taken in 2 ways, 3 can be taken in 2 ways, and 5 can be taken in 3 ways.

Format of Q (= factor) = $2^{0-1} \times 3^{0-1} \times 5^{0-2}$

So, total number of natural number factors = $2 \times 2 \times 3 = 12$

In general, total number of natural number factors of N = $p^x \times q^y$ ..... = $(x + 1)(y + 1)$

Similarly, total number of natural number factors of $N = p^x \times q^y \times r^z$..... = $(x + 1)(y + 1)(z + 1)$

It can be understood that while calculating the factors, we are adding 1 to the power of prime number before multiplication owing to the usage of 'power 0'.

Every natural number (>1) can be factored into the product of prime numbers, and there is only one way to do it for every such number. For instance, $120 = 2^3 \times 3^1 \times 5^1$, and this is only one way to factor 120 into prime numbers. This property is known as "Fundamental Theory of Arithmetic". Since there is just one way of factorizing any natural numbers, there is unique total number of factors that can be obtained.

Worked-out Example 2 How many natural number factors of N = 720 are there?

☝ ***Solution*** $720 = 2^4 \times 3^2 \times 5^1$

Hence, number of factors = $(4 + 1)(2 + 1)(1 + 1) = 5 \times 3 \times 2 = 30$

Finding the Even/Odd Number Factors of a Natural Number

We have already discussed previously that to calculate the number of factors of a number, we need to take care of the different powers that a prime number can take.

For example, let us see the following example:

Worked-out Example 3 Find the odd and even factors of 720.

☝ ***Solution*** $720 = 2^4 \times 3^2 \times 5^1$

Different powers of 2, 3 and 5 used are:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^1	3^1	5^1
2^2	3^2	
2^3		
2^4		

To construct the even factors (factors that are even numbers), we cannot use 2^0 ($2^0 = 1$ = odd number), so to construct even number, total different powers of 2 = 4, total different powers of 3 = 3, and total different powers of 5 = 2

Hence, total number of even factors = $4 \times 3 \times 2 = 24$

To construct the odd factors (factors that are odd numbers), the only power of 2 that can be used = $2^0 = 1$, total different powers of 2 = 4, total different powers of 3 = 3, and total different powers of 5 = 2

Hence, total number of even factors = $1 \times 3 \times 2 = 6$.

Alternatively, once either of odd factors or even factors has been calculated, the other one can be calculated by subtracting it from total number of factors.

Alternatively, it can also be seen that there are five powers of 2 to be used to construct the total number of factors ⇒ each of the powers from 2^0 to 2^4 will be used for exactly same time. Hence, 2^0 will occur in 30/5 = 6 factors (total factors = 30/different powers of 2 to be used). Similarly, 2^1 will also occur in 30/5 = 6 factors = 2^1 will also occur in 30/5 = 6 factors = and so on...

Generalizing the above discussion:

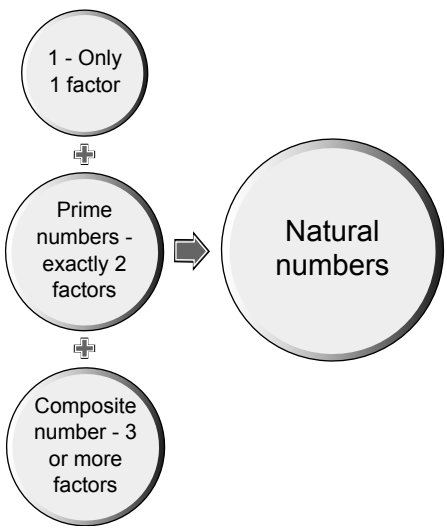
If, $N = 2^x \times p^y \times r^z$

Total number of even factors = (x) (y + 1) (z + 1)

Total number of odd factors = (1) (y + 1) (z + 1)

Finding Prime Factors and Composite Factors

We know that natural number line (starting from 1, 2, 3, ...) can be classified on the basis of number of factors to the natural number.



Above graphics also shows that:

(i) On the basis of number of factors, natural number line can be categorized in three parts—a. 1, b. Prime Number, c. Composite Factors
(ii) Lowest composite number = 4.

Essence of the whole discussion lies in the fact that total number of factors of any natural number = 1 (number 1 is a factor of all the natural numbers) + prime factors + composite factors

So once we have done the prime factorization, to find out the number of prime factors, we just need to count the number of prime factors present. To calculate

the number of composite factors, we will subtract the number of prime factors and 1 from the total number of factors.

Worked-out Example 4 Find the number of prime factors and composite factors of N = 420?

☝ ***Solution*** $420 = 2^2 \times 3^1 \times 5^1 \times 7^1$

Number of prime factors = 4 (namely 2, 3, 5, 7).

Total number of factors = $(2 + 1)(1 + 1)(1 + 1)(1 + 1) = 3 \times 2 \times 2 \times 2 = 24$

So, total number of composite factors = Total number of factors – Prime factors – 1 = 24 – 4 – 1 = 19

Finding Factors which are Perfect Squares or Cubes or Higher Power

A number will be perfect square only if all the prime factors of this number will have even powers. So a number of the format 2^x will be a perfect square only if x = 0, 2, 4, 6, 8 etc.

And similarly, a number will be perfect cube only if all the prime factors of this number will have powers divisible by 3. So a number of the format 2^y will be a cube only if x = 0, 3, 6, 9 etc.

Worked-out Example 5 How many factors of the number N = 720 will be (a) perfect square, (b) cube, (c) a perfect square and cube both?

☝ ***Solution*** $N = 720 = 2^4 \times 3^2 \times 5^1$

(a) For a factor of N = 720 to be a perfect square, it should have only the following powers of its prime factors:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^2	3^2	
2^4		

Number of powers of 2 used = 3

Number of powers of 3 used = 2

Number of powers of 5 used = 1

Hence, total number of factors of N = 720 that are perfect square = $3 \times 2 \times 1 = 6$

(b) For a factor of N = 720 to be a cube, it should have only the following powers of its prime factors:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^3		

Number of powers of 2 used = 2

Number of powers of 3 used = 1

Number of powers of 5 used = 1

Hence, total number of factors of N = 720 that are cubes = $2 \times 1 \times 1 = 2$

(c) For a factor of N = 720 to be a cube and a square both, it should have only the following powers of its prime factors:

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0

Number of powers of 2 used = 1

Number of powers of 3 used = 1

Number of powers of 5 used = 1

Hence, total number of factors of N = 720 that are cubes = $1 \times 1 \times 1 = 1$

Finding Factors which are Co-prime to Each Other

Two numbers are said to be co-prime to each other if their HCF = 1. This can happen only if none of the factors of first number (other than 1) is present in the 2nd number.

Worked-out Example 6 How many sets of two factors of N = 72 will be co-prime to each other?

☝ ***Solution*** $N = 72 = 2^3 \times 3^2$

Let us assume that a and b are two factors of 72 in such a way that HCF (a, b) = 1. Now make the combinations:

Case 1—If HCF of two numbers = 1, then one of the numbers can be taken as 1, and value of other number can be taken as any factor of 72. For example, (1, 1), (1, 2), (1, 3), ...(1, 72) = 12 sets.

Case 2—One number comprises powers of 2 only, and other number comprises powers of 3 only.

A	B
2^1	3^1
2^2	3^2
2^3	

Hence, number of sets = $3 \times 2 = 6$

Since question is asking for "set of two factors", we will consider (2, 3) same as (3, 2).

So total number of sets = 12 + 6 = 18

In general, if a number can be represented as $N = P_1^x \times P_2^y$, then number of sets of factors which are co-prime to each other = $(x + 1)(y + 1) + xy$.

Worked-out Example 7 How many sets of two factors of N = 360 will be co-prime to each other?

☝ ***Solution*** Difference between last question and this question is only the number of prime factors.

$N = 360 = 2^3 \times 3^2 \times 5^1$

We have already counted the number of sets for $2^3 \times 3^2$. This is equal to 18. To calculate the number of sets for $2^3 \times 3^2 \times 5^1$, we will just extend the situation given in the last question.

Sets = $A^{18} \times 5^1 = (18 + 1)\ (1 + 1) + 18 \times 1 = 38 + 18 = 56$

Finding the Sum of Factors of a Natural Number N

We can derive this from the method used to find out the number of factors. Let us understand this with the following question:

Worked-out Example 8 Find the sum of factors of N = 720.

☝ ***Solution*** $720 = 2^4 \times 3^2 \times 5^1$

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^1	3^1	5^1
2^2	3^2	
2^3		
2^4		

We have already seen that factors of this number will be obtained when one of the numbers from each of the sets [(2^0 to 2^5), (3^0 to 3^2) and (5^0 to 5^1)] will be multiplied.

Format of the factors will be = $2^x \times 3^y \times 5^z$. We also know that as long as we will keep on getting a different product value, distinct factors will keep on coming.

So summation of the factors = $(2^0 + 2^1 + 2^2 + 2^3 + 2^4)\ (3^0 + 2^1 + 2^2)\ (5^0 + 5^1)$

$$\text{So, sum} = \frac{(2^5 - 1)}{2 - 1} \times \frac{(3^3 - 1)}{3 - 1} \times \frac{(5^2 - 1)}{5 - 1} = 31 \times 13 \times 6 = 2418$$

In general, if N is any number which can be factorized like $N = a^p \times b^q \times c^r \times \ldots\ldots$, where a, b and c are prime numbers.

$$\text{Sum of factors} = \frac{(a^{p+1} - 1)(b^{q+1} - 1)(c^{r+1} - 1)}{(a - 1)(b - 1)(c - 1)}$$

This formula can be extended for any number of prime numbers. Understand that this formula is using the formula of sum of n terms of a GP.

Worked-out Example 9 Find the sum of (a) prime factors, (b) composite factors, (c) even factors, (d) Odd factors of N = 720.

☝ ***Solution*** $N = 720 = 2^4 \times 3^2 \times 5^1$

(a) For prime factors

There are only 3 prime factors namely 2, 3 and 5. Sum = 10

(b) For composite factors

We have already calculated sum of all the factors = 2418 (in worked-out example 8)

Composite factors are all the factors other than (prime factors and 1). So, sum of composite factors = Sum of all the factors – (sum of prime factors) – 1 = 2418 – 10 – 1 = 2407

(c) For even factors

Powers of 2	Powers of 3	Powers of 5
2^1	3^0	5^0
2^2	3^1	5^1
2^3	3^2	
2^4		

So sum of even factors $= \frac{2(2^4-1)}{2-1} \times \frac{(3^3-1)}{3-1} \times \frac{(5^2-1)}{5-1} = 2 \times 15 \times 13 \times 6 = 2340$

(d) For odd factors

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
	3^1	5^1
	3^2	

Sum of odd factors = $(2^0)\ (3^0 + 3^1 + 3^2)\ (5^0 + 5^1)$

So sum of odd factors $= 2^0 \times \frac{(3^2-1)}{3-1} \times \frac{(5^2-1)}{5-1} = 1 \times 13 \times 6 = 78$

Alternatively, summation of odd factors could have been calculated by actually finding the factors too: 1, 3, 5, 9, 15, 45 = 78.

Note Summation of GP series: a, ar, ar^2, ar^3, ...n terms $a \times \frac{(r^n-1)}{r-1}$ $(r > 1)$

Worked-out Example 10 How many factors of N = 720 are divisible by (a) 10, (b) divisible by 10 but not divisible by 20, (c) divisible by 20 but not divisible by 10?

☝ ***Solution*** $720 = 2^4 \times 3^2 \times 5^1$

Powers of 2	Powers of 3	Powers of 5
2^0	3^0	5^0
2^1	3^1	5^1
2^2	3^2	
2^3		
2^4		

(a) For a factor to be divisible by 10, minimum power of 2 to be used = 1, and minimum power of 5 is to be used = 1.

Hence, all the factors of 720 that are divisible by 10 will be of the format = $2^{1-4} \times 3^{0-2} \times 5^1$. Hence, number of factors = $4 \times 3 \times 1 = 12$

(b) We will 1st calculate the factors that are divisible by 10 = 12

Factors that are divisible by 20 will be of the format $2^{2-4} \times 3^{0-2} \times 5^1 = 3 \times 3 \times 1 = 9$

Hence, factors divisible by 10 but not divisible by 20 = 12 – 9 = 3

Alternatively, we can do this directly too in the following way:

For factor to be divisible by 10:

Powers of 2	Powers of 3	Powers of 5
2^0 – Should not be taken	3^0	5^0 – Should not be taken
2^1	3^1	5^1
2^2	3^2	
2^3		
2^4		

For factor to be not divisible by 20, factor should not be divisible by atleast one of 2^2 or 5^1 ($20 = 2^2 \times 5^1$). But for the factor to be divisible by 10, 5^1 has to be there. So we can conclude that the required factor should not contain 2^2 or any power of 2 more than 2 (in that case too, it will become divisible by 2^2).

Hence, different powers that can be taken are:

Powers of 2	Powers of 3	Powers of 5
2^0 – Should not be taken	3^0	5^0 – Should not be taken
2^1	3^1	5^1
2^2 – Should not be taken	3^2	
2^3 – Should not be taken		
2^4 – Should not be taken		

Total different powers of 2 = 1, Total different powers of 3 = 3, Total different powers of 5 = 1.

So number of factors divisible by 10 but not divisible by 20 = $1 \times 3 \times 1 = 3$

(c) There is no such number divisible by 20 but not divisible by 10. Hence, answer = 0.

Worked-out Example 11 $N = 2^7 \times 3^5 \times 5^6 \times 7^8$. How many factors of N are divisible by 50 but not by 100?

☝ ***Solution*** All the factors which are divisible by 50 but not divisible by 100 will have atleast two powers of 5, and one power of 2.

And its format will be $2^1 \times 5^{2+y}$.

So, number of divisors = $1 \times 6 \times 5 \times 9 = 270$

Worked-out Example 12 How many factors of N = 420 will have factors of the format

(a) 2n + 1, (b) 4n + 1, (c) 6n + 1 [where n is any whole number]

☝ ***Solution*** $420 = 2^2 \times 3^1 \times 5^1 \times 7^1$

(a) 2n + 1 format numbers are nothing but Odd Numbers.

Total number of factors of (2n + 1) format = (1) (1 + 1) (1 + 1) (1 + 1) = 8

Alternatively, total number of factors = (2 + 1) (1 + 1) (1 + 1) (1 + 1) = 24, and even number of factors = (2 + 0) (1 + 1) (1 + 1) (1 + 1) = 16. So odd factors = 24 – 16 = 8

(2) (1 + 1) (1 + 1) (1 + 1) = 16. Hence, odd number of factors = 24 – 16 = 8.

(b) 4n + 1 format $\Rightarrow$ Remainder obtained when divided by 4 = 1

Odd factors are 1, 3, 5, 7, 15, 21, 35, 105.

Out of these 4n + 1 factors, where n is any whole number = 1, 5, 21, 105 = 4 factors

(c) 6n + 1 format $\Rightarrow$ Remainder obtained when divided by 6 = 1

Odd factors are 1, 3, 5, 7, 15, 21, 35, 105.

Out of these, number of 6n + 1 factors, where n is any whole number = 1, 7 = 2 factors.

TEST YOUR LEARNING 1

1. For N = 1440. Find the number of

1. Total factors
2. Even Factors
3. Odd Factors
4. Prime factors
5. factors of the format 2n + 1, where n is natural number
6. factors which are perfect square
7. factors which are cubes

8. composite factors
9. factors of the format 4n + 1, where n is natural number
10. factors which are divisible by 6 but not divisible by 12
11. Sum of all the factors

SOME GENERALIZATIONS

1. Number of ways in which a composite number N can be resolved into two factors such that N = product of two factors

Case 1—when N is not a perfect square.

$$\text{Number of sets of two factors} = \frac{\text{Number of factors of N}}{2}$$

Case 2—when N is a perfect square.

$$\text{Number of sets of two factors} = \frac{(\text{Number of factors of N})+1}{2}$$

Worked-out Example 13 For N = 560. How many factors of N are such that product of these two numbers = 560?

☝ ***Solution*** $N = 560 = 2^4 \times 5^1 \times 7^1$, Number of factors = (4 + 1)(1 + 1) (1 + 1) = 20

Set of two numbers product of which is 560 = 20/2 = 10

Worked-out Example 14 For N = 324. How many factors of N are such that product of these two numbers = 324?

☝ ***Solution*** $N = 3^4 \times 2^2 \Rightarrow$ Total number of factors = (4 + 1) (2 + 1) = 5 × 3 = 15

Number of sets of two factors such that product of these two factors is equal to

$$324 = \frac{(\text{Number of factors of } 324)+1}{2} = \frac{15+1}{2} = 8$$

2. Number of ways in which a composite number can be resolved into two factors, which are prime to each other, is 2^{n-1} where n is the number of distinct prime factors in the expression for N.

Worked-out Example 15 In how many ways N = 720 can be written as the product of two factors which are prime to each other?

☝ ***Solution*** $N = 720 = 2^4 \times 3^2 \times 5^1$

n = number of prime factors = 3

Using the formula given above, $2^{n-1} = 2^2 = 4$

3. **Euler's Totient Function:** Let N be any positive integer > 1, the number of all positive integers less than N and prime to it is denoted by $\phi(N)$. It is obvious $\phi(2) = 1$, $\phi(3) = 2$, $\phi(5) = 4$, $\phi(6) = 2$ ... The function ϕ is called **Euler's Totient function**.

If a, b... are prime to each other, then,

$$\phi(ab) = \phi(a).\phi(b)$$

or $$\phi(abcd\ldots) = \phi(a)\phi(b)\phi(c)\ldots$$

If, $$N = a^p b^q c^r \ldots$$

where a, b, c are distinct primes and p, q, r are positive integers, then,

$$\phi(N) = N\left(1-\frac{1}{a}\right)\left(1-\frac{1}{b}\right)\left(1-\frac{1}{c}\right)\ldots$$

Worked-out Example 16 How many numbers less than N = 720 are prime to 720?

☝ ***Solution*** $720 = 2^4 \times 3^2 \times 5^1$

Number of numbers less than 720 and prime to 720 = 720 (1–1/2) (1–1/3) (1–1/5) = $720 \times \frac{1}{2}\times\frac{2}{3}\times\frac{4}{5} = 192$.

Note This question is same as: How many natural numbers less than 720 are not divisible by any of the numbers 2, 3 or 5?

Number of numbers not divisible by 2 or 3 or 5 = 720 (1–1/2) (1–1/3) (1–1/5) = 192

CONDITIONAL QUESTIONS ON FACTORS

Beyond the regular questions of factors, as we have seen above, we get good questions when one of the ends is set loose giving rise to the conditionality. Although these questions are not typical, and hence do not have any standard method of solving these questions, but going through the questions given below will give exposure.

Worked-out Example 17 A natural number N is having a total of 21 composite factors. What is the (a) minimum number of prime factors, (b) maximum number of prime factors of N?

☝ ***Solution*** These 21 factors of N will comprise of three types:

1. 1 is a factor of this number
2. Some prime factors (assume × factors)
3. Some composite factors (given = 21 factors)

Total number of factors = 1 + x + 21

(a) Since we have to find out the minimum number of prime factors, let us verify if having just one prime factor satisfies it?

If there is one prime factor P, then total number of factors = 1 + 1 + 21 = 23

23 factors can be generated if $N = P^{22}$. Hence, minimum number of prime factor = 1.

We also generalize here that it is possible to make any number of factors with the help of just one prime factor. (For example, if we have to make total 1000 factors, we will take P^{999} where P is a prime number.)

(b) We can see that in any case, number of prime factors has to be less than or equal to 4. (Why? Reason given at the end of this question).

So, let us go by two prime factors P and Q.

Total number of factors = 1 + 2 + 21 = 24

To make 24 factors using two prime numbers P and Q, format of N can be = $P^5 \times Q^3$ or $P^3 \times Q^5$

Or $P^7 \times Q^2$ and so on. So we see that it is possible for two prime factors.

Let us check it now for 3 prime factors:

Total number of factors = 1 + 3 + 21 = 25 factors

To make 25 factors using three prime numbers P, Q and S (assume) format of N can be = $P^0 \times Q^4 \times S^4$

However, in this case, $P^0 = 1$, and hence this prime number will not exist. So, N cannot have three prime numbers. (Since, $25 = 5 \times 5$, it can be distributed only to two prime factors).

Let us check it now for 4 prime factors:

Total number of factors = 1 + 4 + 21 = 26 factors

To make 26 factors using four prime numbers P, Q, Rand S (assume) format of N should be = $P^0 \times Q^1 \times R^0 \times S^{13}$ or may be $P^{13} \times Q^1 \times R^0 \times S^0$

However, in this case, $R^0 = S^0 = 1$, and hence these prime numbers will not exist. So N cannot have four prime numbers.

(Since, $26 = 2 \times 13$, it can be distributed only to two prime factors).

Answer to "why" (in b given above)—If number of prime factors = 5, then minimum number of total factors = (1 + 1) (1 + 1) (1 + 1) (1 + 1) (1 + 1) = 32. This is all because—when we say that P is a prime factor, then minimum power of P has to be equal to 1. So if there are five prime factors, each of the prime factors will have minimum 1 as the power.

Worked-out Example 18 Find the smallest odd positive integer that has the same number of divisors as the number 360.

☝ ***Solution*** Number of divisors of $360 = 2^3 \times 3^2 \times 5^1$

Number of divisors of 360 = (3 + 1) (2 + 1) (1 + 1) = 24

To have 24 factors, we may spread it in various ways:

$24 = 8 \times 3$ or $24 = 6 \times 4$, or $24 = 6 \times 2 \times 2$ or $24 = 4 \times 3 \times 2$ or $24 = 2 \times 2 \times 2 \times 3$

Ideally we should start with the way that spreads it in maximum number of prime factors. However, this may not give us the final result, and we will be required to check the number thus obtained is actually the smallest one.

So if we take four prime factors (which is also the maximum possible number of prime factors):

$N = p^1 \times q^1 \times r^1 \times s^2$ where, p, q, r and s are odd prime numbers.

To have the lowest possible value of N, we should take the lowest possible value of p, q, r and s.

Four lowest prime numbers = 3, 5, 7, 11

So, $N = 11^1 \times 7^1 \times 5^1 \times 3^2$

Now the trade-off that we can do is possible by increasing the power of 3 and eliminating 11.

So now we would go for three prime numbers:

So, $M = 7^1 \times 5^2 \times 3^3 \Rightarrow$ we are multiplying by 3×5 after removing 11, so this number is going to be greater than N.

So lowest value = $11^1 \times 7^1 \times 5^1 \times 3^2 = 3465$

Worked-out Example 19 Number A has 24 factors and number $A \times B$ has 105 factors. What is the least possible number of factors that B can have?

Solution For least number of factors of B, we should try to take common factors in each of A and B as much as possible.

We are required to decide two things:

(i) How many prime factors are there in A and B individually

(ii) What are the powers of those prime factors.

Let us start with number A:

If there is only one prime factor, then number A has to be of format P^{23} $\Rightarrow$ total number of factors of $A \times B$ would be divisible by 24, which is not the case. So we can conclude that A does not have only one prime factor.

If there are two prime factors P and Q, then number A may have different formats:

$P^1 \times Q^{11}$, or $P^3 \times Q^5$ or $P^2 \times Q^7$ (or interchangeably $P^5 \times Q^3$ so on).

We will now verify this one by one for the factors of $A \times B$:

$A = P^1 \times Q^{11}$

$A \times B = P^6 \times Q^{14} \Rightarrow$ Total number of factors = 105. And, $B = P^5 \times Q^3$ $\Rightarrow$ Total number of factors = 24.

Let us now check for $A = P^3 \times Q^5$

$A \times B = P^6 \times Q^{14} \times B = P^3 \times Q^9 \Rightarrow$ total number of factors of $B = 4 \times 10 = 40$

$A \times B$ can also be written as $= P^4 \times Q^6 \times R^2 \Rightarrow$ total number of factors of $A \times B = 5 \times 7 \times 3 = 105$. In this case $B = P \times Q \times R^2 \Rightarrow$ Total number of factors of $B = 2 \times 2 \times 3 = 12$

Thus, the minimum number of factors of B = 12.

PRACTICE EXERCISE 1

1. Which of the following numbers will have lowest total number of factors?

1. 48 **2.** 60 **3.** 30 **4.** 42

2. How many factors of 84 is the product of only two prime, distinct or same, factors?

1. 4 **2.** 6 **3.** 3 **4.** 2

Directions for questions (3–5): Read the information below and solve the questions based on it.

$N = A^2 \times B^3 \times C^4$, where, A, B and C are prime numbers

3. How many factors of N are perfect squares?

1. 6 **2.** 12 **3.** 8 **4.** None of these

4. How many factors of N are cube of any natural number

1. 2 **2.** 3 **3.** 4 **4.** 6

5. How many factors of N will be having exactly 3 factors?

1. 2 **2.** 3 **3.** 4 **4.** None of these

6. A natural number N has four factors. Sum of the factors of N excluding N is 31. How many values can N take?

1. 2 **2.** 3 **3.** 4 **4.** 6

Directions for questions (7–9): Read the information below and solve the questions based on it.

Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4 as the answer—if question cannot be answered even by using both the statements.

7. Is the integer y greater than 4?

A. 2^y is a factor of 8!.

B. 2^4 is a factor of y!.

8. Product of a, b, c, and d is 1003 (a, b, c and d are all natural numbers). Which one is the largest among them?

A. $a > b > c$

B. $a > b > d$

9. How many factors will N have?

A. N^2 is having 81 factors.

B . N is having two prime factors.

10. For how many natural numbers N less than 12, N! + N + 1! will be divisible by 72?

1. 5 **2.** 6 **3.** 7 **4.** 8

11. What is the sum of all the prime factors of (summation of cubes of 1st 2020 natural numbers)?

1. 197 **2.** 198 **3.** 151 **4.** None of these

12. N is a natural number greater than 1 and less than 100. F1, F2, F3,.....Fn are the factors of N in such a way that,

1 = F1 < F2 < F3 <.....Fn = N.

Further, D = F1 . F2 + F2 . F3 + ...+ F(n – 1) . Fn

If D is a factor of N^2, then how many values of N will be there?

1. 0 **2.** 10 **3.** 24 **4.** 25

5. None of these

13. x, y and z are three natural numbers in such a way that 1 < x < y < z. For how many set of values of (x, y, z), (x – 1) (y – 1) (z – 1) is a factor of (xyz – 1)?

1. 0 **2.** 1 **3.** 2 **4.** 5

5. None of these

14. If $n = 2^{29} \times 3^{17}$, how many positive integer divisors of n^2 are less than n but do not divide n?

1. 39 **2.** 493 **3.** 439 **4.** 593

5. 539

15. An army general wants to place the soldiers into a formation that includes all of them and has no unfilled positions. If soldiers are arranged in a square formation, there are 12 soldiers leftover. The general realizes that if he arranges the group in a formation with 9 more rows than columns, there are no soldiers leftover. What could be the maximum number of soldiers?

1. 872 **2.** 1036 **3.** 1125 **4.** 1296

5. 1236

16. If a is the smallest positive integer that is a multiple of 147 and has exactly 147 positive integral divisors, including 1 and itself, the value of $\frac{a}{147}$ is

1. $3^6 \times 2^5$ **2.** $2^4 \times 3^6$ **3.** $2^6 \times 3^5$ **4.** $3^4 \times 2^6$

5. None of these

17. A natural number has total 19 composite factors. How many values of prime factors of N are possible?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. None of these

18. For how many natural number values of x, x (x + 1) (x + 2) (x + 3) has exactly three different prime factors?

1. 0 **2.** 1 **3.** 2 **4.** 4

5. None of these

19. A natural number 'X' has 24 factors and 'X × Y' has 165 factors. What is the least possible number of factors natural number 'Y' can have?

1. 18 **2.** 28 **3.** 25 **4.** 20

5. 16

20. Let $N = 2^{15} \times 3^{12}$. How many factors of N^2 are less than N but do not divide N?

1. 180 **2.** 387 **3.** 207 **4.** 90

5. 194

21. How many factors of 1080 will be divisible by 10 but not divisible by 30?

1. 2 **2.** 3 **3.** 5 **4.** 6

5. None of these

22. How many factors of 1920 will satisfy the following properties:

i. It should be a perfect square

ii. It should be a multiple of 10

iii. It should not be divisible by 30.

1. 0 **2.** 1 **3.** 3 **4.** 5

5. None of these

23. A three digit number is such that number of factors of the original number = Number of factors of the new number made by reversing the digits (left one at the right one and vice versa). If the middle number (tens place digit) = 5, how many such numbers will exist?

1. 0 **2.** 3 **3.** 6 **4.** 9

5. None of these

24. How many natural numbers less than 200 will have 12 factors?

1. 2 **2.** 4 **3.** 6 **4.** 8

5. None of these

25. How many natural numbers from 100 to 200 will have exactly 2 factors?

1. 14 **2.** 16 **3.** 19 **4.** 21

5. 22

26. How many three digit natural numbers will have a total of 3 factors?

1. 8 **2.** 10 **3.** 12 **4.** 14

5. None of these

27. In the above question, how many numbers will be even?

1. 4 **2.** 5 **3.** 6 **4.** 8

5. None of these

28. $N = 2^8 \times 3^{10} \times 5^8 \times 7^2$. What is the number of factors of this number N which are factors of 360 but not the factors of 540?

1. 120 **2.** 210 **3.** 114 **4.** 144

5. 198

29. How many natural numbers less than 100 will have exactly three prime factors?

1. 2 **2.** 4 **3.** 5 **4.** 8

5. None of these

30. How many divisors of 10^6 will have atleast two zeroes at its end?

1. 13 **2.** 22 **3.** 33 **4.** 25

5. None of these

31. Find the product of all the factors of 3^{16}.

1. 3^{33} **2.** 3^{68} **3.** 3^{136} **4.** 3^{128}

5. 3^{142}

32. For how many natural numbers less than 100, total number of factors will be even?

1. 88 **2.** 78 **3.** 67 **4.** 75

5. None of these

33. N is a natural number. 2N is having 28 factors, and 3N is having 30 factors. How many factors 6N will have?

1. 36 **2.** 35 **3.** 32 **4.** 40

5. None of these

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING 1

1. $N = 1440 = 2^5 \times 3^2 \times 5^1$

1. Total factors = $(5 + 1)(2 + 1)(1 + 1) = 6 \times 3 \times 2 = 36$

2. Even factors = $(5)(2 + 1)(1 + 1) = 5 \times 3 \times 2 = 30$

3. Odd Factors = Total – even = 36 – 30 = 6

Alternatively, Odd factors = 1 (number of ways power of 2 can be used) $\times 3 \times 2 = 6$

4. Prime factors = 3 (namely 2, 3 and 5).

5. 2n + 1 is used to represent odd numbers. Since, here n = a natural number (1, 2, 3, ……..), lowest value of 2n + 1= 3.

So, factors of the format 2n + 1 (where, n is a natural number) = 6 – 1 = 5. (except 1, all the other odd factors).

6. Perfect square factors = $(2 + 1)(1 + 1)(0 + 1) = 6$

7. Cube factors = $(1 + 1)(0 + 1)(0 + 1) = 2$

8. Composite factors = Total factors – 1 – prime factors = 36 – 1 – 3 = 32

9. 4n + 1 format of factors will be odd factors, but this will not contain all the odd factors.

Following are the Odd factors = 1, 3, 5, 9, 15, 45.

Out of this 4n + 1 format numbers (where, n is a natural number) are 5, 9, 45.

10. Method 1

Factors which are divisible by 6 will use following powers of the prime numbers:

Minimum power of 2 used = 1

Minimum power of 3 used = 1

2^1	3^1	5^0
2^2	3^2	5^1
2^3		
2^4		
2^5		

Total number of factors = $5 \times 2 \times 2 = 20$

Factors which are divisible by 12 ($2^2 \times 3^1$)will use following powers of the prime numbers:

Minimum power of 2 used = 1

Minimum power of 3 used = 1

2^2	3^1	5^0
2^3	3^2	5^1
2^4		
2^5		

Total number of factors = $4 \times 2 \times 2 = 16$

Number of factors which are divisible by 6 but not divisible by 12 = 20 – 16 = 4

Method 2

Factors that are divisible by 6 but not divisible by 12 will use only one power of 2^1 (one way)

Powers of 3 to be used = 3^1 and 3^2 (2 ways)

Powers of 5 to be used = 5^0 and 5^1(2 ways)

Hence, total number of factors that is divisible by 6 but not divisible by 12 = $1 \times 2 \times 2 = 4$

11. Sum of all the factors $= \frac{(2^6-1)}{2-1} \times \frac{(3^3-1)}{3-1} \times \frac{(5^2-1)}{5-1} = 63 \times 13 \times 6 = 4914.$

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
3	**1**	**2**	**3**	**2**	**3**	**1**	**3**	**4**	**3**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
2	**4**	**3**	**2**	**2**	**3**	**3**	**3**	**4**	**1**
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
2	**1**	**4**	**2**	**4**	**5**	**5**	**5**	**3**	**4**
31.	32.	33.							
3	**5**	**2**							

1. Can be done through the options.

2. There are only four factors of 84 which are the product of only two primes (same or distinct) = 4, 6, 14, 21.

3. $N = A^2 \times B^3 \times C^4$, where, A, B and C are prime numbers.

Perfect Square will be obtained for A^0 and A^2, B^0 and B^2, C^0 and C^2 and C^4.

Number of ways powers of A is used = 2

Number of ways powers of B is used = 2

Number of ways powers of C is used = 3

Hence, number of perfect square factors = $2 \times 2 \times 3 = 12$

4. $N = A^2 \times B^3 \times C^4$, where A, B and C are prime numbers.

Cubes will be obtained for A^0, B^0 and B^3, C^0 and C^3

Number of ways powers of A is used = 1

Number of ways powers of B is used = 2

Number of ways powers of C is used = 2

Hence, number of perfect square factors = $1 \times 2 \times 2 = 4$

7. Using statement A, y can have any integer value from 0—7, hence, cannot be determined using statement A alone.

Using statement B, minimum value of y can be 6. And hence y is greater than 4.

Hence, answer is—Option 1.

8. $1003 = 1 \times 1 \times 17 \times 59$

Using both the statements together, 'a' is the largest one.

9. Using statement A alone, N^2 will not answer this question as it leads to different possibilities for N = (prime number 1)^4 × (prime number 2)^4 ⇒ Number of factors of N = 25

OR, N = (prime number 1)^1 × (prime number 2)^1 × (prime number 3)^1 × (prime number 4)^1 ⇒ Number of factors of N = 16.

Hence, statement 1 alone is not sufficient.

Using statement 2 alone, N = (prime number 1)^4 × (prime number 2)^4 ⇒ Number of factors of N = 25 OR

N = (prime number 1)^{13} × (prime number 2)^1 ⇒ Number of factors of N = 28

Hence, unique answer is not obtained.

Even by using both the statements together, we do not get the unique answer. Hence, option 4 is the answer.

10. $N! + N + 1! = N! (N + 1)$

This is divisible by 72 for N = 5, 6, 7, 8, 9, 10, 11.

11. $1^3 + 2^3 + 3^3 + \ldots\ldots\ldots\ldots + n^3 = (1 + 2 + 3 + \ldots\ldots\ldots\ldots + n)^2 = [(n(n+1))/2]^2$

So sum of $1^3 + 2^3 + 3^3 + \ldots\ldots\ldots\ldots + 2020^3 = (2021 \times 1010)^2$

To find out the prime factors, we will disregard the square sign.

$2021 \times 1010 = 2 \times 5 \times 43 \times 47 \times 101$ ⇒ Sum of prime factors = 198

12. D is a factor of N^2 only if N is a prime number. This can be checked by taking some values of N and correspondingly obtained value of D. And we already know that there are 25 prime numbers from 1 to 100.

14. If $n = 2^{29} \times 3^{17}$, then $n^2 = 2^{58} \times 3^{34}$ ⇒ Number of factors of $n^2 = 59 \times 35$

We can say that $[(59 \times 35) - 1]/2 = 1032$ factors are less than n (if factors of n^2 make pairs, then one will be larger than n and other will be smaller than n). Similarly, there are $30 \times 18 - 1 = 539$ factors of n less than n itself. These are also the factors of n^2.

Hence, answer = 1032 – 539 = 493

15. Number of soldiers = $a^2 + 12 = b (b + 9)$ for some value of a and b.

Multiplying both the sides by 4, we can write $4a^2 + 129 = (2b + 9)^2$

Or, $(2b + 9 + 2a) (2b + 9 - 2a) = 129$. Our objective is to maximize b. For this to happen set the first part equal to 129, and second part equal to 1.

This gives b = 28. Answer = 28 × 37 = 1036. Hence, option 2 is the answer.

16. For a to have 147 divisors, let us assume $a = p^n \times q^m \times r^s \ldots..$

Number of factors = (n + 1) (m + 1) (s + 1)……..= 147. Two of the prime factors of a are 3 and 7, as a is divisible by 147.

Now, $147 = 3^1 \times 7^2$ ⇒ This hints towards having three prime factors with powers 3, 6 and 6. We have already concluded that there are 2 prime factors, let us introduce the third prime factor 2 so as to minimize the value of a.

Hence, $a = 2^6 \times 3^6 \times 7^2$

So the answer = $\frac{2^6 \times 3^6 \times 7^2}{147} = 2^6 \times 3^5$.

17. Total number of factors of a natural number = 1 + Prime factors + Composite factors

Or, total number of factors = 1 + 19 + prime factors = 20 + prime factors

For one prime factor, total number of factors = 21 $\Rightarrow$ this is possible for any p^{20} format number where, p is a prime number.

For two prime factors, total number of factors = 22 $\Rightarrow$ this is possible for any $p^1 \times q^{10}$ format number where p and q are prime numbers.

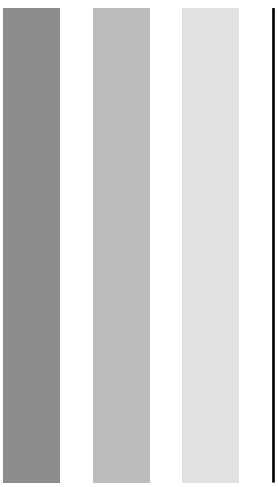
For three prime factors, total number of factors = 23 $\Rightarrow$ this is not possible for three prime factors

For four prime factors, total number of factors = 24 $\Rightarrow$ this is possible for $p^1 \times q^1 \times r^1 \times s^2$ format number where p, q, r and s are prime numbers.

For five prime factors, total number of factors = 25 $\Rightarrow$ this is not possible for five prime factors. We can also say now that with a minimum of five prime factors minimum number of factors = 32. Hence, we cannot have five or more than five prime factors.

So different values of prime factors = 1, 2, 4

18. It can be verified by putting the values that it is possible only for 2 and 3. Hence 2 values.



Chapter 7

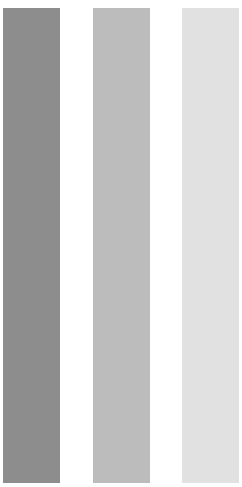
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DEMYSTIFYING NUMBER SYSTEM

DIVISIBILITY

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- Divisibility rules of numbers
- Seed number method—Method to obtain the divisibility rule
- Divisibility rules of the number on base other than base 10
- Types of questions asked from this chapter in the CAT

Essence of divisibility rules lies with the fact that with the help of these rules, we can find out if a number is divisible by other number or not without going through actual division.

Divisibility rules of numbers are specific to that particular number only. It simply means that divisibility rules of different numbers will be different. We shall now see a list of divisibility rules for some of the natural numbers:

DIVISIBILITY RULES

For 2—If unit digit of any number is 0, 2, 4, 6 or 8, then the number will be divisible by 2.

For example—120, 152, 188 etc.

For 3—If sum of digits of any number is divisible by 3, then the number will be divisible by 3.

For example, 123, 456

Worked-out Example 1 How many values of N are possible if 34586N is divisible by 3, where N is the unit digit of the number?

☝ ***Solution*** Sum total of the number = 26 + N

For this number to be divisible by 3, A can take three values namely 1 or 4 or 7. (No other values are possible since N is the unit digit of the number)

Alternatively, instead of adding all the digits of the number, we can eliminate the digits divisible by 3, and add the remaining digits only. In this case, we will eliminate 3 and 6. 34586N

So sum of remaining digits = 4 + 5 + 8 + N = 20 + N ⇒N = 1/4/7

Worked-out Example 2 How many set of values of N and M are possible if 34586NM is divisible by 3, where M is the unit digit and N is the tens place digit of the number?

☝ ***Solution*** **Method 1**

Sum total of the number = 26 + N + M

Now, N + M can take different values such that 34586NM is divisible by 3.

N + M = 1 ⇒ Sum total of the number = 26 + 1 = 27 ⇒ Set of values of N, M possible are = (0, 1), (1, 0) ⇒ 2 sets possible.

N + M = 4 ⇒ Sum total of the number = 26 + 4 = 30 ⇒ Set of values of N, M possible are = (0, 4), (4, 0) (1, 3) (3, 1) (2, 2) ⇒5 sets possible.

N + M = 7 ⇒ Sum total of the number = 26 + 7 = 33 ⇒ Set of values of N, M possible are = (0, 7), (7, 0) (1, 6) (6, 1) (2, 5) (5, 2), (3, 4) (4, 3) ⇒ 8 sets possible.

N + M = 10 ⇒ Sum total of the number = 26 + 10 = 36 ⇒ Set of values of N, M possible are = (1, 9), (9, 1) (2, 8) (8, 2) (3, 7) (7, 3), (4, 6) (6, 4) (5, 5) ⇒ 9 sets possible.

N + M = 13 ⇒ Sum total of the number = 26 + 13 = 39 ⇒ Set of values of N, M possible are = (4, 9), (9, 4) (5, 8) (8, 5) (6, 7) (7, 6) ⇒ 6 sets possible.

N + M = 16 ⇒ Sum total of the number = 26 + 16 = 42 ⇒ Set of values of N, M possible are = (7, 9), (9, 7) (8, 8) ⇒ 3 sets possible.

N + M = 19 ⇒ Not possible as the maximum sum of two digits can be equal to 18.

Hence total number of sets possible = 33.

Method 2

Largest possible value of 34586NM = 3458699

Smallest possible value of 34586NM = 3458600

These are 3458600 ≤ 34586NM ≤ 3458699 ⇒ These are 100 numbers. Among any 100 consecutive natural numbers, there are going to be either 33 or 34 natural numbers divisible by 3 depending upon 1st number taken, which is either divisible by 3 or not. Since 1st number 3458600 is not divisible by 3, hence there are 33 numbers.

So we can conclude that total number of sets possible = 33.

For 4—If last two digits of a number is divisible by 4, then that number will be divisible by 4.

For example, 3796, 248, 1256, etc.

For 5—If last digit of the number is 5 or 0, then the number will be divisible by 5.

For 6—If last digit of the number is divisible by two and sum of all the digits of number is divisible by 3, then that number will be divisible by 6.

Worked-out Example 3 How many of the following numbers will be divisible by 6?

(a) 45794 (b) 326804 (c) 378927 (d) 46924

☝ ***Solution***

(a) Sum of digits of 45794 = 4 + 5 + 7 + 4 (we can neglect 9 since 9 is divisible by 3) = 20 ⇒ Sum of digits is not divisible by 3, so the number is not divisible by 6.

(b) Sum of digits of 326804 = 2 + 8 + 0 + 4 (neglect 3 and 6) = 14 ⇒ Sum of digits is not divisible by 3, so the number is not divisible by 6.

(c) We will not check the sum of digits of 378927, as this number is not divisible by 2.

(d) Sum of digits of 46924 = 4 + 2 + 4 (neglect 6 and 9) = 12 ⇒ Sum of digits is divisible by 3, and the number is even. So the number is divisible by 6.

For 7—The integer is divisible by 7 if and only if the difference of the number of its thousands and the remainder of its divisible by 1000 is divisible by 7.

'OR'

—If the difference between the numbers of tens in the number and twice the unit digits divisible by 7 then the given number is divisible by 7.

For example, Let us take the number 795. The units digit is 5 and when it is doubled, we get 10. The remaining part of the number (i.e., the tens) is 79. If 10 is subtracted from 79 we get 69. Since this result is not divisible by 7, the original number 695 is not divisible by 7.

For 8—If last 3 digits of a number is divisible by 8, then the number itself will be divisible by 8.

For example, 128, 34568, 76232 etc.

For 9—If sum of digits of the number is divisible by 9, then the number will be divisible by 9. For example, Let us check it for 129835782. Sum of digits of 129835782 = 1 + 2 + 8 + 3 + 5 + 7 + 8 + 2 = 36 (neglect 9). Since 36 is divisible by 9, so number will be divisible by 9.

Worked-out Example 4 How many pairs of A and B are possible in number 89765A4B if number is divisible by 9, given that last digit of number is even?

Solution Sum of digits of number is 8 + 9 + 7 + 6 + 5 + A + 4 + B = 39 + A + B.

So, A + B should be 6 or 15. Next value should be 24 but since A and B are digits so it cannot be more than 18. Possible pairs of A and B are:

A	B
0	6
1	5
2	4
3	3
4	2
5	1
6	0
7	8
8	7
9	6
6	9

Since B is even, six possible set of values of A and B are there.

For 10—Unit digit of no. should be zero (0).

For 11—A number is divisible by 11, if the difference between the sum of the digits at the even places and the sum of the digits at the odd places is divisible by 11(zero is divisible by 11).

For example, 6595149 is divisible by 11 as the difference of 6 + 9 + 1 + 9 = 25 and 5 + 5 + 4 = 14 is 11.

For 12—If the number is divisible by 3 and 4, then the number will be divisible by 12. e.g., 144, 348.

For 13—(A + 4B), where B is the unit's place digit and A is all the remaining digits.

For example, Checking the divisibility of 1404 by 13. Here, A = 140 and B = 4, then A + 4B = 140 + 4 × 4 = 156. This 156 is divisible by 13, so 1404 will be divisible by 13.

For 14—If the number is divisible by 2 and 7 both, then the number will be divisible by 14.

For 15—A number is divisible by 15, if the sum of the digits is divisible by 3 and unit digit of the number is 0 or 5.

For example, 225, 450, 375 etc.

For 16—A number is divisible by 16, if the number formed by the last 4 digits of the given number is divisible by 16.

For example, 12578320 is divisible by 16, since last 4 digits of the number, 8320 is divisible by 16.

Process to Define the Divisibility Rule of any Prime Number

We will use "Seed Number" method to define the divisibility rule of any prime number greater than 5. Ideally, any prime number greater than 5 will end up either in 10N + 1 or 10N – 1 within its 1st 10 multiples. In other words, any prime number greater than 5 will give either unit digit 1 or 9 within its 1st 10 multiples. Now this multiple is written in the format of 10N + 1 or 10N – 1 and N is known as the seed of that prime number. Seed numbers are taken negative in the case of +1 and positive in the case of –1. For example, In case of 13, multiples of the format $10N \pm 1 = 39, 91$

$39 = 4 \times 10 - 1 \Rightarrow$ Seed Number = 4

$91 = 9 \times 10 + 1 \Rightarrow$ Seed Number = –9

So the seed number of 13 = 4 and (–9).

How to Use Seed Number to Test Divisibility

If the seed number is N, then the divisibility rule = A + NB and if the seed number = (–N), then the divisibility rule = A – NB

Worked-out Example 5 Check if the number 24635 is divisible by 19.

Seed number of 19 = 2 and – 17. So the divisibility rules for 19 are: A + 2B and A – 17B. We can use either of these two to check the divisibility.

Unit digit of the given number is B, and all the remaining digits = A.

So in the case of 24635, B = 5, and A = 2463 and we are using now A + 2B divisibility rule.

Rule says that if 24635 is divisible by 19, then A + 2B = 2463 + 2 × 5 = 2473 will also be divisible by 19 and vice versa. That is, if A + 2B = 2463 + 2 × 5 = 2473 is divisible by 19, then original number 24635 will also be divisible by 19. And if A + 2B = 2463 + 2 × 5 = 2473 is not divisible by 19, then original number 24635

will also not be divisible by 19. It can be checked that 2473 is not divisible by 19 $\Rightarrow$ 24635 will not be divisible by 19.

We can go a stage further. Instead of dividing 2473 by 19, we can take 247 = A, and 3 = B.

So, A + 2B = 247 + 6 = 253. It can be seen that 253 is not divisible by 19 $\Rightarrow$ 2473 is not divisible by 19 $\Rightarrow$ 24635 is not divisible by 19.

Instead of A + 2B, we could have used A – 17B also as the divisibility rule. Although it can be seen very clearly that it would have led to slightly difficult calculation than the current one.

Seed Number of some of the prime numbers:

Prime Number	Multiple Ending in 9	Seed No.	Multiple Ending in 1	Seed No.
3	9	1	21	–2
7	49	5	21	–2
11	99	10	11	–1
13	39	4	91	–9
17	119	12	51	–5
19	19	2	171	–17
23	69	7	161	–16
29	29	3	261	–26

Divisibility Rule of the Numbers on Base Other 10

The rules that we have discussed till now is valid only for the numbers on base 10. If we take any other base, divisibility rule of the number might change. For example, divisibility rule of 9_{10} is not same as the divisibility rule of 9_{12}. Similarly, divisibility rule of 5_{10} is not same as the divisibility rule of $5_{6.}$

Although, seed number method exists for base other than 10 too. Only difference that occurs is the fact that now we will find out the multiple KN $\pm$ 1, where K is the new base, instead of finding 10N $\pm$ 1. So for example, on base 7, we will now find out the multiples in the format of 7N $\pm$ 1.

[I am not getting into too much details of divisibility rule of the numbers on the base other than 10, because I do not see CAT and other B School examinations moving in that direction.]

SOME OF THE GENERALIZATIONS

1. Divisibility rule of 9_{10} = Sum of the digits should be divisible by 9
This rule is applicable for N_{N+1} (For N on base N + 1).
So divisibility rule of 8_9 = Sum of the digits of the number written on base 9 should be divisible by 8.
Divisibility rule of 7_8 = Sum of the digits of the number written on base 8 should be divisible by 7.

Divisibility rule of 2_3 = Sum of the digits of the number written on base 3 should be divisible by 2.

Worked-out Example 6 Following numbers are written on base 6. Which of the following numbers are divisible by 5_6?

(i) 1234_6 (ii) 21534_6 (iii) 315_6

☝ ***Solution*** Divisibility rule of 5_6 = Sum of the digits of the number written on base 6 should be divisible by 5.

Let us verify the numbers one by one:

(i) 1234_6

Sum of digits = 1 + 2 + 3 + 4 = 10 which is divisible by 5 (we are not required to convert 10 on base 6). Hence, divisible by 5.

Alternatively, we can convert the number on base 10.

$1234_6 = 1 \times 6^3 + 2 \times 6^2 + 3 \times 6^1 + 4 \times 6^0 = 310$ = Divisible by 5.

(ii) 21534_6

Sum of digits = 2 + 1 + 5 + 3 + 4 = 15 which is divisible by 5. Hence, the number is divisible by 5.

(iii) 315_6

Sum of digits = 3 + 1 + 5 = 9 which is not divisible by 5. Hence, the number is not divisible by 5.

We can see that sum of digits should be divisible by 5 on base 6 for the number to be divisible by 6 by writing first few natural numbers on base 6:

1st few natural numbers on base 6:

0	1	2	3	4	**5**	10	11	12
13	**14**	15	20	21	22	**23**	24	25
30	31	**32**	33	34	35	40	**41**	42
43	44	45	**50**	51	52	53	54	**55**
100	101	102	103	**104**	105	110	111	112

Starting from 0, every 5th number has to be divisible by 5. Numbers are = 0, 5, 14, 23, and so on (in grey).

One common theme across all these '**grey**' numbers is the fact that summation of digits is divisible by 5.

Worked-out Example 7 Which of the following numbers are even numbers (Answer in Yes or No)?

(a) 122_3 (b) 143_5 (c) 232_7 (d) 442_7 (e) 268_{99}

☝ ***Solution***

(a) No (b) Yes (c) No (d) Yes (e) Yes

2. Divisibility rule of 11_{10} = Difference between the sum of the digits at the even places and the sum of the digits at the odd places is divisible by 11.
This rule is applicable for $N + 1_N$ (For N + 1 on base N).

PRACTICE EXERCISE 1

Directions for questions (1–11): Find the value of x in the following questions, where 'x' is a digit from 0–9.

1. 4×610 is divisible by 41.

2. 6×0951 is divisible by 31.

3. 6×8947 is divisible by 29.

4. 3×2961 is divisible by 31.

5. $\times 565544$ is divisible by 37.

6. $\times 30749$ is divisible by 43.

7. 9×32815 is divisible by 71.

8. a, b, c and d are natural numbers. Which of the following will always divide N = (a – b) (a – c) (a – d) (b – c) (b – d) (c – d)?

(i) 2 (ii) 3 (iii) 4 (iv) 6

1. Only (i)

2. Only (i), (ii) and (iv)

3. Only (ii)

4. Only (iii) and (i)

5. All of these will divide N.

9. A number N is divisible by 6 but not divisible by 4. Which of the following will not be an integer?

1. N/3 **2.** N/2 **3.** N/6 **4.** N/12

5. None of these

10. A and B are natural numbers such that AB + 1 is divisible by 24. Which of the following will definitely divide A + B?

(i) 12 (ii) 24 (iii) 36 (iv) 48

1. (i) (ii) and (iii) only **2.** (ii) and (iii) only

3. (i) and (ii) only **4.** (i), (ii) and (iv) only

5. All four

11. Following numbers are written on base 2. How many of the numbers are divisible by 2?

(i) 1100101001 (ii) 100100011

(iii) 10100010010 (iv) 1010111010

1. (i) and (iii) only **2.** (i) and (ii) only

3. (iii) and (iv) only **4.** (i), (ii) and (iv) only

5. All four

Directions for questions (12–13): Read the passage below and solve the questions based on it.

A = Set of first N positive numbers. There are 16 numbers in A which are divisible by both X and Y. There are 50 numbers in A which are divisible by X but not by Y and 34 numbers in A divisible by Y but not by X.

12. How many numbers in A are divisible by any of the two numbers?

1. 100 **2.** 50 **3.** 200 **4.** 150

5. None of these

13. How many numbers in N are divisible by X?

1. 42 **2.** 56 **3.** 66 **4.** 72

5. None of these

14. A natural number N satisfies following conditions:

1. Number is having all the 9's.

2. It is divisible by 13.

How many digits are there in N?

1. 5 **2.** 6 **3.** 7 **4.** 8

5. 9

15. N is a prime number when written in the decimal system. If S is the sum of the digits of N, when it is written in base 7, we can surely say that S would always be

1. even **2.** odd **3.** composite **4.** prime

5. None of these

16. On base 7, 1st 100 natural numbers are written. How many of these numbers are divisible by 6?

1. 15 **2.** 16 **3.** 17 **4.** 18

5. None of these

Directions for question (17): Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4 as the answer—if question cannot be answered even by using both the statements.

17. Is 5X + 25 divisible by 50?

A. X is divisible by 5.

B. X is divisible by 10.

18. What is the remainder when 42345346_7 is divided by 6_7?

1. 0 **2.** 1 **3.** 2 **4.** 4

5. 5

SOLUTIONS AND EXPLANATIONS

1. 9

2. 6

3. 1

4. 7

5. 1

6. 5

7. 5

8. 5

9. 4

10. Option 3

11. We are not required to convert these numbers on decimal system. We can simply see that if the unit digit = 0, then the number will be divisible by 2, or else not. Hence, (iii) and (iv) are divisible. Option 3.

12–13—Use set theory formula for intersection and union of two sets.

12. Option 1

13. Option 3

14. Go through option. 999999 is divisible by 13. Hence option 2.

15. If a number is prime in one base, it will be prime in all the other bases too. Hence, option 4 is the answer. We cannot be sure of other options. Hence, answer is option 4.

16. It does not matter which base we are talking about. If there are 100 consecutive natural numbers, then there will be 100/6 = 16 numbers divisible by 6. Hence, answer is option 2.

17. 1. Using statement A alone, assuming values of X (= 5 or = 10) gives that 5x + 25 is sometimes divisible and sometimes not divisible.

Using statement B alone, whatever values of X we take, we see that 5X + 25 = 5 (X + 5) is not divisible by 10.

Answer—option 1

18. Sum of digits of 42345346 = 31 which gives remainder 1 when divided by 6. Hence, net remainder = 1. Option 2 is the answer.

CAT QUESTIONS FROM THIS CONCEPT

CAT 2005

1. The digits of a three-digit number A are written in the reverse order to form another three-digit number B. If B>A and B – A is perfectly divisible by 7, then which of the following is necessarily true?

1. 100<A<299 **2.** 106<A<305 **3.** 112<A<311 **4.** 118<A<317

CAT 2003

2. Consider the sets $T_n = \{n, n + 1, n + 2, n + 3, n + 4\}$, where n = 1, 2, 3……..96. How many of these sets contain 6 or any integral multiple thereof (i.e., any one of the numbers 6, 12, 18, ...)?

1. 80 **2.** 81 **3.** 82 **4.** 83

CAT 2000

3. Let a, b, c, d and e be integers such that a = 6b = 12c, and 2b = 9d = 12 e. Then which of the following pairs contains a number that is not an integer?

1. $\left(\frac{a}{27}, \frac{b}{e}\right)$ **2.** $\left(\frac{a}{36}, \frac{c}{e}\right)$ **3.** $\left(\frac{a}{12}, \frac{bd}{18}\right)$ **4.** $\left(\frac{a}{6}, \frac{c}{d}\right)$

4. Let S be the set of integers x such that

I. $100 \leq \times \leq 200$

II. x is odd and

III. x is divisible by 3 but not by 7

How many elements does S contain?

1. 16 **2.** 12 **3.** 11 **4.** 13

5. Let $N = 55^3 + 17^3 - 72^3$. N is divisible by

1. both 7 and 13 **2.** both 3 and 13

3. both 17 and 7 **4.** both 3 and 17

SOLUTIONS AND EXPLANATIONS

1. 2. Let A = abc, then B = cba

Given, B>A which implies c>a … (1)

As B – A = (100c + 10b + a) – (100a + 10b + 1)

B – A = 100 (c – a) + (a – c)

B – A = 99 (c – a) and (B – A) is divisible by 7; and 99 is not divisible by 7 (no factor like 7 or 7^2) therefore, (c – a) must be divisible by 7 {i.e., (c – a) must be 7, 7^2 etc.) as c and a are single digits. (c – a) must be 7 only, the possible values (c, a) {with c>a} are (9, 2) and (8, 1) with this we can write A as abc = 1b8 or 2b9

As b can take values from 0 to 9, the smallest and largest possible value of are:

$A_{min} = 108$

and $A_{max} = 299$

Only 2 satisfies so 2 is the ans.

2. 1. We can observe that 6 will appear in 5 sets T_2, T_3, T_4, T_5 and T_6. Similarly, 12 will also appear in 5 sets and these sets will be distinct from the sets in which 6 appears, i.e., T_8, T_9, T_{10}, T_{11} and T_{12}. Thus, each multiple of 6 will be appear in 5 distinct sets. Till T_{96} there will be 16 multiplies will appear in $16 \times 5 = 80$ sets.

3. 4. a = 6b = 12c and 2b = 9d = 12e.

Dividing the first equations by 12 and second by 36.

We get, $\frac{a}{12}=\frac{b}{2}=\frac{c}{1}$ and $\frac{b}{18}=\frac{d}{4}=\frac{e}{3}$

$\therefore \quad \frac{a}{108}=\frac{b}{18}=\frac{c}{9}=\frac{d}{4}=\frac{e}{3}$

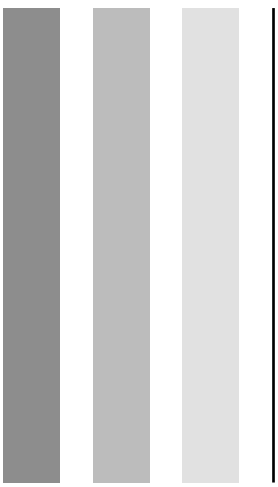
$\therefore$ a:b:c:d:e = 108:18:19:4:3

$\therefore \quad \frac{c}{d}=\frac{9}{4}$ is not an integer

4. Numbers which are divisible by 3 (between 100 and 200) are 33. Numbers which are divisible by 21, i.e., LCM of 7 and 3 (between 100 and 200) are 5. Out of the 33 numbers divisible by 3, 17 are even and 16 are odd. Out of the 5 numbers divisible by 7, three are odd. Hence, the numbers of odd numbers divisible by 3 but not by 7 is (16 – 3) =13. Hence, option 4 is the answer.

5. Break 72^3 into the format $(a + b)^3$.

$N = 55^3 + 17^3 - 72^3 \Rightarrow 55^3 + 17^3 - (55 + 17)^3 = 55^3 + 17^3 - 55^3 - 17^3 - 3 \times 55^2 \times 17 - 3 \times 55 \times 17^2 = -(3 \times 55^2 \times 17 + 3 \times 55 \times 17^2) \Rightarrow$ This is obviously divisible by 3 and 17. Hence, option 4 is the answer.



Chapter 8

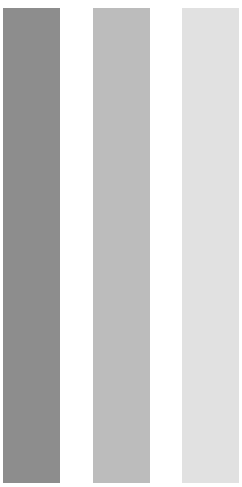
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DEMYSTIFYING NUMBER SYSTEM

CYCLICITY

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LEARNING OBJECTIVES



After completion of this chapter, you should have a thorough understanding of the following:

- Unit digit—Methods to find out the unit digit
- Tens digit—Methods to find out the unit digit
- Cyclicity of digits/numbers
- Process to find out hundred digits and beyond
- Types of questions asked from this chapter in the CAT

CYCLICITY

In this chapter, we are going to learn the methods to find out the unit digit/tens digit/hundreds digit and beyond. Before we move ahead, let us understand some of the basics:

1. Place value of a number—Consider the number 5201862:

Digit	5	2	0	1	8	6	2
Place Value	Ten lacs/ Million	Lacs	Ten Thousand	Thousand	Hundred	Tens	Unit

2. In multiplication and addition, to calculate the unit digit, we should focus only upon unit digit. It also means that we should not focus upon the tens digits and other digits on the left hand side.

For example—Unit digit of AB2 × CDE4 = 8. It is independent of the values of A, B, C, D and E.

Unit digit of AB2 + CDE4 = 6. It is independent of the values of A, B, C, D and E.

Similarly, while calculating tens digit, we should focus only upon the tens place and unit digit, and not on any digit to left hand side of tens digit.

For example—Tens digit of AB2 × CDE4 = cannot be calculated unless we have the value of B and E known to us. It is independent of the values of A, C and D and even if we do not have the value of A, C and D known to us, we can calculate the tens place digit.

Tens place digit of AB34 + CDE56 = 9, and it is independent of the values of A, B, C, D and E.

CALCULATING THE UNIT DIGIT

Let us start with calculating the unit digit of 2^5. We can simply right down $2^5 = 32$, hence, unit digit of $2^5 = 2$.

However, had this been a larger number, it would not have been so easy to calculate the exact value of the number, and hence, not so easy to calculate the unit digit. Here comes the need to define an alternative method to find out the unit digit without actually finding the exact value.

Method 1—Using Remainder Method

When we divide a number by 10, remainder obtained is going to be the unit digit of that number.

For example,

(i) Unit digit of 68 = 8 = Remainder obtained when 68 is divided by 10.
(ii) Unit digit of 21× 47 × 18 = unit digit of 17, 766 = Remainder obtained when $\frac{21\times47\times18}{10}$ = Remainder obtained $\frac{1\times7\times8}{10} = 6$

[To understand this method of finding remainder more, go to the remainder chapter in this book.]

Let us calculate the unit digit of (a) 78^{78} (b) 47^{470}

(a) 78^{78}

Remainder obtained when 78^{78} is divided by 10 = remainder obtained $\frac{8^{78}}{10}$ = Remainder obtained $\frac{64^{39}}{10}$ = Remainder obtained $\frac{4^{39}}{10}$ = Remainder obtained $\frac{64^{13}}{10}$ = Remainder obtained $\frac{4^{13}}{10}$ = Remainder obtained $\frac{64^4 \times 4}{10}$ = Remainder obtained $\frac{4^4 \times 4}{10}$ = Remainder obtained $\frac{256 \times 4}{10}$ = Remainder obtained $\frac{24}{10} = 4$

Above division might have been done in some other way too, you may have your own combinations.

(b) 47^{470}

Remainder obtained when 47^{470} is divided by 10 = Remainder obtained $\frac{7^{470}}{10}$ = Remainder obtained $\frac{2401^{117} \times 7^2}{10}$ = Remainder obtained $\frac{1 \times 7^2}{10} = 9$

Method 2—Using Cyclicity of Digits

Unit digits display certain cyclicity:

Digit ↓/Power →	1	2	3	4	5	6	7	8	Cycle Length
1	1	1	1	1	1	1	1	1	1
2	2	4	8	6	2	4	8	6	4
3	3	9	7	1	3	9	7	1	4
4	6	4	6	4	6	4	6	4	2
5	5	5	5	5	5	5	5	5	1
6	6	6	6	6	6	6	6	6	1
7	7	9	3	1	7	9	3	1	4
8	8	4	2	6	8	4	2	6	4
9	9	1	9	1	9	1	9	1	2
0	0	0	0	0	0	0	0	0	1

(a) 78^{78}

Unit digit of 78^{78} = Unit digit of 8^{78} Cycle length of 8 is 4.

So unit digit of $8^4 = 8^8 = 8^{12} = 8^{16} = 8^{4n}$

Unit digit of $8^1 = 8^5 = 8^9 = 8^{13} = 8^{17} = 8^{4n+1}$

Unit digit of $8^2 = 8^6 = 8^{10} = 8^{14} = 8^{18} = 8^{4n+2}$

Unit digit of $8^3 = 8^7 = 8^{11} = 8^{15} = 8^{19} = 8^{4n+3}$

8^{78} is a number of the format 8^{4n+2}. [We can see that when 78 is divided by 4, remainder obtained = 2]

Hence, unit digit of 8^{78} = unit digit of $8^2 = 4$

(b) 47^{470}

Unit digit of 47^{470} = Unit digit of 7^{470}

Cycle length of 7 is 4.

So, unit digit of $7^4 = 7^8 = 7^{12} = 7^{16} = 7^{4n}$

Unit digit of $7^1 = 7^5 = 7^9 = 7^{13} = 7^{17} = 7^{4n+1}$

Unit digit of $7^2 = 7^6 = 7^{10} = 7^{14} = 7^{18} = 7^{4n+2}$

Unit digit of $7^3 = 7^7 = 7^{11} = 7^{15} = 7^{19} = 7^{4n+3}$

7^{470} is a number of the format 7^{4n+2}. [We can see that when 470 is divided by 4, remainder obtained = 2]

Hence, unit digit of 7^{470} = unit digit of $7^2 = 9$

Method 3—Using Generalization

Maximum cyclicity of unit digit of any number = 4. This can be seen through the above table too.

We can also see that cyclicity of:

Digits	Cycle length
0, 1, 5, 6	1
4, 9	2
2, 3, 7, 8	4

Above result can be generalized in following way:

(i) (Any digit)$^{4n+1}$ = Same digit will be obtained

Example—Unit digit of $8^{53} = 8$, or, Unit digit of $7^{53} = 7$, or, Unit digit of $4^9 = 4$.

(ii) (Any odd digit)^{4n} – Unit digit obtained = 1, where, n is any natural number.

Exception—if the odd digit = 5, then the unit digit obtained for $5^{\text{any natural number}} = 5$.

(iii) (Any even digit)^{4n} – Unit digit obtained = 6, where, n is any natural number.

Understand that all the above points (i), (ii) and (iii) are only necessary conditions, and not sufficient conditions.

It means that if power is 4n, odd digit (except 5) will give 1 at the units place; and if power is 4n, even digit will give 6 at the units place; and 0 will give 0 at the unit place. However, if the unit digit of (odd number)^x = 1, then we cannot say that the power x has to be equal to 4n (n being a natural number). We will discuss it further with the help of following questions.

Worked-out Example 1 In the following questions, on the right hand side, unit digit obtained is given. For how many of the following, can we exactly find out the remainder when x is divided by 4?

(a) (Odd digit)^x = 1

(b) (A natural number)^x = 5

(c) (Even digit)^x = 4

(d) (Even digit)^x = 6

(e) (Even digit)^x = 8

(f) (Odd digit)^x = 7

☝ ***Solution***

(a) x can take more than one value. For example, 9^2 will give unit digit 1 as well as 9^4 will also give unit digit 1. Hence, no unique remainder can be calculated when x is divided by 5.

(b) If the cycle = 1, we cannot find out the answer to above question. Unit digit 5 is possible only if natural number taken = 5. We can also see that x can take any natural number value since 5^N will always give unit digit 1 (for all natural number value of N). Hence, no unique remainder can be calculated when x is divided by 5.

(c) x can take more than one value. For example—2^2 or 4^1 will give unit digit = 4. Hence, no unique remainder can be calculated when x is divided by 5.

(d) x can take more than one value. For example—6^1 or 4^2 will give unit digit = 6. Hence, no unique remainder can be calculated when x is divided by 5.

(e) x can take more than one value. For example—2^3 or 8^1 will give unit digit = 8. Hence, no unique remainder can be calculated when x is divided by 5.

(f) x can take more than one value. For example—3^3 or 7^1 will give unit digit = 7. Hence, no unique remainder can be calculated when x is divided by 5.

In my perception, method 2 (Using cyclicity of digits) and method 3 (Using Generalization) are of maximum use to calculate the unit digit. We will discuss the usefulness of method 1 (Using remainder method) while discussing the process to find out tens place digit.

Worked-out Example 2 Find the unit digit of 512^{2487}.

☝ ***Solution*** 1st of all, unit digit of 512^{2487} = unit digit of 2^{87}.

We know that 2^{4n} will give 6 as the unit place. So unit digit of 2^{84} = 6

$2^{87} = 2^{84} \times 2^3$ = ... 6 × 8 = ... 8

Note It can be seen that in such simple cases, using generalization makes the whole solution very simple. I recommend this method for solving the questions that involve powers only at one level.

Worked-out Example 3 Find the unit digit of (a) $7^{11^{22^{33^{44}}}}$ (b) $5^{11^{22^{33^{44}}}}$ (c) $8^{11^{22^{33^{44}}}}$

☝ ***Solution***

(a) Its difficult to write down $7^{11^{22^{33^{44}}}}$ as $7^{4n} \times 7^x$ as we have done in the last question directly.

So to solve any such question we would go by a 3 step process:

Step 1—Identify the cyclicity of the base of the power.

Step 2—Calculate the remainder obtained when power is divided by the length of the cycle.

Step 3—Assign the unit digit finally.

Applying these steps to solve this question:

Step 1—In this case, base = 7. Cyclicity of 7 = 4

Step 2—Divide the power $11^{22^{33^{44}}}$ by 4 to find out the remainder.

Remainder obtained = $(-1)^{\text{even number}} = 1$. [to know more about remainders, visit the chapter Remainder].

Hence, $7^{11^{22^{33^{44}}}} = 7^{4n+1}$.

Step 3—Unit digit of $7^{11^{22^{33^{44}}}}$ = Unit digit of 7^{4n+1}.

Unit digit of $7^{4n+1} = 7^{4n} \times 7^1 =$ ..... $1 \times 7 =$ 7

Note It can be seen that in these cases of multiple powers, its better to use the cyclicity method of finding the unit digit.

(b) Unit digit of $5^{11^{22^{33^{44}}}}$.

☝ ***Solution*** Length of cycle for the unit digit of 5 = 1. In other words, for all the natural number powers of 5, unit digit obtained = 5. Hence, 5 is the unit digit.

(c) $8^{4n+1} = 8^{4n} \times 8^1 = \ldots . 6 \times 8 = \ldots . 8$ as unit digit.

Worked-out Example 4 What is the unit digit of 7^{7^n}?

Step 1—In this case, base = 7. Cyclicity of 7 = 4

Step 2—Divide the power 7^{7^n} by 4 to find out the remainder.

Remainder obtained = -1^n. Now it depends upon the value of n that if the remainder is 1 or –1 (=3).

And depending upon this, if remainder = 1 then $7^{7^n} = 7^{4n+1} \Rightarrow$ unit digit = 7 or

If remainder = 3 then $7^{7^n} = 7^{4n+3} \Rightarrow$ unit digit = 3.

Hence, unique answer cannot be determined.

In some of the cases the expression for which unit digit is to be calculated may not be given directly. Let us see this with the help of following worked-out example:

Worked-out Example 5 What is the last non-zero digit of 28!?

☝ ***Solution*** Unlike the previous questions, to solve this question we need to identify that which prime number is coming up with what power? At the onset, it can be seen that there are going to be few zeroes. So 1st we will calculate the number of zeroes, and then the 1st digit coming after the zeroes (at the end) will the last non-zero digit.

$28! = 2^{25} \times 3^{13} \times 5^6 \times 7^4 \times 11 \times 13 \times 17 \times 19 \times 23$.

Since there are 6 sets of 5 and 2, hence, this number will have 6 zeroes at the end. So we would remove these six zeroes from the end of 28! And whatever digit is obtained at the end will be the last non zero digit. So, we would remove 6 2's and 6 5's from the prime factorization of the number.

So, now we write $28! = 2^{19} \times 3^{13} \times 5^0 \times 7^4 \times 11 \times 13 \times 17 \times 19 \times 23$ [this number will not have any zero at its end] $\times [2^6 \times 5^6]$

Unit digit of $2^{19} \times 3^{13} \times 5^0 \times 7^4 \times 11 \times 13 \times 17 \times 19 \times 23$ = ...8 × ...3 × 1 × ..1 × 1 × 3 × 7 × 9 × 3 = 8

Worked-out Example 6 What is the unit digit of 1! + 2! + 3! + 4! +.....+100!?

Solution It can be seen that all the terms 5! onwards will have unit digit = 0.

So the net unit digit will come only from 1st four terms:

1! + 2! + 3! + 4! = 1 + 2 + 6 + 4 = 13 ⇒ Hence 3 is the unit digit.

TEST YOUR LEARNING 1

1. Find the unit digit of the following expression:

1. 784^{345}

2. 637^{56934}

3. $1!^{1} + 2!^{2} + 3!^{3} + 4!^{4} + 5!^{5} + \ldots\ldots\ldots\ldots 100!^{100}$

4. $1!^{1!}! + 2!^{2!}! + 3!^{3!}! + 4!^{4!}! + 5!^{5!}$ ---------- $+ 1000!^{1000!}$

5. $3^{332}/50$ = (abcd.....). xy; where number in the bracket is before the decimal, and xy are the 1st two digits obtained after division. Find the value of y.

6. $1^{2^3} + 2^{3^4} + 3^{4^5} + 4^{5^6} + 5^{6^7}$

7. 81 × 82 × 83 × .....× 89

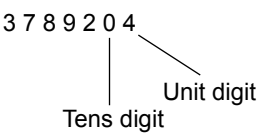
8. 100 × 200 × 300 × 400 × 1900 × 2000

9. $734^{235} \times 378^{457} \times 2156^{545}$

10. $246^{457} \times 347^{397} \times 378^{6254734}$

CALCULATING THE TENS PLACE DIGIT

In any number, 2nd digit from the right is known as tens place digit. For example,



Method to find out the tens place digit:

There are several methods of finding out the tens place digit. We will discuss three of those:

Method 1—Using Remainder Method

When we divide a number by 100, remainder obtained is going to be the last two digits of that number.

For example,

(i) Last two digits of 568 = 68 = Remainder obtained when 568 is divided by 100.

(ii) Last two digits of 5621× 1047 × 218 = Last two digits of 21× 47 × 18 = Remainder obtained when $\frac{21\times47\times18}{100}=66$

Worked-out Example 7 Find the tens place digit of 7^{1000}.

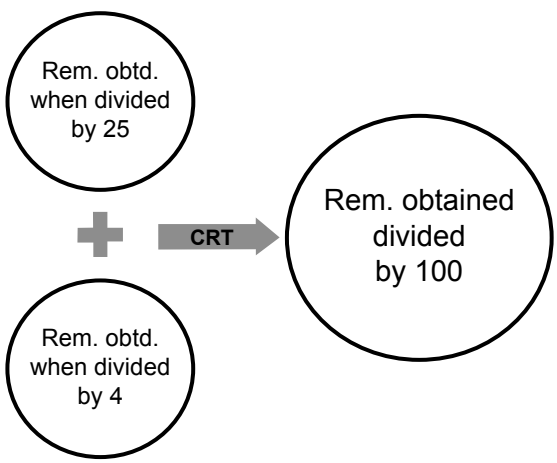
☝ ***Solution*** Remainder obtained when 7^{1000} is divided by 100 = Remainder obtained $\frac{7^{1000}}{100}=$ Remainder obtained $\frac{49^{500}}{100}=$ Remainder obtained $\frac{2401^{500}}{100}=01.$

Hence, tens place digit = 0, and unit place digit = 1.

Worked-out Example 8 Find the tens place digit of 517^{100}.

☝ ***Solution*** Remainder obtained $\frac{517^{100}}{100}=$ Remainder obtained $\frac{17^{100}}{100}$

Instead of dividing a number by 100, we can divide the same number by 25 and 4 individually, and then use Chinese Remainder Theorem to find out the remainder when divided by 100. [Prime factorization of 100 = $5^2 \times 2^2$. Instead of taking 5^2 and 2^2, we could have taken any two parts or more than two parts of 100 such that LCM of all the parts = 100.]



Remainder obtained when divided by $4=\frac{17^{100}}{4}=1$ Remainder obtained $\frac{17^{100}}{25}=$ Remainder obtained $\frac{289^{50}}{25}=$ Remainder obtained $\frac{-11^{50}}{25}=$

Remainder obtained $\frac{1331^{16}\times11^2}{25}=$ Remainder obtained $\frac{6^{16}\times121}{25}=$

Remainder obtained $\frac{1296^4 \times (-4)}{25}$ = Remainder obtained $\frac{-4^5}{25}$ = Remainder obtained $\frac{1024}{25} = 24$

Remainder obtained when divided by 25 = 24

Remainder obtained when divided by 4 = 1

Now we will use Chinese Remainder theorem to find out the number that will satisfy the above conditions.

Number that gives remainder 24 when divided by 25= 24, 49, 74, 99. Out of these numbers, 49 is the number that will give remainder 1 when divided by 4.

Hence Remainder obtained $\frac{17^{100}}{100} = 49$. So the last two digits of $17^{100} = 49$. So the tens place digit = 4.

Note 1 You will get only one number from 01 to 100 that will satisfy the conditions when divided by 25 and 4.

Note 2 If you have done the division right for the parts (as in this case 25 and 8), then you will definitely get one number from 01 to 100 satisfying the remainders. And if you do not get a number from 01 to 100 satisfying the remainders, you must have calculated the remainder/s wrong.

Method 2—Using Generalization

We have seen the cyclicity of unit digit. Similarly, exists cyclicity of tens place digit.

Following is the list of cyclicity of digits for tens place:

Digits	Cyclicity
2, 3, 8	**20**
4, 9	10
5	1
6	5
7	4

Worked-out Example 9 Find the tens place digit of 7^{1000}.

☝ ***Solution*** Using the tens place digit table given above, cyclicity of 7 = 4. Hence, tens place digit of 7^4 = tens place digit of 7^8 = tens place digit of $7^{4n} = 01$.

Worked-out Example 10 Find the tens place digit of 5^{100}.

☝ ***Solution*** 5^N (N > 1; N being a natural number) will have 25 as the last two digits. Hence, last two digits of $5^{100} = 25$.

Besides the above table, we have some more generalizations that will help us in finding the tens place digit of any number. 1st we will see it for odd numbers and then for even numbers.

For Odd Numbers

1. (Any odd number with unit digit 1, 3, 7, 9)^{20N} = Will have 01 as the last two digits.
   For example, last two digits of 37^{20} = last two digits of 37^{40}= last two digits of 37^{60} = 01
   Last two digits of 39^{20} = last two digits of 63^{40} = last two digits of 81^{100} = 01
2. (Any odd number with unit digit 5)$^{N>1}$= will have 25 as the last two digits.
3. Last two digits of following numbers are same:
   $21^5 = 41^5 = 61^5 = 81^5 = 1^5 = 01$ (last two digits)
   $11^5 = 31^5 = 51^5 = 71^5 = 91^5 = 51$ (last two digits)
4. $9^{10} = 19^{10} = 29^{10} = 39^{10} = 49^{10} = 59^{10} = 69^{10} = 79^{10} = 89^{10} = 89^{10} = 01$
5. Let us re-write point number 1 and 4 given above:
   Unit digit = 1 $\Rightarrow$ $(A1)^{10N}$ where, A is any digit at the tens place will give last two digits = 01
   (If A is even, then even for 5N power, tens place digit = 01)
   Unit digit = 3 $\Rightarrow$ $(A3)^{20N}$ where, A is the any digit at the tens place will give last two digits = 01
   Unit digit = 7 $\Rightarrow$ $(A7)^{20N}$ where, A is the any digit at the tens place will give last two digits = 01
   Unit digit = 9 $\Rightarrow$ $(A9)^{10N}$ where, A is the any digit at the tens place will give last two digits = 01
6. Following is pattern of tens place digit of 11:

Power of 11	1	2	3	4	5	6	7	8	9	10
Last two digits	11	21	31	41	51	61	71	81	91	01

Worked-out Example 11 Find the tens place digit of 741^{463}.

☝ ***Solution*** Tens place digit of 741^{463} = tens place digit of 41^{463}.

We know that last two digits of 41^{5N} (point number 3 given above) = 01

Tens place digit of 41^{463} = Tens place digit of $41^3 = 41 \times 41 \times 41 = 1681 \times 41 =$ .....$81 \times 41 = 21$ (we do not have to multiply these actually to find the last two digits, simply add (8 + 4) removing 1 from both the numbers. Unit digit of 8 + 4 = 2 = tens place digit of 81×41. This is exactly the same way we find last non zero digit—by neglecting 0).

So, tens place digit = 2.

Worked-out Example 12 Find the tens place two digits of 31^{886}.

☝ ***Solution*** Last two digits of 31^{886} = Last two digits of $31^{880} \times$ Last two digits of $31^6 = 01 \times 81 = 81$.

Hence, tens place digit = 8

Worked-out Example 13 Find the tens place digit of 42^5.

☝ ***Solution*** Tens place digit of 42^5 = Tens place digit of $2^5 \times 21^5 = 32 \times 01 = 32$.

For Even Numbers

1. (Any even number with unit digit 2, 4, 6, 8)^{20N} = Will have 76 as the last two digits.
   For example, last two digits of 12^{20} = last two digits of 12^{40}= last two digits of 12^{60} = 76
   Last two digits of 34^{20} = last two digits of 36^{40} = last two digits of 82^{100} = 76
2. If 6 is the unit place of any number, for example, 26, 36 etc. Then this number raised to every power that is a multiple of 5 will give 76 as the last two digits.
   Last two digits of $(M6)^{5N}$, where, M is any digit at the tens place = 76
3. If 4 is the unit place of any number, for example, 24, 44 etc., then this number raised to every power that is a multiple of 5 but not a multiple of 10 (in other words, odd multiples of 5) will give 24 as the last two digits.
   Last two digits of $(M4)^{5N}$, where, M is any digit at the tens place = 24. It can also be concluded from here that $(M4)^{10N}$ will give 76 as the last two digits.
   For example, last two digits of 38^5 = last two digits of 38^{15} = 24
   Last two digits of 38^{10} = Last two digits of 38^5 × Last two digits of 38^5 = 24 × 24 = 76
4. Last two digits of 76^N, where, N is any natural number = 76.
5. Last two digits of 24^{2N}, where, N is any natural number = 76 (N = 1, 2, 3, ...).
6. Last two digits of 24^N, where, N is any odd natural number = 24 (N= 1, 3, 5.....).
7. Last two digits of 26^{N+1}, where, N is any natural number = 76.
8. Last two digits of $76 \times 2^{N+1}$, where, N is any natural number = Last two digit of 2^{N+1}.
   For example, Last two digit of 76 × 04 = 04, Last two digit of 76 × 32 = 32, Last two digit of 76 × 16 = 16.

Worked-out Example 14 Find the tens place digit of 784^{1000}.

☝ ***Solution*** Tens place digit of 784^{1000} = Tens place digit of 84^{1000} = Tens place digit of $4^{1000} \times 21^{1000}$
Let us first find the tens place digit of each of the parts individually, and then we will multiply to find out the net tens place digit.
Last two digit of 21^{1000} = Last two digit of 21^{5N} = 01
Last two digit of 4^{1000} = 76
Hence, net Last two digit of 784^{1000} = 76 × 01 = 76 ⇒ Tens place digit = 7
Alternatively, as we have seen above, (any even number)^{20N}, will give 76 as the last two digits.

Worked-out Example 15 Find the tens place digit of 2^{663}.

☝ ***Solution*** Last two digits of 2^{663} = Last two digits of 2^{660} × Last two digits of 2^3 = 76 × 08 = 08 (for this multiplication, look at point 6 above)
Hence, tens place digit = 0

Worked-out Example 16 Find the tens place digit of 586^{1536}.

Solution Tens place digit of 586^{1536} = tens place digit of 86^{1536}

Tens place digit of 86^{1536} = Tens place digit of 86^{1520} × Tens place digit of 86^{16} = ...76 × Tens place digit of 86^{16}.

Now, we will find out the tens place digit of 86^{16} separately.

We have already seen that Last two digits of $(M6)^{5N}$, where, M is any digit at the tens place = 76.

Hence, Last two digits of 86^{15} = 76

So last two digits of 86^{1536} = Tens place digit of 86^{1520} × Tens place digit of 86^{15} × Tens place digit of 86^{1} = 76 × 76 × 86 = 36

Hence, tens place digit = 3

SUMMARY:

Now, we will summarize the whole discussion that we had in Method 2 and define a general approach to find out the tens place digit:

Step 1—1st see if the number is even or odd

Step 2—Take only last two digits of the number.

Step 3—If it is an odd number, let it be and if it is an even number write it in two parts – purely even part and purely odd part.

Step 4—Write the power in the format of 20N + K, where, N and K are natural numbers.

Last two digits of (odd number ending with 1, 3, 7, 9)^{20N} = 01.

Last two digits of (odd number ending with 5)^{20N} = 25.

Last two digits of (even number ending with 2, 4, 6, 8)^{20N} = 76.

Last two digits of (even number ending with 0)^{20N} = 00.

For remaining 'K' powers, go through the actual calculations or take help of the other points given above.

Worked-out Example 17 Find the tens place digit of $582^{456} \times 375^{352}$.

Solution It is an absolutely no-brainer. There are enough number of 2's and 5's that the last two digits are going to be "00". Hence, tens place digit = 0.

Worked-out Example 18 Find the tens place digit of $582^{456} \times 377^{342}$.

We will find out the tens place digits of each of the numbers individually.

Tens place digit of 582^{456} = Tens place digit of 82^{456} = Tens place digit of 82^{440} × Tens place digit of 82^{16} = 76 × Tens place digit of 82^{16}

Let us now find out for 82^{16}: Last two digits of 82^{2} = 24

Hence, last two digits of 82^{4} = 76 Hence, last two digits of 82^{8} = 76 Hence, last two digits of 82^{12} = 76

Hence, last two digits of 82^{16} = 76 (since last two digits of 24^{2} = 76).

So last two digits of 582^{456} = Last two digits of 582^{440} × Last two digits of 582^{16} = 76 × 76 = 76.

Now, we will find out the last two digits of 377^{342}.

Tens place digit of 377^{342} = Tens place digit of 77^{342} = Tens place digit of 77^{340} × Tens place digit of 77^2 = 01 × Tens place digit of 77^2.

Tens place digit of 77^2 = 29.

Last two digits of 377^{342} = last two digits of 377^{340} × last two digits of 377^2 = 01 × 29 = 29.

Last two digits of $582^{456} \times 377^{342}$ = 76 × 29 = 04.

Tens place digit of $582^{456} \times 377^{342}$ = 0.

TEST YOUR LEARNING 2

1. Find the tens place digit of the following numbers:

1. 27^{514} **2.** 46^{225}

3. 107^{543} **4.** 117^{543}

5. 312^{543} **6.** 369^{312}

7. $503^{108} + 426^{253}$ **8.** $247^{14!} + 248^{15!} + 249^{16!} + 250^{17!}$

HUNDREDS PLACE DIGIT

We are going to see and understand the process of finding out hundreds place digit or 1000's place digit and beyond.

Using Remainder Method

When we divide a number by 1000, remainder obtained is going to be the last three digits of that number.

For example,

(i) Last three digits of 53458 = 458 = Remainder obtained when 53458 is divided by 1000.

(ii) Last three digits of 5621 × 1047 × 218 = Last three digits of 621 × 47 × 218

= Remainder obtained when $\frac{621 \times 47 \times 218}{1000} = 766$

So unit digit of 5621 × 1047 × 218 = 6.

Tens digit of 5621 × 1047 × 218 = 6.

Hundreds digit of 5621 × 1047 × 218 = 7.

Worked-out Example 19 Find hundreds digit of 1! × 2! × 3! × .....× 10!.

☝ ***Solution*** We can see that there are going to be more than 3 zeroes at the end. Hence, hundreds digit = 0

PRACTICE EXERCISE 1

1. What is the unit digit of N =
$1^{2^3} + 2^{3^4} + 3^{4^5} + 4^{5^6} \ldots\ldots\ldots\ldots\ldots\ldots + 100^{101^{102}}$? ?

1. 0 **2.** 1 **3.** 2 **4.** 3
5. None of these

Directions for questions (2–3): Read the information below and solve the questions.

Consider the following pyramid of natural numbers:

1
2 3
4 5 6
7 8 9 10

As it can be seen that 1st row has only 1 number = 1, 2nd row has two numbers = 2, 3 and so on.

2. What is the unit digit of last number in the 50th row?

1. 0 **2.** 5 **3.** 4 **4.** 8
5. 2

3. What is the unit digit of 1st number in the 50th row?

1. 0 **2.** 6 **3.** 5 **4.** 4
5. 1

4. Find the unit digit of 5^{5^n}

1. 0 **2.** 1 **3.** 3 **4.** 5
5. 7

5. What is the unit digit of $11^{(5!)!} - 1$?

1. 1 **2.** 0 **3.** 2 **4.** 3
5. None of these

6. What is the unit digit of $12^{(5!)!} - 13^{(5!)!} - 17^{(5!)!}$?

1. 9 **2.** 4 **3.** 0 **4.** 3
5. None of these

7. What is the unit digit of $23^{(5!)!} + 27^{(5!)!} - 12^{(5!)!}$?

1. 9 **2.** 4 **3.** 0 **4.** 3
5. 6

8. Unit digit of $34^{5!} + N^{5!} - 7^{5!} = 1$. Also given that $21 < N < 41$, and N is a natural number. How many values of N are possible?

1. 9 **2.** 4 **3.** 12 **4.** 14
5. 8

9. Unit digit of $72^{5!} + N^{5!} - M^{5!} = 7$, where M and N are natural numbers and $32 < M < 47$ and $64 < N < 74$. How many set of values of M and N are possible?

1. 81 **2.** 25 **3.** 10 **4.** 4

5. 6

10. What is the unit digit of 12! + 13!?

1. 0 **2.** 1 **3.** 2 **4.** 3

5. None of these

Directions for questions (11–12): Go through the following information given below and solve the questions based on it.

Given that:

(An odd number)^x gives unit digit = 9

(An even number)^y gives unit digit = 8.

11. How many different values of unit digit are possible for x + y?

1. 6 **2.** 7 **3.** 8 **4.** 9

5. 10

12. How many different values of unit digit are possible for xy?

1. 6 **2.** 7 **3.** 8 **4.** 9

5. 10

Directions for questions (13–16): Go through the following information given below and solve the questions based on it.

Given that:

(An odd number)^{2N} gives unit digit = D

(An even number)^{2K} gives unit digit = E

Where K an N are natural numbers.

13. Which of the following can never be equal to the unit digit of xy?

1. 6 **2.** 4 **3.** 0 **4.** 2

5. None of these

14. How many different values of unit digit are possible for xy?

1. 2 **2.** 3 **3.** 4 **4.** 5

5. None of these

15. $\frac{ab}{cd} = .04444444 \ldots \ldots \ldots \ldots \ldots \ldots \ldots$ till infinity

All the digits a, b, c and d are non zero. (ab) and (cd) are two digit natural numbers. What is the unit digit of numerator (ab)?

1. 2 **2.** 4 **3.** 8 **4.** 6

5. More than one value possible

16. The product of two 2 digit number is 1938. If the product of their unit's digit is 28 and that of their tens digit is15, what is the sum of two numbers?

1. 91 **2.** 81 **3.** 71 **4.** 77

5. 87

Directions for questions (17–18): Read the information below and solve the questions based on it.

Take any three digit number. Write it in descending order of its digits and then ascending order of its digits. Now take out the difference between the two numbers. Whatever result is obtained, repeat the same process again. Keep repeating the same process till infinity.

17. What is the unit digit of the number obtained at the end of 200th operation?

1. 1 **2.** 0 **3.** 5 **4.** 9

5. No unique value can be achieved

18. What is the tens place digit of the number obtained at the end of 201th operation?

1. 1 **2.** 0 **3.** 5 **4.** 9

5. No unique value can be achieved

19. Find the tens place digit of 7^{7^7}.

1. 1 **2.** 3 **3.** 4 **4.** 6

5. 8

20. Find the tens place digit of $24^{48^{72}}$.

1. 0 **2.** 3 **3.** 7 **4.** 5

5. 9

21. Find the tens place digit of $39^{48^{72}}$.

1. 1 **2.** 2 **3.** 4 **4.** 6

5. 8

22. Find the tens place digit of $49^{50^{51}}$.

1. 0 **2.** 4 **3.** 5 **4.** 6

5. 9

23. Given that m and n are natural numbers. What is the unit digit of $215^{n^m} + 1$?

1. 2 **2.** 4 **3.** 5 **4.** 6

5. More than one value possible

24. Given that m and n are natural numbers. What is the unit digit of $217^{n^m} + 1$?

1. 8 **2.** 0 **3.** 4 **4.** 6

5. More than one value possible

25. $P = x^{x^y}$, where x is any digit from 0 – 9, and y is any natural number. For how many values of x is it possible to find the unit digit of P?

1. 10 **2.** 7 **3.** 5 **4.** 3

5. 0

26. Find hundreds digit of 1! + 2! + 3! + ...+ 10!.

1. 1 **2.** 7 **3.** 9 **4.** 3

5. 0

27. What will be the ten's place digit in the following expression: 1! + 2! + 3! + ……+ 70!?

1. 1 **2.** 3 **3.** 5 **4.** 6

5. 2

28. Unit digit of the number $N = x^2 + xy + y^2 = 0$ (where x and y are natural numbers). What is the tens place digit of this number N?

1. 0 **2.** 1 **3.** 2 **4.** 8

5. Cannot be determined

29. Digits of two three digit numbers are coded with the help of alphabets. It is given that

$$\begin{array}{r} A\,B\,C \\ +\,A\,C\,B \\ \hline B\,C\,A \end{array}$$

All the alphabets represent unique digits.

What is value of unit digit of B–C?

1. 4 **2.** 6 **3.** 5 **4.** 2

5. 1

30. Find the thousands place digit of $1 \times 1! + 2 \times 2! + 3 \times 3! + ...+100 \times 100!$.

1. 2 **2.** 3 **3.** 7 **4.** 9

5. 6

SOLUTIONS AND EXPLANATIONS

TEST YOUR LEARNING 1

1. 1. 4

2. 9

3. Since 5! = 120, it can be seen that 5th term onwards ($5!^5$), all the terms will have zero at the unit place.

Hence, we are supposed to calculate the unit digit only for the 1st four terms = 1 + 4 + 6 + 6 = 17

So, 7 is the unit digit.

4. It can be seen that 5th term onwards ($5!^5$), all the terms will have zero at the unit place.

Hence, we are supposed to calculate the unit digit only for the 1st four terms = 1 + 4 + 6 + 6 = 17

5. Unit digit of 3^{332} = Unit digit of 3^{32} = 1

So, 3^{332} is going to be a large number with the unit digit = 1.

$$\frac{3^{332}}{50} = \frac{\ldots\ldots\ldots

To find out last non-zero digit of 20!, we will find $10! \times (11 \times 12 \times 13 \times \ldots\ldots\ldots \times 19) \times 20 = (\ldots 800) \times (\ldots 80) \times (20)$.

Hence, last non = zero digit = 8

9. Unit digit of $734^{235} = 4$, unit digit of $378^{457} = 8$ and unit digit of $2156^{545} = 6$.

10. Unit digit of $246^{457} = 6$.

Unit digit of 347^{397} = Unit digit of $7^{97} = 7$.

Unit digit of $378^{6254734}$ = Unit digit of 8^{34} = Unit digit of 8^{32+2} = Unit digit of $8^{32} \times 8^2$.

Unit digit of $8^{32} = 8$ and Unit digit of $8^2 = 4$.

Hence, net unit digit = $8 \times 4 = 2$

TEST YOUR LEARNING 2

1.

1. Tens place digit of 27^{514} = Tens place digit of 27^{14} = Tens place digit of 3^{42} = Tens place digit of 3^2 = Tens place digit of 09 = 0.

2. Tens place digit of 46^{225} = Tens place digit of $2^{225} \times 23^{225}$ = Tens place digit of $2^5 \times 23^5 = 32 \times \ldots 43$.

Last two digits of $2^5 = 32$.

Last two digits of $23^5 = 43$.

Net last two digits = 76 $\Rightarrow$ Tens place digit = 7.

Alternatively, we know that last two digit of $(B6)^{5N}$, where B is any digit at the tens place = 76.

3. Tens place digit of 107^{543} = Tens place digit of $7^3 \Rightarrow 7^3 = 43 \Rightarrow$ Tens place digit = 4.

4. Tens place digit of 117^{543} = Tens place digit of 17^{20K+3} = Tens place digit of $17^{20K} \times 17^3 = \ldots\ldots 01 \times 13 = 13$.

Hence, tens place digit = 1

5. Tens place digit of 312^{543} = Tens place digit of $(8^{543} \times 39^{543})$

Let us calculate the last two digits one by one:

Last two digits of 39^{543} = Last two digits of 39^{20N+3} = Last two digits of $39^{20N} \times$ last two digits of $39^3 = \ldots\ldots 01 \times \ldots 19 = \ldots\ldots 19$.

Last two digits of 8^{543} = Last two digits of 2^{1629} = Last two digits of $2^{1620} \times$ Last two digits of $2^9 = 76 \times 12$ (through actual calculation) = 12.

Hence, tens place digit = 1.

6. Last two digits = 61. Hence, tens place digit = 6.

7. Last two digits of 503^{108} = Last two digits of 3^{108} = Last two digits of $3^{100} \times 3^8$.

Last two digits of $3^{100} = 01$.

Last two digits of $3^8 = 61$.

Hence, last two digits of $3^{108} = 61$.

Last two digits of 426^{253} = Last two digits of $26^{253} = 76$.

Hence, net last two digits = 61 + 76 = 37.

Tens place digit = 3

8. $247^{14!} = 247^{20N} \Rightarrow$ Last two digits = 01 $\Rightarrow$ Tens place digit = 0.

$248^{15!} = 248^{20M} \Rightarrow$ Last two digits = 76 $\Rightarrow$ Tens place digit = 7.

$249^{16!} = 249^{20K} \Rightarrow$ Last two digits = 01 $\Rightarrow$ Tens place digit = 0.

$250^{15!} = 250^{20B} \Rightarrow$ Last two digits = 00 $\Rightarrow$ Tens place digit = 0.

So, net tens place digit = 0 + 7 + 0 + 0 = 7

PRACTICE EXERCISE 1

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
1	**2**	**2**	**4**	**2**	**2**	**5**	**5**	**3**	**1**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
5	**5**	**4**	**2**	**5**	**1**	**3**	**4**	**3**	**3**
21.	22.	23.	24.	25.	26.	27.	28.	29.	30.
4	**1**	**4**	**5**	**2**	**3**	**1**	**1**	**1**	**4**

1. Digits will follow a cycle and consequently each of those will occur 10 times. Hence unit digit = 0.

2. It can be seen that unit digit Last term of 1st row = 1.

Unit digit of Last term of 2nd row = 3 = 1 + 2.

Unit digit of Last term of 3rd row = 6 = 1 + 2 + 3.

Unit digit of Last term of 4th row = 0 = 1 + 2 + 3 + 4 = 10.

So, last term of 50th row = 1 + 2 + 3 + 4 + ……+ 49 + 50 = $\frac{50\times 51}{2} = 51\times 25 = 1275$.

Hence, unit digit = 5.

3. Taking a cue from 2, we can see that there are 50 terms in the 50th row. Hence, 1st term of 50th row = 1275 – 49 = 1226

Hence, unit digit = 6.

4. Cycle of unit digit of 5 = 1, so its independent of the value of n. Hence, unit digit of $5^{5^n} = 5$.

5. Unit digit of $11^{(5!)!} = 1$, hence, unit digit of $11^{(5!)!} - 1 = 0$.

6. Unit digit of $12^{(5!)!} = 6$.

Unit digit of $13^{(5!)!} = 1$.

Unit digit of $17^{(5!)!} = 1$.

Hence, net unit digit = 6 – 1 – 1 = 4

7. Unit digit of $12^{(5!)!} = 6$.

Unit digit of $23^{(5!)!} = 1$.

Unit digit of $27^{(5!)!} = 1$.

Hence, net unit digit = 1 + 1 – 6 = 6 (and not –4, for this subtraction to happen 10 will be borrowed from tens digit place).

8. Unit digit of $34^{5!} = 6$ and unit digit of $7^{5!} = 1$.

So unit digit of N has got one value – = 6.

All the even number in the given range of $21 < N < 41$ will give unit digit either 0 or 6 (so we are not required to count number of numbers for unit digit 6 and 0 separately). Our answer dont need zero.

So number of values of N possible = 10 – 2 = 8.

9. Unit digit of $72^{5!} = 6$.

Since the power to all the numbers are 5! = 4n, hence, only unit digits possible are 0, 1, 5, 6.

Now values of N and M should be taken in such a way that the net unit digit becomes = 1.

So, the unit digit of $N^{5!} - M^{5!} = 1 \Rightarrow$ This is possible in two ways:

(i) If unit digit of $N^{5!} = 6$ and unit digit of $M^{5!} = 5$.

Possible values of N = 3 values (66, 68, 72) and possible values of M = 2 (35, 45)

Number of sets = $3 \times 2 = 6$.

(ii) If unit digit of $N^{5!} = 1$ and unit digit of $M^{5!} = 0$

Possible values of N = 4 values (67, 69, 71, 73) and possible values of M = 1 (40)

Number of sets = $4 \times 1 = 4$

So, total number of sets = 6 + 4 = 10

10. Each of 12! and 13! are having two zeroes at the end. Hence, unit digit = 0.

Solutions for questions (11–12):

(An odd number)^x gives unit digit = 9 $\Rightarrow$ Only odd digits that will give 9 at the unit place are 3, 7, 9.

If unit digit of $3^x = 9$, then x can take 2, 6, 10, 14, 18, and so on values. We are concerned with the unit digits of these numbers which are—0, 2, 4, 6, 8.

If unit digit of $9^x = 9$, then x can take 1, 3, 5, 7, 9 and so on values. We are concerned with the unit digits of these numbers which are—1, 3, 5, 7, 9.

Combining these two gives us all the digits from 0 – 9, so we will not look further for the powers of 7^x.

So, different values of x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

(An even number)^y gives unit digit = 8 $\Rightarrow$ Only even numbers that will give 8 at the unit place are 2, 8.

If unit digit of $2^y = 8$, then y can take 3, 7, 11, 15, 19, and so on values. We are concerned with the unit digits of these numbers which are—3, 7, 1, 5, 9.

If unit digit of $8^y = 8$, then y can take 1, 5, 9, 13, 17, and so on values. We are concerned with the unit digits of these numbers which are—3, 7, 1, 5, 9.

So, different values of y = 1, 3, 5, 7, 9.

11. For x+y, all the digits from 0–9 are possible at unit place. Hence, answer = option 5.

12. xy can take any value from 0–9 (if we take y = 1, then for each value of x from 0–9, we will keep on getting the values from 0–9). Hence, answer = option 5.

Solutions for questions (13–16):

(Odd number)^{2n} can have only following unit digits—1, 9, 5.

(Even number)^{2K} can have only following unit digits—6, 4, 0.

13. It can be seen that 6×1 = unit digit of $9 \times 4 = 6$.

Similarly, unit digit of 4×1 = unit digit of $9 \times 6 = 4$.

Unit digit of 1×0 = Unit digit of 9×0 = Unit digit of 5×0 = Unit digit of 5×6 = Unit digit of $5 \times 4 = 0$

These are the only unit digits possible—0, 4, 6. Hence, answer = Option 4.

14. As can be seen above, three different unit digits are possible for xy. Hence, answer = option 2.

15. Value of (cd) = 99, value of (ab) has to be equal to 44. So one of the values of $\frac{ab}{cd} = \frac{44}{99}$.

However, this is not the only value of $\frac{ab}{cd} = 0.4444444 \ldots \ldots \ldots \ldots \ldots \ldots \ldots$ till infinity.

$\frac{44}{99} = \frac{12}{27}$ and many other ratios are possible. Hence, option 5 is the answer.

16. Product of Unit's digits = $28 \Rightarrow 28 = 7 \times 4$, so the unit digits are 7 and 4

Product of Tens digits = 15, $\Rightarrow 15 = 5 \times 3$, so the ten's digits are 5 and 3.

Hence, the 2 numbers are either 57 and 34, or 54 and 37. Multiplying these together we get,

$57 \times 34 = 1938$

$54 \times 37 = 1998$

So 57 and 34 must be the 2 numbers.

Solutions for questions (17–30):

Whatever number is taken, ultimately it will boil down to 495. Hence option 3 is the answer.

321 – 123 = 198

981 – 189 = 792

972 – 279 = 693

963 – 369 = 594

954 – 459 = 495

19. Last two digits of 7^{7^7} = Last two digits of 7^{20N+3} = Last two digits of $7^{20N} \times 7^3$ = $01 \times 43 = 43$.

Hence, tens place digit = 4.

20. Last two digits of $24^{\text{even power}} = 76$.

Hence, tens place digit = 7.

21. 1st of all we are required to find out the format of power of this number (20N + x format).

Remainder obtained when 48^{72} is divided by 20 = 16.

So, $39^{48^{72}} = 39^{20N+16}$

Last two digits of 39^{20N+16} = Last two digits of $39^{20N} \times 39^{16}$.

Last two digits of $39^{20N} = 01$.

Last two digits of 39^{16} = Last two digits of $39^{10} \times 39^{6} = 01 \times 61 = 61$.

Hence, net last two digits = $01 \times 61 = 61$.

22. We will use here the "method of finding the remainder when divided by 100" to find out the tens place digit.

$49^{50^{51}}$ can be written as 49^{4n}. Put n = 1, and value obtained = 2401. It can be seen that remainder obtained when 2401 is divided by 100 = 01. Hence, 0 is the tens place digit.

Alternatively, Last two digits obtained from $51^{\text{even number}} = 49^{\text{even number}} = 01$.

Last two digits obtained from $49^{\text{odd number}} = 49$.

Last two digits obtained from $51^{\text{odd number}} = 51$.

We can say it using different set of words—that tens place of 51 alternates between 51 and 01 and tens place of 49 alternates between 49 and 01.

23. Unit digit of any power raised to 215 = 5 (since cyclicity of unit digit of 5 = 1). Hence, resultant unit digit = 5 + 1 = 6.

24. Cyclicity of unit digit of 7 = 4. Hence, more than one value is possible.

25. 1st of all, its possible to identify the unit digit for x = 0, 1, 5, 6 = 4 values (since their cyclicity of unit digit = 1)

For x = 2, not possible to find out because for y = 1 and for y = 2, we obtain different unit digits.

For x = 3, 7, its not possible to find out (since cyclicity of unit digit = 4).

For x = 4, $P = 4^{4n} \Rightarrow$ So unit digit = 6. So can be calculated for x = 4.

For x = 8, $P = 8^{4n} \Rightarrow$ So unit digit = 6. So can be calculated for x = 8.

For x = 9, since cycle of unit digit of 9 = 2, hence, can be calculated that unit digit = 9.

So possible for 0, 1, 4, 5, 6, 8, 9 = 7 values.

26. This question can be done without using remainder method. We will go through the actual calculation method of find out the last three digits of each of these numbers.

27. Tens place digit of all the numbers 10! Onwards = 0.

So we are required to calculate tens place digit till 1! + 2! + 3! +

……+ 9!.

1! + 2! + 3! + 4! +.....+9! will end with 13 as the last two digits. Hence tens place digit = 1.

30. $1 \times 1! = (2-1) \times 1! = 2! - 1!$

$2 \times 2! = (3-1) \times 2! = 3! - 2!$

$3 \times 3! = (4-1) \times 3! = 4! - 3!$

: : : :

: : : :

$100 \times 100! = (101-1) \times 100! = 101! - 100!$

Adding all these, we obtain = 101! – 1!.

Last four digits of 101! = 0000 $\Rightarrow$ Last four digits 101! – 1! = 9999.

CAT QUESTIONS FROM THIS CONCEPT

CAT 2008

1. What are the last two digits of 7^{2008}?

1. 21 **2.** 61 **3.** 01 **4.** 41

5. 81

CAT 2005

2. The rightmost non-zero digits of the number 30^{2720} is

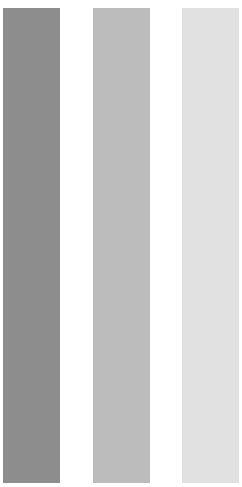
1. 1 **2.** 3 **3.** 7 **4.** 9

SOLUTIONS AND EXPLANATIONS

1. $7^{2008} = (7^4)^{502}$

Last two digits of $7^4 = 01$. Hence last two digits of $(7^4)^{502} = (01)^{502} = 01$. So, option 3 is the answer.

2. 1. $((30)^4)^{680} = (8100)^{680}$. Hence the right most non-zero digit is 1. So, option 1 is the answer.



Chapter 9

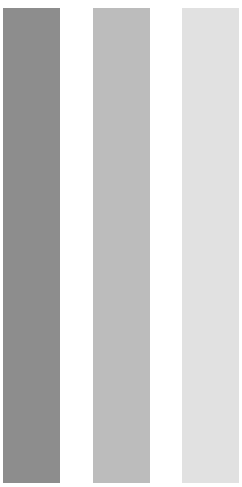
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DEMYSTIFYING NUMBER SYSTEM

SECTION TESTS

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GENERAL INSTRUCTIONS



- Each of the section tests contains 20 questions.
- Each question carries 4 marks.
- There is a negative marking of 25%.
- Some of these test papers contains 5 options, and remaining 4 options each.

SCORE TRACKER

SECTION TEST 1

Section Test 1	Attempts	Right	Wrong	Marks	%Accuracy

SECTION TEST 2

Section Test 2	Attempts	Right	Wrong	Marks	%Accuracy

SECTION TEST 3

Section Test 3	Attempts	Right	Wrong	Marks	%Accuracy

SECTION TEST 4

Section Test 4	Attempts	Right	Wrong	Marks	%Accuracy

SECTION TEST 5

Section Test 5	Attempts	Right	Wrong	Marks	%Accuracy

SECTION TEST 1

1. $f(x) = x^{x!}$ And f(x) is defined for all the natural number values of x.

Find the tens place digit of $\sum_{1}^{20} f(x)$.

1. 0 **2.** 1 **3.** 3 **4.** 5

5. 7

2. f(x) is a real valued function with following properties:

f(1) = 3 and f(3a) = a × f(a) for any positive integer a, what is the remainder when $f(3^{90})$ is divided by 10?

1. 0 **2.** 1 **3.** 3 **4.** 9

5. None of these

3. If the tens digit of a perfect square number is 7, how many units digits are possible?

1. 1 **2.** 2 **3.** 3 **4.** 4

5. 5

4. First 250 natural numbers on base 9 are written. How many of these numbers will be neither divisible by 2 nor divisible by 5?

1. 50 **2.** 100 **3.** 150 **4.** 200

5. None of these

5. N is natural number such that N divides (N – 1)!. For what kind of values of N, will this division occur?

1. Any natural number **2.** Any prime number

3. Any composite number **4.** Any composite number >4

5. No definite answer can be given

6. Given that n is a natural number, then the number (n+1)(n+2).....(n+10) cannot be a

(i) Prime number (ii) Composite Number

(iii) Perfect Square

1. (i) and (iii) only **2.** (ii) and (iii) only

3. (i) only **4.** (iii) only

5. None of these

7. In an examination a student was asked to find (3/14) of a certain number. By mistake he found (3/4) of it. His answer was 150 more than the correct answer. The given number is

1. 180 **2.** 240 **3.** 280 **4.** 290

5. None of these

8. Find the remainder when $7^{21} + 7^{22} + 7^{23} + 7^{23}$ is divided by 25.

1. 0 **2.** 2 **3.** 4 **4.** 6

5. 19

9. The number $3^9 + 3^{12} + 3^{12} + 3^n$ is a perfect cube of an integer for natural number n equalling:

1. 12 **2.** 13 **3.** 14 **4.** 15

5. 17

10. Of the two-digit numbers (those from 10 to 99, both inclusive), how many have unit digit greater than the tens place digit?

1. 38 **2.** 40 **3.** 36 **4.** 41

5. 35

11. Which of the following cannot be the difference of two four-digit numbers formed by the four digits 4, 6, 7 and 9, where each digit can be repeated?

1. 5085 **2.** 5095 **3.** 5795 **4.** 2095

5. All are possible

12. The difference between the squares of two consecutive odd integers is always divisible by

1. 8 **2.** 7 **3.** 6 **4.** 3

5. None of these

13. Find a positive number which when increased by 17 is equal to 60 times the reciprocal of the number.

1. 20 **2.** 10 **3.** 3 **4.** 17

5. 21

14. A number consists of two digits whose sum is 7. If the digits are reversed, then the number is increased by 27. The number is

1. 25 **2.** 34 **3.** 16 **4.** 52

5. 36

15. Find the first and last digits of 2^{43} respectively.

1. 8 and 8 **2.** 2 and 8 **3.** 4 and 8 **4.** 8 and 2

5. 8 and 4

16. Given that n is an odd positive integer, then what is the remainder when $2269^n + 1779^n + 1730^n - 1776^n$ is divided by 2001?

1. 1 **2.** 0 **3.** 19 **3.** 1999

5. 2000

17. Each family in a locality has at most two adults, and no family has fewer than 3 children. Considering all the families together, there are more adults than boys, more boys than girls, and more girls than families, then the minimum possible number of families in the locality is

1. 4 **2.** 5 **3.** 2 **4.** 3

5. 1

18. A book contains 30 stories. Each story has a different number of pages under 31. The first story starts on page 1 and each story starts on a new page.

What is the largest possible number of stories that can begin on odd page numbers?

1. 20 **2.** 23 **3.** 29 **4.** 25

5. None of these

19. Assuming that p, q, r and s are distinct prime numbers greater than 10, find the number of common multiples of $14 \times p^7 \times q^5 \times r^4$ and $98 \times p^3 \times q^{15} \times s^7$ which are factors of the product of the given two numbers?

1.192 **2.** 96 **3.** 48 **4.** 24

5. None of these

20. A number when written to the base 16 contains only 3 zeros and three 1's and no other digits. Find the maximum number of zeros in the number when it is represented in base 2.

1. 16 **2.** 18 **3.** 20 **4.** 24

5. None of these

ANSWERS AND EXPLANATIONS

ANSWER GRID

1	2	3	4	5	6	7	8	9	10
3	**4**	**1**	**2**	**4**	**1**	**3**	**1**	**3**	**3**
11	12	13	14	15	16	17	18	19	20
3	**1**	**3**	**1**	**1**	**2**	**4**	**2**	**2**	**2**

1. $F(1) = 1^{1!}$

$F(2) = 2^{2!}$

$F(3) = 3^{3!}$

$F(4) = 4^{4!}$ And so on.

$\sum_{1}^{20} f(x) = f(1) + f(2) + f(3) + ...+ f(19) + f(20).$

Instead of finding the actual sum, we would find out the last two digits of all these numbers and find the sum of tens place digit, later on.

Last two digits of $F(1) = 1^{1!} = 01$

Last two digits of $F(2) = 2^{2!} = 04$

Last two digits of $F(3) = 3^{3!} = 61$

Last two digits of $F(4) = 4^{4!} = 56$

Last two digits of $F(5) = 5^{5!} = 25$

Last two digits of $F(6) = 6^{6!} = 76$

We can see that 5! Onwards, all the factorials like 6!, 7!, ...are the multiples of 20.

Hence, we are not required to actually calculate the tens place digit, we can simply keep on writing the last two digits depending upon number being even or odd.

Last two digits of $F(7) = F(9) = F(11) = F(13) = F(17) = F(19) = 01$.

Last two digits of $F(8) = F(12) = F(14) = F(16) = F(18) = 76$.

Last two digits of $F(10) = F(20) = 00$.

Last two digits of $F(15) = 25$.

Hence, sum of last two digits $= 7 \times 01 + 04 + 61 + 56 + 2 \times 25 + 6 \times 76 + 2 \times 00 = 634$.

Hence, tens place digit = 3

3. We are required to take the square till 99 only, because last two digits of the numbers beyond 100 will be repetitive as that of starting from square of 0.

Further, it can be seen that last two digits of $(50+N)^2$ will be same as the last two digits $(50 - N)^2$.

So, now we will look at the numbers from 1 to 50 only. It can be seen again that last two digits of $(25 + N)^2$ is same as last two digits of $(25 - N)^2$. Hence, we will now look at the squares till 25.

Visualizing upon, we can see that tens place digit 7 occurs in case of 24^2 = 576, and this is the only case less than 25 when tens place digit of a perfect square = 7. So unit digit = 6, and this is the only possibility.

4. It does not matter if its base 9 or base 10. Number of numbers divisible in any range will remain same.

Hence, out of 1st 250 natural numbers, number of numbers not divisible by 2 or 5 = 250. (1 – 1/2) (1 – 1/5) = $250 \times \frac{1}{2} \times \frac{4}{5}$ = 100.

5. Can be checked using options.

Alternatively, this is one of the standard derivations from Wilson's Theorem (discussed in Remainder Chapter).

7. Ans. 3

$$\frac{3}{4} \times -\frac{3}{14}x = 150 \text{ or, } \frac{15}{28}x = 150$$

$$\text{So, } x = \frac{150 \times 28}{15} = 280.$$

8. Ans. 1

Given expression = $7^{21}(1 + 7 + 7^2 + 7^3) = 7^{21} \times 400$ which is divisible by 25.

Hence, remainder is zero.

9. Ans. 3

For n = 14, $3^9(1 + 3^3 + 3^6 + 3^5) = 3^9(1 + 27 + 729 + 243) = 3^9 \times 10^3$.

10. Ans. 3

12 to 19 = 8, 23 to 29 = 7,

34 to 39 = 6, 45 to 49 = 5,

56 to 59 = 4, 67 to 69 = 3,

78 to 79 = 2, 89 = 1.

Total = 36.

11. Ans. 3

12. Ans. 1

$(2x + 3)^2 - (2x + 1)^2$

$= 4x^2 + 12x + 9 - (4x^2 + 4x + 1)$

$= 8x + 8 = 8(x + 1)$.

13. Ans. 3

Let the number be x

$$x + 17 = \frac{60}{x} \Rightarrow x^2 + 17x - 60 = 0$$

$\Rightarrow (x + 20)(x - 3) = 0 \Rightarrow x = 3 (x > 0)$.

Alternatively, ideally these questions should be done using options.

14. Ans. 1

Let the number be $10x + y$, then number obtained by reversing the digits = $10y + x$. Given is $9(x - y) = 27 \Rightarrow x - y = 3$.

$x + y = 7$

$\Rightarrow y = 5, x = 2$

So, number = $10x + y = 25$.

16. Hint: $2001 = 3 \times 23 \times 29$. Now divide the given expression one by one by each of these three prime numbers.

17. Go through the options.

18. Let us designate stories with an odd number of pages odd stories and stories with an even number of pages even stories. There are 15 odd stories and 15 even stories. Odd stories change the starting page of the next story (if odd page is the starting, then even page will be the starting of the next story, and if even page is the starting, then odd page will be the starting of the next story too), whereas, even stories do not change the starting page of the next story (if odd page is the starting, then odd page will be the starting of the next story too, and if even page is the starting, then even page will be the starting of the next story too). So, the odd stories must start alternately on odd and even pages. Hence, 8 of them must start on odd pages and 7 on even pages (irrespective of how the stories are arranged). We can, however, control the even stories. In particular, if we put each of them after an even number of odd stories, then they will all begin on odd pages. So, number of odd stories = 15 + 8 = 23.

Hence, option 2 is the answer.

20. Number contains exactly 3 zeros and 3 ones. Maximum and minimum of such numbers is $(111000)_{16}$ and $(100011)_{16} = (16^5 + 16^4 + 16^3)$ and $(16^5 + 16^1 + 16^0) = (2^{20} + 2^{16} + 2^{12})$ and $(2^{20} + 2^4 + 2^0)$.

In any case, the number on any base 10 will be the sum of three powers of 2 with the maximum power of 2 being 20.

Hence, in base 2 the number will have exactly 21 digits with exactly three 1's and 21 – 3 = 18 zeros.

SECTION TEST 2

1. What are the last three digits of $2003^{2002^{2000}}$?

1. 307 **2.** 921 **3.** 467 **4.** None of these

2. When a natural number N is divided by 19, remainder obtained is 8. Which of the following can never be the remainder when N is divided by 19K, a natural number multiple of 19?

1. 65 **2.** 27 **3.** 103 **4.** 85

3. How many of the following numbers will divide $1980^{1981^{1982}} + 1982^{1981^{1980}}$ completely?

(i) 1981^{1980} (ii) 1981^{1981} (iii) 1981^{1979}

1. (i) and (ii) only **2.** (ii) and iii only

3. (i) and (iii) only **4.** (i), (ii) and (iii)

4. A large number is formed by writing 1st 60 natural numbers side by side as follows:

1234567891011121314…….5960. Mr Hari, a sleepwalker, comes and erases 100 digits out of this large number. Which of the following could be the largest possible sum of leftover digits?

1. 79 **2.** 80 **3.** 81 **4.** 82

Directions for questions (5–20): Read the directions given below and solve the questions based on it.

Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4 as the answer—if question cannot be answered even by using both the statements.

5. P is an integer. Is $(P+a) \times (P + b)$ an even integer?

Statement 1: a is an even integer.

Statement 2: b is an even integer.

6. In a row of 15 players, who is standing at the middle position?

Statement 1: Gagan is standing 5th from right end of the row and Mohini is standing 5th from the left end of the row.

Statement 2: Abhishek is standing equidistant from Gagan and Mohini.

7. What is the value of N, where N is an integer?

Statement 1: $N^4 = 1$

Statement 2: $N^3 = 1$

8. What is the number of factors of $4 \times N$ (where N is a natural number)?

Statement 1: 2N is having 20 factors.

Statement 2: 3N is having 15 factors.

9. P is a prime number > 100. Which of the following is the largest number dividing (P – 1)(P – 2)(P + 1)(P + 2) always?

1. 120 **2.** 240 **3.** 360 **4.** None of these

10. LCM of 10, 20 and N = 60. How many values of N will be there (N is a natural number)?

1. 1 **2.** 4 **3.** 6 **4.** 8

11. A person was adding the page numbers of a book starting from 1. After some time, he found the sum to be equal to 1000 and realizes that he has missed out one leaf (and consequently both the page numbers on it on either side). What is the sum of page numbers that he missed out?

1. 23 2.53 **3.** 11 **4.** 35

12. How many times does the digit 4 occur from 1 to 1000?

1. 199 **2.** 200 **3.** 299 **4.** 300

13. How many numbers from 4 to 499 are neither divisible by 2 nor 4 nor 5?

1. 198 **2.** 199 **3.** 200 **4.** 201

14. A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. Which of the following number of rows is not possible?

1. 3 **2.** 4 **3.** 5 **4.** 6

15. Let D be a recurring decimal of the form D = 0. a_1a_2 a_1a_2 a_1a_2... where digits a_1 and a_2 lie between 0 and 9. Further, at most one of them is zero. Which of the following numbers necessarily produces an integer, when multiplied by D?

1. 18 **2.** 108 **3.** 198 **4.** 288

16. A shipping clerk has five boxes of different but unknown weights each weighing less than 100 kg. The clerk weighs the boxes in pairs. The weights obtained are 110, 112, 113, 114, 115, 116, 117, 118, 120 and 121 kg. What is the weight of the heaviest box?

1. 60 kg **2.** 62 kg **3.** 64 kg

4. Cannot be determined

17. The number 311311311311311311311 is:

1. Divisible by 3 but not by 11.

2. Divisible by 11 but not by 3.

3. Divisible by both 3 and 11.

4. Neither divisible by 3 nor by 11.

18. In a school drill, a number of children are asked to stand in a circle. They are evenly spaced and the 6th child is diametrically opposite the 16th child. How many children are made to stand in the circle?

1. 16 **2.** 20 **3.** 22

4. None of the above

19. A census taker approaches a house and asks the woman who answers the door, "How many children do you have, and what are their ages?"

Woman: "I have three children, the product of their ages is 36, the sum of their ages is equal to my house number (census taker knows this number).

The census taker walks next door, comes back and says, "I need more information."

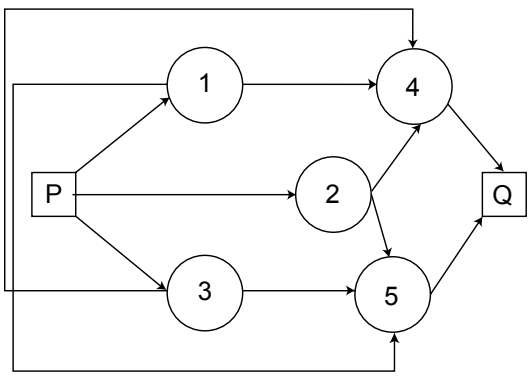
The woman replies, "I have to go, my youngest child is sleeping upstairs."

Census taker: "Thank you, I now have everything I need."

What is her house number?

1. 21 **2.** 13 **3.** 16 **4.** 24

20. What is the number of routes from P to Q?



1. 5 **2.** 6 **3.** 8 **4.** None of these

ANSWERS AND EXPLANATIONS

ANSWER GRID

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	**4**	**4**	**2**	**4**	**3**	**1**	**3**	**3**	**3**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
4	**4**	**1**	**4**	**3**	**2**	**4**	**2**	**2**	**2**

1. Remainder obtained when $2003^{2002^{2000}}$ is divided by 1000 = Remainder obtained when $3^{2002^{2000}}$ is divided by 1000.

Hence, last three digits of $2003^{2002^{2000}}$ = Last three digits of $3^{2002^{2000}}$

Last three digits of 3^N = Last three digits of 3^{100K+N}, where, K and N are natural numbers.

Remainder obtained when 2002^{2000} is divided by 100 = Remainder obtained when 2^{2000} is divided by 100 = Remainder obtained when $(2^{10})^{200}$ is divided by 100 = Remainder obtained when $(24)^{200}$ is divided by 100.

Now we know that last two digits of $24^{1+N} = 76$.

Hence, remainder obtained = 76.

So last three digits of $2003^{2002^{2000}}$ = Last three digits of $3^{76} = 921$.

2. All the options except option 4 has a number which when divided by 19 gives remainder = 8. When 85 is divided by 19, remainder obtained = 9.

3. Since the powers are odd, it will be divisible by 1981^{1981}. Hence, also divisible by 1981^{1980} and 1981^{1979}.

5. Option 4—Without knowing about P, we cannot conclude anything for sure.

6. Option 3. 1st statement gives the position of 11th and 5th person. Middle of these two is the person standing at middle position.

7. Option 1. Question can be done by 2nd statement alone, but not by the 1st statement alone.

8. Option 3.

9. Consider the language of the question—'greatest' number that will always 'divide'.

P is a prime number > 100 $\Rightarrow$ P is an odd number. Hence, (P – 1) (P + 1) are even and (P – 2) (P + 2) are odd. (P – 1) and (P + 1) are consecutive even natural numbers, hence, their product will be divisible by 8.

Secondly, (P – 2), (P – 1), P, (P + 1), (P + 2) are 5 consecutive natural numbers, hence, one of these five has to be a multiple of 5. Since P is not a multiple of 5 (else this will not be a prime number). Hence, one of (P – 2), (P – 1), (P + 1), (P + 2) has to be a multiple of 5 (Rather one of P – 2 or P + 2).

Thirdly, all the prime numbers are of the format $6N \pm 1$, so two of these four numbers will be multiple of 3, and hence, product of 4 numbers will be a multiple of 9.

So (P – 2) (P – 1) (P + 1) (P + 2) will be a multiple of $8 \times 5 \times 9 = 360$.

10. Lowest value of N = 3. For the other numbers, we will check the factors of 60 which are divisible by 3 and can make LCM = 60. These numbers are 3, 6, 12, 15, 30, 60.

12. It is a fact—all the digits from 1 to 9 appear 300 times (100 times at unit place, 100 times at tens place and 100 times at hundreds place) from 1 to 999 (including both the limits).

13. We are not required to consider 4—this question is same as "How many numbers from 4 to 499 are neither divisible by 2 nor 5"?

Let us calculate 1st for "from 1 to 500" $\Rightarrow$ Numbers divisible neither by 2 nor by 5 = 500 (1 – 1/2) (1 – 1/5) = 500 × 1/2 × 4/5 = 200.

We are required to subtract 1 and 3 from this list = 200 – 2 (namely 1 and 3) = 198.

15. $99 \times D = a_1a_2$. Hence, $D = \frac{a_1a_2}{99}$. So, D must be multiplied by 198 as 198 is a multiple of 99.

16. Going through options—60 is wrong because then to arrive at a total of 121, the other box will have to weigh 61 kg which will be obviously not the highest. 64 is wrong too, because then to add up to 121, the other weight will have to be 57 and to make up to a total of 120, the next box shall have a weight 63 which obviously makes the maximum possible total as 64 + 63 = 127. 62 is the correct answer because the other boxes shall be 59, 54, 56. These will give all the totals given above.

SECTION TEST 3

1. What are the last two digits in the following expression:
$12^{(5! \times 10! \times 15! \times \ldots\ldots\ldots \times 95! \times 100!)}$?

1. 56 **2.** 76 **3.** 46 **4.** 02
5. 52

2. What is the remainder obtained when $\frac{30^{10000000000}}{54000}$?

1. 300 **2.** 1 **3.** 0 **4.** 3000
5. None of these

3. There are 37 balls numbered from 1 to 37 placed in a vertical pipe one after the other, not necessarily in order. Balls are put in such a way that number on each ball divides the sum of numbers on all the previous balls. If the number on the first ball is 37, find the number on the 3rd ball.

1. 2 **2.** 19 **3.** 2 or 19 **4.** 23
5. 19

4. What is the number of zeroes in the following expression:
$2^{11} \times 125^3 \times 7^{11} + 3^{12} \times 7^{11} \times 2^{10} \times 5^1 + 11^{12} \times 2^{13} \times 5^{5^5}$?

1. 10 **2.** 5 **3.** 2 **4.** 1
5. None of these

5. N = 100! + 102! + 104! + ...+ 206! + 208!. If N is divisible by 4^x, find the maximum value of x.

1. 35 **2.** 42 **3.** 44 **4.** 56
5. None of these

6. What is the unit digit of N = 120! + 70! + 35! + 25! + 10! + 4! + 3!?

1. 2 **2.** 4 **3.** 6 **4.** 0
5. 8

7. How many times digits 1and 2 will occur when we write down 1st 1000 natural numbers?

1. 401 **2.** 601 **3.** 300 **4.** 600
5. 301

8. Ram and Shyam have got certain number of red and white balls with them. Total number of balls is 90. It is also given that ratio of red and white balls that Ram has is 7:1 and for that of Shyam is 9:1. Which of the following can be the number of red balls that Shyam can has?

(i) 10 (ii) 9 (iii) 45 (iv) 40

1. (i), (ii) or (iii) but not (iv)
2. Only (ii) or (iii) but not (i)
3. Only (i) or (ii) but not (iii)
4. (i), (iii), or (iv) but not (iii)
5. Only (iii)

9. A and B are two natural number such that $A \times B = 160000$. If none of A and B are the multiples of 10, find the value of $(A + B)$.

1. 881 **2.** 661 **3.** 781 **4.** 284

5. None of these

10. Find the last two digits of following expression:

$7^{(1! + 2! + 3! + ...+99! + 100!)}$.

1. 01 **2.** 49 **3.** 43 **4.** 07

5. None of these

11. $N = 2^5 \times 7^3 \times 5^3 \times 11^2$. How many factors of N are multiples of 50 but not the multiples of 100?

1. 71 **2.** 62 **3.** 48 **4.** 67

5. None of these

12. P be the set of prime factors of 7040 and Q be the set of prime factors of 4180. X is an element of set P and Y is an element of set Q. Find the maximum value of $|Y - X|$.

1. 11 **2.** 13 **3.** 17 **4.** 16

5. 21

13. What is unit digit of following expression:

$1432^{40! \times 30! \times 20!} \times 2427^{4! \times 5! \times 10!} \times 2319^{10! \times 20! \times 30!} \times 1552^{1234}$?

1. 1 **2.** 6 **3.** 0 **4.** 9

5. 5

14. How many factors of N = 125000 will have three zeroes at its end?

1. 25 **2.** 15 **3.** 12 **4.** 20

5. None of these

15. Which is the lowest natural number which when divided by 5 gives remainder 3, when divided by 6 gives remainder 4?

1. 63 **2.** 58 **3.** 40 **4.** 43

5. None of these

16. What is the maximum possible difference of 2 give digit number formed using same digits in which one number is formed by reversing the digits of other?

1. 88911 **2.** 88902 **3.** 88909 **4.** 89002

5. None of these

17. $N = 1 \times 1! + 2 \times 2! + 3 \times 3! + ...+100 \times 100!$. What is the remainder obtained when (N+11) is divided by 9900?

1. 11 **2.** 0 **3.** 1 **4.** 10

5. 111

18. Which of the following expressions is not a perfect square (${}^nc_r$ and ${}^nP_r$ are having its standard meaning)?

1. ${}^4C_2 \times 47! \times 50!$ **2.** ${}^{22}P_1 \times 97! \times 100!$

3. $2 \times {}^7C_4 \times 124! \times 126!$ **4.** ${}^3P_2 \times 79! \times 81!$

5. ${}^7C_5 \times 26! \times 28!$

19. Given that 34! = X95232799ab960414084761860964 35cd000000 (X is the leading digit and a, b, c and d are the missing digits from the actual value of 34!).

What is the sum of (a+b+c+d)?

1. 0 **2.** 1 **3.** 4 **4.** 6

5. None of these

20. What is the remainder when $4^{12^{14^{16^{10!}}}}$ is divided by 6?

1. 2 **2.** 1 **3.** 3 **4.** 4

5. None of these

ANSWER AND EXPLANATIONS

ANSWER GRID

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
2	**3**	**1**	**4**	**5**	**4**	**2**	**2**	**1**	**4**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
5	**3**	**1**	**5**	**5**	**2**	**4**	**4**	**5**	**4**

SECTION TEST 4

1. What is the remainder obtained when $^{200}C_{100}$ divides LCM (1, 2, 3, 4, …….,199, 200)?

1. 101 **2.** 50 **3.** 200 **4.** 0

2. How many digits will be used in writing all the natural numbers from 11 to 201?

1. 481 **2.** 484 **3.** 486 **4.** 487

3. The number of positive integer valued pairs (x,y) satisfying $4x-17y = 1$ and $x < 1000$ is

1. 59 **2.** 57 **3.** 55 **4.** 58

4. Let a, b, c be distinct digits. Consider a two digit number 'ab' and a three-digit number 'ccb', both defined under the usual decimal number system, if $(ab)^2 = ccb$ and $ccb > 300$, then the value of b is

1. 1 **2.** 0 **3.** 5 **4.** 6

5. Let X be a four digit number with exactly three consecutive digits being same and is a multiple of 9. How many such X s are possible?

1.12 2.16 3.19 **4.** 20

6. What is the remainder when $1 + 22 + 333 + 4444 + \ldots\ldots + \underbrace{NNNN\ldots\ldots N}_{\text{(N times)}}$ written in base N + 1, is divided by N – 1?

1. 0 **2.** 11 **3.** 22 **4.** None of these

Directions for questions (7–20): Read the directions given below and solve the questions based on it.

Mark 1 as the answer—if question can be answered by one of the statements alone, but not by the other statement alone.

Mark 2 as the answer—if question can be answered by either of the statements alone.

Mark 3 as the answer—if question can be answered only by using both the statements together, and not be any of the statements alone.

Mark 4 as the answer—if question cannot be answered even by using both the statements.

7. If $x = y^3$, then what is the value of x, where x is a natural number?

Statement 1: $xy = 16$.

Statement 2: $x + y = 4$.

8. A is a natural number. Is A + 3 odd?

Statement 1: (A+1)! is even.

Statement 2: (A–1)! is odd and A is not prime.

9. N is a prime number greater than 3. What is the remainder when $7^N - 6^N - 1$ is divided by 43?

1. 1 **2.** 0 **3.** 42

4. Cannot be determined

10. When a certain number is multiplied by 7, the product consists entirely of fives. Which of the following is the least value of such a number?

1. 79365 **2.** 78365 **3.** 77365 **4.** 79265

11. In a division question, the divisor is 10 times the quotient and five times the remainder. What is the dividend, if the remainder is 46?

1. 5636 **2.** 5536 **3.** 5336 **4.** 5436

12. The H.C.F. of two expressions m and n is 1. Their L.C.M. is

1. (m + n) **2.** (m – n) **3.** mn **4.** $\frac{1}{mn}$

13. In an examination, a student attempted 15 questions correctly and secured 40 marks. If there were two types of questions (2 marks and 4 marks questions), how many questions of 2 marks did he attempt correctly (Assume no negative marking)?

1. 5 **2.** 10 **3.** 20 **4.** 8

14. There are 300 seats in an exhibition hall. The price of each ticket when the hall is full is 60. For every Rs 5 increase in the ticket, the number of tickets sold goes down by 10. Find the price of ticket for which the exhibition hall owner earns the maximum revenue.

1. 100 **2.** 110 **3.** 120 **4.** 105

15. A 4 digit number has 15 factors. Which of the following is a possible value of number of factors of N^2?

1. 36 **2.** 44 **3.** 74 **4.** 88

16. Product of N consecutive integers is always divisible by A!. What is the maximum possible value of A?

1. N **2.** N/2 **3.** 2N **4.** None of these

17. A three digit number xyz is written as another three digit number zyx made by interchanging the hundreds place digit and unit digit. Which of the following number will always divide the difference of these two numbers?

(i) 3 (ii) 9 (iii) 11 (iv) 17

1. (i), and (ii) but not (iii)

2. (ii) and (iii) but not (iv)

3. (iii) and (iv) but not (i)

4. All four

18. If n divides (n – 1)! completely, then n can take any value which is a:

1. a prime number

2. a composite number

3. natural number

4. a composite number greater than 4

19. Consider the following statements:

Statement 1—If n is a natural number, then the number $(n + 1)(n + 2) \ldots\ldots\ldots\ldots..(n + 10)$ can be a perfect square.

Statement 2—$2^n + 1$ can be an odd prime only if n is a power of 2.

Now choose the answer from the given options:

1. Statement 1 is true but statement 2 is not true.

2. Statement 2 is true but statement 1 is not true.

3. Both the statements 1 and 2 are true.

4. Neither statement 1 nor statement 2 is true.

20. a and b are natural numbers in such a way that unit digit of $N = a^2 + ab + b^2$ is zero. How many different values of remainders are possible when N is divided by 100?

1. 1 **2.** 3 **3.** 12 **4.** 18

ANSWERS AND EXPLANATIONS

ANSWER GRID

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	**2**	**1**	**1**	**4**	**4**	**1**	**1**	**2**	**1**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
3	**3**	**2**	**4**	**3**	**1**	**2**	**4**	**2**	**1**

1. Using the property that ${}^{2n}C_n$ divides LCM(1, 2, 3, ……, 2n) where, n is a natural number, remainder obtained = 0.

3. The difference between two integers will be 1, only if one is even and the other one is odd. 4x will always be even, so 17y has to be odd and hence, y has to be odd.

Moreover, the number 17y should be such a number that is 1 less than a multiple of 4. In other words, we have to find all such multiples of 17, which are 1 less than a multiple of 4. The first such multiple of 17 goes on increasing. The difference between it and its closest higher multiple of 4 is in the following pattern, 0,3,2,1, e.g., 52–51= 1.

68–68 = 0, 88–85 = 3, 104–102 = 2, 120–119 = 1, 136–136 = 0

So, the multiples of 17 that we are interested in are 3, 17, 11, 15.

Now, since, $x < 1000$, $4x < 4000$.The multiple of 17 closest and less than 4000 is 3995 (17×235). And incidentally, 3996 is a multiple of 4, i.e., the difference is 4.

This means that in order to find the answer, we need to find the number of terms in the AP formed by 3, 7, 11, 15… 235, where, a = 3, d = 4. Since, we know that $t_n = a + (n-1)\,d$, so $235 = 3 + (n-1) \times 4$. Hence, n = 59.

4. $(ab)^2 = ccb$, the greatest possible value of 'ab' to be 31, Since $31^2 = 961$ so $18 < ab < 31$. So, the possible value of ab which satisfies $(ab)^2 = ccb$ is 21. so $21^2 = 441$, a = 2, b = 1, c = 4.

7. Statement 1 alone is sufficient, but statement 2 alone is not sufficient.

From statement 1 alone, we obtain x = 8 or –8. Since x is a natural number, x = 8 is the only solution.

8. Clearly statement 1 is true for any natural number value of A. Therefore, statement 1 is not sufficient alone.

In statement 2, if A is not prime then for (A–1)! To be odd, A must be = 1.

Hence, A+3 is not odd.

So, statement 2 alone is sufficient.

11. Remainder = 46 $\Rightarrow$ Divisor = 5 × 46 = 230

Next, 10 × quotient = 5 × remainder $\Rightarrow$ Remainder = 2 × quotient i.e., quotient = 23.

Dividend = 23 × 230 + 46 = 5336.

12. L.C.M. $= \dfrac{\text{Product of expressions}}{\text{H.C.F.}} = \dfrac{mn}{1} = mn$

13. Suppose number of 2-mark questions = x

Suppose number of 4-mark questions = y

$x + y = 15$

$2x + y = 40 \Rightarrow x = 10, y = 5$

14. Go through the options.

15. N is of the form $N = P_1^{14}$ i.e., $N^2 = P_1^{28}$ Hence, Factors = 29.

Or $N = P_1^2 \times P_2^4$ i.e., $N^2 = P_1^4 \times P_2^8$ Hence, Factors = $5 \times 9 = 45$

Hence, answer is 29 + 45 = 74.

17. zyx – xyz will be always divisible by 99. Hence, divisible by 9 and 11 but not by 17.

18. Go through the options by assuming some values and eliminating the options.

20. In this case, since unit digit is zero, hence, last two digits will be 00. So there will be just one remainder when divided by 100 = 00.

SECTION TEST 5

1. Find LCM of 1000! and (1001 × 1002 × 1003 × 1004 × 1005 × 1006).

1. 1001! **2.** 1003 × 1000! **3.** 1006! **4.** 1000!

2. If K! has n zeroes at the end, where n is the smallest four digit number. Find smallest possible sum of digits of K.

1. 8 **2.** 3 **3.** 7 **4.** None of these

3. A =1! + 2! + 3! +……+ 100000!, what is the remainder obtained when A^A is divided by 5?

1. 1 **2.** 3 **3.** 0 **4.** 7

4. If A = 10! + 11! + 12! +……+ 100! , find hundreds place digit of 'A'.

1. 0 **2.** 2 **3.** 3 **4.** 7

5. Find the units digit of the LCM of $(7^{3001} - 1)$ and $(7^{3001} + 1)$.

1. 0 **2.** 8 **3.** 6 **4.** 4

6. If the sum of two natural numbers and their LCM is 89, then how many such pairs of numbers are possible?

1. 7 **2.** 6 **3.** 5 **4.** 4

7. N^2 = abb……..(b written consecutively for n times at the end), where b is a non-zero digit. What is the maximum number of times b can occur at the end of any perfect square?

1. 2 **2.** 3 **3.** 4 **4.** None of these

8. What is the highest power of 2 in the product 1005 × 1006 × 1007 × ------- × 2009?

1. 1004 **2.** 995 **3.** 997 **4.** 1008

9. What is the remainder when $(98^5 + 99^5 + 101^5 + 102^5)$ is divided by 400?

1. 1 **2.** 100 **3.** 200 **4.** 0

10. N = (1 × 2) × (2 × 3) × (3 × 4) × …….. × (99 × 100). What is the remainder when N is divided by 101?

1. 1 **2.** 2 **3.** 3 **4.** 0

11. $A = 1!^{99} + 2!^{98} + 3!^{97}$……+ $99!^1$, find unit digit of 'A'.

1. 1 **2.** 3 **3.** 6 **4.** 7

12. A 50-digit number has all 7's. Find the remainder when the number is divided by 74.

1. 3 **2.** 13 **3.** 23 **4.** 33

13. Let $N = 2^{33} \times 3^{21}$. How many factors of N^2 are less than N but do not divide N?

1. 693 **2.** 132 **3.** 523 **4.** 833

14. For n>0, $(2n)!/(n!)^2$ is divisible by

1. n+1 **2.** n+5 **3.** 4n+5 **4.** 4n–1

15. A number N^2 is such that exactly 3 of its factors are less than N. Find the number of factors of N^3.

1. 6 **2.** 10 **3.** 15 **4.** 20

16. What is the sum of all the two digit natural numbers which when divided by 7 give remainder 3?

1. 676 **2.** 1028 **3.** 576 **4.** None of these

17. Consider the following statements:

Statement 1—If difference between two perfect squares is even, then this difference has to be divisible by 4.

Statement 2—If difference between two perfect squares is odd, then this difference has to be divisible by 3.

Now choose the answer from the given options:

1. Statement 1 is true but statement 2 is not true.

2. Statement 2 is true but statement 1 is not true.

3. Both the statements 1 and 2 are true.

4. Neither statement 1 nor statement 2 is true.

18. In a group of 120 persons there are 80 Bengalis and 40 Gujratis. Further, 70 persons in the group are Muslims and the remaining are Hindus. Then the number of Bengali Muslims in the group is

1. between 10 and 14. **2.** between 15 and 19;

3. exactly 20; **4.** 25 or more

19. 300 persons are participating in a meeting in India, out of which 120 are foreigners, and the rest are Indians. Out of the Indians, there are 110 men who are not judges 160 are men or judges, and 35 are women judges. There are no foreign judges. How many Indian women attended the meeting?

1. 60 **2.** 45 **3.** 55 **4.** 40

20. There are six persons: A, B, C, D, E and F.

A has 3 items more than C.

D has 4 items less than B.

E has 6 items less than F.

C has 2 items more than E.

F has 3 items more than D.

Which one of the following figures cannot be equal to the total number of items possessed by all the six persons?

1. 41 **2.** 53 **3.** 47 **4.** 58

ANSWERS AND EXPLANATIONS

ANSWER GRID

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
4	**4**	**2**	**2**	**4**	**3**	**2**	**1**	**3**	**4**
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
4	**1**	**1**	**1**	**2**	**1**	**1**	**4**	**3**	**2**

1. $1001 = 11 \times 13 \times 7$

$1002 = 2 \times 501$

$1003 = 17 \times 59$

$1004 = 2 \times 502$

$1005 = 5 \times 201$

$1006 = 2 \times 503$; Since all factors lie in 1000!, LCM = 1000!

3. $A = (\ldots\ldots.13)^{(\ldots\ldots13)}$

Clearly unit digit of $A = (\ldots\ldots\ldots\ldots 3)^{(\ldots.13)} \Rightarrow$ Unit digit of $A = (\ldots\ldots\ldots\ldots 3)^{1} = 3$

Clearly remainder = 3 when divided by 5.

4. 10! = 3628800

11! = ….800

12! = ….600

13! = ….800

14! = ….200

15! = ….000

100! = ….000

On adding .....200 (All the numbers 15! Onwards will have last three digits as 000).

Hence, hundreds place digit is 2.

5. Since, $(7^{3001} - 1)$ and $(7^{3001} + 1)$ are two even numbers with the difference of 2 hence, their HCF = 2.

LCM = $[(7^{3001} - 1) \times (7^{3001} + 1)]/2 = (7^{6002} - 1)/2$

Unit digit of $(7^{6002} - 1)/2$ of $(9 - 1)/2 = 8/2 = 4$.

7. $38^2 = 1444 \Rightarrow$ It gives us the largest number of times a non-zero digit can occur at the end of a perfect square = 4 occurring 'thrice.'

10. $(1 \times 2) + (2 \times 3) + (3 \times 4) + \ldots\ldots\ldots\ldots\ldots\ldots + (99 \times 100)$.

General term = n(n+1)

Sum of n terms = $\Sigma n^2 + \Sigma n = n(n + 1)(2n + 1)/6 + n(n + 1)/2$.

$= n(n + 1)(n + 2)/3$

For n + 99 the sum is divisible by 101.

11. A = ...1 + ... 4 + … 6 + … 6 + ……. 0 + …. 0 + …. all other have unit digit = 0
A = …….7

12. N = 7777777777.....50 times

N/2 leaves remainder 1.....(a)

N/37 leaves remainder 3 (as 777 is divisible by 37). .....(b)

Smallest number satisfying (a) and (b) is 3.

13. Total number of factors of N = 34 × 22 = 748.

Now, Total number of factors of $N^2 = 67 \times 43 = 2881$. Number of factors of N^2 less than N = (2881 – 1)/2 = 1440. Now, all the factors of N will also be factor of N^2. Hence, all the factors of N except N are to be excluded. Therefore, answer is 1440 – 748 +1 = 693

14. Put n = 1 we get $(2n)!/(n!)^2 = 2$

Put n = 2 we get $(2n)!/(n!)^2 = 6$,

Put n = 3 we get $(2n)!/(n!)^2 = 20$,

Hence, option 1 is the answer.

15. Since 3 factors of N^2 are less than N, so 3 factors of N^2 must be greater than N and 1 factor = N (since N^2 is a perfect square). Hence, total number of factors of n^2 is 7. Hence, $N^2 = P^6$ where, P is a prime number.

Hence, $N^3 = P^9$. Hence, number of factors of $N^3 = 9+1 = 10$.

17. If the difference between two perfect squares is even, then either both numbers are odd or both are even.

$x^2 - y^2$ = Even $\Rightarrow$ (x + y) (x – y) = Even $\Rightarrow$ Either Both x and y are even or both are odd.

Hence, each of (x + y) and (x – y) will be even. So, (x + y) (x – y) = 4K.

So statement 1 is true.

To verify statement 2, we can check values.

$7^2 - 6^2 = 13 \Rightarrow$ Not a multiple of 3. Hence, statement 2 is not necessarily true.